

KISS: Migraine in Pregnancy

[BMJ 2018;360:k80](#) [BASH 2019 Drug and Therapeutics Bulletin 2021;59:152-156](#) [SIGN 155](#)

Background:

- **Reassure women migraine often improves in pregnancy** - 50-75% will have improvement mainly in the second/third trimester, although those with aura have a more variable course.
- There is an increased risk of secondary headaches in pregnancy e.g. cerebral venous thrombosis and pre-eclampsia, particularly after the first trimester.

Assessment: Rule out red flags/secondary causes (see Headache chapter also):

- In women with a history of migraine check if the current headache is similar to previous ones or are there any differences, particularly relating to aura.
- Do neuro exam, specifically assessing for neck stiffness; test eye movements, visual fields, and pupillary responses; perform fundoscopy to rule out papilloedema.
- Check blood pressure and dip urine.

Management: ([click here for good info from UK Teratology Information Service](#))

- **Non-pharmacological** - keep well hydrated, don't skip meals; reduce caffeine but avoid abrupt withdrawal; good sleep hygiene; regular exercise; consider acupuncture.
- **Acute pharmacological treatments:**
 - Paracetamol first line; ibuprofen can be considered in 1st/2nd trimester (avoid in 3rd trimester when risks of premature ductus closure and oligohydramnios increase).
 - Sumatriptan can be considered in all trimesters in refractory cases - reasonable safety evidence for sumatriptan. Meta-analysis showed not associated with major congenital malformation or prematurity. Conflicting data on spontaneous abortion rates- some small studies showed increase, recent large cohort study (432 patients) showed no increase. Avoid in those with hypertension/ pre-eclampsia
 - Use **anti-emetic together with analgesia** (even in absence of nausea/vomiting - can improve gastric motility/drug absorption) - cyclizine or prochlorperazine 1st line; ondansetron, domperidone metoclopramide may be considered 2nd line.
- **Prophylaxis (should not be initiated in primary care - seek specialist guidance):**
 - Consider stopping prophylactic treatments before pregnancy
 - If required aspirin 75mg OD up to 36/40, B-blockers (e.g. propranolol 10-40mg TDS) or amitriptyline 10-25mg nocte may be considered.
 - Do NOT use valproate or topiramate (high teratogenicity risk), or ARBs (high risk in 2nd/3rd trimester).

What about breastfeeding?:

An important question given that it is relatively common for women to get migraine flares after delivery (due to abrupt fall of oestrogen). Paracetamol, NSAIDs, sumatriptan, propranolol and amitriptyline (low dose) are all thought to be safe with breastmilk concentrations not thought to be high enough to affect babies. The [breastfeeding network](#) has a handy summary.



KISS: General/Acute Management of Migraine

[SIGN 155, 2022](#), [NICE 2021](#), [CG150](#), [BASH 2019](#)

Triggers/exacerbating factors:

- Aim for regular routine with sleep, meals, hydration and exercise. Promote relaxation e.g. yoga
- Many [potential triggers](#) but evidence for trigger avoidance limited
- Review **medications that may be contributory**:
 - Oestrogen-containing contraceptives, HRT, PPIs, SSRIs
 - **Overuse of acute migraine medication**- risk of medication overuse headache if taking acute medications on ≥ 2 days/week, triptans >10 days per month, simple analgesia (e.g. paracetamol/ibuprofen) >15 days per month. It can limit effect of prophylactic medication

General information and Patient information:

- SIGN have produced some really helpful summary documents:
 - click [here](#) for a really useful treatment pathway summary
 - click [here](#) for a quick reference guide to the medications
- Patient information:
 - [NHS Choices: Migraine triggers](#)
 - [Migraine Trust, Coping with migraine](#)

Acute treatment:

- Take early in the HEADACHE phase, not the aura phase. Always discuss risk of medication-overuse headache when starting acute treatment
- Take a flexible approach depending on migraine severity; options include:
 - **Aspirin** 900mg (NNT for pain free at 2 hrs = 8.1), other **NSAID** e.g. **ibuprofen** 400-600mg (NNT 7.2 for 400mg), Naproxen 500mg (NNT 11); **adding anti-emetics** (e.g. **prochlorperazine** 10mg, **metoclopramide** 10mg) improves effectiveness of analgesia even in the absence of nausea and vomiting (improves gastric motility/drug absorption).
 - **Triptans**:
 - **Sumatriptan 50-100g PO first line** (NNT 6.1-4.7), repeat at 2hrs if partial response
 - Lack of effect at 2 hours = treatment failure; lack of response to one triptan does not predict response to other triptans so try alternatives (or different delivery mode) if 2 treatment failures
 - If patient suffers with early vomiting/ acute severe migraine consider nasal zolmitriptan (5mg) or subcut sumatriptan. NB a significant proportion of nasal triptan is gastrically absorbed.
 - Combining long-acting NSAID (e.g. naproxen) with triptan is more effective than triptan monotherapy
 - **Note triptans contraindicated in uncontrolled hypertension and symptomatic cardiovascular/ cerebrovascular disease.**
 - Few case reports of serotonin syndrome when co-prescribed with SSRIs, however risk thought to be low and unclear if risk any higher than with SSRIs alone. Therefore not contra-indication
- Paracetamol can be considered if unable to take other options, first line in pregnancy (NNT 12)
- Do NOT use opioids - generally ineffective and high risk of medication overuse headache



KISS: Prophylactic Management of Migraine

[SIGN 155, 2022](#), [NICE 2021 CG150](#), [BASH 2019](#)

Prophylaxis:

- Consider preventative treatment if ≥ 4 migraines/month, but take a **flexible approach** depending on both severity as well as the frequency of attacks, as well considering **individualised treatment** depending on co-morbidities (e.g. amitriptyline if insomnia etc.)
- Titrate slowly to maximum effective tolerated dose for minimum 6-8 weeks before deciding on effectiveness (ideally with [headache diary](#))
- Consider gradual withdrawal after 6-12 months of effective treatment.
- **Recommended drug treatments:**
 - **Propranolol** 80-160mg daily. First line, reduced migraine frequency $>50\%$ NNT 4
 - **Topiramate** 50-100mg daily. Reduced migraine frequency $>50\%$ NNT 4. Advise all women who may become pregnant on the risk of **foetal malformation** during pregnancy, the need for effective contraception (e.g. IUS/ depo injection/ combined hormonal contraception with barrier method) as there is a risk of **reduced effectiveness of hormonal contraception** on topiramate, and to seek advice on alternative prophylaxis if planning a pregnancy; note also [new concerns](#) regarding **topiramate use and neuro-developmental disorders**.
 - Candesartan target dose 8mg BD (reminder - also C/I in pregnancy)
 - Amitriptyline 10-150mg ON
 - Sodium valproate 400-1500mg daily (NB foetal malformation/pregnancy risk, see [MHRA guidance](#))
- **CRGP monoclonal antibodies:** fremanezumab, galcanezumab and erenumab are options for people with migraine when at least 3 preventive drug treatments have failed ([TA659](#), [TA682](#), [TA764](#))
- Note that **gabapentin is NOT effective** and should not be offered.
- **Supplements:** Riboflavin (400mg/day), Co-enzyme Q10 (150mg/day), Magnesium (400-600mg/day)
- **Non-drug treatment:**
 - **Acupuncture** (10 sessions) is recommended as an option (including by NICE if topiramate/propranolol unsuitable/ ineffective)
 - **CBT** added to drug therapy can reduce migraine-associated disability
 - **Botulinum toxin A:** recommended for chronic migraine after failure of 3 or more oral prophylactic treatments (NB not effective for episodic migraine)
- **Menstrual migraine:** Consider short term preventative strategy - Zolmitriptan 2.5mg BD/TDS or frovatriptan 2.5mg BD for 2 days before until 3 days after bleeding starts

NB Learning Points & Quality Improvement Ideas: Migraine

- 2/3 people with chronic migraine have associated medication overuse which adds significant disability and renders prophylactic treatments less effective - specifically ask about acute treatment use (including over-the-counter analgesics) in all migraine sufferers and if taking acute medications ≥ 2 days/week warn them they are at high risk of concurrent medication overuse headache.
- If using simple analgesics for acute treatment add an anti-emetic (e.g. prochlorperazine, metoclopramide) even in the absence of nausea/vomiting - improves gastric motility/drug absorption.
- Triptans are most effective if taken in the headache phase, NOT the aura phase.
- Consider starting prophylaxis if ≥ 4 migraines/month and consider withdrawing prophylaxis after 6-12 months of effective treatment.
- **Quality Improvement Idea:** Review all patients receiving ≥ 10 doses of triptan/month; consider medication overuse headache and withdrawal of all acute medications.

HEADACHES

"The truth is never pure and rarely simple". Thus starts the new [2019 headache guideline from BASH](#) (British Association for the Study of Headache, not to be confused with BASHH - British Association for Sexual Health and HIV). This is an Oscar Wilde quote and nicely encapsulates the difficulty we have in diagnosing and managing headaches, particularly in primary care. This guideline is honest and open about the variation in the presentation of headaches and the fact that many headache types can overlap, but aims (with some success) in providing *'a simple, safe and standardised approach which can be used in real-time to help manage the majority of common headache conditions'* and is very much aimed at the 'jobbing' clinician. BASH are keen that this guideline will *'largely replace the multiplicity of different regional headache algorithms'* and reduce variation in care. The [NICE guideline](#) on headaches was published in 2012, and although updated in 2015, is still looking a bit old. The latest offering from BASH gives a much more up-to-date, evidence-based oversight of headache management.

► [Assessing for serious secondary causes of headache BASH 2019](#)

The majority of headaches are primary (e.g. migraine, tension-type headache), but as always an **assessment of headache starts with trying to identify the serious secondary causes of headache** (i.e. precipitated by another disorder), sometimes referred to as 'red flag' headaches. BASH reiterate that serious secondary causes of headache are uncommon but as we know the concern about a serious underlying cause is a common factor in headache presentation. Whilst the NICE guideline had a long list of potential symptoms associated with serious causes for headache, BASH point out that ***'very few secondary causes of headache have been reliably shown to have unique headache symptoms'***. For example, there is no typical 'brain tumour type headache'. Brain tumours most commonly have headaches which are like episodic tension-type headache or migraine without aura. They focus more on features that significantly increase the likelihood of a serious secondary cause, and those features that do not help differentiate primary from secondary headaches. Whilst a brief neurological examination is useful, they stress that history is the key to the diagnosis of headaches.

KISS: Assessing for serious secondary causes of headache [BASH 2019](#)

The four **most consistent indicators*** for serious secondary headache are:

- Thunderclap headache.
- Associated neurological deficit - [click here](#) for useful video reminder on simple neuro examination.
- Associated systemic features.
- Patients over the age of 50.

Use **temporal patterns** to help differentiate serious secondary causes:

- Sudden onset headache reaching maximum intensity within 5 minutes = Thunderclap headache; **this has the greatest probability of a secondary precipitant.**
- Recent onset and progressive headache - evolution of headache over days to weeks; if associated with systemic features and/or neurological signs there is an increased risk of a secondary precipitant.
- Recurrent episodic headache - if in isolation is most likely due to primary headache disorder.
- Headache occurring on the majority of days in a month - if headache present for ≥ 15 days/month over 3 months in isolation it is most likely due to primary headache disorder.

Features that do NOT help differentiate primary from secondary headaches:

- Severity
- Response to treatment

Reassuringly patients with typical migraine or tension-type headache and a normal neurological examination **do NOT have an increased risk of a secondary precipitant.**

So, as has been quite consistently advised in previous guidelines (including NICE), we can (and should) be making a positive diagnosis of primary headache when appropriate and looking to rule in benign primary headache disorders rather than focussing on ruling out secondary causes by running through a long list of red flag symptoms - BASH point out *'there is currently poor evidence that different disease processes associated with headache have unique or specific clinical presentations not covered by the clinical indicators* above'*.

► Thunderclap Headache and SAH

Thunderclap headache (acute onset headache reaching maximum intensity within 5 minutes) is the **most common isolated headache associated with a secondary cause**. Subarachnoid haemorrhage (SAH) is present in 19-25% of thunderclap presentations, and so our **role in primary care is to refer anyone with thunderclap headache to hospital immediately for urgent CT scanning**. If CT is done within 6 hours of onset, it has a sensitivity for SAH of 98-99%, but this drops to 90% after more than 6 hours, thus reinforcing the importance of us referring these patients urgently without delay. If CT negative, lumbar puncture (LP) will be performed. Thunderclap headaches can be associated with other serious secondary causes e.g. intracranial haemorrhage, venous sinus thrombosis, arterial dissection, so even if CT/LP are normal further investigations may be warranted, but will obviously be done through secondary care.

Recent NICE guidance on SAH ([NG 228 Nov 2022](#)) highlights the importance of recognising thunderclap headaches as a 'red flag' for SAH, whilst balancing this by noting that thunderclap headaches are also associated with other conditions such as migraine, cough, coitus (see below) or exertion. Although most people with a thunderclap headache do not have SAH they reinforce the importance of immediate referral to secondary care if people present with thunderclap headache for CT to be done ideally within 6 hours.

What about a thunderclap headache associated with sexual activity?

This is not an uncommon presentation...to scan or not to scan? [Primary headache associated with sexual activity](#) is a recognised primary headache type in the international classification. This term replaces previous commonly used terms such as coital cephalgia, benign sex headache or orgasmic headache. It is recognised now as a single entity but with variable presentation e.g. it may be bilateral or unilateral, it may be localised to the occiput or it may be generalised. It is defined as:

- At least two episodes of pain in the head and/or neck with the following criteria
 - Brought on by and occurring only during sexual activity
 - Increasing in intensity with sexual excitement and/or abrupt explosive intensity just before/with orgasm
 - Lasting from 1 minute to 24 hours with severe intensity, and up to 72 hours with mild intensity

Note that it is NOT associated with other features such as disturbance of consciousness, vomiting or other neurological symptoms. It can also only be diagnosed by definition if $\geq$ two attacks. The guideline stresses that *'on the first onset of headache with sexual activity it is mandatory to exclude SAH'*.

Primary headache with sexual activity can occur at any age, in men and women (although more common in men 2:1) and occurs independently of the type of sexual activity. Some patients may only have this once or twice in their lives, but recent studies show that up to 40% of patients run a chronic course over a year.

Imaging in headache

BASH have reviewed the evidence and broadly speaking stuck with previous guidance when it comes to imaging in headaches in that for **most people imaging is unlikely to help diagnosis and comes with a substantial risk of incidental findings**:

- Outside the emergency setting (i.e. relevant to us in primary care) the evidence confirms that the **risk of finding a serious secondary pathology in patients with isolated headache and normal neurological examination is similar to that in people with no headache**; i.e. in primary care isolated headache (and normal neurological exam) does not increase your risk of having serious underlying brain pathology.

- **Incidental findings are present in 6-15%** of people having brain imaging, which can increase anxiety and can affect insurance/premiums. BASH have put together a PIL explaining the risks of incidental findings on brain imaging [click here](#).
- Normal imaging may reduce healthcare use in the short term (<1 year) but this effect is not sustained in patients with anxiety or depression.

In summary, as per NICE guidance, we should be making a **positive diagnosis of people with primary headache disorder** and that the presence of headache in this situation does not increase their risk of serious brain pathology; if considering brain imaging we should discuss the risks of incidental findings and that it may not alleviate anxiety around underlying causes in the longer term. NICE recommend neuroimaging should not be used 'solely for reassurance' in those with primary headache disorders.

► **Differentiating between common primary headache disorders** [BASH 2019](#)

There are no features that are universally specific to particular primary headache disorders, but there are features that increase/decrease the likelihood of particular disorders:

Feature	Migraine	Tension-Type (TTH)	Cluster
Laterality	Unilateral or bilateral (unilateral in ~10-20%)	Bilateral	Unilateral (right or left but not bilateral)
Quality	Pulsating, throbbing	Pressing, tightening	
Severity	Moderate or severe, often disabling	Mild-moderate but not disabling	Very severe
Relationship to activity	Motion sensitive (prefer to remain still during attack)	Not aggravated by activity	Restless during attack
Associated features	Nausea +/- vomiting, photophobia, phonophobia	No nausea/vomiting, photophobia or phonophobia	Ipsilateral to pain: Conjunctival injection, lacrimation, rhinorrhoea, eyelid swelling/drooping
Longevity	Attacks last 4-72 hours	Attacks last hours-days	Attacks last 15mins-3 hours
Frequency	Typically 1-2 attacks/month		1-3 attacks per day (up to 8) usually daily for 2-3 months

- A few further points on primary headache features:
 - **Autonomic features** (lacrimation, conjunctival injection, rhinorrhoea) are characteristic of trigeminal autonomic cephalgias (TAC, see below) but **can occur in up to 25% of migraine sufferers**.
 - **Aura** typically occurs with migraine but can be present in all headache disorders.

► **Migraine** - See separate chapter.

► **Medication overuse headache (MOH)** [BASH 2019](#) [BJGP 2020; 70 \(691\)](#)

Medication overuse headache is a very common condition, often under-recognised, and is seen in many (if not the majority) of patients with chronic migraine or tension-type headache.

- **Background points:**
 - It is classified as a chronic headache disorder with headaches occurring on ≥ 15 days/month for ≥ 3 months and the exact cause of MOH remains unknown.
 - It affects 1-2% of the population and is present in **20-50% of the chronic headache population**; it is present in about **two-thirds of chronic migraine sufferers**.
 - ALL medications used to treat acute headache can result in MOH, but it is more likely with triptans, opioids or combination analgesics (at risk if treatment taken on ≥ 10 days/month) than simple analgesics (at risk if treatment taken on ≥ 15 days/month).
 - In patients with migraine or TTH taking analgesics for other reasons (e.g. MSK pain, fibromyalgia) CAN result in MOH.
 - The prevalence of co-morbid psychiatric disorders is greatly increased in patients with MOH.
 - Dependence-related behaviour is noted in 60-70% of cases and relapses are common.

■ Management:

- The **most important step is to consider and identify the diagnosis**.
- Specifically, ask all patients with chronic headache how often they use acute headache medication (including over-the-counter analgesics), preferably correlated with a headache diary ([click here](#)).
- MOH is not easy to treat as the **only treatment for MOH is to withdraw the offending medication**. Patients often report being unable to cope without their analgesics and rather continue as they are.
- Timing of withdrawal is important and should be determined by home/work circumstances and probably best attempted during a period of leave.
- There is no difference in outcome with either abrupt or gradual withdrawal, although opioids may need to be withdrawn gradually.
- Warn patients that **headaches will improve gradually but may take up to 12 weeks** and they may also experience nausea, sleep disturbance and increased anxiety during withdrawal phase, but if successful there benefits are substantial.
- If the underlying headache disorder is known, appropriate prophylactic treatment can be started during or after withdrawal, and preventative treatments become more effective once overused medications are withdrawn, but for many simply withdrawing the medication may treat the condition.
- Addition of corticosteroids or NSAIDs are not helpful during withdrawal.
- Once withdrawn restrict the use of acute medications to maximum 2 days/week.
- **The most important 'treatment' is prevention** - warning patients early in presentations of headaches to limit analgesics to maximum 2 days/week, especially those with migraine, significantly reduces MOH risk.

■ Prognosis:

- 22-45% relapse back into MOH within 1 year and 40-60% within 4 years.

► Tension-Type Headache (TTH) [BASH 2019](#)

TTH is the commonest primary headache disorder with a mean global lifetime prevalence of 42%.

■ Clinical features:

- Characterised by mild-moderate (NOT severe), headache. It is bilateral and often described as pressing or tightening like a vice or band.
- It is **NOT aggravated by routine physical activity which is a key criterion for distinguishing it from migraine**.
- Disabling TTH is rare and most with disabling 'TTH' actually have migraine.

■ Management:

- Reassurance may suffice in the majority of patients.
- Acute attacks - Paracetamol, aspirin (300-900mg) or alternative NSAID.
- **Warn patients about risks of MOH** - use acute medications no more than 2 days/week.
- **Preventative treatments** rarely needed but can be considered if symptoms causing disability:
 - Acupuncture - 6 sessions.
 - Amitriptyline 10-150mg nocte.

[Cervicogenic headache](#) is now classified as an independent headache type and increasingly recognised. It may overlap with TTH and migraine and may share features of both e.g. it may be unilateral and associated with migrainous features such as nausea, vomiting and photophobia although these are generally not as severe as in migraine. It is a headache causally associated with cervical myofascial trigger points as pain sources (causing pericranial tenderness on examination) and although it is usually accompanied by a painful neck there may be a headache with no neck pain. The distinguishing features are reduced cervical range of motion, side-locked pain (always on one side), provocation of typical pain by digital pressure on neck muscles and by head movement and posterior to anterior radiation of the pain. Imaging is not routinely recommended as degenerative changes in the cervical spine are almost universal over the age of 40 and such changes are equally widespread people with or without headaches.

► **Trigeminal Autonomic Cephalgias (TACs)** [BASH 2019](#)

The TACs are a rare group of primary headache disorders with prominent autonomic features and shared pathophysiology. There is a **high misdiagnosis rate** and little evidence to base management on.

■ **Cluster Headache:**

- Probably the most well-known of the TACs; it has a prevalence of ~0.1%, peaks in the 3rd/4th decades and is 4 times more common in men
- **Attacks are characterised by:**
 - Excruciating strictly unilateral headaches (although the headache can move from side to even during bouts, but very rarely bilateral)
 - **Ipsilateral cranial autonomic features** including lacrimation conjunctival injection, rhinorrhoea, nasal congestion, drooping/swelling of the eyelid
 - **Restlessness:** patients typically walk up and down or rock to/fro, unlike migraine sufferers who tend to prefer to remain still
 - Duration 15 mins to 3 hours, 1-2 times/day (as high as 8/day), in clusters of 2 weeks to 3 months.
 - Clusters may be predictable in some e.g. seasonal or at set times of day, including at night
- There are a **number of features that overlap with migraine:**
 - Nausea, vomiting, photophobia and phonophobia can be present in both
 - Up to 25% of migraine sufferers can experience autonomic features
 - Aura can be experienced in cluster headaches
- **Referral?** Not mentioned by BASH, but given TACs are rare and the high rate of misdiagnosis it would seem sensible to refer to a specialist (as per [NICE guidance](#) - neuro-imaging may be required)
- **Acute treatment:**
 - Best evidence for sumatriptan subcutaneously 6mg (max 2 in 24 hours).
 - Alternatively nasal sumatriptan/zolmitriptan
 - High flow oxygen 7-15L/min is also effective
- **Preventative treatments** (all likely to be initiated by a specialist):
 - Verapamil is effective but carries a risk of cardiac conduction abnormalities; ECG recommended at baseline, following each dose increase and every 6 months once on a stable dose
 - Oral corticosteroids may also be effective, but only in the short-term
 - Other options include occipital nerve block, lithium, melatonin, vagal nerve stimulator

■ **Paroxysmal hemicrania:**

- Very rare condition with similar features to cluster headaches, but with short attacks (2 to 30 minutes) occurring frequently (up to 50 times) through the day; it may become chronic
- The pain is strictly unilateral and associated with ipsilateral autonomic features, such as conjunctival inflammation and eye-watering, nasal congestion or rhinorrhoea, sweating, miosis
- It has a clear therapeutic response to indomethacin, in a dose range of 25mg to 300mg daily. Patients may take up to a week to respond
- Start with indomethacin 25mg tds for 7 days, then increase to 50mg tds and then 75mg tds if needed
- Once patients respond, dose reduction is recommended

■ **Hemicrania Continua:**

- As it says on the tin when paroxysmal hemicrania becomes continuous for > 3 months
- The pain is strictly unilateral and associated with ipsilateral autonomic features, such as conjunctival inflammation and eye-watering, nasal congestion or rhinorrhoea, sweating, miosis
- Like a cluster headache, patients may be agitated and restless
- Responds absolutely to therapeutic doses of indomethacin, as for paroxysmal hemicrania

■ **SUNA**

- SUNA = short-lasting neuralgiform attacks with cranial autonomic features. This term now replaces the formerly used term SUNCT (short-lasting unilateral neuralgiform attacks with conjunctival injection and tearing). whilst conjunctival inflammation and lacrimation are the most common autonomic symptoms in all TACs, it does not always occur hence the preferred term SUNA.

- It is rare, the mean age of onset is 48 and it is slightly more common in men than women
- The attacks are the shortest and most frequent of all the TACs, and it is characterised by very short (mean duration is one minute) and frequent attacks (up to 30 attacks per hour)
- Attacks may be spontaneous, but attacks are often triggered by sensory stimuli e.g. eating, brushing teeth, cold air etc
- It is often misdiagnosed as trigeminal neuralgia, due to the episodic 'stabbing' nature of the pain and the fact that it may be triggered by sensory stimuli; however, with TN pain is in the distribution of the V2 maxillary and V3 mandibular divisions of the trigeminal nerve and it should not involve the V1 ophthalmic division and there are no autonomic features; SUNA pain is always primarily in the distribution of V1
- Due to the short-lived nature of attacks, no acute treatments are effective; unfortunately, it does not respond absolutely to indomethacin unlike hemicrania
- Lamotrigine is the most effective preventative option with a dose range of up to 400mg. Carbamazepine and gabapentin may also be effective.

NB Learning Points & Quality Improvement Ideas: Headache

- The **four most consistent indicators for a serious secondary cause of headache** are thunderclap headache (acute onset reaching maximum intensity within 5 minutes), focal neurological deficit, systemic features, patients > 50 years old.
- **Thunderclap headache** (acute onset headache reaching maximum intensity within 5 minutes) is the most common isolated headache associated with a secondary cause. Subarachnoid haemorrhage (SAH) is present in 19-25% of thunderclap presentations, and so our role in primary care is to refer anyone with thunderclap headache to hospital immediately for urgent CT scanning. If CT is done within 6 hours of onset, it has a sensitivity for SAH of 98-99%, but this drops to 90% after more than 6 hours, thus reinforcing the importance of us referring these patients urgently without delay.
- Primary headache associated with sexual activity is recognised as a primary headache type **HOWEVER** by definition it needs to have occurred ≥ 2 times, so on the first presentation, it must be treated as a thunderclap headache and urgently referred
- Need a reminder on a brief neurological examination? Useful [video here](#)
- Patients with **migraine or TTH and a normal neurological exam do not have an increased risk of a secondary precipitant** in this situation brain imaging generally not recommended as it does not aid diagnosis and runs a high risk of incidental findings (up to 15%) - [click here for PIL](#).
- **Cervicogenic headache** is increasingly recognised, and may overlap with TTH and migraine; examining for neck stiffness and pericranial myofascial tender trigger points is very worthwhile
- **Medication overuse headache (MOH) is common**, affecting 20-50% of those with chronic headache:
 - Specifically, ask all patients with a chronic headache about acute medication use.
 - Withdrawal of the overused acute medication is the only treatment effective for MOH.
- **The TACs trigeminal autonomic cephalgias** (cluster headache, hemicrania and SUNA) are rare but increasingly recognised. SUNA is often misdiagnosed as trigeminal neuralgia.
- **Audit:** Patients on triptans prescribed ≥ 10 tablets/doses per month - call in for review as they are at high risk of having developed medication overuse headache.

KISS: Headache

[BASH Headache guideline 2019](#) [NICE CG 150 2015](#)

■ Serious secondary causes of headache:

- The most consistent indicators for a serious secondary cause of headache are:
 - **Thunderclap headache** (acute onset reaching max intensity within 5 minutes) subarachnoid haemorrhage present in 19-25% of presentations; refer immediately to a hospital for CT +/- LP.
 - **Focal neurological deficit** (e.g. idiopathic intracranial hypertension).
 - **Systemic features** (e.g. CNS infection).
 - **Patients > 50 years old** (e.g. giant cell arteritis).
- Severity of headache & response to treatment doesn't help differentiate primary/secondary causes.

■ Imaging:

- Not indicated in the context of typical migraine or tension-type headache and normal neurological exam - in primary care patients with isolated headache and normal neurological exam have the same risk of serious secondary pathology as those with no headache.
- Should not be done solely for reassurance - may only help anxiety in the short term and high risk of incidental findings (6-15%) which can cause further anxiety; [click here for PIL](#).

■ Migraine: See separate chapter

■ Medication Overuse Headache (MOH):

- Common and under-recognised - present in 20-50% of people with chronic headache and as many as 2/3 of those with chronic migraine.
- More common with triptans/opioids/combo analgesics (at risk if taken on ≥ 10 days/month) than simple analgesics (at risk if taken on ≥ 15 days/month).
- **Management:**
 - Assess for underlying mental health problems (common) and dependence-related behaviour.
 - The **only effective intervention is withdrawal of the offending medication** no difference if done abruptly or gradually - warn patients improvement may take as long as 12 weeks.
 - For many withdrawal is all that is needed, but some will need appropriate prophylaxis for the underlying headache disorder - can be started during or after withdrawal.
- Prognosis poor - 25-45% relapse at 1 year, 40-60% relapse at 4 years.

■ Tension-Type Headache (TTH):

- **Clinical features:**
 - Mild/moderate, often bilateral, pressing/tightening like a vice/band.
 - Rarely severe **disabling 'TTH' is often migraine.**
 - Not aggravated by routine physical activity - key distinguishing feature from migraine.
- **Management:**
 - Reassurance may be enough in many.
 - Acute treatment - aspirin 300-900mg, paracetamol (warn about MOH - max use 2 days/week).
 - Preventative treatment rarely needed but if so consider amitriptyline (10-150mg) or acupuncture.

■ Cluster Headache:

- Excruciating, strictly unilateral headache with ipsilateral autonomic features (e.g. rhinorrhoea, lacrimation, conjunctival injection) with episodes lasting 15min-3hrs, multiple times/day.
- Autonomic features, aura, nausea/vomiting, photophobia **can be present in migraine and cluster.**
- Generally **restless during an attack** (c.f. migraine when people want to remain still).
- Consider referral for new suspected cases - high misdiagnosis rate and may need neuro-imaging.
- Acute treatment - sumatriptan 6mg S/C; prophylaxis guided by a specialist but verapamil effective.

SUSPECTED NEUROLOGICAL CONDITIONS IN ADULTS OVER 16

Neurological presentations account for about 10% of GP consultations and must be one of the most difficult areas we have to navigate. Presentations are often vague or non-specific, patients and parents often have (disproportionate) anxieties about serious underlying causes, and the spread of potential causes ranges from the benign and functional through to the serious and life-threatening! 'I think I need a brain scan Doc' is often an underlying expectation, whether verbalised or not. So a recent NICE guideline ([Suspected neurological conditions: recognition and referral](#)) published in May 2019 is a welcome help for us in primary care. NICE have recognised that while there are plenty of neurological disease-specific guidelines, patients don't present to primary care with a clear diagnosis in many cases, but come to see us with a group of symptoms which could indicate a variety of neurological conditions. This guideline gives an over-arching view of a large number of symptoms/signs or conditions which are generally either under or over-referred, and gives **useful information on not only when we should refer, but also importantly when we don't need to refer**. We do however need to be aware that the guideline notes that the vast majority of the recommendations are not evidence-based (as there simply is no evidence available), and are based on committee knowledge and experience, but there was a good spread of members in the committee including GPs and laypeople. We have put together a series of easy-to-reference KISS tables to summarise the guidance, but it's worth pointing out a few of the recommendations in a bit more detail as well as a few specific learning points and reminders. We will deal with the adult section to the guideline here, and discuss the children's section in the next chapter.

► *Suspected neurological conditions in adults >16: recognition and referral* (NICE NG 127 May 2019)

■ Sudden-onset acute vestibular syndrome

- For adults with a sudden onset vestibular syndrome (vertigo, nausea or vomiting and gait unsteadiness) we need to exclude hypoglycaemia in patients with diabetes and also postural hypotension as possible causes, but we also need to remember that this **can be the presentation of a posterior circulation stroke**
- We should refer immediately if there is any other focal neurological deficit. If there is not, NICE recommend we consider using the **HINTS test** (head-impulse, nystagmus, test-of-skew), which if negative makes a diagnosis of stroke extremely unlikely. NICE quote evidence that the HINTS test is broadly similar to MRI in ruling out stroke, however, it needs to be done by someone with expertise and experience in its use and if this is not available referral is necessary
 - For the HINTS test, all 3 facets (head impulse test, nystagmus test, test-of-skew) have to be satisfied to rule out a stroke/central cause for the vertigo and we should also note this test is only valid for people who are currently symptomatic and have persistent symptoms
 - If any aspect of the test is a positive referral to exclude a central cause is needed
 - [Click here for written information on HINTS test](#) and [click here for a video explanation](#)
- If there is no gait unsteadiness, then in the absence of any other neurological deficit then it is very unlikely that there is any serious underlying neurological condition
- If there is vertigo on head movement then benign paroxysmal positional vertigo is the most likely diagnosis, and we should offer a Hallpike manoeuvre for confirmation and a canalith repositioning manoeuvre (Epley) for treatment
- If there is episodic dizziness associated with headaches consider vestibular migraine as a cause
- If there is episodic dizziness associated with altered consciousness refer to exclude epilepsy

■ Facial pain

- We see a lot of people with facial pain, usually benign TMJ-related pain. However, we need to consider serious pathologies that can present with facial pain, such as
 - Trigeminal neuralgia, if the history is consistent and the pain is triggered by touching the affected part of the face
 - Temporal or giant cell arteritis, if accompanied with scalp tenderness and/or jaw claudication
 - Intracranial mass lesions may present with facial pain and persistent numbness or other neurological signs, so pain + persistent numbness is a red flag and needs urgent referral

• Dense amnesia:

- Dense amnesia is a strange symptom, ironically rarely forgotten by both patient and GP once it has presented! It can be very alarming for patients and GP alike.



- If the dense amnesia (that is the inability to recall the recent past or form new memories) lasts less than 8 hours, there are no epileptic features and there is complete recovery then likely diagnosis is transient global amnesia (TGA)
- TGA is a benign condition and very unlikely to recur and a referral is not required and we can reassure the patient. If there are recurrent episodes referral is indicated as it could be epileptiform in nature
- **Sensory symptoms, including tingling and numbness**
 - If rapidly progressive and associated with weakness or imbalance we would obviously refer immediately. Otherwise, we see a lot of patients just with tingling and numbness. Do you always check deep tendon reflexes? Diagnoses to consider include:
 - peripheral neuropathy, if persistent predominantly distal altered sensation with depressed deep tendon reflexes
 - central brain or spine disease, consider if associated with brisk reflexes and refer if present
 - stroke and TIA with acute unilateral numbness
 - migraine with aura in patients with recurrent sensory symptoms with or without headaches that last for 5 to 60 minutes
 - patients who have a brief (<2 minutes) fixed pattern disturbances of sensation may have epilepsy and should be referred
 - meralgia paresthetica is tingling or pain in the distribution of the lateral cutaneous nerve of the thigh, and is a surprisingly common condition; reassure it is benign and it may spontaneously improve
 - sensory symptoms and tingling on waking last < 10 minutes are common and benign and do not need a referral
- **Loss of sense of smell or taste**
 - This is a surprisingly common presentation in primary care, and NICE noted that this was a common reason for referral to neurology but it is usually idiopathic and rarely has a serious neurological cause. They recommend:
 - Not to routinely refer for neurology review or neurological imaging unless the symptoms are persistent beyond 3 months and they are unexplained
 - Not to routinely refer if associated with head injury; it is common and does not indicate further brain injury and cannot be treated but we need to warn the patient, it is likely to be permanent
 - If they present with repetitive smell or taste hallucinations we should refer routinely as this could be a symptom of temporal lobe epilepsy
- **Tics and involuntary movements**
 - Tics are common and benign, and treatment options are limited so they do not need a referral unless they are very troublesome or accompanied by additional progressive neurological symptoms; if patients are very troubled by them then consider referring for psychological therapy to help with associated anxiety or distress
 - With small involuntary muscle twitches (fasciculations) these are benign and only need a referral if they are accompanied by muscle wasting, weakness or rigidity
 - With involuntary movements, a key factor is if they can be suppressed with concentration. If not and they are severe and persistent referral is indicated to exclude neurological causes such as chorea. However involuntary movements of the face, neck, limbs or trunk that can be suppressed by mental concentration are benign and do not need a referral unless they are accompanied by muscular weakness, wasting or
- **Patients with functional neurological disorders:**
 - One of the themes of the guideline was regarding people with known functional neurological disorders. As we know they commonly present with recurrent symptoms and knowing when to refer back to neurology is difficult. They recommend that for people who have been diagnosed with a functional neurological disorder by a specialist and present with recurrent (i.e. the same) symptom type (e.g. dizziness, limb weakness, memory problems, word-finding difficulties, sensory symptoms) and no new signs, a referral back to the specialist is not necessarily needed
 - If they present with new symptoms, they need to be re-assessed on an individual basis as per the symptom type in the guideline

NEUROLOGY

KISS: Suspected neurological conditions: recognition and referral in adults aged >16

NICE NG 127 May 2019

Presenting Symptoms	Associated features/assessment	Possible diagnosis	?Refer
Blackouts	If associated features suggest epilepsy	Epilepsy	Urgent
Blackouts	If features suggestive of vasovagal (even if brief limb jerks)	Vasovagal	Not routinely
Dizziness	Acute vestibular syndrome - consider HINTS test if symptoms ongoing; click here for further detail and here for video	Posterior circulation stroke	Immediate
Dizziness	Vertigo on head movement - check Hallpike manoeuvre - click here for video	BPPV	Only if unable to perform Epley in primary care
Dizziness	Vestibular symptoms lasting 5mins-72 hours associated with headaches	Vestibular migraine	Not routinely
Dizziness	Fixed pattern dizziness with altered consciousness	Epilepsy	Routine
Dizziness	If no imbalance or focal neurological deficit, or associated with a functional neurological disorder	Unlikely serious	Not routinely
Facial pain	Persistent facial numbness or neurological signs	Tumour	Urgent
Facial pain	Paroxysmal neuropathic pain in trigeminal distribution, often physical triggers e.g. touch	Trigeminal neuralgia	Only if refractory to usual treatment
Facial pain	Scalp tenderness, jaw claudication; raised ESR	Temporal arteritis	Urgent (if visual symptoms refer immediately)
Gait unsteadiness	Rapidly progressive gait ataxia (unsteadiness)	Tumour or paraneoplastic	Urgent
Gait unsteadiness	Gradually progressive ataxia - check alcohol, thyroid function, B12/folate, coeliac screen	Multiple	Routine
Gait unsteadiness	Gait apraxia (difficulty initiating or coordinating walking)	Normal-pressure hydrocephalus	Routine
Handwriting difficulties	If acute onset consider stroke (although rare)	Stroke	Yes - speed will depend on presentation
Handwriting difficulties	Slow and small handwriting	Parkinson's disease	Routine
Handwriting difficulties	If specific to handwriting and no other abnormalities	Focal dystonia - click here for info and PIL	Routine
Memory problems	If <50, no neurological signs and normal brief memory testing; address underlying causes e.g. drugs, alcohol, stress, affective disorder		Not routinely
Memory problems	Concentration difficulties associated with CFS, fibromyalgia, anxiety disorder or functional neurological disorder		Not routinely
Memory problems	Dense amnesia - if single episode lasts <8 hours and complete recovery	Transient global amnesia	Not routinely
Memory problems	Recurrent dense amnesia	Epilepsy	Routine

NB: Referrals - Immediate = same day/within hours; Urgent = within 2 weeks

KISS: Suspected neurological conditions: recognition and referral in adults >16 CONT.

NICE NG 127 May 2019

Presenting symptoms	Associated features/assessment	Possible diagnosis	?Refer
Limb/facial weakness	Sudden onset and focal	Stroke/TIA	Immediate or urgent
Limb/facial weakness	Rapidly progressive (within 4 weeks) symmetrical limb weakness	Neuromuscular disorder or cervical myelopathy	Immediate
Limb/facial weakness	Back pain with perineal numbness, or new-onset bladder/bowel/sexual dysfunction	Cauda equina syndrome	Immediate
Limb/facial weakness	Rapid onset (within hours/days) progressive single limb or hemiparesis	Malignancy	2-week rule
Limb/facial weakness	Slowly progressive (weeks-months) limb or neck weakness	Motor neurone disease	Routine (If swallowing difficulty refer urgently; If breathing difficulty refer immediately)
Limb/facial weakness	Uncomplicated Bell's palsy	Bell's palsy	Not routinely
Limb/facial weakness	Bell's palsy with aberrant reinnervation (inc. sweating or jaw-winking) after >5 months	Bell's palsy	Routine
Limb/facial weakness or sensory symptoms	Recurrent weakness or recurrent numbness/tingling in person with a known functional neurological disorder, with no new neurological signs	Functional neurological disorder	May not need re-referral
Posture distortion	Persistent abnormalities of head or neck posture, +/- head tremor	Cervical dystonia Click here for more info/PIL	Do NOT refer for cervical imaging or orthopaedic opinion Routine referral to neurology
Sensory symptoms	Rapidly progressive (hours-days) symmetrical numbness	GBS or transverse myelitis	Immediate
Sensory symptoms	Recurrent brief (<2 mins) fixed pattern sensation disturbance	Epilepsy	Urgent
Sensory symptoms	Persistent distal altered sensation with brisk reflexes	Brain/spinal cord disease	Routine
Sensory symptoms	Fully reversible develop over at least 5 mins, and last 5-60 minutes +/- headache	Migraine	Not routinely
Sensory symptoms	Persistent distal altered sensation, depressed reflexes; check B12/thyroid/coeliac/U&E/glucose/ESR	Peripheral neuropathy	Depends on the underlying cause
Sensory symptoms	Tingling/pain lateral thigh	Meralgia paraesthetica	Not routinely (only if severe symptoms)
Sensory symptoms	Tingling/sensory disturbance present on waking from sleep and < 10 mins	Sleep-related compression	Not routinely

NB: Referrals - Immediate = same day/within hours; Urgent = within 2 weeks

KISS: Suspected neurological disorders: recognition and referral in adults >16 CONT.

NICE NG 127 May 2019

Presenting symptoms	Associated features/assessment	Possible diagnosis	?Refer
Sleep disorder	Insomnia +/- jerks on falling asleep or brief sleep paralysis	Benign	Not routinely
Sleep disorder	Excessive sleepiness - use Epworth score/consider sleep apnoea, but if not this consider	Narcolepsy or cataplexy	Routine
Sleep disorder	Agitated or violent movements (more complex and severe than simple jerking)	Sleep behaviour disorder	Consider routine referral
Smell/taste problems	Distorted or sudden loss of smell or taste	Usually benign	Only if unexplained & > 3/12
Smell/taste problems	After head injury	Common and not treatable	Not routinely
Smell/taste problems	Transient repetitive smell or taste hallucinations	Epilepsy	Routine
Speech/swallowing/ language	Progressive slurred or disrupted speech	Motor neurone disease or myasthenia gravis	Routine (If swallowing difficulty refer urgently; If breathing difficulty refer immediately)
Speech/swallowing/ language	Dysphonia - rule out structural or malignant ENT cause first, then consider	Laryngeal dystonia or Parkinson's	Routine
Speech/swallowing/ language	Minor word-finding difficulties associated with anxiety or functional neurological disorder	Commonest cause of word-finding difficulty	Not routinely
Tics/involuntary movements	Tics - consider psychological therapies first		Not routinely but consider if persistent, severe & causes distress
Tics/involuntary movements	Involuntary eyelid movements	Blepharospasm	Only if bilateral or persistent >3/12
Tics/involuntary movements	Involuntary movements of face, neck, limbs, trunk (that can't be suppressed by temporary concentration i.e., not Tics)	Chorea	Routine
Tics/involuntary movements	Small involuntary muscular twitches (fasciculations)	Usually benign	Not unless associated with muscle wasting, weakness or rigidity
Tremor	Asymmetric, stiffness, slowness, gait disturbance, balance problems	Parkinson's disease	Routine
Tremor	Bilateral, no problems with muscle tone or speed of movement; check medications, thyroid function, alcohol	Essential tremor	Not routinely, unless disabling and no response to 1st line tx e.g. propranolol

NB: Referrals - Immediate = same day/within hours; Urgent = within 2 weeks

SUSPECTED NEUROLOGICAL CONDITIONS IN CHILDREN UNDER 16: RECOGNITION AND REFERRAL

As discussed in the introduction to the previous chapter, NICE has recently produced a comprehensive guideline to help us with the myriad neurological presentations we come across in primary care ([NG 127 Suspected neurological conditions: recognition and referral May 2019](#)). There is often a degree of underlying anxiety when any person presents with possible neurological symptoms, and this anxiety (for both patient and doctor!) is probably accentuated if the person is a child - *'could this be a brain tumour'* is often an underlying concern for both parties. For a summary of the full guideline, we have put together some easy-to-reference KISS tables at the end of this chapter, but it is probably worth highlighting a few areas with a bit more detail first.

Suspected neurological conditions in Under 16s: recognition and referral [NICE NG127 May 2019](#)

■ Dizziness:

- NICE wants to reassure us that dizziness without any accompanying symptoms or signs is very unlikely to be caused by a brain tumour and **referral is often not needed**
- However, we should consider other causes if recurrent (and possible referral depending on cause):
 - Migraine
 - Postural hypotension (usually in the over 8s)
 - If associated with transient loss of consciousness or there is a family history of cardiomyopathy/unexplained sudden death consider cardiology referral for cardiac cause
 - If dizziness has fixed symptom pattern consider epilepsy

■ Headaches:

- Refer immediately (same-day assessment) for headaches with any of the following features (**red flags for significant intracranial pathology, including brain tumour**):
 - Headaches that wake them at night, are present in the morning or progressively worsening
 - Triggered by coughing, sneezing or bending down
 - Associated with vomiting, ataxia, change in conscious level or pervasive lethargy
 - Associated with squint or failure of upward gaze
 - Within 5 days of head injury
 - Associated fever and features of meningism
- Refer urgently children under 4 with headaches (or features that may suggest headache e.g. excessive crying, high-pitched cry, excessive irritability)
- Recurrent headaches/migraine:
 - Check for/address emotional stress - a common trigger for migraine and tension headaches
 - Check no features of medication overuse headache
 - Be aware high blood pressure can cause headaches in children
 - Children with migraine do not routinely need a referral unless affecting school, family or social life

■ Abnormal neck posture:

- Check for painful lymphadenopathy as a possible cause as this can often be managed in GP
- If abnormal head tilt present this **could be due to posterior fossa tumour** - NICE note this symptom is often present before the more well-known symptoms become apparent (ataxia, vomiting, headaches) and are often dismissed so encourage referral for this symptom

■ New-onset night terrors in children over 5:

- Could indicate epilepsy so referral is indicated
- If the child has had night terrors under the age of 5 which then persist, referral not recommended unless they continue beyond age 12 (most will settle by then)

■ Tics:

- Children with simple motor tics do not need a referral unless they are bothersome - reassure parents and emphasise they should not be reprimanded for them either at home or school
- Be aware they are more common in children with autism and learning difficulties
- If tics having a significant impact on life refer depending on possible underlying cause to:
 - Mental health service if associated with anxiety or OCD
 - Neurodevelopmental team if possible autism or ADHD
 - Neurology if severe tics

KISS: Suspected neurological conditions in children <16: recognition and referral

NICE NG 127 May 2019

Presenting symptoms	Associated features/assessment	Possible diagnosis	?Refer
Attention, concentration	Mid-activity vacant spells	Epilepsy	Urgent to paed
Blackouts/vacant spells	Behavioural outbursts with altered consciousness or amnesia	Epilepsy	Urgent paed
Blackouts	Any children <12 with blackouts	Various	Urgent paed
Blackouts	If >12 and clear features of vasovagal syncope (even if brief jerking of limbs)	Vasovagal	Not routinely
Dizziness	Isolated dizziness - unlikely to be a symptom of a brain tumour; check ears and also consider migraine as a cause	Usually benign but consider migraine	Use clinical judgement
Dizziness	If >8 and related to posture	Postural hypotension	Not routinely
Dizziness	If recurrent and associated with blackout, palpitations, or FH cardiomyopathy/sudden unexplained death	Cardiac	Routine cardiology
Dizziness	Fixed symptom pattern	Epilepsy	Urgent
Headaches	Red flags: - Headaches that wake them at night, are present in the morning or progressively worsening - Triggered by coughing, sneezing or bending down - Associated with vomiting, ataxia, change in conscious level or pervasive lethargy - Associated with squint or failure of upward gaze - Within 5 days of head injury - Associated fever and features of meningism	Various	Immediate
Headaches	Any child <4 (may present with excessive crying or high pitched crying or excessive irritability)	Significant intracranial disease	Urgent neurology
Headaches	Recurrent headaches consistently worsened by upright posture and relieved by lying down	Intracranial hypotension	Routine
Headaches	Features of migraine and no red flags - check for medication overuse and emotional stress	Migraine	Only if interfering with daily life
Head shape/size abnormality	Children <4 - head circumference differs by ≥2 centile lines from the previous measurement; check for signs of ↑ ICP (tense fontanelle, 6th nerve palsy, failure of upward gaze, vomiting, ataxia, headache)	Microcephaly, hydrocephalus, raised ICP	Routine to paed; immediate if signs ↑ICP
Head shape/size abnormality	Head circumference >98th centile that has not changed by ≥2 centiles and no other concerning features	Familial macrocephaly	Not routinely
Head shape/size abnormality	Babies <1 with positional plagiocephaly and developing normally - advise changing baby's lying position	Positional plagiocephaly	Not routinely

NB Referrals - Immediate = same day/within hours; Urgent = within 2 weeks

KISS: Suspected neurological conditions in children <16: recognition and referral continued

NICE NG 127 May 2019

Presenting symptoms	Associated features/assessment	Possible diagnosis	?Refer
Hypotonia ('floppiness')	Babies <1 with acute hypotonia	Liver, renal, cardiac disease	Immediate
Hypotonia ('floppiness')	Babies <1 and symptoms present for weeks/months	Underlying neurological disease	Urgent to paed's if weak (e.g. feeding/breathing difficulty) otherwise routine
Limb/facial weakness	Sudden onset or rapidly progressive (hours-days) limb or facial weakness	Various	Immediate
Limb/facial weakness	Progressive limb or facial weakness	Various	Urgent neurology
Limb/facial weakness	Limb weakness as part of developmental disorder	Cerebral palsy	Routine to paed's
Gait abnormality	New-onset gait abnormality	Infection, appendicitis, hip disease	Immediate
Motor delay	If not sitting unsupported by 8 months, not walking independently by 15 months (girls) or 18 months (boys) or show early hand preference (age <1)	Cerebral palsy	Routine child development service +/- physio/OT
Motor regression	Motor developmental regression - if boy check CK levels	Various (inc muscular dystrophy)	Routine (neurodevelopment or neurology depending on local pathways)
Posture distortion	Abnormal neck posture after head/neck trauma	Spinal trauma	Immediate to ED
Posture distortion	Abnormal neck posture/abnormal head tilt - if no underlying MSK cause or painful lymphadenopathy	Posterior fossa tumour	Yes neurology - use clinical judgement on urgency
Posture distortion	Abnormal limb posture with no apparent MSK cause	Various	Routine neurology
Sensory symptoms	Tingling with other symptoms e.g. weakness, bladder or bowel dysfunction	Spinal cord disease or GBS	Urgent neurology
Sensory symptoms	Episodic isolated tingling, altered sensation or paraesthesiae - rule out nerve compression first	Epilepsy	Routine neurology
Sensory symptoms	Temporary tingling/numbness triggered by either hyperventilation or posture-related nerve compression		Not routinely
Speech problems	New-onset slurred or disrupted speech not caused by medicines, alcohol or drugs	Various	Urgent neurology
Speech problems	Children >2 with abnormal or regressive speech/language	Wide, inc. autism	Consider routine referral

NB Referrals - Immediate = same day/within hours; Urgent = within 2 weeks

KISS: Suspected neurological conditions in children <16: recognition and referral continued

NICE NG 127 May 2019

Presenting symptoms	Associated features/assessment	Possible diagnosis	?Refer
Sleep disorder	New-onset early-morning headaches or sleep disturbance in children with neuromuscular disorders	Respiratory failure/hypoventilation	Urgent respiratory
Sleep disorder	Symptoms suggestive of narcolepsy +/- cataplexy	Narcolepsy	Neurology
Sleep disorder	Symptoms suggestive of night seizures - risk of SUDEP	Epilepsy	Urgent neurology
Sleep disorder	Symptoms suggestive of sleep apnoea	Sleep apnoea	Routine ENT
Sleep disorder	Night terrors, sleep talking or walking in children <5 - common and rarely a sign of neurological disorder; offer sleep hygiene advice		Not routinely
Sleep disorder	New-onset night terrors in children >5 or night terrors that persist in children >12	Epilepsy	Routine
Squint	Squint with loss of red reflex	Retinoblastoma	Immediate ophthalmology
Squint	Squint with ataxia, vomiting, or headache	Raised ICP	Immediate acute paed
Squint	Paralytic squint	Intracerebral tumour or inflammation	Urgent neurology
Squint	Non-Paralytic squint (retains the ability to move both eyes fully in all directions)	Non-Paralytic squint	Routine ophthalmology
Involuntary movements	Sudden onset chorea, ataxia, dystonia	Space occupying lesion, metabolic cause, degenerative condition, para-infectious (e.g. rheumatic fever)	Immediate neurology
Tics	Simple tics that are not troublesome to the child are common and usually resolve - reassure parents and emphasise that tics should not be reprimanded at home or school; do not offer medication	Usually benign but be aware more common in autism or learning disability	Not routinely
Tics	If causing a significant impact on the quality of life and associated with anxiety/OCD	Anxiety/OCD	Mental health services
Tics	If causing a significant impact on the quality of life and associated with symptoms suggestive of autism or ADHD	Autism, ADHD	Neuro-developmental team
Tics	Severe tic disorder		Neurology
Tremor	If any additional neurological symptoms/signs present (e.g. unsteadiness) or if a sudden onset	Space occupying lesion	Urgent
Postural Tremor	Rule out drug side effect (e.g. valproate, beta-agonist) and hyperthyroidism (check thyroid function)		Refer to OT if affecting ADLs e.g. writing, eating, dressing

NB Referrals - Immediate = same day/within hours; Urgent = within 2 weeks

FUNCTIONAL NEUROLOGICAL DISORDERS

Functional neurological disorders (FNDs) are a fascinating group of conditions, but at the same time intimidating and scary for both clinician and patient alike. Despite the complex nature of the conditions they are, as we know, conditions that frequently present in primary care. Most GPs will see presentations of functional neurological disorder over an average year, yet we have very little in the way of guidance to help us. Given the often lengthy waits for review or assessment by specialists we (and our patients) often feel somewhat left in the dark. So a recent BMJ review recognising that most clinicians have little formal education in FNDs ([BMJ 2020;371:m3745](#)) and which focussed on education for the generalist, was most welcome.

What are they?

- FNDs are conditions *'whose origin arises primarily from a disorder of nervous system functioning rather than clearly identifiable pathophysiological disease'*.
- They are the second commonest cause for new neurology consultations.
- FNDs are **disorders of the voluntary motor or sensory system** with *genuine* symptoms including paralysis, tremor, dystonia, sensory disturbance, speech symptoms and seizures.
- The hallmark is that they are **incongruent with recognised pathophysiological disease**.
- They often co-exist with other persistent symptoms including dizziness, pain and fatigue, and patients may have other functional disorders e.g. IBS, fibromyalgia, chronic pelvic pain.

What causes FNDs?

- There has been a big shift over the last 2 decades in terms of the putative mechanisms of FNDs.
- Previously they were considered a consequence of adverse life events but increasing evidence supports the theory that in FND early pre-conscious phases of motor planning are corrupted by involuntary predictions about body state, and interference from emotionally orientated areas (e.g. limbic system).
- The **brain effectively prioritises strong predictions about what it expects to do (e.g. leg weakness) over the actual incoming sensory input**.
- This model helps to explain that previous adverse life events *can* be a risk factor for FND, but don't have to be present, and that other recognised pathophysiological triggers e.g. migraine, epileptic seizures can also trigger FNDs.

How are the features of FNDs assessed and how is a diagnosis made?

- One of the **key learning points from this paper is that FNDs should not be thought of as a diagnosis of exclusion** - as with IBS making a positive diagnosis is possible (and desirable).
- **History:**
 - Symptoms are likely to be wide-ranging including motor and sensory symptoms, as well as associated symptoms such as pain, fatigue, sleep and memory disturbance.
 - Ask about a typical day - this helps build a picture of disability and any co-morbid mental health illness.
 - Ask about onset and course and any physical triggers e.g. minor trauma, migraine, syncope.
 - Ask about **dissociative symptoms** e.g. **depersonalisation** (a feeling of being disconnected from our body) and **derealisation** (a feeling of being disconnected from the world around you). These are common symptoms and recognising them as such can be a huge relief to patients.
 - Consider home video footage to aid diagnosis.
 - **Should we ask about life events?** Not necessarily, this review concludes. As it is accepted they do not need to be present to make the diagnosis pursuing this line of enquiry may not help, and may even hinder assessment. The paper argues that people that don't have such life events present may feel this questioning intrusive and unnecessary, and those who have had such events may feel they are being blamed for their symptoms due to past events.

- **Clinic features** (often multiple present) - [click here](#) for summary infographic:
 - Functional limb weakness or movement disorders - in principle these features are distractible i.e. can be alleviated by performing other tasks or movements.
 - Functional dystonia - typically presents with fixed positions e.g. clenched fist, inverted ankle, in contrast to other dystonias which are usually mobile (i.e. not fixed).
 - Functional or dissociative seizures:
 - Subjective descriptions often include autonomic arousal e.g. palpitations, warmth, sweating as well as dissociative experiences (see above).
 - Objective features include eyes being tightly closed shut, tearfulness, duration > 5 minutes, hyperventilation, side-to-side head shaking; ~30% will have events looking like syncope - sudden motionless unresponsiveness with eyes closed for >2 minutes is rarely due to another cause.
 - Functional visual loss - may be tubular (rather than conical) or spiralling visual fields.

Are further investigations and referral needed?

- The simple answer is yes, we should refer all suspected cases of FND for specialist assessment.
- However, discussing the possibility of the diagnosis in primary care, having some knowledge of the condition and how it is diagnosed can greatly improve the outcome of further management.
- The paper also highlights the importance for us to consider co-existent conditions e.g. migraine, hypermobility disorders or epilepsy (~20% of people with functional seizures also have epilepsy), as well as psychological co-morbidities.
- The paper recommends we arrange some simple blood tests including B12, thyroid function etc whilst awaiting neurological review.
- Neuro-imaging is likely to be undertaken, but it is useful to warn patients that ~1:6 will have incidental findings to reduce the risk of harm or worry. Neurophysiological investigation is also likely to be done.

How do we discuss FND with patients?

- As with many other conditions in medicine we need to validate the symptoms, recognise they are real, and that this is not a condition of exclusion (which might lead to the perception that it is 'not a problem').
- Demonstrating positive clinical findings can be helpful to gain insight into FND e.g. showing them that leg weakness can be alleviated by distracting activities helps the idea that you can **'retrain the brain'**.
- Give it a name and explain the mechanism - Using the analogy that it is a **'problem with the software rather than the hardware'** can be helpful.
- Be very wary of turning a risk factor into the 'cause' of the disorder.
- Be positive that treatment can help - 'retraining the brain' with physiotherapy and/or psychological therapies can help and give insight into triggers and how to manage them.

NB Learning Points: Functional Neurological Disorders (FNDs)

- FNDs should not be seen as a 'diagnosis of exclusion' but can be based on positive clinical findings.
- Although adverse life events or psychological comorbidities may be present, they are neither necessary nor sufficient to make the diagnosis.
- Ask about **dissociative symptoms** e.g. depersonalisation (the feeling that you are not connected to your body) or derealisation (the feeling that you are not connected to the outside world).
- We need to be aware that pathophysiological conditions can co-exist with FNDs e.g. migraine, epilepsy.
- Although the diagnosis should be made by a specialist, we have a crucial role in primary care to discuss the possibility of the diagnosis and what further investigations may be done - this can greatly help future management.
- Language is important - using the analogy that it is a **'problem with the software, not the hardware'** can help; we also need to support positive messages that treatment can help by 'retraining the brain' through physiotherapy and/or psychological therapies.
- **Patient information:** [Neurosymbols.org](#) and [non-epileptic attack](#) are both good.
- **Patients support groups:** [FNDHope](#); [FNDAction](#); [FNDDimensions](#); [FNDFriends](#)

RESTLESS LEGS SYNDROME

Restless legs syndrome (RLS), also known as Willis-Ekbom disease, is increasingly recognised as a common neurological disorder which has a significant impact on health. We certainly seem to be seeing it much more often, so new research [Cochrane 2019](#) and a new review of the evidence [BMJ 2017;356:j104](#) is welcome. These papers have multiple learning points. For example, iron can be beneficial even in people with normal iron levels and until recently dopaminergic drugs were the first-line treatment but we now know these drugs can lead to augmentation of the symptoms in many patients and should no longer be first-line choices.

► **Who gets RLS?** [BMJ 2017;356:j104](#)

RLS is common. It is estimated to affect 5-10% of adults, with 2-3% experiencing moderate to severe symptoms. It is twice as common in women than men and mean age of onset is the third and fourth decade. A positive family history is common. The later it comes on in life the more likely it is that it is secondary to another condition e.g. peripheral neuropathy, iron deficiency or renal disease. RLS is highly familial and the striking improvement in symptoms with dopaminergic drugs points to a dopaminergic dysfunction as primary cause, but the mechanism remains unclear. Iron deficiency has repeatedly been associated with RLS.

► **How does it present?** [BMJ 2017;356:j104](#)

The key symptom is an overwhelming urge to move the legs during periods of rest or inactivity, especially at night. It is associated with restlessness and distressing paraesthesia-like sensations in the lower legs. The symptoms follow a circadian pattern and are worse in the evening. Periodic leg movements can also occur, whilst asleep and also during the day. Symptoms and severity vary enormously and symptoms may remit for periods of time. There are 5 **essential diagnostic criteria** that must be met and simple blood tests are recommended (see KISS box below). The paper recommends referral for specialist investigation if we suspect associated neurological diseases, such as peripheral neuropathy, radiculopathy or myelopathy.

► **How should patients with RLS be managed?** [BMJ 2017;356:j104](#)

- **Lifestyle guidance** for all patients is clearly key, but patients have often tried and tested many non-drug options by the time they seek medical help.
- **Review exacerbating medications.** Many commonly prescribed medications exacerbate symptoms of RLS, these include antihistamines, anti-nausea drugs, dopamine antagonists, antidepressants including SSRIs, beta-blockers, some antiepileptics and lithium.
- **Ensure serum ferritin is > 50 ng/ml** using iron supplementation
- **Medication should be considered when symptoms are having a significant negative impact on the quality of life, sleep and function.**
 - **Dopamine agonists** (e.g. pramipexole, ropinirole) and **gabapentinoids** (gabapentin and pregabalin) have been extensively studied and found to be effective at treating the sensory symptoms. The dopamine agonists are more effective for the periodic leg movements and the gabapentinoids for sleep. **However, long-term studies show a progressive loss of response and augmentation of symptoms over 5 to 10 years with dopamine agonists.** Current recommendations are therefore to start treatment with a gabapentinoid and dopamine agonists should be avoided unless absolutely necessary, and then if needed to use the lowest effective dose for the shortest time.

► **What is the evidence for iron therapy?** [Cochrane 2019, CD007834](#)

This new systematic review of trials of iron versus placebo has just published. There was a mix of intravenous and also oral iron preparations used:

- Overall the studies showed that iron was superior to placebo, although the benefit was low to moderate
- *Iron was helpful, even if blood iron levels were normal at the start of the study*
- Subgroup analysis showed no significant difference between oral or intravenous iron preparations, or in patients with or without a low ferritin at baseline



KISS: Restless Legs Syndrome, Willis-Ekbom Disease

Based on [Cochrane 2019](#) & [BMJ 2017;356:j104](#) CKS 2022

1 Make the diagnosis

- The 5 essential diagnostic criteria (patients must have all 5) are:
 - An urge to move the legs usually accompanied by unpleasant sensations in the legs:
 - Patients may find symptoms difficult to describe and may include tingling, burning, itching, throbbing, 'creepy-crawly' sensations (like ants crawling), fizzing/fizzy sensations.
 - This usually occurs during periods of rest or inactivity.
 - The urge to move and accompanying sensations are eased by movement.
 - Symptoms are worse in the evening and night than during the day.
 - The features are not explained by another condition e.g. leg oedema, arthritis etc.
- Assess severity with [RLS international rating scale](#)

2 Is it primary or secondary (associated conditions or drugs)?

- In most it is idiopathic, 50% have a positive family history.
- The most common secondary causes are iron deficiency, renal disease and peripheral neuropathy.
- **Commonly prescribed drugs can precipitate or exacerbate** it: antihistamines, anti-emetics inc metoclopramide and prochlorperazine, dopamine antagonists, antidepressants including SSRIs, beta-blockers, some antiepileptics and lithium.

3 Investigations: Check FBC & iron studies, especially ferritin, in all. Consider U&E, LFT, TSH, glucose, B12/folate, calcium studies, inflammatory markers depending on clinical presentation.

4 Management

- Review medication and stop/change exacerbating drugs.
- **Treat with iron if ferritin <50mcg/ml, and aim to keep ferritin > 50-75 long-term:**
 - new evidence suggests that iron may be beneficial even in patients with normal ferritin levels.
- **MILD RLS: advice and reassurance** are generally sufficient:
 - Good [sleep hygiene](#), reduce caffeine and alcohol especially before bedtime, stop smoking.
 - Increase daytime exercise. Walking, stretching, relaxation exercises.
- **MODERATE TO SEVERE RLS**, consider drug treatment if significant impact on the quality of life:
 - Dopamine agonists and gabapentinoids are both effective at helping the sensory symptoms, but given the risks of augmentation with dopamine agonists gabapentinoids generally preferred 1st line.
 - Start with gabapentin (600 to 1200mg daily) or pregabalin (150mg to 450mg daily).
 - Dopamine agonists are highly effective but long-term have a diminishing response and can cause augmentation of symptoms. Avoid unless absolutely necessary, and then use the lowest effective dose for the shortest possible time e.g. pramipexole or ropinirole.
 - A weak opioid e.g. codeine can be considered as an alternative for PRN use if RLS is painful.
- **Referral to a neurologist?**
 - Refer if doubt about the diagnosis, or symptoms refractory to treatment or if augmentation occurs with a dopamine agonist.

[NHS Patient information](#)

[Restless Legs Syndrome, UK support & information charity](#)

PARKINSON'S DISEASE

With the ageing population, Parkinson's is becoming increasingly prevalent. *Historically recognised as a movement disorder, it is becoming increasingly clear that many non-motor symptoms are often associated, which can often be as distressing as the primary motor symptoms and are often modifiable.* NICE have recently updated their guidance ([NICE Parkinson's disease in adults, 2017](#)) to reflect this. A good proportion of the guideline is aimed at secondary care, with only minimal updates from the 2006 guideline on the diagnosis, assessment and treatment of motor symptoms, but there is a large amount of new information to help guide managing the non-motor symptoms of PD e.g. to consider modafinil with daytime sleepiness.

The BMJ have done a good summary of the guidance ([BMJ August 2017](#)) including an excellent colourful infographic for those who like the visual reviews.

The recent NICE guideline [NICE 2019, NG127](#) on recognition and management of suspected neurological conditions gives guidance on referral.

Parkinson's Disease, Diagnosis & Management [NICE 2017, NG71](#) & [BMJ 2017;358;j1951](#)

► How should we diagnose Parkinson's Disease?

- Suspect PD in people presenting with tremor, stiffness, slowness, balance or gait problems
 - consider referral with handwriting difficulty, specifically new gradual onset of small and slow handwriting
 - Parkinson's tremor is usually asymmetrical (unilateral or predominantly unilateral) and more prominent at rest, and associated with the slowness of movement, stiffness or balance problems
 - IF PD suspected **refer people quickly and untreated** to a specialist (NB this is one of many recommendations where the guidance may not match the reality of current service provision)
- **Diagnosis of PD is still clinical** and based on the UK Parkinson's Disease Society Brain Bank, which requires 3 steps to confirm the diagnosis ([click here for full criteria](#)):
 - Step 1: **Bradykinesia and at least one of:**
 - **Muscular rigidity**
 - **Rest tremor, or**
 - **Postural instability** unrelated to primary visual, cerebellar, vestibular, or proprioceptive dysfunction
 - (NB note that **tremor is not essential for the diagnosis** but bradykinesia is)
 - Step 2: Exclusion criteria (e.g. repeated strokes or head injury, drug cause, no response to levodopa, exposure to neurotoxins, sustained remission etc.)
 - Step 3: Three or more supportive criteria required (e.g. unilateral onset, progressive course, rest tremor, good response to levodopa, persistent asymmetry etc.)

► What should happen once the diagnosis is made?

- NICE recommend all patients with PD should have an 'accessible point of contact with specialist services' e.g. with direct access to a specialist nurse
- Patients should be reviewed every 6-12 months
- **All patients who drive must inform the DVLA** - [click here for DVLA link for patients](#)
- As always, patient information and support is crucial - click here for [Parkinson's UK website](#)

► How should motor symptoms be treated with drugs?

- It remains the case that no treatments are neuroprotective and all treatment is symptomatic
- NICE says that any alterations to motor drugs should be done by or under the advice of a specialist (NB fine in theory, but often not realistic in practice so the following guidance is useful for us...)



- **First-line (should be initiated early):**
 - **Levodopa** if motor symptoms affecting the quality of life
 - If motor symptoms not affecting the quality of life consider the choice of:
 - **Dopamine agonists** (non-ergot derived formulations should be used e.g. pramipexole, ropinirole), **Levodopa** or **MAO-B inhibitors** (e.g. selegiline, rasagiline)
- **Comparison of first-line drugs**

	Levodopa	Dopamine agonists	MAO-B inhibitors
Motor symptoms	More improvement	Less improvement	Less improvement
Improvement in daily activities	More improvement	Less improvement	Less improvement
Motor complications	More complications	Less complications	Less complications
Adverse events e.g. sleepiness, hallucinations, impulse control disorder	Fewer events	More events	More events

- **Adjuvant therapy:**
 - **If dyskinesia or motor fluctuations develop** despite optimal levodopa consider adding dopamine agonists, MAO-B inhibitors or COMT inhibitors (e.g. entacapone)

► **What about impulse control disorders?**

- These are increasingly recognised as a problem in PD, and can occur with any dopaminergic therapy at any stage of the disease but are **particularly associated with dopamine agonists**. Patients and family should be warned of this before starting dopamine agonists.
- There are a variety of presentations e.g. **compulsive gambling, hypersexuality, binge eating, obsessive shopping**; they can happen at any stage in the disease course
- Smoking and alcohol can increase the risk
- Seek advice from a specialist if this develops - drug modification is likely to be needed

► **What about non-pharmacological therapy?**

- Physiotherapy and physical activity are key interventions
 - **Physical activity is very important in the management of PD** it can help with many of the motor and non-motor symptoms
 - **Offer referral to specialist physiotherapy**, especially if experiencing balance or motor function problems
 - Consider the Alexander Technique for people with balance or motor problems (NB and a great RCT a few years ago showed that Tai Chi also helps [NEJM2012](#))

- **Occupational therapy**, consider referral especially if patients are having difficulty with activity of daily living (ADL)
- **Speech and language therapy**
 - Consider referral especially if problems with communication, swallowing or saliva
- **Nutrition**
 - Consider referral to a dietician
 - Advise patients with PD to take vitamin D supplements

► *How should non-motor symptoms be managed?*

- **Daytime sleepiness**
 - Patients with daytime sleepiness and/or sudden onset sleep should not drive and **should inform the DVLA** - [click here for patient link for DVLA notification](#)
 - Can happen with any medications, but particularly with dopamine agonists
 - Consider **modafinil** if no reversible causes and symptoms excessive
- **Rapid eye movement sleep disorder**
 - Consider **clonazepam** or **melatonin** if no medication cause
- **Orthostatic hypotension**
 - Review drugs (esp antihypertensives, dopaminergic, anticholinergics, antidepressants)
 - Consider **midodrine** or **fludrocortisone** (if midodrine C/I, not tolerated or not effective)
- **Depression**: NICE have not given any specific recommendations for the treatment of depression in PD, which is probably due to a lack of good evidence, however, it is frustrating as this is a **common problem**. However, **CKS** gives some good pointers ([click here for good PIL from Parkinson's UK](#)):
 - Can be difficult to diagnose as **features may be wrongly attributed to the PD**; If motor fluctuations assess during an 'on' time
 - **CBT should be considered**
 - **Medications** - SSRIs can be considered (but limited evidence); TCAs have slightly better evidence but may be limited due to side effects e.g. cognitive impairment, falls
- **Psychotic symptoms**
 - Do not treat hallucinations and delusions if they are well tolerated
 - Medications may need to be adjusted under specialist supervision
 - Consider **quetiapine** if no cognitive impairment present
 - Clozapine (under strict specialist supervision) if standard treatment not effective
- **Dementia**
 - Review potential causative drugs (e.g. TCAs, anticholinergics and some PD treatments)
 - Consider **cholinesterase inhibitor** or **memantine** (if cholinesterase inhibitor not tolerated or C/I)
- **Drooling of saliva**
 - Only consider drugs if non-drug treatment (e.g. speech & language therapy) not available/not effective
 - Consider **glycopyrronium bromide**

Palliative care

- We all know how important good palliative care is for any neurodegenerative condition and NICE quite rightly highlight this as an area that we need to consider, including the importance of considering DNACPR orders and LPAs among other things
- NICE also recommends we **consider referring to palliative care services at any stage of PD** to discuss palliative care and end-of-life care

KISS: Parkinson's Disease

Based on [NICE 2019, NG127](#) & [NICE July 2017](#) & [BMJ 2017;358;j1951](#)

Diagnosis:

- Diagnosis depends on the presence of **bradykinesia** plus **one of rigidity, rest tremor or postural instability** ([click here for full diagnostic criteria](#)).
- Refer all suspected cases early and **untreated** for specialist diagnostic confirmation.
- All patients who drive must inform the DVLA ([click here for DVLA notification link](#)).

Non-pharmacological management:

- Exercise - **physical activity** has been shown to be very important in reducing motor and non-motor symptoms; consider referring to **physio** especially if balance or motor problems.
- Consider referral to **Occupational Therapy** if difficulty with activities of daily living & to **Speech & Language therapy** if problems with communication, swallowing or saliva.
- Patient information and support: lots of good information from [Parkinson's UK](#).

Drug treatment of motor symptoms:

- All drugs are for the symptomatic benefit and none influence the long-term progression.
- Initiation and alteration of all drugs should be done under specialist supervision.
- **First-line drugs:**
 - **Levodopa** - better for motor symptoms with fewer adverse effects but higher long-term motor complications.
 - **Dopamine agonists** (non-ergot derived e.g. pramipexole, ropinirole) - less good for motor symptoms but fewer motor complications, but higher adverse effects.
 - **MAO-B inhibitors** - less good for motor symptoms but fewer motor complications, but higher adverse effects).
- **Adjuvant** therapy: consider adding dopamine agonist, MAO-B inhibitor or COMT inhibitor to levodopa if dyskinesia or motor fluctuations despite optimal levodopa therapy.
- **Impulse control disorders** (e.g. hypersexuality, gambling, binge eating) can occur with any dopaminergic therapy, but particularly dopamine agonists; warn patients and family about this potential complication as can be distressing.
 - If it occurs seek specialist advice, but we should not alter/stop medications without advice - medications usually need to be slowly reduced due to risk of dopamine withdrawal.

Non-motor symptom treatment (review potential causative/contributory drugs in all cases):

- Daytime sleepiness, particularly associated with dopamine agonists; consider **modafinil**.
- Rapid eye movement sleep disorder - consider **clonazepam** or **melatonin**.
- Orthostatic hypotension: meds review important; consider **midodrine** or **fludrocortisone**.
- Depression ([CKS 2018](#)) - can be difficult to diagnose as features may be wrongly attributed to the PD; consider **CBT**; the best evidence is for **TCA**s but use may be limited by side effects (cognitive impairment and falls) so **SSRIs** may be more appropriate.
- Psychosis: don't treat if well tolerated; consider **quetiapine** or clozapine (specialist only).
- Dementia: consider **cholinesterase inhibitor** or **memantine**.
- Drooling: refer to **speech & language therapy** or consider **glycopyrronium bromide**.

Palliative care - consider referring at any stage to consider end-of-life care.



KISS: Mental Health Complications in Parkinson's Disease

[CKS 2022](#)

Sleep Disturbance: check for underlying cause:

- Depression - see below.
- Restless legs syndrome - see [CKS](#) for further management.
- Rapid eye movement (REM) sleep behaviour disorder - clonazepam or melatonin may be considered under specialist supervision.
- Nocturnal akinesia (inability to turn over in bed) - levodopa or oral dopamine agonists or rotigotine may be considered by a specialist.
- Nocturia.
- Vivid dreams, nightmares, or hallucinations - may be caused by PD medication (e.g. selegiline and dopamine agonists). Seek specialist advice about altering drug treatment.

Depression and Anxiety:

- Can be difficult to diagnose as **features may be wrongly attributed to the PD.**
- Ask family/carers for supplementary history.
- If motor fluctuations present assess during an 'on' time.
- CBT should be considered.
- **Medications** (consider liaising with specialist due to potential drug interactions):
 - SSRIs can be considered but they can worsen motor symptoms e.g. restless legs, periodic limb movements and REM sleep behaviour disorder.
 - TCAs have slightly better evidence and maybe more effective but the risk of side effects e.g. cognitive impairment falls.

Impulse control disorders:

- These are complex, compulsive, repetitive behaviours which affect 1–14% of people with PD.
- Features may include hypersexuality, compulsive gambling, binge eating, obsessive shopping, or punding (stereotyped behaviours such as repetitive assembling or collecting).
- They may occur with any dopaminergic medication but are **particularly associated with the dopamine agonists**, and may develop at any stage in the course of Parkinson's disease.
- Symptoms may not be reported by the person and may be concealed.
- Smoking and/or alcohol consumption increase the risk.
- **Specialist input is required** - medication adjustment is likely to be needed but **we should not alter medications in primary care** and await specialist advice.

Psychotic symptoms:

- Have a low threshold for suspecting psychosis and arrange a specialist review if this is suspected.
- Consider possible **underlying causes**, such as:
 - Parkinson's disease itself and medication (particularly dopamine agonists).
 - Delirium - infection, constipation, pain, dehydration, electrolyte disturbance.
 - Dementia.
 - Depression.
- Mild psychotic symptoms do not need treatment if they are well tolerated.
- More severe psychotic symptoms may require a gradual withdrawal of medication which may be causing symptoms or the use of an atypical antipsychotic e.g. quetiapine under specialist supervision. Rarely, clozapine drug treatment may be initiated by a specialist.

Excellent **patient information** on non-motor symptoms, including mental health issues [Parkinson's UK](#)

NB Learning Points: Parkinson's Disease

- Tremor is not essential for the diagnosis; bradykinesia is plus one of rigidity, rest tremor and postural instability
- None of the drug treatments are neuro-protective - they are for symptomatic treatment only
- Physical activity is very important in treating motor and non-motor symptoms - consider referring all patients to physio
- First-line drugs for motor symptoms are levodopa, dopamine agonists or MAO-B inhibitors; levodopa best for motor symptoms and improved ADLs but higher motor complications
- Impulse control disorders can occur at any stage with any dopaminergic therapy
- Non-motor complications are common and distressing - review potential causative or contributory drugs; NICE have given good guidance on treatment options ([click here](#))

FACIAL PALSY

The most common cause of facial palsy is Bell's palsy but as this new BJGP paper ([BJGP 2019;69:579](#)) explains there are over 50 aetiologies. Nevertheless, facial palsy will fall into one of three groups, each with their distinct challenges and management strategies.

Facial Palsy: when and why to refer for specialist care [BJGP 2019;69:579](#)

This helpful article defines facial palsy in three groups:

- **Acute flaccid facial paralysis (AFFP)**
 - Majority of cases will be Bell's palsy; 83% make a full recovery with prompt oral steroids, 72% without
 - We should exclude Ramsay Hunt syndrome by assessing for vesicular rash in/around the ear/mouth/scalp
 - AFFP is differentiated from LFFP (see below) by the presence of viable facial muscles
- **Longstanding flaccid facial paralysis (LFFP)**
 - Persistent facial palsy of >12 months; congenital or acquired; AFFP becomes LFFP after 12/12 of palsy
- **Post-paralytic facial palsy (PPFP)**
 - This is a potential complication of recovery after facial palsy due to abnormal neural regeneration
 - Effects include: synkinesis (unwanted facial movements), hypertonia (tightness of muscles) and hyperkinesis (exaggerated movements).
 - All can affect facial symmetry and is associated with psychological morbidity.

The three key points from the paper are:

1. *Understanding when we should refer patients with ongoing facial palsy.*
2. *In patients with LFFP, there is minimal chance of meaningful improvement, which is why earlier referral in patients with AFFP is recommended.*
3. *Ideally, patients should be referred to a centre with a specific facial palsy service.*

The latter is important because management is increasingly interventionalist an MDT approach involving facial physiotherapists, psychologists and surgeons. Surgery aims to protect from ocular damage, restore nasal opening and restore facial symmetry, at rest and with movement. Surgical techniques such as cross-facial nerve grafting with muscle transplantation can provide meaningful improvements in severe cases.

KISS: When to Refer Patients with Acute Flaccid Facial Paralysis [BJGP 2019;69:579](#)

- **IMMEDIATELY**
 - Facial palsy with other neurology, slow onset (>3D), palpable pre-auricular mass, systemic symptoms, bilateral palsy (ENT/neuro)
 - Persistent dry eye, corneal irritation or ulceration (ophth)
- **3 WEEKS AFTER ONSET**
 - No improvement in facial movements (ENT)
- **3 MONTHS AFTER ONSET**
 - Incomplete resolution of facial palsy

NB Learning Points & Quality Improvement Ideas: Facial Palsy

- Oral pred 50mg for 10 days started within 72h improves resolution of palsy (NNT = 8-10)
- Patients with no improvement in palsy at 3 weeks or incomplete resolution at 3 months should be referred
- Examine carefully for any sign of vesicles around the external auditory meatus, suggestive of zoster (Ramsay Hunt Syndrome). With facial palsy and severe pain, consider adding in high dose acyclovir as early treatment of Ramsay Hunt significantly improves prognosis
- Patient support can be found at [Facial Palsy UK](#)

KISS: Bell's Palsy

[BJGP2017;67:e329](#) [Clinical Evidence 2014;04;1204](#), [BJGP2014;64;651](#) & [DTB2013;12:138](#)

■ Definition

- An acute, unilateral partial or complete paralysis of the face. It is an idiopathic unilateral lower motor neurone facial nerve palsy of rapid onset often accompanied with mild pain, numbness or disturbed taste on the side of the tongue
- Peak incidence aged 15 to 40
- It accounts for 2/3 of facial nerve palsies and it is a diagnosis of exclusion; diagnosis involves elimination of other causes such as stroke, infections, parotid tumours, middle ear disease etc
 - A search for other causes should not delay treatment with steroids
 - **Lyme disease** can present with facial palsy; ask about possible exposure & consider testing
- Severe pain, vesicles in ear or mouth, hearing loss and imbalance suggest **Ramsay Hunt syndrome** due to herpes zoster and require immediate antivirals and specialist guidance
- Children are more likely to have an underlying cause (50%) and should be referred, but overall their prognosis is excellent with 90% full recovery

■ Prognosis

- 70% recover spontaneously, often within 3 weeks
- 30% have delayed recovery, and one in six end up with persistent facial weakness
 - This is more likely in those with complete paralysis of the nerve
- Failure to improve at all after 3 weeks suggests an alternative diagnosis and referral for MRI scan recommended

■ Eye protection

- Reduced tear production and difficulty closing the eye puts people at risk of corneal dryness, ulceration and even blindness. Appropriate eye care is crucial (artificial tears in the daytime, eye ointment at night, taping eyelids at night)

■ Steroids

- *Evidence is convincing that prednisolone is effective*
- It is most effective if given within the first 72 hours, but there is some evidence to support treatment if patients present within 7 days
- Prednisolone 50mg daily for 10 days most effective dose (NNT 8 for facial weakness)

■ Antivirals

- Antiviral therapy in idiopathic Bell's palsy remains controversial. The latest [Cochrane review 2019](#) did not find any convincing evidence that anti-virals give any benefits above and beyond corticosteroids, but the evidence was limited both in terms of quantity and quality.
- If patients have vesicles and/or severe pain, hearing loss or vertigo this is diagnostic of **Ramsay Hunt syndrome**; refer and treat with 'shingles dose' of antivirals AND prednisolone

■ Physiotherapy

- There is a lack of evidence, but some evidence to support 'facial re-training' to improve facial motor function

■ Surgery

- Consider referral all cases with a residual disability after a few weeks to a plastic surgical facial reanimation service for consideration of surgical nerve grafting

EPILEPSY

As far as national guidelines go, there haven't been any substantial changes over the last few years. Excellent guidelines were produced by SIGN back in 2015, which were revised in 2018 ([SIGN 143, updated 2018](#)). The bulk of the NICE guidelines date back to its inception in 2004, although it has had a very recent update in 2021 ([NICE CG 137, updated May 2021](#)) based on the new MHRA information on pregnancy and antiepileptic drugs (see our subsequent chapter on epilepsy, contraception and pregnancy for more detail on that and other issues relating to women of childbearing age with epilepsy). One **issue with epilepsy that remains as important as ever is SUDEP** (sudden unexplained death in epilepsy) - although a few years old now a useful BMJ article explores this problem and what we can do in primary care to help reduce the risk of SUDEP ([BMJ 2015;350:h718](#)). We do a brief review on the use of medical cannabis in epilepsy as the much anticipated NICE technology appraisals [TA614](#) and [TA615](#) for cannabidiol (Epidyolex) were published in December 2019, and the issue of access to these potential treatments remains in the national press. Finally, we review the important topic of **depression and antidepressants in epilepsy and seizures**, based on a DTB review in 2020 ([Drug and Therapeutics Bulletin 2020;58:137-140](#)) and recent [updated guidance from the Specialist Pharmacy Service](#) in May 2021.

► *Sudden unexpected death in epilepsy (SUDEP)* [BMJ2015;350:h718](#)

- An estimated 1.16 per 1,000 people with epilepsy die suddenly and prematurely each year.
- Sudden death in epilepsy peaks in young adults, especially men.
- A 20-year-old man with epilepsy has a 1% chance of dying before the age of 30.
- Epilepsy is second only to stroke in years of life lost to neurological disease.
- 30% of deaths are accidents, mostly from drowning or burns.

Many of these deaths are unavoidable, but there is **evidence to suggest that improved care may result in fewer untimely sudden deaths**. Risk factors for death in epilepsy have been discovered including:

- Alcohol problems.
- Failure to collect anticonvulsant prescriptions in the preceding 3 to 6 months.
- A history of injury in the previous year.
- Depression (due to poorer seizure control, often through non-adherence to treatment) - see next section for more discussion on this.
- Tonic-clonic seizures have increased in the frequency in the last 6 months.

The paper argues that prevention needs to start in primary care and much can be done:

- Patients with worsening control can be referred back to specialist clinics.
- Patients who do not collect prescriptions could be identified and contacted.
- Patients with alcohol and substance misuse can be referred to local addiction services.
- Patients who have had injuries could be identified as being at increased risk.
- Patients could be screened for depression.

► *Seizures, epilepsy and antidepressants* [DTB 2020;58:137-140](#) [SPS May 2021](#)

As discussed in the paper above, a well-recognised risk factor for poor seizure control is depression, often thought due to poor concordance with antiepileptic drugs (AEDs). However, there are other factors we need to consider linking seizures and depression, including the **(potential) risk of lowering seizure thresholds with antidepressant drugs (and which ones)**, and the interesting concept that depression and epilepsy may even share linked underlying pathophysiologies. These issues were discussed in the above DTB article in 2020. The article highlights that there is a real mish-mash of heterogenous data regarding seizures and antidepressants with much of it low quality and short term, however, there are some important themes and messages for us to be aware of. First a **few background points**:

- The prevalence of depression is estimated at 23% in people with epilepsy compared to 8.4% in the general adult population - this **contributes significantly to lower quality of life for people with epilepsy**, and can have a greater effect on quality of life than the seizures themselves.
- Seizure rates are higher in those with depression than in the general population, and there is evidence that depression and seizure disorders could share underlying mechanisms.
- We should still consider psychological therapies for people with epilepsy and depression, but antidepressants will need to be considered in some.
- Although overall there may be an increased risk of seizures with antidepressant use, it is a **myth that antidepressants should be avoided in people with epilepsy**, as the risks are low if commonly used antidepressants are used at standard doses, and the seizure risks differ between drugs.

So, what are the risks of seizure with antidepressants and which should we consider using?

- Overall, there may be an increased risk of seizure with antidepressant use, but the data is conflicting, and interestingly studies show an increased risk of seizure in those on placebo in depression trials, supporting the **theory that depression in its own right may be a risk factor for seizures**, with possible shared underlying mechanisms.
- Given the limitations of the data, it is difficult to draw definitive conclusions about **which antidepressants are safest with regards to seizures and epilepsy**, and even which classes are safest, but what data we have suggests:
 - **Citalopram or sertraline are reasonable first line options.**
 - Venlafaxine, duloxetine and mirtazapine also appear to be relatively safe.
 - Tricyclics should generally be avoided as they can potentially lower seizure threshold, but doxepin appears relatively safe.
 - There is evidence that both clomipramine and bupropion increase the risk of seizures.
- **Is there a dose-response effect?** There is evidence of a dose-response effect however this is based on intentional overdose data (which clearly shows increased seizure risk) and for antidepressant doses way above usual; if standard doses are used there appears to be minimal increased risk.
- **Combinations of antidepressants?** There is (limited) data that combining antidepressants may increase seizure risk and should generally be avoided with monotherapy preferred.
- **Interactions between AEDs and antidepressants?**
 - This is complex, with both antidepressants and AEDs potentially affecting each other's plasma levels via interaction of the cytochrome P450 enzyme system - the SPS recommend we check the [BNF drug interaction checker](#) before prescribing to assess risks of interactions.
 - The other important issue to remember is negative synergistic side effects with some AEDs and antidepressants e.g. weight gain with valproate and mirtazapine and hyponatraemia with carbamazepine/oxcarbazepine and SSRIs.

For useful **pragmatic guidance on antidepressants and epilepsy** see KISS below:

KISS: Antidepressants for People with EpilepsyDTB 2020 [Specialist Pharmacy Service May 2021](#)**Background:**

- Depression is significantly more common in people with epilepsy than the general population, and the two conditions may share underlying mechanisms/causes.
- If present **depression contributes significantly to lower quality of life**, and may be more of a factor than the seizures themselves for people with epilepsy.
- **Depression is also linked to poor seizure control**, thus identifying and treating it is important for both quality of life, seizure control, and may reduce the risk of sudden unexpected death in epilepsy.
- Although psychological therapies should still be used in people with epilepsy, antidepressants may still be required and it is a myth that antidepressants cannot be used for people with epilepsy.
- Although there may be an increased risk of seizures with antidepressant use, the risk is small, especially if the commonly used antidepressants are used at standard doses.
- The **relatively low risk of antidepressants causing increased seizures rarely outweighs the risk of leaving depression untreated**.

Which antidepressants to use:

- **SSRIs are generally considered the safest and first line options** - sertraline or citalopram probably safest, followed by escitalopram and fluoxetine.
- **Other low to moderate-risk antidepressants:**
 - SNRIs (duloxetine or venlafaxine)
 - Mirtazapine
- Tricyclics (TCAs) should generally be avoided but if needed doxepin has the best safety profile.
- **High risk antidepressants (to avoid)** include other TCAs including amitriptyline, and especially clomipramine, as well as bupropion.
- **Avoid concurrent antidepressant use** - combining antidepressants does seem to increase the risk of seizures.
- Use a low dose first and gradually increase to the lowest effective dose.
- If one antidepressant is ineffective switch to an alternative (do not combine). For guidance on switching antidepressants see good information from CKS [click here](#).

Other considerations:

- If depression is present antiepileptic drugs (AEDs) with mood stabilising effects can be considered (e.g. carbamazepine, lamotrigine) but would obviously have to be initiated by specialists.
- **Drug interactions with AEDs:**
 - Complex potential interactions can occur between AEDs and antidepressants due to effects on the cytochrome P450 enzyme - before prescribing use the BNF drug interaction checker [click here](#).
 - Carbamazepine/oxcarbazepine and SSRIs can increase risk of hyponatraemia.

If seizures occur or increased seizure frequency:

- Check sodium levels for hyponatraemia, especially if using SSRIs.
- Consider switching to alternative antidepressant.
- Consider advice from a specialist.

Consider specialist input:

- If the AED may be causing/contributing to depression (e.g. levetiracetam, phenytoin, topiramate).
- If the patient may benefit from a mood stabilising AED.
- If repeated treatment failures with standard antidepressants at usual doses.



► *Cannabis and cannabinoids for epilepsy* [Journal of Neurology, Neurosurgery & Psychiatry 2018;89](#)

We include this subject briefly as it has been such a hot topic in both medical and lay press. In some ways, this subject will not affect GPs, as it is important to confirm that **GPs cannot prescribe these cannabis products**. As discussed in the [BMJ in April 2019](#) Cannabis-based products that were previously listed in Schedule 1 can now be prescribed by doctors on the General Medical Council Specialist Register in the UK, on a named-patient basis. We have seen the stories of individuals that have been severely affected by the change in the regulations, but is there any evidence for their use in epilepsy in general? This systematic review, which can be seen via the BMJ Journals, is the best evidence we have to date on the subject:

- There have been very few high-quality RCTs and those that have been conducted have only used CBD (cannabidiol) in paediatric samples with rare and serious forms of drug-resistant epilepsy
- Of those studies reviewed a reasonable proportion of patients experienced a decrease in seizure frequency when using pharmaceutical-grade CBD in addition to anti-epileptic drugs, however complete seizure freedom was unlikely, and minor adverse events were likely

Subsequent to this review **NICE produced its much-anticipated guidance on medical cannabis NG144 in November 2019**. and shortly after 2 technology appraisals [TA614](#) and [TA615](#) on the use of cannabidiol (Epidyolex) for 2 rare severe treatment-resistant forms of paediatric epilepsy - Dravet and Lennox-Gastaut syndromes. Although this won't affect us directly in primary care, it's useful to be aware of the guidance:

- NICE accepted there was enough evidence to recommend CBD alongside clobazam as an option for both these conditions if seizures were not controlled with usual anti-epileptic drugs
- Prescribing needs to be done by tertiary paediatric specialists

NB Learning Points and Quality Improvement Ideas: Epilepsy

- Sudden unexpected death in epilepsy (SUDEP) could be reduced by **identification of high-risk people in primary care** - those that don't pick up prescriptions regularly, those with depression or substance misuse and those with worsening seizure frequency may all be a risk.
- **Depression is highly prevalent in people with epilepsy** and may contribute more to worse quality of life than the seizures themselves, and is a risk factor for poor seizure control and SUDEP.
- Although there may be small increased risks of seizures with antidepressant drugs, for the majority of people needing antidepressants, these small risks rarely outweigh the risks of untreated depression.
- If using antidepressants in people with epilepsy **SSRIs (particularly sertraline and citalopram) are probably safest**, with venlafaxine, duloxetine and mirtazapine also considered low/moderate risk; consider using the BNF drug interaction checker before prescribing to assess interactions with AEDs.
- Be aware carbamazepine/oxcarbazepine and SSRIs can both increase risk of hyponatraemia - if seizures worsen on this drug combination check a sodium level.
- **Routine blood testing including AED levels is not needed**; however, consider bloods including AED levels if toxicity concerns or unexplained loss of seizure control.
- There is very little evidence to guide the use of cannabis-based products in epilepsy in general, however, NICE have recommended use of CBD (Epidyolex) as an option alongside clobazam for two rare syndromes if seizures are not controlled with usual anti-epileptic drugs but prescribing needs to be done by tertiary paediatric specialists.
- **Quality Improvement Ideas:**
 - Actively look for and follow up patients with epilepsy with these risk factors above for SUDEP.
 - Consider setting up an alert system for those who do not collect their medication.
 - Are you still doing annual routine blood tests for people on AEDs? These are not needed (in adults) and are not recommended by either SIGN or NICE. Save some work and phlebotomy time!

KISS: Epilepsy

Based on [SIGN 143, updated 2018](#) & [NICE CG137, updated May 2021](#)

Diagnosis and classification:

- Refer all suspected seizures to a specialist-dedicated first-seizure clinic urgently (ideally to be seen within 2 weeks).
- History from patient and eye-witness give the most important diagnostic information and are the mainstay of the diagnosis.
- EEG is not routinely indicated and a normal EEG cannot exclude epilepsy.
- MRI is the imaging modality of choice but is not routinely required.
- Blood tests should be done to identify potential causes and co-morbidity e.g. U&E, glucose, calcium.
- The seizure type and epilepsy syndrome should be identified, including the important distinction between **focal** and **generalised** epilepsy - EEG plays an important role in this distinction.

Treatment with antiepileptic drugs (AEDs):

- **Specialist initiated.** The aim is 'complete seizure freedom' wherever possible.
- AEDs should be started after a first tonic-clonic seizure if there is a history of previous absence or focal seizures, a structural brain disorder, high risk of recurrence or the EEG shows unequivocal epileptic changes.
- **Choice of monotherapy (monotherapy preferred initially):**
 - Focal seizures: lamotrigine first choice; carbamazepine or levetiracetam alternatives /2nd line.
 - Generalised or unclassified: sodium valproate most effective; If not tolerated or contraindicated, various options including lamotrigine, carbamazepine or topiramate.
 - In women of childbearing age, avoid valproate and consider levetiracetam or lamotrigine.
- **If epilepsy resistant to monotherapy:**
 - Review the diagnosis and treatment adherence.
 - Consider (specialist guidance) combination therapy if treatment with two first-line AEDs has failed.
- **Bloods?** Routine AED monitoring not routinely needed; may be useful in certain situations e.g. assessment of adherence or toxicity, or unexplained loss of seizure control; 'Routine' bloods e.g. FBC/LFT are not required for monitoring.
- **Women with epilepsy:** (NB refer women to [Women & Epilepsy](#) guidance)
 - Need appropriate advice about contraception and pre-pregnancy counselling, but be reassured that most will have a normal pregnancy and delivery.
 - See [Epilepsy Action guidance on AEDs and contraception](#).
 - See the next chapter on epilepsy, contraception and pregnancy for more detail.
- **Older people with epilepsy:**
 - The presentation may be different; commonly seen with neurodegenerative disorders, dementia and vascular disease.
 - Older people with new-onset seizures should be screened with cognitive function tests and have a CVD risk assessment.
 - Lamotrigine or possibly levetiracetam is preferable with focal seizures.
 - Gabapentin is an alternative monotherapy.

Patient and clinician information:

- Epilepsy action has great resources for patients and GPs, including guidance on driving, benefits and employment. Also a seizure diary and information on avoidable seizure triggers. [Click here](#)

EPILEPSY, CONTRACEPTION AND PREGNANCY

Whilst the [Valproate Pregnancy Prevention Programme](#) has been an important piece of work for patient safety, another key area of patient safety is addressing the risk of women with epilepsy stopping their (other) AEDs due to their **concerns over AED drug safety in pregnancy**. Which begs the question, **which AEDs are safest?** Thankfully we have now had a comprehensive review of AEDs in pregnancy conducted by the Commission on Human Medicines, reported in the [MHRA in 2021](#), which was subsequently reviewed in the [Drug and Therapeutics Bulletin in July 2021](#). Subsequent to this we have had further data on the risks of both topiramate [DTB 2022;60:165](#) and pregabalin [MHRA Drug safety Update April 2022](#).

Linked to this is the important question of which contraception is most effective/safe to reduce the risks of unplanned pregnancy for women with epilepsy, which is dependent on the AED they are taking. An excellent paper [BMJ2017;357:j2010](#) provides useful guidance for this question, summarised in the KISS below.

► **Anti-epileptic drugs (AEDs) in pregnancy** [MHRA Jan 2021](#) [Drug and Therapeutics Bulletin 2021;59:99](#)

Although we are rightly concerned about the risks of AEDs in pregnancy (particularly valproate), probably the bigger risk to the mother (and baby) is poor seizure control, particularly if AEDs are stopped without guidance. As the DTB highlights there is concern that due to the known risks of some AEDs in pregnancy, this may lead to poor concordance with AEDs in general, resulting in increased seizure frequency, with the subsequent increased morbidity and mortality. Sadly **epilepsy remains a leading cause of maternal mortality in the UK**, and **sudden unexpected deaths in epilepsy (SUDEP) in pregnancy has actually increased over the past 10 years** with 8 deaths in 2013-2015 more than doubling to 18 deaths in 2016-18. Thus good data on AED safety is critical, and we have a crucial role in general practice to disseminate this information, as well as to give contraception advice and pre-pregnancy counselling. So what does the data show:

- Based on large cohorts of pregnancies exposed to lamotrigine (>12,000) and levetiracetam (>1,800), **there was NO increased risk of major congenital malformations for either drug** (when used at standard maintenance doses).
- There is also NO increased risks of foetal loss, prenatal growth restriction or preterm birth with either lamotrigine or levetiracetam.
- The data on neurodevelopmental disorders for lamotrigine and levetiracetam was more limited and not adequate to definitively rule out an increased risk, but what data was available did not show an increase in neurodevelopmental disorders.
- Based on data for exposure to carbamazepine (~9000 cases), phenobarbital (~1,800 cases), phenytoin (~2000 cases) and topiramate (~1000 cases) there was increased risk of major congenital malformations for all these drugs. However, the risks were relatively small; based on a background risk of major congenital malformations in the general population of 2-3/100 babies, the risks were:
 - Carbamazepine 4-5/100 babies (dose-dependent)
 - Phenobarbital 6-7/100 babies (dose-dependent)
 - Phenytoin 6/100 babies
 - Topiramate 4-5/100 babies (dose-dependent)
 - To put this in context the risks of major congenital malformation with valproate is 10/100 babies (with a 30-40% increased risk of neurodevelopmental disorders).
- Data regarding the risks of neurodevelopmental disorders for carbamazepine, phenobarbital, phenytoin and topiramate was somewhat limited, and no definitive conclusions can be made, but what data is available suggests:
 - Carbamazepine - no increased risk
 - Phenobarbital and phenytoin possible increased risk (but inconsistent findings)
 - Topiramate possible increased risk but very limited data
- The available data for pregabalin suggests it may be associated with a slightly increased risk of congenital malformations, but this data is under review and no conclusions have been reached yet.
- There is limited data for gabapentin (but see new data below), oxcarbazepine and zonisamide so risks are uncertain.

So, the messages highlighted by the MHRA review and the DTB for us and women with epilepsy are:

- **Make sure women not planning pregnancy have appropriate contraception** (see KISS below).
- Discuss pregnancy planning at every opportunity.
- **Refer *urgently* for specialist care if pregnancy is planned** (ideally to a joint epilepsy/obstetric clinic) - as the MHRA reminds us seizure thresholds/frequency can vary depending on which AED the woman is on, and which trimester of pregnancy she is in, so it is crucial these women have specialist-led care.
- **Do NOT stop or reduce doses of AEDs until decisions made with specialist teams.**
- If pregnancy is being considered start folic acid 5mg OD.
- We should **reassure women that current data does not show any increased risks with lamotrigine and levetiracetam**, and although there may be risks with some of the other AEDs, these risks can be mitigated with careful pre-pregnancy planning with their specialist.
- The MHRA have put together a useful summary of their findings in a [patient safety information leaflet](#)

KISS: Risks of Antiepileptic Drugs in pregnancy

MHRA Jan 2021 DTB July 2021

Drug	Major congenital malformations (background risk 2-3 cases/100 babies)	Neurodevelopmental disorders	Other effects in pregnancy
lamotrigine	NO increased risk	NO increased risk	NO ↑ risk of foetal loss, prenatal growth restriction or preterm birth
levetiracetam	NO increased risk	NO increased risk	NO ↑ risk of foetal loss, prenatal growth restriction or preterm birth
valproate	10/100 babies	high risk 30-40% ↑	
carbamazepine	4-5/100 babies (dose dependent)	no increased risk	risk uncertain
phenobarbital	6-7/100 babies (dose dependent)	possible ↑ risk	↑ risk intrauterine growth restriction
phenytoin	6/100 babies	possible ↑ risk	
topiramate	4-5/100 babies (dose dependent)	possible ↑ risk	↑ risk intrauterine growth restriction
pregabalin	possible small ↑ risk	risk uncertain	risk uncertain
gabapentin	risk uncertain	risk uncertain	risk uncertain
oxcarbazepine	risk uncertain	risk uncertain	risk uncertain
zonisamide	risk uncertain	risk uncertain	↑ risk intrauterine growth restriction

NB level/strength of evidence:

Red = increased risk with strong/convincing evidence

Green = NO increased risk based on strong/convincing evidence

Amber = data not definitive/slightly limited data

Grey = very limited data or limitations on data available - unable to draw any conclusions

- Useful patient information leaflet on AEDs - [click here](#)

Risks of congenital malformations with pregabalin [DTB 2022;60:117](#) [MHRA April 2022](#)

Subsequent to the reviews above we have had new data, **strengthening the concerns over the congenital malformation risks with pregabalin**. Reported by the MHRA in April 2022 and subsequently reviewed in the DTB they report new data from a Nordic study reviewing **2700 pregnancies exposed to pregabalin in the first trimester**, and compared the risks of congenital malformations against other comparator drugs:

- Compared to both lamotrigine monotherapy and duloxetine monotherapy, pregabalin monotherapy showed increased risks of congenital malformations with adjusted prevalence ratios of:
 - 1.29 vs lamotrigine (95% CI = 1.01-1.65)
 - 1.39 vs duloxetine (95% CI = 1.07-1.82)

There were limitations, as noted by the MHRA, in that information on confounding factors was limited as was a comparison of the data by pregabalin indication, but the irony will not be lost that the vast majority of women on pregabalin were not on it for anti-epileptic purposes, but mostly for anxiety and pain. However, the **MHRA has strengthened its advice on pregabalin and pregnancy** as follows:

- Provide counselling to patients using pregabalin on:
 - The potential risks to an unborn baby.
 - The need to use effective contraception during treatment.
- Continue to avoid the use of pregabalin during pregnancy unless clearly necessary and only if the benefit to the patient clearly outweighs the potential risk to the fetus.

As part of the **structured medication reviews** we are being encouraged to do for people on gabapentinoids, this is a factor we need to discuss with women of childbearing potential.

Risks of neurodevelopment disorders with topiramate [Drug and Therapeutics Bulletin 2022;60:165](#)

As discussed in the data above there are suggestions that topiramate may increase the risk of neurodevelopment disorders and further data from a very large Scandinavian cohort study and reported in the DTB, gives further data on this concern:

- This huge cohort of ~4.5 million women analysed neurodevelopmental outcomes with and without exposure to AEDs in pregnancy, including >16,000 women with epilepsy.
- For **women with epilepsy exposed to topiramate in pregnancy vs those not exposed to AEDs there was a significant increase in neurodevelopmental disorders**:
 - Adjusted HR for autistic spectrum disorder = 2.8 (95% CI = 1.4-5.7)
 - Adjusted HR for intellectual disability = 3.5 (95% CI = 1.4-8.6)
- The risks were dose-dependent
- The adjusted HR for any neurodevelopment disorder for topiramate was 2.1 (95% CI = 1.1-4.0) compared to those exposed to valproate HR = 2.4 (95% CI = 1.9-3.1)

This data **supports the concern that topiramate, as well as valproate, do cause an increased risk of neurodevelopment disorders**. This data is being reviewed by the MHRA and EMA and reinforces the current recommendation that **women of childbearing potential on topiramate should be on very reliable forms of contraception** (see below).

KISS: Epilepsy and Contraception

[BMJ2017;357;j2010](#)

- Discourage barrier methods alone.
- Choice is based on patient choice and potential interactions with anti-epileptic drugs, which depends on whether or not she is taking an enzyme-inducing drug.
- *Is she on an enzyme-inducing drug?*
 - **Enzyme inducers:**
 - Carbamazepine, eslicarbazepine, oxcarbazepine
 - Phenytoin, phenobarbital, primidone, perampanel
 - Topiramate in doses >200mg
 - These drugs increase the metabolism of oestrogen and progestogen
 - **Safe to use:** copper IUD, levonorgestrel IUS and medroxyprogesterone acetate depot (Depo-Provera) are all first-line choices
 - **Contraceptive failure:** can occur when using the combined pill, patch, ring, progestogen-only pill and progestogen implants (Nexplanon). If a woman still chooses to use these methods, advise barrier contraception as well. If she chooses the COCP, increase the oestrogen dose to 50mcg to 70mcg, tricycle packets and reduce the pill-free interval to 4 days
 - **Emergency contraception (EC):** copper IUD is recommended, alternatively a 3mg dose of levonorgestrel (ulipristal is NOT recommended)
 - **Non-enzyme inducers**
 - Clonazepam, clobazam, levetiracetam, lamotrigine
 - Sodium valproate (NB this should be avoided in all women of childbearing age due to high teratogenic potential)
 - Gabapentin, pregabalin, piracetam
 - Ethosuximide, lacosamide, tigabine, vigabatrin, zonisamide
 - Avoid lamotrigine with the COCP, as the COCP reduce blood levels of lamotrigine and may result in poor seizure control
 - Otherwise, **no restrictions** can choose any method for contraception and EC
- Advise all women of the importance of pre-conception planning
- There is a great, interactive online tool where women can see choices for their particular AED at [Epilepsy Action Contraception](#)

NB Learning Points & Quality Improvement Ideas: Epilepsy, contraception and pregnancy

- We have a crucial role to play in general practice to support women of childbearing age who have epilepsy - take every opportunity to discuss contraception and pregnancy plans.
- Although there are risks with anti-epileptic drugs (AEDs) in pregnancy, **poor concordance with AEDs contributes to significant morbidity and mortality in pregnancy**, and sudden unexpected death in epilepsy is still a leading cause of maternal death.
- We have had a recent comprehensive review of AED safety in pregnancy:
 - **Lamotrigine and levetiracetam are the safest** with the data showing NO increased risks of congenital malformations, foetal loss, preterm birth or growth restriction. There are increased risks of congenital malformations with carbamazepine, phenytoin, phenobarbital and topiramate.
- We have new data also confirming increased risks of neurodevelopmental disorders with topiramate.
- **Valproate should be avoided in all women of childbearing age.**
- Remember to prescribe folic acid 5mg daily for all women with epilepsy considering pregnancy.
- **Quality Improvement Idea:** Review women of childbearing age with epilepsy on carbamazepine, phenytoin, phenobarbital and topiramate - have they had an informed discussion about risks, contraception options (use KISS above) and any pregnancy plans.

MULTIPLE SCLEROSIS

The new NICE guideline on MS ([NG220](#)) has recently been published June 2022 with few significant changes to the previous iteration and continues to encourage a holistic approach, although the implication is that most of the management decisions should be specialist-led. A more recent [BMJ 2016;354:i3518](#) paper has looked at the evidence around the many *new disease-modifying therapies* (DMTs). There is increasing interest in the **possibility that early diagnosis and treatment with DMTs may prove beneficial**, but the conclusion from the BMJ article above is that 'the trend towards earlier treatment and the availability of powerful new therapies may prove beneficial for the long-term health of patients, but more convincing data are needed'. Intriguingly we also have recent evidence suggesting that **simvastatin may be beneficial in reducing disability in secondary progressive MS** ([CNS Drugs. 2015 Apr;29\(4\):277-91](#)), with a large phase 3 trial launched in the summer of 2017.

Although diagnosis & management of MS are very much specialist secondary care based, the NICE guidance and associated papers stress that **we as GPs have a vital role** in referring suspected cases, detection of possible relapses, managing symptoms and also **ongoing monitoring and support of this unpredictable illness**. We cover diagnosis, management of symptoms and relapses in the following KISS summaries based on NICE guidance and latest BMJ reviews. But first, a brief bit of **background on MS** and a review of the medically and politically hot topic of **cannabinoids in MS** with the latest systematic review on the subject, and the much anticipated [NICE guideline on medical cannabis published in November 2019](#).

► Background:

- MS is the most common cause of serious physical disability in working-age adults and affects >125,000 people in the UK - Prevalence is more common in more northerly populations (~120/100,000 in England/Wales and ~190/100,000 in Scotland); almost twice as common in women, typically presenting aged 20 to 40.
- The cause remains unknown. We know it is an acquired immune-mediated inflammatory disease believed to be due to an abnormal immune response to environmental triggers in people who are genetically predisposed. The initial inflammatory phase is followed by a progressive degenerative phase due to neuronal damage.
- The most common pattern is relapsing-remitting MS (RRMS) - 85% have RRMS at the onset - Disease-modifying therapy (DMT) reduces the relapse rate.
- About two-thirds with RRMS go on to develop secondary progressive MS (SPMS) - DMT does not reduce secondary progression, however, early data suggests simvastatin may reduce disability progression.
- 10-15% have primary progressive MS (PPMS) at the onset, where symptoms do not remit but gradually progress and worsen from the onset

► *Assessment of efficacy and tolerability of cannabinoids in MS* [JAMA October 2018](#)

This latest systematic review and meta-analysis did its best to assess the current state of play with the evidence base for the use of cannabinoids in MS. Currently, there is only one licensed cannabinoid product in the UK for use in spasms with MS (Sativex) but this is not recommended by NICE (see below) due to lack of convincing data and its exorbitant cost. However, cannabinoids are commonly used by patients with MS - a [survey in 2014 by the MS Society](#) suggests 1:5 do so. So does this latest study help us?

- The review included 17 RCTs including 3161 patients with MS and compared cannabinoid products to placebo for a range of symptoms and adverse events
- The data was of low quality, with significant heterogeneity and lack of objective endpoints
- What evidence there was suggested possible small subjective improvements in pain, spasticity and bladder dysfunction, but at a cost of increased adverse events (although no serious adverse events)

This data is by no means convincing due to the low quality of the original studies and so **we really can't draw any firm conclusions yet as to whether cannabinoids do help MS symptoms or not**, and as the linked editorial suggests '[Benefits to patients still unclear](#)' and that high-quality studies '*are still required to inform on the benefits of cannabinoids for patients with MS.*'

► *Sativex for spasticity in MS* [NICE NG144 November 2019](#)

Previous NICE guidance recommended against the use of Sativex spray (a THC:CBD oromucosal spray) for spasticity in MS, however as part of this new guideline, alongside a new economic evaluation, they have reversed this recommendation. Although the quality of the evidence was low, taken in conjunction with the new economic modelling (offsetting drug cost with reduced management costs), NICE now recommend:

- A 4-week trial of Sativex for adults with MS and moderate-severe spasticity that have not responded to other pharmacological treatments for spasticity, and to continue only if $\geq 20\%$ improvement in symptoms.
- Importantly **prescribing should be both initiated, supervised and doses adjusted by a specialist**, although under shared care agreements prescribing may be handed over to GPs

KISS: Diagnosis of Multiple Sclerosis (MS)

[NICE NG220 June 2022](#) [BMJ 2016;354:i3518](#) [CKS 2020](#)

Common clinical presentations/features:

- Loss or reduction of vision in one eye with painful eye movements - If patients have had an isolated episode of optic neuritis refer to a neurologist.
- Diplopia.
- Ascending sensory disturbance and/or weakness.
- Progressive difficulties with balance or gait.
- Lhermitte's symptom: sensory symptoms travelling down the back and into the limbs on neck flexion.
- Typically presents <50 years of age, most commonly in the late 20s, and may have had previous neurological symptoms.
- Evolution of symptoms: Typically evolve over 24 hours, persist over several days or weeks and then improve and are not accounted for by fever or infection.

Do not routinely suspect MS if:

- The main symptoms are fatigue, depression or dizziness, vague sensory phenomena unless focal neurological features present.

Blood tests before referral (to exclude other diagnoses):

- FBC, ESR/CRP, LFT, U&E, calcium, glucose, thyroid function, B12, HIV serology.

Referral/diagnosis:

- All suspected cases of MS should be referred to a consultant neurologist.
- Only a consultant neurologist should diagnose MS, based on the McDonald criteria.
- Diagnosis of MS should NOT be based on MRI findings alone (but there are suggestions that asymptomatic MRI findings typical of MS could present a window of opportunity for early drug therapy to reduce long-term disability, but this is controversial).

Information and support:

- Information and support is obviously really important for patients with MS throughout their life, but particularly in the early stages.
- Patients should have a **single point of contact** to coordinate care and help access to services and should be someone with knowledge of MS services and who has expertise in managing MS.
- There is lots of excellent patient support and information through the [MS Society UK](#) and the [MS trust](#)
- Driving - **all patients with MS must inform the DVLA** - [click here for the patient link](#)

KISS: Management of Multiple Sclerosis (MS) Symptoms

[NICE NG220 June 2022](#) [BMJ 2016;354:i3518](#)

- **Exercise:** Exercise is a crucial part of managing MS - a [review in 2015](#) ([Ther Adv Neurol Disord. 2015 May; 8\(3\)](#)) concluded it improves symptoms, can prevent complications and associated comorbidities and may be neuroprotective; it concluded that both aerobic and resistance exercises should be recommended and that exercise is safe and well-tolerated. [Click here for info from MS society.](#)
- **Fatigue:**
 - Don't assume fatigue is always caused by MS - consider sleep problems, pain, spasticity, bladder dysfunction, medication side effects, other illnesses eg infection/anaemia/thyroid.
 - Advise balance, aerobic and stretching exercises including yoga.
 - Assess for emotional stress, anxiety/depression - consider mindfulness and CBT.
 - Drug options for fatigue should be specialist-led - options include amantadine, modafinil, SSRIs; Do not use Vitamin B12.
- **Mobility and balance problems:**
 - Consider supervised exercise programmes, exercise referral schemes, physio or rehab specialists, but importantly support ongoing exercise after programmes finished; If balance problems are present consider vestibular rehab ([click here for free internet-based balance retraining programme](#))
- **Spasticity:**
 - Encourage patients to self-manage their spasticity and adjust drug doses accordingly.
 - Consider and **treat triggers** - infections, bladder/bowel dysfunction, poor posture, pain.
 - **Drugs** - baclofen (1st line) or gabapentin (2nd line) or a combination of those 2 - increase doses slowly over ≥ 2 -week increments. NB be aware of recent [MHRA advice October 2017](#) on gabapentin - risk of severe respiratory depression in higher-risk individuals inc. neurological disease and other CNS drugs (many with MS) as well as risks of abuse and dependence ([MHRA April 2019](#)).
 - If baclofen +/- gabapentin don't adequately control spasticity refer to MDT for further treatment.
 - **Sativex** (THC:CBD oromucosal spray) IS now an option under new [NICE guidance Nov 2019](#) for those with moderate-severe spasticity not responding to other pharmacological treatments - **Initiation, monitoring and dose alteration need to be done by a specialist**, but under shared care agreements prescribing can be handed over to GPs.
- **Pain:** Assess for and treat **MSK pain** - commonly secondary to problems with mobility, spasticity and posture. Treat **neuropathic pain** as per [NICE CG173](#) (noting concerns above over gabapentinoids).
- **Psychological problems:**
 - **Emotional lability** - consider amitriptyline
 - **Cognitive problems** - consider anxiety, depression and fatigue as causes and treat appropriately; consider referring to OT and neuropsychologist if ongoing.
- **Oscillopsia:** The subjective sensation of horizontal and/or vertical movement of the visual field; consider gabapentin (1st line) or memantine (2nd line); refer if not improving.
- **Other treatments:**
 - **Vitamin D?** NOT recommended; there is an association between low Vit D and MS, but there is no evidence treating with Vit D improves symptoms as per latest [Cochrane review September 2018](#).
 - **Statins?** A [meta-analysis](#) ([CNS Drugs. 2015 Apr;29\(4\):277-91](#)) concludes **statins may be beneficial for secondary progressive MS** which could provide a novel and cost-effective treatment; a phase 3 trial was launched summer 2017. [Click here for info from MSSoc](#)
 - **Disease-modifying therapies** (DMTs) will be specialist led/prescribed ([click here](#) for NICE summary).
 - Good info and support on DMTs from [MSSociety](#) and [MS Trust](#)

KISS: Managing Relapse in Multiple Sclerosis (MS)

[NICE NG 220 June 2022](#) [BMJ 2015;350:h1765](#)

■ **Is this a relapse?**

- New symptoms, or worsening of existing symptoms, after a stable period of > 1 month, and typically develop over hours/days and improve over days/weeks (must last for ≥24 hours).
- Most relapses (~70%) are monofocal (i.e. involve one site):
 - the commonest relapse phenotype is sensory (48%)
 - followed by weakness (34%) and
 - problems with visual acuity (20%).
- Differentiate between **true relapse** and **other symptom changes** (often difficult):
 - **Pseudo-relapse** - exacerbation of existing symptoms (i.e. *not* new symptoms) triggered by heat or systemic inflammation, typically transient.
 - **Day-to-day fluctuations** - often stereotypical in nature and disproportionately reported when considering disability progression.
 - **Functional relapse** - common in general practice, challenging to differentiate and functional overlay can also complicate true relapses.

■ **Rule out infection** before considering treatment:

- commonest sources UTI and respiratory.

■ **ALL relapses should be discussed with a specialist before treatment:**

- Not all relapses need treating and can be difficult to diagnose in primary care.
- Even if not treated relapse rate may affect disease-modifying treatment.
- Ideally, assessment/treatment for relapse should be done within 14 days of onset.

■ **Treatment** if appropriate is high-dose steroids:

- Either oral methylprednisolone 0.5mg (500mg) OD 5 days or
- IV methylprednisolone 1g daily for 3-5 days (if oral not tolerated or they need to be admitted).
- Be aware that high-dose steroids may cause cognitive or other mental health side effects eg insomnia, depression, confusion, and agitation.
- Do NOT give people with MS a supply of steroids to self-administer at home for future relapses.

ESSENTIAL TREMOR

Although often thought of as being entirely benign, **essential tremor is a disabling, progressive neurological disorder**. It is the most common movement disorder, with an estimated prevalence of up to 5% in older people. It causes anxiety and embarrassment and although most people are only mildly affected it can cause significant disability. The evidence has been recently reviewed in [BMJ Best Practice 2019](#), and the recent NICE guideline [NICE 2019, NG127](#) on recognition and management of suspected neurological conditions gives guidance on referral.

What is essential tremor (ET)?

ET is a persistent bilateral tremor of the hands and forearms or an isolated tremor of the head, without abnormal posturing and with no evidence that it arises from another cause. The diagnosis cannot be made if there are abnormal neurological signs.

ET is one of the most common movement disorders with an estimated overall prevalence from 1% to 2%, rising to 5% in people aged over 65. Men and women are equally affected. It usually begins during early adulthood and slowly progresses, but it can come on in later life. There is often a family history and in 60% of people, it is inherited as an autosomal dominant. In these patients, it is more likely to come on at an earlier age, and in sporadic cases in later life.

KISS: Assessment of Tremor

[BMJ Best Practice 2019](#) & [NICE 2019, NG127](#) & [Patient Plus: Tremor](#)

- Essential tremor is usually bilateral and does NOT involve muscle tone or speed of movement
- Parkinson's tremor is usually asymmetrical (unilateral or predominantly unilateral) and more prominent at rest, and associated with the slowness of movement, stiffness or balance problems
- **Examine** patients with hands in lap (rest or static tremor), outstretched (postural tremor) and then 'finger-nose' (kinetic or intention tremor)
- **Rest or static tremor** occurs in a relaxed, supported limb at rest
 - suggests Parkinson's Disease (often unilateral at onset and improves with intention), Parkinsonism, other extrapyramidal disease or multiple sclerosis
- **Postural tremor** when the body part is held in a fixed position against gravity (e.g. arms outstretched)
 - ET is usually a distal, symmetrical postural tremor of the upper limbs, of low amplitude but a rapid frequency.
 - As well as essential tremor, this type is also seen with physiological tremor, anxiety, thyrotoxicosis, medication side effect (especially antidepressants) and alcohol. It is also seen in peripheral neuropathy
- **Kinetic or intention tremor** occurs during voluntary movement, which worsens as the target is approached, indicates a cerebellar cause e.g. brainstem or cerebellar disease, MS, vascular disease etc

What is the clinical significance of ET?

Often misunderstood, ET is a clinically significant movement disorder. Although it is a benign condition in terms of life expectancy, it is persistent and slowly progressive and may cause significant embarrassment and disability. Although most people are only mildly affected, it can be physically and psychologically disabling; it interferes with essential activities, such as writing and using a computer, which makes work and day-to-day life difficult.

What does the evidence say about treatments?

Reassurance and explanation remain of course the mainstay of treatment, as well as guidance about avoiding exacerbating factors (NB including drugs, such as long-acting beta-agonists or SSRIs). If drugs are needed, there is evidence for the drugs in the KISS box below.

So, when should we refer? [NICE 2019, NG127](#)

The 2019 guidance states

- suspect essential tremor in adults with *symmetrical postural tremor* with no features of Parkinsonism
- in adults with suspected ET review medication, check thyroid function, assess alcohol consumption and refer only if they have disabling symptoms and first line drug treatments are ineffective
- consider referral if there is troublesome head tremor to a movement disorder clinic

KISS: Drug Management of Essential Tremor

Based on [BMJ Best Practice 2019](#) and [Cochrane 2015 CD009681](#)

- **First-line: Beta-blockers**
 - Propranolol effective compared to placebo, insufficient evidence to judge the efficacy of other beta-blockers e.g. metoprolol, atenolol etc
- **Second-line option: Anti-epileptic drugs**
 - Primidone and topiramate have evidence of short-term efficacy compared to placebo
 - We do not know if other AEDs, such as gabapentin or levetiracetam, are effective
- **Third line: Benzodiazepines**
 - Alprazolam, very limited evidence from one small study that reduces tremor but at high risk of side effects such as sedation. Cochrane conclude: 'evidence insufficient for assessment of relative benefits and harms'. Similarly for clonazepam, and a lack of evidence for diazepam and lorazepam
- **Specialist option:**
 - Limited evidence that botox injections improve symptoms scores, but at a cost of hand weakness. A specialist option for severely affected cases.
 - There is fascinating emerging evidence for non-invasive neurosurgery, using focused ultrasound thalamotomy [NEJM2016](#)

NB Learning Points: Essential Tremor

- Essential tremor is the most common movement disorder. Most cases are genetic and it is autosomal dominant.
- Although benign in terms of life expectancy, it is NOT clinically insignificant, is progressive and can cause considerable morbidity
- Management consists of reassurance and explanation with appropriate lifestyle advice, including avoiding exacerbating factors [NHS Choices: Essential Tremor](#)
- Review exacerbating drugs e.g. beta-agonists and SSRIs
- If drugs are needed, propranolol is the first choice. If this is ineffective, contraindicated or not tolerated consider topiramate or primidone. Referral for consideration of botox is also an option in difficult cases.

NEUROPATHIC PAIN

NICE updated their *neuropathic pain guideline in 2020* [NICE CG173](#). The 2020 update of the guideline brings it into line with the NICE guideline on back pain and sciatica [NICE NG59](#), with a very clear recommendation ***NOT to prescribe gabapentinoids or other antiepileptics for sciatica as there is no overall evidence of benefit and there is evidence of harm.***

Neuropathic pain affects 16% to 26% of people with diabetes and 8-19% of people post herpes zoster. Chronic neuropathic pain after surgery is being increasingly recognised and may be present in as many as 10-50% of people. The guidelines emphasise the importance of regular review, dose titration of drugs, assessing severity and impact of function and strongly encourages non-pharmaceutical approaches such as psychological and physical therapies. ***For patients with painful diabetic neuropathy, we have new evidence for combination therapy from the OPTION-DM trial.***

► ***Combination therapy for patients with painful diabetic neuropathy*** [OPTION-DM Trial, Lancet 2022](#)

- Diabetic peripheral neuropathy (DPNP) affects up to 50% of people with diabetes and about half of these present to us with neuropathic pain
- In line with NICE guideline, see KISS below, first line agents are amitriptyline, duloxetine, gabapentin or pregabalin; combinations are not recommended by NICE
- Unfortunately only a third of patients get a good response to monotherapy; this trial set out to see if a combination of these drugs may be beneficial if patients do not respond to single agents
- They looked at patients (n=140) who had daily pain that they rated at least 4/10 in severity, who had not responded to monotherapy after 6 weeks of treatment; they were then offered combination therapy with a supplementary drug for 16 weeks
- The options were amitriptyline then supplemented with pregabalin, pregabalin then supplemented with amitriptyline and duloxetine supplemented with pregabalin
- At the end of the 16 weeks, the average pain scores were reduced from 6.6 to 3.3 in all 3 combination groups - a 50% reduction in pain score is considered clinically significant
- **Conclusion:** combination therapy was well tolerated and led to improved pain relief in patients with suboptimal pain control compared with a monotherapy

NB Tips for patients based on [BMJ 2020;369:m1315](#)

- There are no 'magic bullets' for treating neuropathic pain.
- The aim is to get your pain 50% better (i.e. if you currently rate it as 8/10, a reduction in pain to 4/10 will be a good result) or to give you enough pain control to help you sleep, work and play.
- There is probably a drug which will work well for you, but people have very individual responses, so you may have to try a few before you find the one which works best for you in the long term.
- You may need to take these drugs for as long as 6-8 weeks before noticing any improvement.
- Do not stop the drugs abruptly, you may get side effects.

NB Learning Points - Neuropathic Pain

- We need to adopt a flexible approach; Amitriptyline, Duloxetine, Gabapentin & Pregabalin are all appropriate first-line choices
- Do not use gabapentinoids for sciatica or spinal stenosis.
- Warn patients that pain will not disappear; use rating scales and aim to achieve a 50% reduction and functional 'quality of life' improvement.
- The drugs take at least 2 to 4 weeks to have any beneficial effect (6-8 weeks for gabapentinoids).
- Slowly titrate up every 2 to 4 weeks; if no benefit at therapeutic dose, switch drugs but taper the withdrawn drug slowly (esp with gabapentinoids) to reduce risk of withdrawal side effects.
- Opioids should NOT be used.

KISS: Neuropathic Pain

[NICE CG173, 2020](#) & [CKS 2020 SIGN 136 2019](#)

[Lancet Neurology 2015;14;162 \(1a\)](#) & [BMJ 2020;369:m1315](#)

- For trigeminal neuralgia, offer carbamazepine as initial treatment.
- For sciatica, see separate sciatica guideline [NICE NG59](#), but do NOT offer gabapentinoids, other antiepileptics or benzodiazepines for sciatica.
- For all other types of neuropathic pain offer a choice of the following as initial treatment, taking into account levels/impact of pain, as well as benefits AND adverse effects of proposed medications:
 - **Amitriptyline** (25mg to 125 mg daily) (A) - Start at 10-25mg 2 hours before bed.
 - **Duloxetine** (60-120mg daily) (A) - Start at 60mg, 30mg in elderly.
 - **Gabapentin** (1200mg daily) (A) - Start at 300mg in the evening, 100mg in the elderly.
 - **Pregabalin** (300-600mg daily) (A) - Start at 75mg in the evening, 25mg in the elderly.
- If initial treatment is ineffective or not tolerated, offer one of the remaining 3 drugs and consider switching again if necessary
- **Gabapentinoids:**
 - Before prescribing these drugs **carefully assess for those with evidence of, or at risk of, drug abuse or dependence behaviour**
 - Do not prescribe for sciatica or spinal stenosis; we have good evidence these drugs do not work ([BMJ 2020;369:m1315](#)) for these indications.
- Assess after 2 to 4 weeks, and titrate the dose.
- Warn patients that initial side effects will improve with time and they will take 2 to 4 weeks to work
- If ineffective once at therapeutic dose, or not tolerated, switch to an alternative, but taper the withdrawn drug to reduce discontinuation symptoms (particularly common with gabapentinoids).
- **Combination therapies?** Not recommended by NICE due to lack of evidence, although the GDC accepted that combination low doses may improve tolerability, and [SIGN](#) recommend that combination therapies should be considered.
- Consider tramadol for short-term rescue therapy only, or while awaiting referral, but prescribe cautiously and watch for signs of misuse.
- Consider **capsaicin cream (0.075%)** for localised neuropathic pain if wish to avoid oral therapy.
- Consider **capsaicin patches (8%)** for peripheral neuropathic pain if first-line therapies have been ineffective or not tolerated (A). Not recommended by NICE but they are by SIGN.
- **Lidocaine patches?** Controversial (and very expensive!); little evidence available apart from use in post-herpetic neuralgia (PHN); NHSE state they should not be routinely prescribed in primary care, but the British Pain Society recommend they be considered if recommended by specialist pain clinics; SIGN recommend them as an option for PHN if first-line agents ineffective.
- Do NOT use morphine.
- Refer to specialist pain clinic and/or relevant specialist (e.g. neurologist, diabetologist) if pain levels are severe or significantly affecting ADLs.

Non-pharmacological therapies

- Supported self-management is recommended in addition to pharmacological therapies.
- Physical and psychological therapies should also be considered:
- Progressive relaxation exercises (C) - NB Numerous online tools are available, this one is 30 minutes recording and particularly good [Progressive relaxation audio recording from NHS Cov & Warwick](#).
- CBT (C) manual therapies (B), exercise (A) and TENS (B).

CHRONIC PAIN

In 2021 NICE published their first-ever guideline on chronic primary pain [Chronic Pain 2021](#)

Controversial and radical, it was discussed in a BJGP editorial [BJGP2020;70;576](#) and summarised in the [BMJ BMJ2021;373:n895](#). We have also had new evidence published on antidepressants for musculoskeletal pain [BMJ2021;372:m4825](#), with an accompanying editorial on their evidence of lack [BMJ2021;372;n80](#), on gabapentin for chronic pelvic pain in women [Lancet2020;396;909](#) and a Drug Safety Update on Opioids [Drug Safety Update Sept 2020](#).

The NICE guideline is a radical and practice-changing guideline that really hit the headlines with the key recommendation to avoid *all* analgesics but to consider non-pharmacological approaches such as psychological therapy supervised exercise programmes and acupuncture and if drugs are to be prescribed then to consider antidepressants.

Many GPs and pain specialists may criticise NICE for an 'ivory tower' approach in their recommendations which seem far removed from the reality of practice, and do not take into account the complexity of clinical care and the lack of service provision with limited access to non-pharmacological treatments. Others may praise them for a bold, evidence-based approach that will draw a line in the sand and henceforth make it easier for GPs not to prescribe drugs that they have been unwilling to, to open up a more honest dialogue with patients about the limitations and dangers of prescribing and to help lever organisational change to make non-pharmacological methods of treatment more available and accessible to us in primary care. A personal NB view is that it is disappointing that NICE have stuck with the term 'chronic pain' which is so often misunderstood by patients as meaning 'really bad' which can re-enforce a dysfunctional core belief. In our view, long-term or persistent pain would have been better...

Nobody knows better than GPs the huge problem of chronic pain and the secondary dependency on prescription opioids and gabapentinoids. In primary care, we have been aware of this issue for many years now, and evidence has been accumulating of the inefficacy and potential harm of commonly used treatments such as opioids and also of the risk of addiction. However, it was given new impetus by a *report from Public Health England in late 2019 on prescribed medicines* [PHE Prescribed Medicines review 2019](#). This report was a public health evidence review based on prescription data and found that:

- *A staggering 1 in 4 people in England are taking prescription medicines with potential for dependence and withdrawal such as opioids, sleeping pills, gabapentinoids and antidepressants*
- The report stresses that whilst benzodiazepines, z-drugs, opioids and gabapentinoids are associated with a risk of dependence and withdrawal, antidepressants are associated with withdrawal reactions
- Prescribing rates are strongly linked with areas of socio-economic deprivation
- The report concludes with a series of recommendations including making information on medicines that can cause dependence more transparent and accessible
- NICE is developing a [separate guideline on safe prescribing and withdrawal management of prescribed drugs](#) associated with dependence and withdrawal which is due in 2021

A recent [Drug Safety Update Sept 2020](#) reminds us of the *risks of dependence and addiction* with opioids, even at therapeutic doses, and that all patients need to be warned of this before prescribing and there should be a clear agreed plan for when treatment will end. This advice applies to tramadol, codeine and dihydrocodeine as well as to the stronger opioids such as morphine and oxycodone. It has been recommended that the packaging of all these drugs will come with 'contains opioid and can cause addiction' warnings on the labels. Patients should also be warned of the risk of *withdrawal* reactions, and the need for graded withdrawal and dose tapering.

The paper also discusses the problem of *opioid-induced hyperalgesia*. Some patients develop this increased sensitivity to pain with long-term use of opioids. This pain might be quite different from the original underlying pain, but maybe easy to confuse with breakthrough pain leading to an inappropriate escalation of dose. Opioid hyperalgesia resolves with a gradual reduction in dose.

► *Chronic Pain* [NICE 2021, NG193](#)

The guideline refers to *the assessment of chronic primary and secondary pain in the over 16s, but to the management of chronic primary pain*. This is a crucial point in understanding the key recommendation not to prescribe analgesia.

Chronic primary pain is a syndrome defined by the WHO in [ICD-11](#). This NICE guideline is therefore separate to other guidelines on chronic secondary pain due to underlying specific conditions that cause pain such as neuropathic pain, headache, low back pain & sciatica, arthritis, endometriosis etc. This is a crucial point, emphasised in the BJGP editorial [BJGP 2020;70;576](#) and also the DTB [DTB2020;58;162](#). Public reaction has been that this guideline refers to all chronic pain conditions, and this is not the case.

Chronic primary pain ICD-11 is defined as a *pain that cannot be accounted for by another diagnosis*, that persists or recurs for more than 3 months in one or more anatomical regions and is characterised by significant emotional distress (anxiety, anger/frustration or depression) or functional disability. It is recognised as a pain syndrome in its own right, with biological, psychological and social contributory factors. It is separate from *secondary* chronic pain which can be accounted for by another diagnosis such as low back pain and sciatica. Examples include widespread chronic musculoskeletal pain, fibromyalgia and chronic pelvic pain.

However, as the DTB editorial [DTB2020;58;162](#) points out, chronic *primary* pain is a diagnosis that leaves as many questions as answers. For example, many patients will have some underlying conditions (e.g. degenerative changes in the spine, history of migraine etc). *NICE state that if there is a secondary cause of the pain, but the impact of the pain seems out of proportion to the underlying condition to consider whether it should be better managed as chronic primary pain.*

NICE estimate that chronic pain may affect 1 in 3 adults. Almost half of these people have a diagnosis of depression and two-thirds are unable to work because of it. The main headline is that patients should NOT be prescribed paracetamol, NSAIDs, benzodiazepines or opioids for chronic primary pain even for short-term use, based on evidence that there is little or no evidence that they improve pain, psychological distress or quality of life and that there is evidence of long-term harm including dependency.

Key recommendations:

- Patients should receive personalised, holistic care based on good communication and a collaborative, supportive relationship between patient and health professional and have care plans
- **Prescribing**
 - *Patients should NOT be prescribed paracetamol, NSAIDs, benzodiazepines, antiepileptic drugs (including gabapentinoids), opioids, steroids, cannabis-based products or local anaesthetics*
 - Crucially no evidence was identified on the effectiveness of opioids for chronic primary pain but evidence of dependency was identified
 - Even short-term use of opioids may be harmful and therefore NICE recommend against opioids for chronic pain even for short-term use
 - If patients are already taking any of these drugs, explain the risks of continuing and reach a shared decision
 - If a shared decision is made to stop antidepressants, opioids, gabapentinoids or benzodiazepines be aware of the problems associated with withdrawal

- *Antidepressants can be considered for chronic pain* (NB but note new evidence below)
 - Consider duloxetine, SSRI (fluoxetine, paroxetine, citalopram or sertraline) or amitriptyline for people aged 16 or over with chronic pain after a full discussion of the risks and benefits
 - Note that this is an off-label use of antidepressants
 - NICE state that evidence indicated that antidepressants improve pain, quality of life and psychological distress compared to placebo. Most of the evidence related to women with fibromyalgia however NICE felt the response to treatment would be sufficiently similar to allow recommendations across all primary pain conditions.
 - Duloxetine is the only licensed option for chronic primary pain and has a larger evidence base, but NICE state that due to a lack of head-to-head comparisons between different antidepressants duloxetine could not be recommended in preference to other antidepressants

■ Non-pharmacological treatments

- The doctor-patient relationship and shared decision making is crucial
- Acupuncture is recommended as an option for some patients, delivered in a community setting
 - This can be in a Chinese or Western medical approach
 - NICE state that *'there is a large evidence base showing acupuncture to be clinically effective in the short term (3 months) but there is uncertainty over long-term benefits'*
 - To achieve cost-effectiveness it is recommended to a maximum of 5 sessions
- Patients should be encouraged to exercise and be offered, supervised group exercise programme
- Consider psychological therapy including cognitive behavioural therapy (CBT) and acceptance and commitment therapy (ACT)
- Do not offer electrical (TENS) therapy or ultrasound
- There is insufficient evidence for pain management programmes or manual therapy
- There was insufficient evidence for relaxation therapy or mindfulness, but what evidence there was suggested some benefit and a recommendation is made for future research

NB comment: *This guideline has huge implications for GPs.* As [BJGP2020;70;576](#) points out, weaning patients off these drugs will be a major challenge for primary care. The lack of non-pharmacological options and services often mean that prescribing is the only option that GPs can offer patients, and furthermore prescribing is often initiated or driven by specialist pain clinics. Hopefully, the guideline will drive service change, but in the meantime, it highlights the need for a more honest dialogue with patients about expectations from treatment. As every GP knows, managing patient expectations is vital to improve outcomes and hopefully, this guideline will make it easier to do so. In the meantime, we need greater access to non-pharmacological approaches including psychological therapies, exercise programmes and also social prescribing.

As the DTB editorial [DTB2020;58;162](#) states, this guideline will need a complete change of dialogue with our patients as fundamental as the change in narrative around antibiotics for minor self-limiting infections and benzodiazepines for insomnia. We will need a *'shift from a passive paternalistic program of pain management to one that focuses on function and patient's priorities'* with a willingness to reduce patient expectations for analgesia.

► Can we halt the rise in prescribing for pain and distress? [BJGP2020;70;432](#)

This was a really excellent editorial discussing the increase in prescriptions for chronic pain and distress. It articulates what we all know from the front line: psychoactive drugs are often prescribed to try to help symptoms of human distress which are caused by complex psychosocial factors. This has led to the medicalisation of human distress; it may give some short-term relief via the anxiolytic and euphoric effects of drugs such as gabapentinoids and opioids but it leads to longer-term iatrogenic harm.

Over the last decade there have been significant increases in the prescription of antidepressants and gabapentinoids, and although prescriptions for opioids and benzodiazepines have started to fall prescribing remains high. The editorial discusses what we are very familiar with, that social hardship, adverse childhood experiences, chronic pain and human distress are all inexorably linked. Patients in distress seek relief and prescribing 'has become the norm in primary care'.

Can we halt this? The editorial points out that in the past we have succeeded in reducing prescriptions for amphetamines and benzodiazepines and more recently with antibiotics and antipsychotics in people with dementia. But to achieve this it is clear that the nature of the consultation needs to change so that we gain a greater understanding of the complex social and emotional factor leading to an individuals' pain and distress. We also need greater access to a wider, more comprehensive integrated service including social prescribing and psychological support for patients. **The editorial concludes:** *'in order to put a halt to what in 20 years might be seen as the age of mass iatrogenic poisoning of the brain, we need to substantially reduce prescribing and increase deprescribing now'.*

► ***Antidepressants for musculoskeletal pain*** [BMJ2021;372:m4825](#)

So, the NICE guideline says to avoid all analgesics for chronic primary pain but that if a prescription is necessary antidepressants may be considered.

In 2021 the BMJ have published this new systematic review of randomised trials looking at the efficacy of antidepressants compared to placebo in patients with chronic MSK pain including low back and neck pain and osteoarthritis. Conclusions:

- For back pain, there was moderate certainty evidence that the effect of SNRIs (duloxetine) was small and 'not clinically important'
- TCAs and SNRIs 'might' be effective for sciatica, but the certainty of the evidence was low

An accompanying editorial [BMJ2021;372:n80](#) discusses the uncertainty in the evidence base but has a pithy conclusion: *'overall drug treatments are largely ineffective for back pain and osteoarthritis and have the potential for serious harm'.*

► ***Gabapentin for chronic pelvic pain in women*** [Lancet 2020;396:909](#)

This was an excellent, randomised double-blind placebo-controlled trial looking at women with chronic primary pelvic pain. As chronic pelvic pain is so common, this is really important research. Up to 24% of women may experience chronic pelvic pain, and although it can, of course, be caused by serious pelvic pathology such as endometriosis in 55% of women no cause is found at laparoscopy.

- The trial included 306 women aged 18 to 50 with significant pelvic pain for at least 3 months and a laparoscopy with no obvious pelvic pathology.
- They were randomised to gabapentin (titrated to a maximum of 2700mg daily) or placebo for 16 weeks
- **Results**
 - Treatment with gabapentin did NOT result in lower pain scores than placebo
 - Treatment with gabapentin resulted in a higher rate of side effects than placebo; 10% of women in the gabapentin group had a serious adverse event compared to 2% in the placebo group
- **Conclusion:** gabapentin is ineffective for chronic primary pelvic pain and 10% have severe side effects

The authors conclude that this is a definitive study and that no further research is required looking at gabapentin in chronic pelvic pain. It doesn't work.

For the potential role of medical cannabis and chronic pain and a discussion around recent evidence, please see the Medical Cannabis chapter in the Prescribing section.

NB Learning Points & Quality Improvement Ideas

- Explain that doctors are now advised NOT to prescribe analgesic drugs including paracetamol, NSAIDs, opioids and gabapentinoids for chronic primary pain due to a lack of benefit and risk of harm
- Offer alternatives such as exercise programmes, acupuncture and talking therapy
- Use your social prescriber, exercise on referral schemes and refer for psychological therapies
- If a prescription is indicated, consider an antidepressant in preference to analgesics
- **Quality Improvement Idea**
 - Change to in-house prescribing guidelines based on this guidance
 - Audit of patients on long-term opioids and offer of guided withdrawal
 - Audit of patients on long-term paracetamol and opioids for chronic pain
 - The Sheffield Persistent Pain site has a superb set of resources and information to help patients to live better lives [Sheffield Persistent Pain](#), and [Live Well With Pain](#) has great resources for clinicians and patients.
 - See also [Tame the Beast](#) - an excellent set of resources for patients and clinicians which helps re-frame the discussion around persistent pain and how we all need to think about it differently.

KISS: Chronic Primary Pain

Based on [NICE 2021, NG193](#)

Chronic primary pain is defined as pain that persists or recurs for more than 3 months in one or more anatomical regions and is characterised by significant emotional distress or functional disability.

The guideline is separate to other guidelines on chronic *secondary* pain due to underlying conditions that cause pain such as neuropathic pain, headache, low back pain & sciatica, arthritis, endometriosis etc and which have separate guidelines.

► Assessment

- Good communication and a collaborative and supportive relationship between health professional and patient is the key to success
- Develop trust, and understand the patient's experience of pain, it's impact on their life and function including on physical and psychological well-being including sleep, work, relationships and family life and to manage expectations
- Ask about patient's understanding of what causes their pain, their acceptance of it and their expectations about the outcome of treatments
- Validate patients' experience of pain and offer ongoing support
- Communicate sensitively but be honest, help to set realistic expectations and develop a care plan

► Management

- **Non-pharmacological treatments**
 - Acupuncture is recommended as an option for some patients, delivered in a community setting
 - Patients should be offered supervised group exercise programme and encouraged to exercise generally
 - Consider psychological therapy including cognitive behavioural therapy (CBT) and acceptance and commitment therapy (ACT)
- **Prescribing**
 - *Patients should NOT be prescribed paracetamol, NSAIDs, benzodiazepines, antiepileptic drugs including gabapentinoids, opioids, steroids or local anaesthetics*
 - If patients are already taking any of these drugs, explain the risks of continuing and reach a shared decision
 - If a shared decision is made to stop antidepressants, opioids, gabapentinoids or benzodiazepines be aware of the problems associated with withdrawal
 - *Antidepressants can be considered for chronic pain*
 - Consider duloxetine, SSRI (fluoxetine, paroxetine, citalopram or sertraline) or amitriptyline for people aged 16 or over with chronic pain after a full discussion of the risks and benefits
 - NICE state that evidence indicated that antidepressants improve pain, quality of life and psychological distress compared to placebo.
 - Note that this is an off-label use of antidepressants
 - There is no recommendation to favour duloxetine (the only licensed option) over other antidepressants due to lack of head-to-head trials, tailor treatment choice to your patient

GABAPENTINOIDS FOR PAIN

The use (and misuse) of gabapentinoids has been a really hot topic over the past few years. Primarily licensed as anticonvulsants, over the past decade they have been predominantly used for pain. Aggressive marketing and [fierce protection of a secondary patent \(ultimately dismissed by the Supreme Court\)](#) meant that pregabalin was one of the highest-selling drugs globally in 2017. In 2018 more than 14 million prescriptions for pregabalin and gabapentin were issued in England alone. Evidence for their use in specific neuropathic pain conditions, such as diabetic neuropathy, filtered out into much more widespread use in other 'neuropathic' pain conditions, driven by national guidelines when evidence was sparse. Compounded by a desire to avoid the use of opiates in chronic pain, and lack of access to chronic pain clinics, clinicians across the board have rapidly increased prescribing of these drugs, despite mounting evidence on their lack of effect, and side effect profile.

The concerns and risks of gabapentinoids were brought into sharp focus by a [government review](#) into the misuse of these drugs, culminating in a change in the [classification of these drugs to a class C controlled drug](#) in 2019. We deal with the risks, including misuse and addiction, risks in our prescribing section, but here we summarise the evidence (or lack thereof) for gabapentinoid use in pain, which was reviewed in a [fantastic therapeutics article in the BMJ in April 2020 \(BMJ 2020;369:m1315\)](#).

► *Remind me how gabapentinoids work and how I prescribe them...*

Gabapentinoids bind to the α 2-delta subunit of voltage-gated calcium channels, which decreases the release of glutamate, noradrenaline, and substance P. This is believed to contribute to their anticonvulsant, analgesic, and anxiolytic actions.

- Titrate doses every 3-7 days up to maximum 600mg/day in 2 divided doses for pregabalin, and maximum 3600mg/day in 3 divided doses for gabapentin.
- Once at maximum tolerated dose, a **trial for 6-8 weeks is needed to assess if the benefits outweigh the harms**.
- Consider stopping at this stage if no benefits (or harms outweighing benefits), but **do not discontinue abruptly** otherwise withdrawal effects (agitation, dysphoria, fatigue) are common. Reduce by 50-100mg of the daily dose every week for pregabalin and maximum of 300mg of the daily dose per week for gabapentin.

► *So what conditions do gabapentinoids help with?*

We have **moderate-quality evidence that gabapentinoids are effective for people with moderate to severe post-herpetic neuralgia or moderate to severe diabetic peripheral neuropathy**. It is important to highlight that these benefits are for moderate to severe pain, and can't be translated to people with milder pain levels. Based on Cochrane review evidence:

- **Pregabalin** (600mg/day for ≥ 8 weeks) 41/100 people will have $\geq 50\%$ pain relief vs placebo (15/100 with post-herpetic neuralgia, 28/100 with diabetic neuropathy).
- **Gabapentin** (1200mg/day for ≥ 8 weeks) 32/100 will have $\geq 50\%$ pain relief for post-herpetic neuralgia vs placebo (17/100).
- **Gabapentin** (1200mg/day for ≥ 8 weeks) 38/100 will have $\geq 50\%$ pain relief for diabetic neuropathy vs placebo (21/100).
- Overall about 50% of participants experienced pain reduction of a third.

► *How do they compare to other neuropathic agents?*

This review asserts that **other first-line agents** - tricyclic antidepressants (TCAs) and SNRIs (mainly based on duloxetine data) - **have better NNTs** (NNT 3.6 for TCAs 95% CI = 3.0-4.4; NNT 6.4 for SNRIs 95% CI = 5.2-8.4) than gabapentinoids and are *'likely to be more effective'*.

► ***Sounds good, so what are the downsides?***

Well, this review I'm sure supports what we have all seen in clinical practice - although these drugs can help a proportion of people with neuropathic pain or fibromyalgia, **they have a heavy side effect burden**. Although there was a significant nocebo effect in the trials, the highlight is that **almost 2/3 of patients taking these drugs will report adverse events**. See below table for a summary, but a few specifics:

- For gabapentin **dizziness** (19% vs 6.6% placebo), **somnolence** (14% vs 5.2% placebo), and **gait disturbance** (14% vs 2.6% placebo) are commonly reported side effects.
- For pregabalin **dizziness** (24% vs 6% placebo) and **somnolence** (17% vs 5% placebo) are common.
- In the fibromyalgia reviews, a staggering 91% experienced 1 or more adverse effects (albeit with a similarly high nocebo effect of 73% in placebo groups), but this translated to a significant 28% withdrawing from the studies due to adverse effects compared to only 11% from placebo arms.
- Serious adverse events were similar with gabapentinoids compared to placebo.
- Self-reported misuse (which is likely to significantly underestimate true misuse levels) is reported in 0.5-1.1% of people; this is substantially higher in those with opiate use disorders.

Further concerns over pregabalin came in the form of data on congenital malformation risks ([DTB 2022;60:117](#) & [MHRA April 2022](#)). See the Epilepsy, contraception and pregnancy chapter for more detail but in summary, this data showed that **compared to both lamotrigine monotherapy and duloxetine monotherapy, pregabalin monotherapy showed increased risks of congenital malformations**. There were limitations, as noted by the MHRA, in that information on confounding factors was limited as was comparison of the data by pregabalin indication, but it will come as no surprise that the vast majority of women on pregabalin were not on it for anti-epileptic purposes, but mostly for anxiety and pain.

► ***So what conditions do these drugs NOT work in?***

The bottom line is that the only conditions for which we have evidence of effectiveness for these drugs is moderate to severe post-herpetic neuralgia/diabetic neuropathy and moderate to severe fibromyalgia. **We have good evidence of NO benefit in low back pain, sciatica, spinal stenosis and migraine** (i.e. evidence of lack, NOT lack of evidence). The evidence for acute pain, HIV neuropathy, neuropathic cancer pain, and other forms of neuropathic pain is insufficient (i.e. lack of evidence). One of the biggest uses for these drugs is for low back pain and sciatica but **NICE have recently updated their guideline NG59 Dec 2020** to reflect the current evidence, and now **recommend that gabapentinoids should NOT be used for sciatica or low back pain**.

► ***Gabapentin for chronic pelvic pain in women*** [Lancet 2020;396:909](#)

This was an excellent, randomised double-blind placebo-controlled trial looking at women with chronic primary pelvic pain. As chronic pelvic pain is so common, this is really important research. Up to 24% of women may experience chronic pelvic pain, and although it can, of course, be caused by serious pelvic pathology such as endometriosis in 55% of women no cause is found at laparoscopy.

- The trial included 306 women aged 18 to 50 with significant pelvic pain for at least 3 months and a laparoscopy with no obvious pelvic pathology.
- They were randomised to gabapentin (titrated to a maximum of 2700mg daily) or placebo for 16 weeks
- Results
 - Treatment with gabapentin did NOT result in lower pain scores than placebo
 - Treatment with gabapentin resulted in a higher rate of side effects than placebo; 10% of women in the gabapentin group had a serious adverse event compared to 2% in the placebo group
- Conclusion: gabapentin is ineffective for chronic primary pelvic pain and 10% have severe side effects

The authors conclude that this is a definitive study and that no further research is required looking at gabapentin in chronic primary pain. It doesn't work.

► *Summary of benefits and harms of pregabalin and gabapentin* [click here](#) for a visual summary table

Click here for Cates plot of benefits/harms	≥50% pain reduction in post-herpetic neuralgia	≥50% pain relief in diabetic neuropathy	Adverse events	Serious harms	Misuse
Pregabalin	41%	41%	69%	3.4%	0.5%
Placebo	15%	28%	57%	3.4%	0
Benefit/harm above placebo	26%	13%	12%	0	0.5%

Click here for Cates plots of benefits/harms	≥50% pain reduction in post-herpetic neuralgia	≥50% pain relief in diabetic neuropathy	Adverse events	Serious harms	Misuse
Gabapentin	32%	38%	63%	3.2%	1.1%
Placebo	17%	21%	49%	2.8%	0
Benefit/harm above placebo	15%	17%	14%	0.4% (not significant)	1.1%

NB Learning Points: Gabapentinoids for pain

- We have moderate-quality evidence that both pregabalin and gabapentin give meaningful benefits in pain scores for both moderate to severe post-herpetic neuralgia and moderate to severe diabetic neuropathy - ≥50% reduction in pain scores is seen in 13-26% more patients than those on placebo.
- However, this is at a cost of substantial levels of adverse events - 12-14% more than those on placebo will report adverse events.
- Dizziness, somnolence and gait disturbance are common adverse effects and are common causes of discontinuation of these drugs.
- We have high-quality evidence supporting the use of pregabalin in moderate to severe fibromyalgia - 30-50% reduction in pain levels is seen in 10% more patients than those on placebo.
- Titrate doses every 3-7 days and once at maximal tolerated dose, a trial of 6-8 weeks is recommended before being able to assess benefits/harms.
- Do not discontinue these drugs abruptly - there is a high risk of withdrawal side effects (agitation, dysphoria, fatigue).
- We need to be aware gabapentinoids can be used as drugs of abuse, particularly in opioid abuse, by boosting euphoric highs and reducing withdrawal symptoms - exercise extreme caution if considering these drugs in people with, or at risk of, drug misuse or dependency.
- We have good evidence gabapentinoids are NOT effective for low back pain, sciatica, migraine and spinal stenosis.
- Current evidence suggests tricyclic antidepressants and SNRIs (mainly duloxetine) may be more effective than gabapentinoids for neuropathic pain.

NB Quality Improvement Ideas: Gabapentinoids for pain

- Review all your patients on pregabalin or gabapentin for non-evidence-based indications - especially low back pain and sciatica; discuss the evidence suggesting they are ineffective for these indications and consider a slow withdrawal of the drugs.
- Before considering pregabalin or gabapentin for neuropathic pain consider either tricyclic antidepressants or duloxetine - current evidence suggests they may be more effective.
- When starting pregabalin and gabapentin warn patients that they may need to take the drugs for a minimum of 8 weeks before they notice any benefit - if not consider a slow withdrawal at this stage.
- Review all patients on gabapentinoids and opiates - are they at risk of abuse?

FIBROMYALGIA

Fibromyalgia is common (with a prevalence of around 2%), and miserable for patients and its diagnosis & management remain a major challenge for us in primary care. In 2016 the [EULAR Fibromyalgia 2016](#) guideline reassessed the evidence base for the management of fibromyalgia, which sadly concluded that the 'size of effect for most treatment is relatively modest'. Given the lack of good evidence for management, a recent BMJ trial [BMJ2018;360:k851](#) comparing Tai Chi classes with aerobic exercise is welcome. Since the EULAR guideline we have had the NICE guideline on chronic pain ([NG193 2021](#)), which obviously has a much wider remit than just fibromyalgia, but as a chronic primary pain syndrome fibromyalgia is included within this guideline. Most recently we have had a Royal College of Physicians guideline focussing on the diagnosis of fibromyalgia ([RCP diagnosis of fibromyalgia 2022](#)).

We deal with the NICE guideline separately in the chronic pain chapter, but there are some **important general points to take from both the fibromyalgia-specific RCP diagnosis and EULAR management guidelines**:

- **Fibromyalgia syndrome (FMS) is a real and 'valid' syndrome.** The cause is unknown but there may well be genetic factors, and research suggests the condition evolves due to changes in the central and peripheral nervous system and the immunological system.
 - Both guidelines accept that the **diagnosis and management of fibromyalgia is not easy** - presentation is varied, the condition complex, heterogenous, often with overlapping/associated conditions, and the evidence base is limited. However, the result is often long delays to diagnosis (often years) which causes substantial patient distress, health burdens and significant healthcare utilisation with multiple specialist referrals.
 - We are therefore encouraged to make **an early, positive diagnosis** in primary care without a referral (unless diagnostic uncertainty). Importantly **FMS should not be seen as a 'diagnosis of exclusion'** - FMS can (and often does) sit alongside other illnesses or conditions and only a **limited number of tests** of exclusion are required unless directed by the clinical picture.
 - The **ideal locus of care for people with fibromyalgia is primary care**. There is no advantage to referral and indeed it may encourage excessive dependence on health professionals, increase expectations and lead to needless investigations.
 - **Optimal management is non-pharmacological**, with the strongest evidence to support exercise classes (see below for new evidence on Tai Chi). A holistic approach paying close attention to the biopsychosocial context is required.
 - Patients should be **strongly encouraged to continue with normal activity** as much as possible and to remain in the workplace, with adjustments if necessary.
 - **Medication is the last resort** but can have a role, particularly for those with more severe pain. However, the evidence is weak and all the studies are short-term. Furthermore, the drugs may produce side effects which exacerbate some of the symptoms e.g. fatigue and dizziness with amitriptyline and pregabalin or sleep disturbance with duloxetine.
- **Effect of tai chi versus aerobic exercise for fibromyalgia, [BMJ 2018;360:k851](#)**
- 226 patients with fibromyalgia were randomised in a prospective RCT to either supervised aerobic exercise classes or Tai Chi mind-body classes twice weekly over 24 - follow up one year.
 - All groups (i.e. aerobic exercise and Tai chi classes) showed clinically significant improvement compared to their baseline scores.
 - Tai chi was similar or superior to aerobic exercise across a wide range of measures, including the overall impact of fibromyalgia symptoms.
 - The results were consistent across 3 different instructors suggesting generalisability of the findings.
 - More patients attended the Tai Chi classes and the authors postulate that the gentle, low-impact movements of Tai Chi may make patients more likely to enjoy, manage and continue to practise it.
 - **Conclusion:** Tai chi mind-body classes are an effective option in the multidisciplinary management of patients with fibromyalgia.

KISS: Assessment of Fibromyalgia Syndrome (FMS)[RCP 2022](#)**Background:**

- FMS is a condition characterised by persistent and widespread pain, associated with intrusive fatigue, sleep disturbance, impaired cognitive and physical function, and psychological distress.
 - Be aware symptoms can fluctuate and change over time.
- **FMS is NOT:**
 - A condition caused by somatisation.
 - Driven by inflammation of muscles/joints.
 - A reflection of poor coping, deconditioning, or poor fitness.
 - A maladaptive reaction to trauma or distress.
- Although experiencing stressful life events does increase the risk of developing FMS, many people develop FMS without such triggers.
- **FMS pain is 'nocioplastic'** - caused by a disturbance in peripheral and/or central nervous system pain processing
 - c.f. nociceptive pain caused by harmful stimuli and neuropathic pain caused by nerve damage or pain/impingement on nerves.

Assessment:

- Assessment needs to be carried out in the biopsychosocial context of pain.
- **Good consultation skills are key:**
 - listen, understand the breadth of symptoms and discuss in a non-judgemental way.
 - *'Notably where conducted well, a consultation that leads to a diagnosis can have therapeutic and reassuring effects which should not be underestimated'.*
- **DO:**
 - Explain causes that you have ruled out.
 - Be clear this does not mean there is no reason for their pain.
 - Discuss what nocioplastic pain is, and provide written information (see KISS below).
- **DON'T SAY:**
 - There is no reason for the pain.
 - I can't find anything wrong with you.
 - It's psychological.
 - There is nothing we/anyone can do.
- Assessment and diagnosis can be (and ideally should be) done by non-specialists (e.g. primary care) unless in cases of uncertainty, and should be done as early as possible.
- Scoring tools are helpful, but diagnosis *'should not become a tick box exercise'*.
- **What factors should alert us to the possibility of FMS?**
 - **Widespread pain** - however note patients may just report focal pain, and if repeat consultations for focal pain ask directly about the presence of more widespread pain.
 - **Intrusive fatigue** - may be physical, cognitive or emotional.
 - **Hypersensitivity** - may be to light/sound etc but also disproportionate pain during examination.
 - **Longevity** - pain persisting beyond 3 months.
 - **Ineffective treatments so far** - this may be rehab/manual therapy or drug therapy.
 - **Patients feeling overwhelmed by their symptoms.**
 - NB The RCP also highlight that clinicians can also have a sense of being 'overwhelmed' in such complex consultations and that *'This can also be useful information in itself and point towards FMS'.*

KISS: Diagnosis of Fibromyalgia syndrome (FMS)

[RCP 2022](#)

Diagnosis (see also the [FMS diagnostic worksheet on RCP website](#) for more detail and printouts):

- The **RCP recommend we use the ACR 2016 FMS diagnostic criteria (all needed)**:
 - Generalised pain in $\geq 4/5$ body regions (left & right, upper & lower, axial).
 - Widespread pain (use widespread pain index/WPI) together with associated symptoms (use symptom severity scale/SSS). See FMS diagnostic worksheet above.
 - Symptoms for ≥ 3 months.
- **FMS is not a diagnosis of exclusion**, there are no specific tests for FMS, and patients with other conditions (e.g. RA) may also develop FMS.
- Consider **differential diagnosis/conditions that may mimic FMS**:
 - Endocrine disease (e.g. hypothyroidism).
 - Rheumatological conditions (e.g. RA, ankylosing spondylitis, SLE, PMR).
 - Neurological disease (e.g. neuropathies).
 - Drug-induced symptoms - NB important to remember high dose opioids (which patients with pain may have been prescribed) can cause **hyperalgesia** which may mimic the pain/hypersensitivity seen in FMS.
 - Sleep apnoea.
 - Depression.
 - Chronic fatigue syndrome (obviously can be overlap here but pragmatically if the predominant symptom is fatigue/excessive sleep rather than pain assess for CFS, but some patients may fit the criteria for both FMS and CFS.)
- **Investigations**:
 - FBC
 - U&E
 - ESR and CRP
 - LFT
 - TSH/thyroid studies
 - Blood glucose
 - CK
 - Further investigations should only be done based on clinic suspicion of other specific diseases.
- **Other conditions** patients/clinicians may consider in the differential diagnosis/be worried about?
 - MS - can cause generalised pain and fatigue but also has focal neurological deficits.
 - Further investigations e.g. MRI are not recommended unless there is a specific concern about MS or other neurological conditions.
 - Functional neurological disorders - can cause neurological symptoms but do not present with pain.
 - We also need to be aware long covid symptoms can mimic FMS and CFS.

Diagnostic uncertainty/when to refer:

- **Diagnostic uncertainty is a common conundrum**:
 - This may be due to sub-threshold symptom scores, other health conditions having an impact on presentation/symptom scores, symptoms changing or fluctuating over time.
 - Share uncertainty with patients, and if **diagnosis is uncertain a 'watchful waiting' strategy is appropriate** with safety netting for alternate significant diagnoses.
 - But do agree a management plan even if the diagnosis is uncertain.
- **Consider onward referral** (likely to rheumatology or pain services) if:
 - There is concern about other significant diagnosis or assessment is complicated by other known conditions e.g. RA, neurological disorders, or complex multiple health conditions.
 - The FMS symptoms are *persistently* equivocal.

KISS: Management of Fibromyalgia Syndrome (FMS)

[EULAR Recommendations 2016](#) [BMJ 2018;360:k851](#) [RCP 2022](#) [NICE NG193 2021](#)

Management:

- The **focus of management should be on improving health-related QoL** (not pain per se).
- It is appropriate to offer treatments based on a working diagnosis, even if there is diagnostic uncertainty.
- Patient education, participation in management and self-efficacy are key.
- There should be a comprehensive, individualised assessment of pain, function and the psychosocial context.
- Good patient information is important (see below resources).
- ***Ideal management is non-pharmacological***, with an **emphasis on physical therapy** and requires active patient participation
 - **Regular physical activity** has strong evidence to support it, reducing pain and improving function - this should be done in a gradual fashion and may be either aerobic, strengthening or both. NICE recommend group exercise therapy.
 - Recent evidence suggests **Tai Chi classes** are more effective than aerobic exercise classes and are an evidence-based option which some patients may prefer to aerobic exercise.
 - **CBT** reduces pain and disability, with sustained results. **ACT** (acceptance and commitment therapy) is also recommended by NICE.
 - There is some (weaker) evidence to support mindfulness, acupuncture and hydrotherapy.
- **Vitamin D?** Remains controversial and uncertain and is not mentioned in either the RCP or EULAR guidelines. There is evidence of low vitamin D levels associated with fibromyalgia, but no convincing evidence treating with vitamin D has any benefits to symptoms.
 - It's worth noting that testing for vitamin D is not suggested in the RCP guideline.
- **Medication.** Based on a thorough evaluation of the evidence, there is **weak** evidence from **short-term** studies for the following based on EULAR recommendations (but note discrepancies with NICE):
 - Amitriptyline 10mg to 50mg (consider if sleep disturbance).
 - ?Pregabalin? 300mg to 600MG - note NICE actively we should NOT offer gabapentinoids.
 - Duloxetine 60mg to 120 mg
 - ?Tramadol? (for severe pain) - again noting NICE recommendation that opioids should NOT be used in chronic pain conditions.
 - ?SSRIs? not recommended by EULAR due to lack of evidence but NICE does recommend citalopram, fluoxetine, sertraline or paroxetine as options.
 - Do NOT offer NSAIDs.

Patient information:

- [Versus Arthritis Fibromyalgia](#)
- [NHS Choices Fibromyalgia](#)
- [RCP PIL](#)
- [Fibromyalgia action UK](#)
- [Elippinpain](#)
- There are a number of other excellent resources in [Appendix 5 of the main RCP guideline](#) covering other areas including mindfulness resources, employment and education advice and other pain education websites.

DEPRESSION IN ADULTS

In June 2022, NICE published a new update of their depression guideline [NICE NG222 Depression in Adults](#). The original version of this guideline dated way back to 2009, and the update has been mired in controversy and delay. The new guideline adopts a more holistic, individualised and patient-centred approach to depression with less emphasis on prescribing and a much greater choice of non-pharmacological treatment options. All of this is to be welcomed but, as is so often the case, commissioners and GPs will struggle to be able to provide recommendations for patients at the coal face...

Whilst this broader range of treatment options is welcome, given the pressures on service provision antidepressants will remain a part of routine primary care depression management. The issues surrounding antidepressant prescribing (how effective are they, increasing recognition of side effects and withdrawal effects, how/when do we stop them? etc.) are very much a topic in their own right. See the following chapter on antidepressant prescribing for an in-depth review of these issues and practical guidance/tips on how to manage antidepressant prescribing and de-prescribing.

► *NICE Depression Guidelines 2022* [NICE NG222 Depression in Adults](#)

The new guideline was published in June 2022, and has been a long-awaited update from the original guideline back in 2009. The update has been mired in controversy and has had several postponements.

The new guideline is to be welcomed, as it offers much more choice to patients in terms of therapy options, and encourages a more individualised and holistic approach to management with less emphasis on prescribing and a [broad range of treatment options](#) recommended as options for patients to choose from. However, whilst this extensive menu of treatment options looks fantastic for patients and clinicians, it will be a challenge for NHS commissioners to be able to provide these services (note that the most cost-effective option, guided self-help, is recommended first). The guideline though is clear that '*commissioners should ensure that people can express a preference for NICE recommended treatments and that these treatments are available in a timely manner*' so we have to hope that the guideline will lever service delivery. It is likely that social prescribers, mental health support workers and of course primary care psychological services (e.g. the NHS IAPT service) will have a major role in delivering and monitoring treatments.

Major changes to the guideline include a less strict classification into different degrees of severity of depression, which instead of mild, moderate or severe is now considered in the more general categories of less or more severe depression. *In this guideline, less severe depression was defined as scoring < 16* on the [PHQ-9 scale](#), and more severe as ≥ 16 . We should consider using a validated measure, such as PHQ-9, to inform and evaluate treatment but 'conduct a comprehensive assessment' that does not rely solely on a symptom count, and takes previous history and individual factors into account.

Patients can be offered a choice from a very broad menu of treatments for [less severe depression](#) and [more severe depression](#); although clinician and patient can make a shared decision about treatment choice, NICE do list them in order of recommended use, based on clinical and cost-effectiveness with guided self-help as first option. It is interesting to note counselling is well down the list of preferred options.

An important factor that NICE acknowledge when treating people with depression is building a trusting therapeutic relationship, and facilitating [continuity of care](#) by ensuring they can see the same healthcare professional whenever possible. This is clearly beneficial for the patient and also rewarding for the clinician but in the current NHS climate, sadly, continuity can be difficult to achieve.

As far as [prescribing](#) is concerned, for *patients with less severe depression we should not recommend antidepressants as a first-line treatment unless it is the patient's preference*. For more severe depression the first-line option is an antidepressant in combination with individual CBT, but if the patient prefers psychological treatment alone is a recommended option.

There is little change with prescribing from the original guideline in that SSRIs are first choice, but there are strengthened recommendations on discussing the benefits/harms/side effects/withdrawal effects and exploring patient concerns on starting/stopping antidepressants (see antidepressant prescribing chapter for more detail).

If there is no response at 4 weeks there is the option to either increase dose if it is well-tolerated, or to switch to an antidepressant of a different class. We should make individualised choices on the next choice (e.g. another SSRI, SNRI, TCA etc). If TCA's are considered, lofepramine is recommended as the best-tolerated option. When switching, we should not stop and start abruptly but follow cross-tapering advice [CKS, Guidance on Switching Antidepressants](#). If there is still no response to treatment, there is the option of augmentation of the antidepressant with a low dose of an antipsychotic (e.g. quetiapine) or lithium or in combination with another antidepressant. However, this should only be on specialist advice and ideally initiated in specialist settings.

A significant change in the guideline is the guidance is strengthened on the risk of **withdrawal reactions**, and the need to advise patients of the risk and to **slowly taper** medication when stopping. We should explain that whilst withdrawal reactions may be mild and transient, they can be severe and prolonged. When stopping we should take into account pharmacokinetic profile of the drug. For instance, those with a short half-life such as paroxetine and venlafaxine will need slower tapering than those with a long half-life such as fluoxetine. For most drugs they should be slowly reduced to 75% or 50% of the previous dose rather than fixed-dose increments. Once low doses are reached, liquid preparations can be used to allow slower tapering. Fluoxetine can be withdrawn more quickly, and in people taking 20mg daily a period of alternate day, dosing can be sufficient. See our chapter on antidepressant prescribing for more detail on this.

There is an emphasis on **risk assessment**, and the need to ask directly about **suicidal ideation** and intent. People should be advised about the potential risk of increased agitation, anxiety and suicidal ideation in the initial stages of treatment and this needs to be appropriately 'safety netted'. However if a person with depression is considered at risk of suicide, we should not withhold treatment on the basis of their suicide risk but consider referral and increasing the level of support such as more frequent face-to-face or telephone contacts. We should monitor for suicidal ideation in the early weeks of treatment, and anyone at increased risk of suicide and young adults under the age of 25 should be reviewed at one week after starting.

For patients who have responded to therapy, there is new emphasis on **relapse prevention** including group CBT or mindfulness-based cognitive therapy MBCT with or without continuing medication based on patient preference. An important factor that NICE acknowledge when treating people with depression is building a trusting therapeutic relationship, and facilitating **continuity of care** by ensuring they can see the same healthcare professional whenever possible. This is clearly beneficial for the patient and also rewarding for the clinician but in the current NHS climate, sadly, continuity can be difficult to achieve.

KISS: Depression

Based on [NICE NG222 Depression in Adults](#)

Recognition and assessment

- Be alert to the possibility of depression, including in people with chronic physical health problems, and consider asking the two 'screening questions'
 - *During the last month have you often been bothered by feeling down, depressed or hopeless?*
 - *During the last month have you been bothered by having little interest or pleasure in doing things?*
- If they answer yes to either question, perform a comprehensive mental health assessment taking previous history and current context into account. Consider using a validated measure (e.g. PHQ-9) and make an individualised clinical judgement as to whether the patient has
 - **Less severe depression** (includes sub-threshold or mild depression, in this guideline defined as depression scoring < 16 on the [PHQ-9 scale](#)), or
 - **More severe depression** (includes moderate and severe depression, in this guideline defined as depression scoring ≥ 16 on the [PHQ-9 scale](#))
- Ask about any history of mood elevation, to determine if may be part of bipolar disorder
- **Risk assessment:** *Always ask directly about suicidal ideation and intent; arrange help appropriate to need, and if they present considerable immediate risk to themselves or others refer urgently to specialist mental health services*

Treatment choices

- Discuss treatment options and preferences, and encourage shared decision making
- **Treatment options for 'less severe depression'**
 - Discuss the menu of [treatment options for less severe depression](#)
 - Treatment options are listed in order of recommended use with **guided self-help as first choice**
 - Next choices are group CBT; group behavioural activation; individual CBT; individual BA; group exercise; group mindfulness or meditation; interpersonal therapy; SSRI antidepressants; counselling; short-term psychodynamic psychotherapy
 - Match the choice of treatment to clinical needs and preferences; any option can be used as first-line but consider the least intrusive and least resource-intensive treatment first
 - If the person has a clear preference or experience to guide them, support this choice unless clinical concerns about suitability
 - Do not routinely offer antidepressants as a first-line treatment unless it is the person's preference
- **Treatment options for 'more severe depression'**
 - Discuss the [treatment options for more severe depression](#)
 - If the person has no clear preference, offer the options in this order and make a shared decision
 - Individual CBT +/- antidepressants; individual behavioural activation; antidepressant medication; individual problem solving; counselling; short-term psychodynamic psychotherapy; interpersonal therapy; self-help with support; group exercise
- **Psychotic depression**
 - For patients with depression and psychotic symptoms, offer referral to specialist mental health services for a full risk assessment, programme of coordinated multidisciplinary care and psychological treatments after improvement of acute psychotic symptoms. Consider combination treatment with antidepressant and anti-psychotic e.g. olanzapine or quetiapine.

KISS: Depression Continued

Based on [NICE 2022 DRAFT Guideline CG90](#)

Continued management

- For all treatments for people with depression, review how well the treatment is working between 2 to 4 weeks after starting treatment and monitor therapeutic response and side effects
- *Facilitate continuity of care*
- Monitor suicidal ideation, especially in the early weeks of treatment
- Consider routine outcome monitoring e.g. PHQ-9 or similar symptom score
- **Remission:** if treatment achieves full or partial remission, make a shared decision as to whether or not to continue on treatment for relapse prevention
 - If people have been treated with antidepressants and considered higher risk of relapse, consider continuing treatment for up to 2 years to prevent relapse, with or without a course of psychological therapy such as mindfulness-based cognitive therapy MBCT; if they do not wish to continue on medication offer MBCT for relapse prevention
- **No treatment response:** if patients have not responded after 4 weeks of psychological therapy, switch to an alternative therapy with or without an SSRI or mirtazapine alone
 - Note [CKS Guidance on Switching Antidepressants](#) for cross-tapering advice
 - If patients have not responded to an antidepressant drug consider increasing the dose of the current drug if it is well-tolerated, OR switch to another in the same class, OR switch to another medication in a different class (e.g. SSRI, SNRI, TCA or MAOI)
 - If still fail to respond, combination therapy with an additional antidepressant from a different class, an atypical antipsychotic or lithium may be considered *but should be initiated in specialist settings or after consulting a specialist. Antipsychotics should be prescribed under shared care arrangements.*
- **Prescribing considerations**
 - **Initiating treatment**
 - Advise re potential side effects and withdrawal symptoms on stopping, that it may take 4 weeks to see a response, review after 2 to 4 weeks to assess response
 - Review after 1 week if aged under 25 or there is a concern about suicidal risk
 - Advise that treatment may need to be taken for 6 months after remission of symptoms
 - **Stopping medication**
 - Advise patients to consult you before stopping, and the need to reduce the dose gradually over time but that most patients can stop them successfully
 - Advise patients that if they stop abruptly or miss doses, they may get **withdrawal reactions** such as restlessness or agitation, problems sleeping, altered feelings and thoughts, unsteadiness, sweating and abdominal symptoms
 - Advise that these withdrawal symptoms may be mild and go away within 1 or 2 weeks, but they can last longer (weeks or months) and be severe especially if they are stopped suddenly
 - When reducing the drug, take into account the pharmacokinetic of the drug and the duration of treatment
 - Reduce by 75% or 50% of the previous dose rather than fixed dose reduction, and use liquid preparations if necessary to allow slow tapering when small doses have been reached

NB Learning Points & Quality Improvement Ideas

- NICE have produced a **new depression guideline in 2022**. It offers much more choice to patients in terms of therapy options and encourages a more individualised and holistic approach to management with less emphasis on prescribing and a broad range of treatment options
- The 'only way is CBT' mantra of previous guidelines has gone, with a wide range of non-pharmacological therapies available as first-line choices e.g. CBT, behavioural activation, group exercise, group mindfulness or meditation, interpersonal therapy, counselling and short-term psychodynamic psychotherapy
- It is likely that social prescribers, mental health support workers and of course primary care psychological services (e.g. the NHS IAPT service) will have a major role in delivering and monitoring treatments.
- Patients are offered a range of treatment options based on whether they have less or more severe depression
 - Less severe depression (includes sub-threshold or mild depression, in this guideline defined as depression scoring < 16 on the [PHQ-9 scale](#)), or
 - More severe depression (includes moderate and severe depression, in this guideline defined as depression scoring ≥ 16 on the [PHQ-9 scale](#))
- For patients with less severe depression do NOT offer antidepressants unless this is the patient's preference, and guided self-help is the first-line recommendation
- The new guideline has strengthened recommendations to warn patients regarding the risk of withdrawal reactions from antidepressants and the need to slowly reduce medication when stopping
- For patients who fail to respond to two different antidepressants, augmentation therapy (adding in an antipsychotic) or combination therapy (adding another antidepressant from another class) should only be done with specialist advice
- Ask about suicidal ideation and intent at every stage, and monitor carefully in the early weeks of treatment, especially in younger adults

ANTIDEPRESSANTS - SIDE EFFECTS AND PRESCRIBING ISSUES

We are all well aware of the enormous rise in antidepressant prescribing in the UK over the last 10-15 years - a trend that is mirrored throughout the western world and was the subject of a review in a recent DTB article ([Drug and Therapeutics Bulletin 2022;60:7-12](#)). It is estimated that 7.8 million people received a prescription for an antidepressant in England in 2019/20 which represents one in six adults. However, this prescribing trend is increasingly being questioned both in terms of the efficacy of the drugs themselves, but also due to the increasing recognition of side effects (some of which may be long-term) and withdrawal reactions.

Some of these factors have been addressed in the new [NICE NG222 Depression guidelines](#) (see separate chapter on Depression in adults) but in this chapter, we deal with some of these issues in more detail including some of the underlying evidence and guidance on the benefits and risks of antidepressants and also incorporate some 'top tips' to help us when it comes to prescribing these drugs.

1 *How effective are antidepressants?* [Drug and Therapeutics Bulletin 2022;60:7-12](#)

Much of the evidence for antidepressants comes from short-term (6-12 weeks) placebo-controlled trials focussing on the newer antidepressant classes - SSRIs and SNRIs. Several meta-analyses have found that SSRIs/SNRIs improve depression rating scales over placebo by small amounts. Improvements of about 2 points on the 52 point clinician rated Hamilton Depression Rating Scale (HAM-D) have generally been observed, but whether this constitutes a clinically 'significant' improvement is questioned. NICE had previously used 3 points on this scale to define a significant improvement but more recent analyses suggest a change of 3-5 points should be used to define the minimum for a clinically significant change.

Some authors have argued that these smaller improvements still constitute a worthwhile effect, postulating that sub-groups do get a greater than average response which would be more clinically meaningful. However recent analysis suggests there is no significant heterogeneity in antidepressant response ([PLoS One 2020;15:e0241497](#)) with a 'near-constant treatment effect' across all patients, which goes against highly responsive sub-groups. Alongside this, previous data has suggested those with more severe depression show a stronger response to antidepressants, but more recent analyses suggest little difference whether mild, moderate or severe depression is present.

The above information relates to adults, but antidepressant prescribing in 12-17-year-olds is also increasing rapidly and has doubled between 2005-2017. Data for adolescents and children is even less convincing with a recent Cochrane review ([Cochrane Database Syst Rev 2021;5:CD013674](#)) finding no benefit over placebo, with the authors questioning 'whether they should be used at all' especially given the increased suicide risk in this population. Re-analysis of the original data supporting use in this age group has questioned the methodology used.

The treatment effects of antidepressants may have also been inflated by dichotomising the data into artificial categories of 'response' and 'non-response' (partly driven by regulatory requirements), but we know response will be a continuum, and it is increasingly recognised that this technique increases the risk of false positives and creating spuriously inflated effect sizes.

Finally, the data we have has questionable relevance for many of the patients for whom these drugs are prescribed. The rating scales focus narrowly on symptom scores, not outcomes that are likely to be more relevant to patients such as social functioning or quality of life. The trials are generally only 6-12 weeks long, which is uninformative for clinical treatment of depression which often involves treatment for months or years, and we have a real lack of data to assess benefits (and risks) of long-term antidepressant use.

2 What about side effects?

There are a wide range of side effects of the newer generation antidepressants, and one that has become more recognised recently and is a real 'Hot Topic' is sexual side effects:

KISS: Antidepressant Induced Sexual Dysfunction

[Medical Journal Australia 2020](#)

Background:

- Sexual dysfunction is a frequent and distressing adverse effect of antidepressants and a common cause of non-adherence.
- Antidepressant-induced sexual dysfunction can affect all phases of sexual activity including desire, arousal and orgasm in men and women.
- The true incidence of these side effects is hard to determine but has been reported to be as high as 80% with SSRIs. *In one study of women taking an SSRI, 70% reported reduced libido, 80% reduced arousal and 40% problems with orgasm.*
- Depression and anxiety also, of course, causes sexual dysfunction, making it complicated to disentangle. However depression per se is more commonly linked with reduced sexual desire and drive, whilst antidepressants more commonly affect sexual pleasure, ejaculation and orgasm.
- Patients report they frequently worry about sexual side effects, but are rarely asked about them by their prescriber; they value the opportunity to discuss these concerns.

Management strategies:

- Assessing sexual function at baseline before prescribing and again afterwards is important so that we can an idea of what is due to the underlying mental health condition and what due to the drug.
- **Time:** in some cases, sexual dysfunction will improve and resolve
- **Dose:** side effects are dose-dependent, so consider a dose reduction if underlying depression or anxiety is well controlled.
- **Switching antidepressants:** the evidence base is limited but switching to an antidepressant with a lower risk (see table below) seems reasonable if their underlying condition allows.
- **Augmentation:** there is some evidence [Cochrane 2013](#) to support that for women the addition of bupropion to their antidepressant regimen may be effective (NB but unlicensed and very expensive) and for men the addition of sildenafil.

Risk of sexual dysfunction with different antidepressants:

- The evidence base is limited as there are very few good quality trials looking at sexual function as an outcome, but from the available evidence we have we can conclude:

Antidepressants	Risk of sexual dysfunction
SSRIs, SNRIs and clomipramine - Escitalopram & paroxetine possibly higher risk than other SSRIs	HIGH
Tricyclics (excluding clomipramine)	MEDIUM
Mirtazapine	LOW TO MEDIUM
Bupropion, agomelatine, reboxetine	LOW

KISS: Post-SSRI Sexual Dysfunction

[BMJ 2020;368;m754](#)

Background:

- A recent BMJ editorial has highlighted another issue with antidepressants and sexual dysfunction, namely post-SSRI sexual dysfunction, described as *'an important iatrogenic condition'*.
- Although this condition has been reported for years, it has recently become a Hot Topic following the formal recognition in 2019 by the regulatory body the European Medicines Agency of sexual dysfunction as a side effect of SSRIs and SNRIs which may persist despite the discontinuation of the drug.
- The EMA have recommended that product information on relevant antidepressants is updated to warn patients about the risk of long-term sexual dysfunction after treatment (see below).

Symptoms of Post-SSRI sexual dysfunction:

- Include genital numbness, decreased libido, failure to become aroused, failure to orgasm, pleasureless or weak orgasm and premature ejaculation.
- It can be associated with a more general sense of emotional numbing.
- It is a debilitating condition with a significant impact on quality of life.
- The syndrome is similar to post-finasteride syndrome, which causes similar symptoms.
- Although the symptoms of post-SSRI sexual dysfunction are clearly described, more research is needed to disentangle the effects of the medication from those of underlying mental health conditions such as depression.

Who does it affect and what is the prognosis?

- Post SSRI sexual dysfunction has been reported by patients of both sexes of all ages from all ethnic backgrounds.
- Case reports show that it can come on after a few doses or only become apparent after years of exposure and that it can last for years.
- It may appear only when the antidepressant is tapered or stopped.
- Some patients report spontaneous recovery, others have a relapsing-remitting course.
- The prevalence and incidence is unknown.
- Patients are rarely warned about the risk of long-term sexual side effects and controlled studies have not examined this outcome.

Conclusion:

- The editorial concludes that: *'recognition of post-SSRI sexual dysfunction by the European drug regulator means that prescribers must inform patients of the risk when discussing possible treatment with an SSRI or SNRI so that they can make a fully informed choice about their options'*.

Change to the Summary of product characteristics for SSRIs and SNRIs

[EMA 2019](#)

Selective serotonin reuptake inhibitors (SSRIs)/serotonin-norepinephrine reuptake inhibitors (SNRIs) may cause symptoms of sexual dysfunction. There have been reports of long-lasting sexual dysfunction where the symptoms have continued despite discontinuation of SSRIs/SNRI.

► **Other side-effects of antidepressants** [Drug and Therapeutics Bulletin 2022;60:7-12](#)

Adverse effects of antidepressants are well recognised and are often greater when longer-term use is studied. Observational data from Holland ([Eur Neuropsychopharmacol 2013;23:1443–51](#)) looking at longer-term antidepressant use (average use ~1 year) showed that of those on SSRIs (the largest cohort studied):

- 64% had at least 1 adverse effect with 31% having 3 or more.
- About 1 in 5 patients on SSRIs reported daytime sleepiness, dry mouth, profuse sweating or weight gain.
- About 1 in 10 reported restlessness, muscle spasm or twitching, nausea, constipation, diarrhoea or dizziness.
- Profuse sweating was even more common with venlafaxine (>30%) as was weight gain with mirtazapine (~30%).

Note also **serotonin syndrome**, recently reviewed in the [DTB June 2022](#), discussed separately in the prescribing chapter.

③ **Withdrawal effects** [Drug and Therapeutics Bulletin 2022;60:7-12](#) [RCPsych - Stopping antidepressants](#)

Withdrawal effects from SSRIs have been described in the literature for over 30 years, but it is only relatively recently that there has been recognition of the true numbers of people this affects and the impact it has on them. A systematic review in 2019 ([Addict Behav 2019;97:111–21](#)) found that about half of patients experienced withdrawal symptoms when they stopped antidepressants whether they stopped abruptly or tapered their doses. There was slight variation in the rates depending on the type of studies reviewed, but whether RCTs, observational data or patient surveys they all consistently suggest >50% of people suffer withdrawal effects. The systematic review also found many patients experience these effects for several months and occasionally years.

The updated [NICE guideline on depression](#) is much more explicit in its recognition of withdrawal effects and how to manage them (see section below 'How do I stop antidepressants?'). NICE lists the following as common withdrawal symptoms:

- Restlessness or agitation
- Problems sleeping
- Altered feelings (for example, suicidal thoughts, irritability, anxiety, low mood, tearfulness, panic attacks, irrational fears, or confusion)
- Unsteadiness, vertigo or dizziness
- Sweating
- Abdominal symptoms (for example, nausea)
- Altered sensations (for example, electric shock sensations)
- Palpitations, tiredness, headaches, and aches in joints and muscles

The increasing recognition of the burden of withdrawal effects culminated in a position paper by the [Royal College of Psychiatry in 2019](#) as well as an update to the [NICE guidance on depression in 2019](#).

The RCPsych have also recently put together some excellent information on withdrawal symptoms for patients (and clinicians!) - [click here](#).

► **How can we differentiate between relapse and withdrawal effects?**

[Drug and Therapeutics Bulletin 2022;60:7-12](#) [RCPsych - Stopping antidepressants](#)

This is an inherently tricky area, and one also reviewed in the DTB article and is an area much debated. Many of the common withdrawal effects (as outlined by NICE above) e.g. anxiety, depression, appetite changes, insomnia register on depression scales has led to many people questioning the true treatment effects of antidepressants in 'relapse' trials.

Given the data above that withdrawal effects may last for many weeks or months, it is quite possible that many patients labelled as having a 'relapse' may be suffering long-term withdrawal side effects of the drugs themselves (which then resolve once antidepressants are re-started as would be expected in a withdrawal scenario). This is clearly an uncertain area, and one where more robust data is sorely needed, but the Royal College of Psychiatrists have put together some useful 'rules of thumb' to help both us and our patients help differentiate between true relapse and withdrawal side effects ([How can I tell if it is withdrawal symptoms or my anxiety/depression coming back?](#)):

- Withdrawal symptoms normally start soon after medication is reduced or stopped. This may be one or two days for some antidepressants – or even after missing a single dose. Usually, they take a few days to start and then get worse.
- The return of depression or anxiety usually takes longer – typically weeks or months. Some antidepressants, like fluoxetine, take a lot longer to leave the body. So, with these, symptoms can start days or even weeks after stopping or reducing doses. This can make it harder to tell if symptoms are due to withdrawal or the return of anxiety or depression.
- Some withdrawal symptoms do not happen in anxiety or depression – such as 'electric shocks' or 'zaps'. People often say, "I've never felt this before" or "This doesn't feel like my depression."
- Withdrawal symptoms usually improve quickly (in days or even hours) once antidepressants are restarted. This is much quicker than the weeks that antidepressants will normally take to relieve symptoms of anxiety or depression that have returned.

4 *How long do antidepressants take to work?* [Lancet 2019;6:903](#)

The general dogma has always been that it is expected improvements with antidepressants should be apparent by 4 weeks, and indeed the updated [NICE guideline on depression](#) continues this by recommending that if no response is seen at 4 weeks we should consider either alternative treatments, upping the dose of the antidepressant or switching to an alternative antidepressant. However, we have some evidence that improvements with antidepressants may take longer than this:

- The PANDA trial was a primary care-based placebo-controlled RCT done in 179 GP in the UK looking at the effectiveness of sertraline for moderate depression.
- There was no evidence that sertraline was superior to placebo for symptoms of depression at 6 weeks, although some secondary outcomes including anxiety were improved.
- However, there was weak evidence that sertraline was superior to placebo at 12 weeks for symptoms of depression, suggesting improvements in depressive symptoms may take longer to emerge.

This theme was echoed in an interesting paper from the [DTB 2019](#) which discussed research [J Clin Psych 2018](#) looking at trajectories of acute antidepressant efficacy, and asking the crucial question of *how long should we wait for a response?*

The research (a systematic review of randomised placebo-controlled trials) looked at patients who did not initially respond to antidepressants at 4 weeks to see if they may have a therapeutic response if the drug were continued beyond 4 weeks compared to placebo. The primary outcome was a therapeutic response in these previously non-responsive patients over different time frames. They found:

- There was a response rate of 22% in previous non-responders between weeks 5 and 8, compared to 13% with placebo (NNT = 11).
- Between weeks 8 and 12 there was a therapeutic response in 10% with continued treatment compared to 2.4% of controls; beyond 12 weeks there was no further benefit of continuing.
- **Conclusion:** in patients who do not respond to antidepressants after 4 weeks, improvements with antidepressant monotherapy can be expected for up to 12 weeks, however beyond 8 weeks any further improvement with ongoing treatment is small.

The DTB sensibly conclude that these results may be helpful in discussions with patients who have not initially responded that it may be worth continuing with the same monotherapy up until 8 weeks.

5 *How do I switch antidepressants?* [CKS Switching Antidepressants 2022](#)

This is a common question and is nicely addressed by the CKS on depression.

KISS: Switching Antidepressants [CKS 2022](#)

General considerations:

- When switching from one antidepressant to another, be aware of the need for gradual and modest increases of dose, interactions between antidepressants, and the risk of serotonin syndrome when combinations of serotonergic antidepressants are prescribed (see separate KISS in prescribing section on serotonin syndrome).
- Abrupt withdrawal should generally be avoided when switching from one antidepressant to another.
- Cross-tapering is usually preferred, however, in some instances, it may be possible to withdraw the current antidepressant and start the new antidepressant on the next day, e.g. when switching from one SSRI to another SSRI or a SNRI; the exception to this is fluoxetine - be aware that interactions with fluoxetine may still occur for 5 weeks after stopping due to its long half-life.
- Cross-tapering with a TCA is inadvisable with paroxetine and fluvoxamine
- Clomipramine is a potent inhibitor of serotonin reuptake and serotonin syndrome is more likely if co-administered with an SSRI or SNRI - cross-tapering is not recommended except under specialist supervision.

Tapering regimes:

Switch from:	Switch to TCA (except clomipramine)	Switch to SSRI (except fluoxetine)	Switch to fluoxetine	Switch to SNRI	Switch to mirtazapine
TCA (except clomipramine)	Direct switch possible	gradually reduce TCA to 25-50mg daily then start SSRI and slowly withdraw TCA over next 5-7 days	Halve dose of TCA, add fluoxetine and then slowly withdraw TCA	Cross taper cautiously with low dose SNRI	Cross taper cautiously
SSRI (except fluoxetine)	Cross taper cautiously with low dose TCA	Direct switch possible	Direct switch possible	Direct switch possible (caution if paroxetine used)	Cross taper cautiously (see below for example)
Fluoxetine	Stop fluoxetine, start TCA low dose 4-7 days later and increase dose very slowly	Stop fluoxetine, start other SSRI at low dose 4-7 days later	N/A	Stop fluoxetine, start SNRI at low dose 4-7 days later	Cross taper cautiously
SNRI	Cross taper cautiously with low dose TCA	Direct switch possible	Direct switch possible	Direct switch possible	Cross taper cautiously
Mirtazapine	Cross taper cautiously	Cross taper cautiously	Cross taper cautiously	Cross taper cautiously	N/A

- **Example for cross tapering regime from citalopram to mirtazapine:**

	Pre-switch dose	Week 1	Week 2	Week 3	Week 4
Withdrawing citalopram	40mg daily	20mg daily	10mg daily	Nil	Nil
Introducing mirtazapine	Nil	15mg daily	30mg daily	30mg daily	45mg daily (if needed)

6 *Combining and augmenting antidepressant drugs*

So what do we do if monotherapy with standard antidepressants does not work? Can we add in or combine antidepressants or drugs from a different class e.g. an antipsychotic? The term combining is generally used when adding in drugs of the same class (e.g. 2 antidepressants) and augmenting is generally used when a drug from a different class (e.g. antipsychotic) is added to an antidepressant but the terms are often used interchangeably.

► Mirtazapine added to SSRIs or SNRIs for treatment-resistant depression [BMJ 2018;363:k4218](#)

Adding mirtazapine to SSRI and SNRIs is a common strategy used in primary care for people who don't respond to SSRI/SNRI monotherapy. This seems to have been driven by secondary care but is there any evidence for this strategy? This placebo-controlled RCT, which importantly was done in primary care, should question this practice:

- Placebo-controlled RCT with 480 patients recruited from 106 general practices across the UK.
- All patients had been established on monotherapy with either SSRI or SNRI for a minimum of 6 weeks and remained depressed; >90% had been on treatment for > 6 months with many reporting previous episodes of depression; severity of symptoms was evenly spread between mild-moderate, moderate-severe and severe.
- All participants continued SSRI/SNRI and were then randomised to adding either placebo or mirtazapine 30mg.
- Main outcome measure was depressive symptoms at 12 weeks, measured using the Beck depression inventory score; secondary outcomes included measures of anxiety, quality of life, and adverse effects at 12, 24, and 52 weeks.
- There was no clinically significant difference for any outcome measure between the two groups.
- There was a significant increase in adverse effects in the mirtazapine group - 46 participants who reported adverse effects stopped their drug compared with nine in the placebo group.
- Conclusion: *This study did not find evidence of a clinically important benefit for mirtazapine in addition to an SSRI or SNRI over placebo in a treatment-resistant group of primary care patients with depression with a much higher side effect burden.*

► Pharmacological interventions for treatment-resistant depression [Cochrane 2019:CD010557](#)

- This was a systematic review of 10 RCTs (1a), n=2,700 patients published in December 2019.
- Most of the evidence included was considered to be low to moderate quality, and all of it was short-term, but there was some high-quality evidence.
- Augmentation with antipsychotics may be effective in the short term:
 - There was a small body of high-quality evidence that augmenting current antidepressant therapy with quetiapine improves depression scores over 8 to 12 weeks.
 - There was some low-quality evidence that augmentation with olanzapine was effective.
 - However, the antipsychotic augmentation groups had higher drop out rates (10% to 39% vs. 12% to 23% in comparator groups).
- There is high-quality evidence that combining current antidepressant with mirtazapine does NOT produce a clinically meaningful benefit in depression scores.
- Conclusion: *there is high-quality evidence that augmenting antidepressant treatment with quetiapine reduces depression scores at 8 to 12 weeks, but at a cost of greater adverse events and drop-out rates; there is high-quality evidence that combining the antidepressant with mirtazapine is not effective.*

The updated [NICE guideline on depression](#) on depression suggests the following regarding combining/augmenting drug therapy:

- If a person whose depression has had no response or a limited response to antidepressant medication does not want to try a psychological therapy, and instead wants to try a combination of medications, explain the possible increase in their side-effect burden.

- Consider adding an additional antidepressant medication with a complementary mechanism of action.
- Consider combining an antidepressant medication with an atypical antipsychotic (for example, aripiprazole, olanzapine, quetiapine or risperidone) or lithium or lamotrigine.
- However, NICE recommends '*Combination therapy should be initiated in specialist mental health settings or after consulting a specialist.*'

7 ***Long-term antidepressants - continuation versus discontinuation*** [Cochrane 2021:CD013495](#)

We have huge numbers of patients on long-term repeats of antidepressants. Should we advise them to continue or try to withdraw?

Guidelines generally advise that people take an antidepressant for at least 6 months after they start to feel better, and for at least 2 years if they have had ≥ 2 episodes of depression. In the UK, almost half of people using antidepressants (8% of the population) have been taking them for more than two years and there is a concern that despite being initially appropriate, they are often not discontinued after the recommended duration. As discussed above there is a high rate of side effects with long-term use, so what are the risks of discontinuation if we do stop antidepressants in those on long-term therapy? Cochrane addressed this in 2021:

- The researchers wanted to assess the safety and effectiveness of approaches for discontinuation versus continuation of long-term antidepressant use.
- They looked at data from 33 trials comparing different approaches to continuation versus discontinuation.
- Unfortunately, nearly all the studies were done in specialist settings and the quality of the evidence was low. The authors were unable to make any definite conclusions, due to uncertainty and confusion about confounding relapse of depression and anxiety with withdrawal symptoms.
- **Conclusion:** We lack evidence on the most safe and effective way to stop ADs in adults on long-term treatment, and there is a lack of distinction between relapse of symptoms and withdrawal reactions. Since this Cochrane review a further UK primary care study has been published which has also generated some debate - the ANTLER Trial

► **Maintenance or discontinuation of antidepressants in primary care** [NEJM 2021;385:1257](#)

- The Antidepressants to Prevent Relapse in Depression (ANTLER) trial was a randomised double-blind trial in 150 UK general practices to assess the effects of maintenance of antidepressant therapy versus discontinuation, importantly in primary care.
- All patients had a history of ≥ 2 episodes of depression or had been taking antidepressants for ≥ 2 years, and they felt well enough to consider stopping them.
- They were randomised to either maintain their current therapy or to taper and then discontinue use with a matching placebo.
- The primary outcome was the first relapse of depression over the next year.
- Results:
 - At 52 weeks relapse occurred in 56% in the discontinuation group, and in 39% in the maintenance group.
 - Patients with ≥ 3 previous episodes of depression were more than twice as likely to have a relapse than those with fewer episodes.
- Conclusion: *Among patients in UK primary care either on antidepressants for ≥ 2 years or with ≥ 2 previous episodes of depression, those assigned to stop their medication had a higher rate of relapse over the next year; however, almost half the patients (44%) who did discontinue did not relapse.*

So whilst on the face of it this study suggests a higher rate of relapse for people who discontinue long-term antidepressants, the results have been questioned by the recent DTB review ([Drug and Therapeutics Bulletin 2022;60:7-12](#)).

The DTB point to a number of factors that suggest that the higher 'relapse' rate in the discontinuation group may simply be due to confounding of relapse by withdrawal effects of the drugs:

- First, the tapering was done relatively rapidly for people on long-term therapy, increasing the risk of withdrawal effects - doses were halved for 4 weeks then they had half dose and placebo on alternate days for a further 4 weeks and thus discontinuation was completed by 8 weeks (much quicker than the newer recommendations which would suggest tapering long term antidepressants over many months).
- Second many of the symptoms/domains used in the relapse scale overlap or are similar to common withdrawal effects.
- Finally, the vast majority of the excess 'relapses' in the discontinuation group vs maintenance group occurred in the first 12 weeks, which temporally is the time over which one would expect to see the highest rates of drug withdrawal effects.

8 *How do I stop antidepressants?* [Drug and Therapeutics Bulletin 2022;60:7-12](#) [NICE NG222 Depression in adults](#) [RCPsych Stopping antidepressants](#)

This is the million-dollar question, which is hampered by a significant lack of evidence to guide us and there is a clear need for further studies in this area. The Cochrane review above ([Cochrane 2021:CD013495](#).) did not identify any particularly useful data due to the risks of bias/confounding between withdrawal symptoms and relapse of depression and was compounded by the fact that most trials stopped antidepressants quickly within 4 weeks. However, there are observational trials suggesting that tapering antidepressants over months to doses much lower than generally used in clinical practice generally results in better outcomes (but this has practical considerations - see below). The reason for this is likely to be due to the pharmacotherapy of the drugs themselves - the increase in effect on adding antidepressants (and therefore the effect on withdrawing them) is hyperbolic i.e. the effect on target receptors rises steeply for small initial doses then flattens out at clinically employed doses. Thus when reducing doses (in an ideal world) this should be done by exponential reductions - e.g. by reducing by 10% of their previous dose, rather than fixed dose reductions. However, this will require formulations of drugs not currently available to us.

KISS: How do I stop antidepressants?

[DTB 2022](#) [NICE NG222 Depression in adults](#) [RCPsych](#)

Principles:

- Appreciate that stopping antidepressants can be a difficult process and withdrawal symptoms can interfere with daily life - *'withdrawal of antidepressants should not be imposed on patients'* and should be done at a time that suits them. It is crucial this process is patient-led.
- Be upfront and explain the risks of withdrawal symptoms and what these symptoms may feel like, especially if tapering is done too rapidly - forewarned in forearmed.
- However, reassure patients that the majority of people are able to come off their antidepressants if done slowly with minimal/mild withdrawal reactions - they should not be alarmed by withdrawal symptoms and as the RCPsych note *'if you do get withdrawal symptoms it doesn't mean that you can't stop your antidepressant. You will just need to taper more slowly, with smaller reductions in dose, over a longer period of time.'*
- Factor in/balance potential 'upsides' of stopping antidepressants - as discussed above many patients suffer side effects from these drugs which may improve on stopping the medication - sexual dysfunction, emotional numbing, drowsiness, weight gain and agitation are common side effects that are likely to improve on discontinuation, although as discussed above we need to be aware these effects can be long term in some people.
- Consider psychological support/therapies whilst tapering antidepressants e.g. mindfulness, cognitive and acceptance therapy. A recent meta-analysis ([JAMA Psychiatry. 2021;78\(8\):868-875](#)) found that psychological therapies along with tapering antidepressants was no difference in outcomes to continuing antidepressants.
- Taper slowly and be flexible - for a small number of people who have been on antidepressants for short periods (e.g. weeks) they may be able to taper relative quickly over a few weeks, but for most this process will take months.
- Be aware of the pharmacokinetics of the antidepressant they are on - drugs with short half-lives (e.g. venlafaxine, paroxetine) are more likely to have withdrawal side effects (and therefore will need even slower tapering) than those with long half-lives (e.g. fluoxetine).

How do we discontinue antidepressants in practice?

- Warn patients withdrawal effects generally come on quickly (within a few days) of reducing doses but generally go away within 1-2 weeks (but may take longer especially if drug with a long half-life e.g. fluoxetine), and reassure them these are withdrawal symptoms and not a relapse.
- Advise against abruptly stopping the medication and warn patients this process is likely to take many months in most cases.
- Slowly reduce the dose to a proportion of the previous dose rather than fixed dose reductions - NICE suggest reducing by 25-50% but the RCPsych recommend much smaller reductions of 5-10% may be needed unless patients have only been on medication for a short period of time.
- Avoid missing medications on some days - this causes fluctuations in drug levels and increases the risks of withdrawal effects.
- Be flexible with the tapering regime and make sure it is patient led - make sure any withdrawal effects have resolved before making another dose reduction.
- If patients get more severe withdrawal symptoms go back to the previous dose until those have settled then taper with smaller increments.
- Provide/offer/suggest psychological support e.g. mindfulness, referral to local psychological therapies support, 'check ins' with clinical or mental health support workers.
- Direct them to excellent patient information and support from RCPsych - [click here](#)

KISS: Tapering Regimes

[RCPsych](#)

- Example of tapering regime for citalopram:

Current dose	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
40mg daily	20mg daily	10mg daily	5mg daily - 1/2 x 10mg tablet or liquid	2.5mg daily - liquid	1.25mg daily - liquid	0.6mg daily - liquid	Stop

- Reduce doses as above every 2-4 weeks, but some people will need to reduce more slowly and some may need to use smaller incremental changes e.g. 10-25% reductions.
- Note drugs with shorter half-lives (e.g. paroxetine, venlafaxine) are likely to need much smaller incremental reductions e.g. 10%.
- The major barrier we have is the lack of access to liquid preparations:
 - Some are available in liquid form (e.g. citalopram) but some have no liquid formulations available (e.g. sertraline) and some may have liquid forms available but only as 'specials' making them unavailable on most NHS formularies - discussion with local medicines management teams for advice should be considered.

SELF-HARM

NICE released their new guidance '[Self-harm: assessment, management and preventing recurrence NG225](#)' in September 2022. This timely guidance is set against a backdrop of **increasing prevalence**. A recent study published in the [Lancet](#) reported a rise in lifetime non-suicidal self-harm from 2.4% (95% CI 2-2.8) in 2020 to 6.4% (5.8-7.2) in 2014. Worryingly, by 2014, almost **1 in 5 girls and women (19.7%) aged 16-24** reported non-suicidal self-harm.

Self-harm is defined as **intentional self-poisoning or injury**. It can take many forms, including self-cutting, burning, head banging, punching, swallowing things that shouldn't be swallowed (e.g. razor blades) and harmful use of medications. The reasons behind self-harm are varied and unique to the individual. However, many describe self-harm as a **way to release overwhelming emotions**. The term 'deliberate' self-harm is no longer recommended because it implies a calmly planned act for which the individual is to blame, rather than a response to severe distress.

Who self-harms?

Whilst anyone can be affected it is more prevalent in the following groups ([RCPsych](#)):

- Young people
- Prisoners
- Asylum seekers
- Veterans
- Gay, lesbian, bisexual and transgender people (possibly related to the stress of prejudice and discrimination)
- Victims of neglect, emotional, physical or sexual **abuse during childhood**
- Groups/ friends who self-harm together

Common factors include feeling **hopeless, isolated, lacking in control** over their life, and **not being listened to** or understood. Alcohol and drug use problems often co-exist.

Why now? People will often have been in distress for a long time prior to self-harming. Common precipitating events include breakdowns in close relationships or employment difficulties.

► Psychosocial assessment

NICE recommends considering referral to a mental health professional for a **psychosocial assessment** as soon as possible after self-harm. This is a comprehensive assessment looking at the individual's needs, vulnerabilities and safety considerations in order to identify psychological and environmental factors that might explain an act of self-harm. The factors identified can then be **targeted in a personal care plan**. Given that access to mental health services is variable this may be done in primary care.

Factors **particularly relevant to young people**:

- Ask about peer groups, social support, education and home situations
- Use of **social media** and the effects on mental health
- **Child protection**

Factors **particularly relevant to older people**:

- Screen for **depression, cognitive impairment, physical ill health** and frailty
- Recognise increased potential for **loneliness** and isolation
- Recognise that there are **higher rates of suicide after an episode of self-harm for older people**

Check for caring responsibilities in all. Consider historic factors, changeable and current factors, future factors (including specific upcoming events) and protective factors.

► Management

Management should be guided by the psychosocial assessment. Support for social aspects may include signposting to a social prescriber.

Treatment of any co-existing conditions that may be related to self-harm should be optimised. Examples include depression, PTSD, drug and alcohol misuse.

Pharmacological therapy is not recommended purely for self-harm. However, it may form part of the treatment of co-morbidities linked to self-harm, such as anxiety or depression.

Adults may benefit from **CBT-based psychotherapy**. There is some evidence that it has positive effects on the repetition of self-harm at long-term follow-up and on depression, hopelessness and suicidal ideation. Children and teenagers may be offered **Dialectical behaviour therapy adapted for adolescents (DBT-A)**. This is concurrent family and individual therapy aiming to equip young people and their parents and carers with the skills to reduce or stop self-harm and suicidal behaviours, effectively manage their emotions and improve their relationships. Of note, the studies on DBT-A focused on females over the age of 12 who frequently self-harmed, so the benefits to others are unclear.

► Safety planning

The creation of a safety plan is recommended as a tool to reduce self-harm and suicide. The idea is to enable people to recognise times/ situations when they are at greater risk of self-harm or suicide, and to identify self-care strategies and wider resources to keep them safe. As **discussed in the suicide prevention chapter**, evidence from a recent meta-analysis suggests that the simple intervention of creating a safety plan can reduce the risk of suicidal behaviour by a staggering 43% ([BJPsych](#)).

What could be in a safety plan for self-harm?

- Use the individual's words and ideas so it personal to them
- Recognise triggers and warning signs of increased distress, further self-harm or a suicidal crisis
- Identify individualised coping strategies- e.g music/ exercise
- Identify social contacts/ settings as a means of distraction
- Identify family members or friends to provide support/ help resolve the crisis
- Include contact details for the mental health service including OOH emergency contacts
- Keep the environment as safe as possible by working together to remove/ restrict access to lethal means of suicide

This written plan should be held by the person and, with consent, shared with family, carers and the medical team. The RCPsych has useful information and a simple safety plan template [Feeling Overwhelmed](#). Also recommended is [Staying Safe, My Safety Plan](#).

► Harm minimisation

Ways to self-harm safely are sometimes discussed as part of a harm minimisation strategy. The new NICE guidance does not make any recommendations about 'safe' self-harm due to a lack of evidence either for or against. Instead, it recommends that the focus should be on avoiding, delaying or reducing further episodes. Examples of this include a discussion of distraction techniques, providing information on potential complications of self-harm and the impact of drugs and alcohol on the urge to self-harm. Wound care advice should be provided as part of harm reduction.

► What are the risks?

- Without help 1 in 3 will continue to self-harm
- Complications from the self-poisoning/ injury: wound infection, damage to nerves/tendons, scarring
- Suicide: **over a 15 yr period 3% of people who self-harm will die by suicide**

A recent [BJPsych meta-analysis](#) identified previous self-harm as one of four major risk factors for suicide (HR 1.68). Of note, a recent [BMJ paper](#) on suicide risk assessment reported **4% of people who attend ED with self-harm will kill themselves within 5 years**. This leaves clinicians with the difficult balance of recognising that whilst self-harm is common amongst those who complete suicide, the vast majority of people who self-harm will not attempt or complete suicide. **How does that inform practice?**

- Beware of patients who present to ED with self-harm
- Beware of patients with escalating self-harm or using dangerous methods
- Safety plan

Patient resources

- [Royal College of Psychiatrists](#): Patient information leaflet giving information on what self-harm, self-help strategies and treatments are available.
- [National Self-harm network](#): UK charity offering an online forum for peer support
- <https://selfharm.co.uk>: free online course for 11-19yrs olds struggling with self-harm
- [CALM: campaign against living miserably](#). Web chat, helpline and general resources focused around men
- [Calm Harm App](#): free app which aims to help users manage or resist the urge to self-harm. Based on DBT.
- [Samaritans](#): 24/7 phone line, call 116 123

Support for parents/ carers:

- [Royal College of Psychiatrists](#): Parent and carers information leaflet
- [Parents helpline and web chat](#)

Safety plans:

- Simple principals and plan: [Feeling Overwhelmed](#)
- Digital: [Staying Safe, My Safety Plan](#)

NB Learning Points & Quality Improvement Ideas: Self Harm

1. **Safety plans are recommended as a tool for reducing the risk of self-harm or suicide.** Could you audit patients with a history of self-harm to see if they have one in place?
2. **Resources:** a key part of the support of patients and their family/carers is access to peer support and information. Do you have a list of local and national resources for peer support and information on self-harm? Could you compile a template ready to send via message to patients when needed. See resources above

KISS: Self Harm

[NICE NG225 Sept 2022](#)

Primary care assessment

- Assess physical and mental health ASAP after an episode of self-harm:
 - Assess emotional and **mental state**, and level of distress
 - Is there an **immediate concern about the person's safety?** Suicidal ideation?
 - Explore **why** they have self-harmed and what **function** it takes for them
 - Assess any injury/ poisoning:
 - Wounds: give self-care advice/ nursing support/ refer to ED as appropriate
 - Poisoning: consult [Toxbase/ BNF](#) or the [National Poisons Information Service](#)
- **Safeguarding:** assess for domestic abuse, violence, exploitation
- **Establish means** of self-harm - discuss how to remove access to this where possible.
- Seek **consent to liaise** with family/ carers for collateral information and to provide ongoing support
- Assess whether there are any concerns about **capacity**
- Check whether the person has a **care plan** or **safety plan**

Primary care management

- Offer regular appointments for review. If there are ongoing **safety concerns** the first review should be **within 48hrs**.
- **Medication review:** consider the **safety profile** (e.g. TCAs, are they being prescribed any opiate analgesics?) and **quantity** of medication prescribed, especially if the overdose risk is high. Consider potential interaction with ETOH/ recreational drugs.
- Optimise management of any **co-existing mental health conditions**
- Consider **safety plan**
- **Provide advice on scar management**
- Provide **information on self-help resources** (see patient resource box above), local **peer support groups**, online forums, national and local charities
- **Family/carers-** recognise the emotional impact on them, provide advice on how they can support someone who self-harms and what to do if they self-harm again. Signpost to resources (see above)

When to refer

- Consider referral to mental health or social care for a **psychosocial assessment** for anyone who presents with the following self-harm
- **RED FLAGS for prioritising referral to mental health:**
 - The person's levels of concern/ **distress** are **rising**, high or sustained
 - Increasing **frequency/ severity** of self-harm.
 - Increasing **suicidal intent**
 - Person asks for more support from mental health services
 - Clinician concern
 - Rising distress in family members/ carers/ children

EATING DISORDERS

NICE updated their **guideline on the assessment and management of eating disorders** in children, young people and adults in December 2020 [NICE 2017, NG69](#). The BMJ have also published an excellent clinical update [BMJ2017;359:j5245](#), complete with a useful [infographic](#). *Both the guideline and the paper urge prompt referral of patients with suspected eating disorders as early specialist multidisciplinary treatment significantly improves long-term outcomes.*

Eating disorders have a profound long-term effect on psychological, physical and social dimensions of health and have a major impact on patients and their families. So, this guideline is very important. *However, GPs in many areas are likely to find resource-related barriers to implementing some of the recommendations, such as to refer ALL suspected cases of eating disorders 'without delay' to a 'community based eating disorder service'.* Whilst specialist help may be difficult to access in many parts of the NHS, and waiting lists incredibly long, it is an **important learning point that early intervention is needed to improve outcomes**. We often adopt a 'wait and see approach' in primary care and whilst for many conditions this is the best thing to do for people with eating disorders we may be doing them a disservice. **We summarise the assessment and referral of eating disorders in the KISS below.**

More recently in 2022, we have 2 new/updated guidelines. SIGN published their eating disorder guideline in January 2022 ([SIGN 164](#)) which, like NICE, is very much secondary/tertiary care focussed and only deals with management rather than assessment, but it has a few interesting points not addressed by NICE - notably the **importance of addressing dysfunctional exercise**. But more importantly, very recently the new guideline from the RCPsych has been published for **assessing and managing the sick patient with an eating disorder**. Previously known as MARSIPAN, the new guidance has the new acronym MEED ([Medical Emergencies in Eating Disorders](#)), which we discuss below and summarise in the KISS summaries.

► **New Guideline on managing eating disorders - [SIGN164 Jan 2022](#)**

SIGN have recently published a guideline on eating disorders, and whilst the bulk focuses on specialist management, there are some interesting points to note, that are not covered in the NICE guideline. The SIGN guideline **highlights 3 areas to support achieving and maintaining recovery**:

- Refeeding to an optimised healthy weight for those with anorexia, both as a life-saving measure and also as an adjunct to reduce relapse.
- Assessing and treating the acutely ill patient using the MARSIPAN (now MEED) guidelines as discussed below.
- **Assessing for, and treating, dysfunctional exercise/activity** with specialist supervised exercise programmes as part of a comprehensive multidisciplinary treatment programme, using the principles of the Safe Exercise at Every Stage guidance.

Why focus on dysfunctional exercise? [Safe Exercise at Every Stage \(SEES\)](#)

- Dysfunctional exercise is one of the strongest predictors of eating disorder relapse, second only to social isolation.
- It is often the first presenting and last remaining symptom of an eating disorder.
- **~80% of individuals with an eating disorder engage in dysfunctional exercise.**
- The thought is that providing patients with safe movement support and education may improve mental and physical eating disorder symptomatology, treatment compliance and long-term prognosis.

What about medications?

The SIGN guideline (like NICE) very much focuses on psychological therapies as the mainstay for all eating disorders, but whilst NICE simply say medications should not be offered as a sole treatment for any eating disorders, it is worth noting potential options that SIGN recommend alongside psychological therapies and supervised exercise programmes:

- **Anorexia nervosa** - olanzapine is recommended as a potential adjunct to psychological therapies.
- **Bulimia** - Antidepressants (1st line fluoxetine up to 60mg daily) are an option.
- **Binge eating disorder** - medications are NOT recommended either as an adjunct or alternative option.

► **Medical Emergencies in Eating Disorders** - [MEED RCPsych May 2022](#)

An important new document was published in May 2022 by the Royal College of Psychiatrists to give a comprehensive framework for both clinicians and commissioners on how to **manage medical emergencies associated with eating disorders**. The summary of the risk assessment is in the KISS summaries below but some **important general points** for us to be aware of from this document:

- This guidance succeeds the previous MARSIPAN guidance.
- It has been broadened to **include other eating disorders** including bulimia nervosa, avoidant restrictive food intake disorder (ARFID) and binge eating disorder, as well as Anorexia Nervosa.
- The key message is the same as previous guidance - **Patients with eating disorders may be at high risk despite appearing well and having normal blood parameters.**
- Be aware of **suicide risk** as well as physical risk - 1:5 deaths from Anorexia nervosa are due to suicide.
- The guidance also highlights the **key role we have in primary care** *'The GP or school-based health staff may be the first professional to see and assess the patient. They play a critical role in engaging the patient, initiating treatment and referring for specialist care'*.
- The guidance points out the difficulty we may have with a first presentation of someone with a possible eating disorder who may be very ill, as **other differential diagnoses need to be borne in mind**. A few principles are offered here:
 - Eating disorders are relatively common and other conditions that may mimic this presentation (e.g. cytopenias and weight loss seen in haematological diseases) are relatively rare.
 - If the diagnosis is in doubt do not be diverted from the importance of providing nutrition as continued weight loss will increase risk.
 - We should perform a full clinical assessment and basic blood tests (FBC, blood film, coeliac screen, CRP) but don't be sidetracked by abnormal LFTs - raised liver enzymes are common in severe malnutrition both during starvation and refeeding episodes.
 - The guidance states *'It would be reasonable to refer for specialist assessment any patient in whom an eating disorder is considered as part of the differential diagnosis and to withdraw the referral if an alternative diagnosis is confirmed. Most importantly, the search for a diagnosis should never delay nutritional therapy in an undernourished patient.'*
- The guidance highlights the **differences in risk in different populations**:
 - In **children/adolescents** weight loss is often more acute due to lower body fat stores, and medical compromise occurs relatively frequently.
 - **Adults** with more long-standing eating disorders may run higher levels of risk and reach care later and thus be more unwell on presentation, often with depleted intracellular electrolytes (e.g. potassium) which may not be reflected in serum levels.
 - Patients with **bulimia nervosa** can present with severe electrolyte disturbance and GI complications with a normal weight or overweight.
 - **Males** should be specifically asked about excess training and use of anabolic steroids - use of anabolic steroids and vitamin D injections can lead to increased physical risk at higher BMI levels.
 - **People who overtrain** may have extremely low body fat levels and larger muscles.

NB Learning Points & Quality Improvement Ideas: Eating disorders

- We need to lower our threshold for considering the presence of an eating disorder in a broad range of clinical scenarios - think '**could this be an eating disorder**'.
- Remember eating disorders can present in people with normal weight or overweight.
- The recent SIGN guideline alerts us to look for the **presence of dysfunctional exercise** as either a presenting feature or complicating factor for eating disorders - dysfunctional exercise is thought to be present in ~80% of people with eating disorders and is a very strong predictor for relapse.
- The NICE guideline recommends that ALL suspected cases of eating disorders are referred to a specialist, age-appropriate service.
- **GPs have a vital role in detecting really sick patients** who need urgent medical intervention. The recent MEED guidance gives a good traffic light system for risk assessment but it is really important to note this traffic light guidance is for 'impending risk to life' so even people in the green categories are at risk of complications - **do not reassure them that their risk is 'low'** but emphasise the severity of the illness and importance of engagement with treatment.
- Patients with eating disorders can **appear deceptively well despite being very sick** and at high risk of complications (or death) - they can remain energetic and active shortly before collapse and can have normal blood parameters in the face of severe malnutrition.
- **Quality Improvement Ideas:**
 - Practice audit: have all patients identified as having an eating disorder been referred and are they having annual check-ups as per the primary care monitoring KISS below?
 - Familiarise yourself with the MEED guidance below - have you had any significant events or near misses of sick patients with eating disorders in your practice? Consider a group learning session to highlight the risks of eating disorders and the new MEED guidance.
 - If you see patients/family members with suspected eating disorders consider directing them to the resources in the box below for help and support whilst awaiting specialist input.

Eating Disorders - Further resources

Patients and families:

- BEAT:
 - [Main site](#)
 - [For friends and family](#)
 - [Helplines and webchat](#)
- Advice and support for parents from [MindEd](#) and [FEAST-ED](#)
- Information on anorexia and bulimia from the [Royal College of Psychiatrists](#)
- [Comprehensive resources from the CCI](#) in Australia on disordered eating
- [MalevoiceED](#) - resources for males with eating disorders

Primary Care Professionals:

- [RCGP Tools](#)
- [Academy for Eating Disorders](#) - including some useful 1-page leaflets on 9 truths about...
- [Medical Emergencies in Eating Disorders](#) (MEED) for adults and children
- [Books on prescription/book recommendations](#)
- [Self-guided CBT](#)

KISS: Background to Eating Disorders

[DSM-5 Criteria for Eating Disorders](#), [BMJ2017;359;j5245](#), [CKS 2019](#), [NICE 2020](#), [NG69](#)

- Eating disorders have a profound long-term effect on psychological, physical and social dimensions of health and have a major impact on patients and their families.
- The emotional and physical consequences of eating disorders result in a high mortality rate from malnutrition, suicide and electrolyte imbalance, particularly in those with anorexia nervosa.
- It is estimated that >700,000 people in the UK have an eating disorder, ~90% of whom are female.
- But recent community based epidemiological suggest that as many as 25% of people with eating disorders are male and that men and boys remain under-recognised - see [AED information](#).
- Both NICE guideline and BMJ paper urge **prompt referral of patients with suspected eating disorders** as early specialist multidisciplinary treatment significantly improves long-term outcomes.
- People with eating disorders should be managed in specialist multidisciplinary care services, including close monitoring of weight, mental and physical health and risk factors; dietary counselling and nutritional support should be offered as well as a range of psychological treatments.

Diagnostic criteria for eating disorders:

- **Anorexia Nervosa** (approximately 10% of eating disorders):
 - Restriction of energy intake leading to a significantly low body weight in the context of age, sex, development and physical health.
 - Intense fear of gaining weight or becoming fat or persistent behaviour that interferes with weight gain even though at low weight.
 - Disturbance in the way in which one's body weight is experienced e.g. undue influence of body shape on self-evaluation, or persistent lack of recognition on the seriousness of current low body weight.
- **Bulimia Nervosa** (approximately 40% of eating disorders):
 - Recurrent episodes of binge eating, associated with inappropriate compensatory behaviours (e.g. self-induced vomiting, misuse of laxatives, fasting, excessive exercise) to avoid weight gain.
 - Episodes occur $\geq$ once per week for at least 3 months.
 - Self-evaluation is unduly influenced by body shape and weight.
 - The disturbance does not occur during episodes of anorexia.
- **Binge Eating Disorder** (approximately 50% of eating disorders):
 - Recurrent episodes of binge eating, $\geq$ once a week for 3 months.
 - Typically eating much more food more rapidly than normal associated with a sense of a lack of control. Binges are associated with 3 of:
 - Eating much more rapidly than normal.
 - Until uncomfortably full.
 - Large amounts of food when not hungry.
 - Eating alone due to embarrassment about how much one is eating.
 - Feeling disgusted with oneself or guilty afterwards.
 - Marked distress regarding binge eating.
 - Binge eating is not associated with compensatory behaviour (e.g. vomiting or purging) or anorexia or bulimia.

KISS: Assessment and Referral in Eating Disorders

[NICE NG69, 2017](#) [CKS 2019](#)

Could this be an eating disorder?

- To improve our detection of eating disorders we need to be aware that eating disorders occur in either sex at any age (although the risk is highest in boys and girls aged 13 to 17), and be aware they **may present in multiple ways** with physical, mental or social problems.
- Be aware **eating disorders may be present in people with normal, or above normal weight**, behaviours are often covert and some people may have limited insight into their behaviours.
- Eating disorders may present in many ways including:
 - **Concerns directly related to weight or diet**, raised by the patient or their family:
 - Unusually low or high BMI, restrictive diet when already underweight, disproportionate concern about weight (e.g. concerns about weight gain as a side effect of contraception).
 - Rapid weight loss ($\geq 1\text{kg}$ a week is a 'red flag' that needs an urgent intervention).
 - **Behavioural signs:**
 - Social withdrawal related to food, excessive exercise.
 - Participation in high-risk activities e.g. professional sport, dance and modelling.
 - **Physical symptoms and signs:**
 - Menstrual or unexplained gastrointestinal symptoms, including abdominal pain.
 - Unexplained electrolyte imbalance or hypoglycaemia.
 - Physical signs of malnutrition e.g. poor circulation, fatigue, dizziness, palpitations, fainting, pallor.
 - Faltering growth in children and young people.
 - Atypical dental erosions.
 - **Mental health** problems including depression, anxiety and substance misuse.

Assessment:

- **Screening tools?**
 - A simple 'screening tool' does exist for eating disorders called the **SCOFF questionnaire**; however research [Int J Eat Dis 2016](#) shows that whilst it is specific (94%, so few false positives) it is poorly sensitive (54%, so lots of false negatives).
 - NICE state SCOFF, or other **screening tools, should not be used as the sole method to determine whether someone does have an eating disorder**.
- Do not use single measures such as BMI or duration of illness to determine whether to offer treatment for an eating disorder.
- So, ultimately primary care assessment should be based around having a high index of suspicion in people presenting with possible symptoms and signs and sensitively assessing for an eating disorder through further questioning.
- We should also do a **physical assessment and risk assessment** (see KISS below), including consideration of investigations; but CKS state that *'extensive investigation is not usually required in primary care. Most people with an eating disorder will have normal blood results which are a poor indicator of risk, however, some tests may be useful to rule out complications'*.

Refer:

- Refer when we clinically suspect an eating disorder based on our sensitive questioning. **Do not use watchful waiting for managing eating disorders.**
- *'If an eating disorder is suspected after an initial assessment, refer immediately to a community based, age-appropriate eating disorder service for further assessment or treatment'.*
- NICE deliberately do not give specific BMI levels to guide management: *'Do not use single measures such as BMI or duration of illness to determine whether to offer treatment'*.

KISS: Risk Assessment in Eating Disorders

MEED 2022

Background:

- The new guidance accepts that doing risk assessments for people with eating disorders is not easy and that we need to *'balance risks in physical, psychological and social domains while paying attention to matters of insight, motivation, consent and the legal framework for intervention'*.
- The risk assessment combines **clinical assessment with investigations** with a traffic light system indicating risk to life (note point below that 'low risk to life' does NOT mean low risk).
- The guidance highlights that the framework **requires clinical judgement** for interpretation and that *'risk assessment should be used to inform rather than dictate clinical management'*.
- However, the general guidance is that **≥1 red or ≥2 amber ratings should be considered high risk**(with the implication that admission should be considered).

Potential pitfalls for us to be aware of when doing a risk assessment:

- Patients can appear deceptively well with normal blood parameters despite being very sick.
- Be aware that the cognitions, associated distress and fear of weight restoration may limit the capacity of individuals to provide an accurate account of their presentation - there is a risk we can be falsely reassured by this; consider parent/carer information.
- Some patients may have an extremely powerful drive to exercise which overrides their nutritional reserve - they can appear energetic right up to the point of collapse.
- 'Normal' blood parameters are frequently seen in advanced malnutrition - do not be reassured by this.
- **Do NOT reassure patients that their risk is low** - firstly this can compound the dismissive nature of their cognitions and that change is not necessary and secondly even if all the parameters fall in the green criteria ('low impending risk to life') they are potentially still at risk of complications. **Emphasise the severity of the problem and the lack of precision in risk assessments.**

Is compulsory admission and treatment needed?

- The aim will always be to give people with eating disorders treatment voluntarily with their agreement.
- Although a last resort **compulsory admission and treatment may be needed in situations where a patient with an eating disorder refuses treatment and there is an imminent risk to life.**
- In these situations, the person may be highly distressed or very unwell and lack capacity and decisions need to be made in their best interests.
- The MEED guidance covers the **key questions relating to the law** regarding compulsory treatment:
 - Eating disorders are classified as mental disorders so are potentially subject to compulsory treatment under Mental Health legislation.
 - Feeding (including NG feeding) is regarded as a core component of treatment for Anorexia nervosa thus can be enforced (if necessary) under the Mental Health Act.
 - **Capacity** - Anyone over the age of 16 is assumed to have capacity to make their own decisions unless demonstrated otherwise - generally, patients with eating disorders will fulfil most of the criteria for capacity but **often do NOT fulfil the ability to 'use and weigh up the relevant information'** as has been demonstrated in several legal cases in England and Wales (e.g. the severely undernourished patient who states 'I cannot accept tube feeding (or another refeeding plan) because I cannot tolerate weight gain' is making the decision based on the anxiety about weight restoration inherent to the diagnosis, and hence lacks the capacity to accept the need for adequate nutrition).
- In the situation where a **patient requires treatment necessary for their own health and safety and either lacks capacity or refuses treatment we need to call the Mental Health Services for assessment of Mental Health or Capacity Acts** ([click here](#) for flowchart summary - see P117/Figure 5)

KISS: Risk Assessment in Eating Disorders Continued (see above KISS for interpretation)

[MEED 2022](#) & [Click here](#) for a summary checklist (Appendix 3/P182)

Clinical parameter	Red: High impending risk to life	Amber: Alert to high concern for impending risk to life	Green: Low impending risk to life
Weight loss	≥1kg/week for ≥2 weeks or rapid weight loss or calorie intake <500kcal/day for ≥2d	500g-1kg/week for ≥2 weeks	<500g/week or fluctuating weight
BMI (<18s use median % age adjusted BMI)	<18 yrs age BMI <70% >18 yrs BMI <13	<18 yrs BMI 70-80% >18 yrs BMI 13-14.9	<18 yrs BMI >80% >18 yrs BMI >15
HR	<40 bpm	40-50 bpm	>50 bpm
BP	Standing syst. <0.4th centile for age or <90 if >18 yrs with recurrent syncope and post. drop >20mm/Hg or ↑HR >30 bpm (35bpm if <16 yrs)	Standing systolic <0.4th centile for age or <90 if >18 yrs with occasional syncope and post. drop >15mm/Hg or ↑HR >30 bpm (35bpm if <16)	Normal standing systolic BP
Hydration	Fluid refusal or severe dehydration (10%) - reduced urine output, dry mouth, ↓ skin turgor, postural BP drop (above), tachycardia, sunken eyes	Moderate dehydration (5-10%) - reduced urine output, dry mouth, postural BP drop (above), normal skin turgor, mild tachycardia	No more than 5% - may have dry mouth or -ve fluid balance
Temp.	<35.5°C	<36°C	>36°C
SUSS (Sit Up and Squat-Stand) Test -click here	Unable to sit up from lying flat, or to get up from squat at all or only by using upper limbs to help. Score 0-1	Unable to sit up or stand from squat without noticeable difficulty. Score 2	Able to sit up from lying flat and stand from squat with no difficulty. Score 3
ECG F = female M = Male	<18 yrs QTc >460ms (F), >450ms (M) >18 yrs QTc >450ms (F), >430ms (M) - AND other significant abnormality	<18 yrs QTc >460ms (F), >450ms (M) >18 yrs QTc >450ms (F), >430ms (M) - NO other significant abnormality - taking QT ↑ meds	<18 yrs QTc <460ms (F), <450ms (M) >18 yrs QTc <450ms (F), <430ms (M)
Biochem.	Hypokalaemia <2.5mmol/L Hypoglycaemia <3mmol/L ↓albumin/sodium/calcium Transaminases >3x normal		
Haem.	Low WCC Hb <10g/L		
Activity/exercise	Dysfunctional exercise >2 hrs/day with malnutrition	Dysfunctional exercise >1 hr/day with malnutrition	Dysfunctional exercise <1 hr/day
Purging	Multiple daily episodes of vomiting/laxative abuse	Regular episodes (≥3x/week) vomiting/laxative abuse	
Other	Severe cognitive slowing, severe haematemesis, significant alcohol, mod/high suicide risk	Low suicide risk but self harming behaviours e.g. cutting	

KISS: Primary Care Monitoring and Management [CKS 2019](#) [NICE NG69, 2020](#)

- Ensure there is a clear agreement between primary and secondary or tertiary care about the responsibility for monitoring a person with an eating disorder.
- **Annual check (at least) of physical and mental health:**
 - For those with chronic anorexia nervosa not receiving ongoing treatment from secondary care. This should include:
 - Weight or BMI, pulse and blood pressure.
 - Problems with daily functioning.
 - Assessment of risk, related to physical and mental health.
 - ECG if purging and/or significant weight changes, or if on medication that can compromise cardiac functioning e.g. electrolyte imbalance, bradycardia, hypokalaemia, or a prolonged QT interval.
 - Monitor growth & development in children and young people.
 - 'Relevant blood tests'; Depending on the clinical situation this might include:
 - FBC (?anaemia, leucopenia, thrombocytopenia - associated with malnutrition or GI loss).
 - U&E (?hypokalaemia associated with laxative abuse or vomiting, ?hyponatraemia associated with excess water intake, ?elevated electrolytes associated with dehydration).
 - LFTs - may be elevated with dehydration.
 - Blood glucose.
 - Consider FSH, LH, oestradiol, prolactin in women presenting with amenorrhoea.
 - Further tests may be indicated on specialist advice or if alternative/concurrent diagnoses are suspected e.g. TFTs, Ca, Po, Mg, B12, folate, ferritin.
- **Compensatory behaviours:**
 - **Vomiting** - give advice on the importance of regular dental and medical review:
 - Avoid brushing their teeth immediately after vomiting; this can brush acid onto the teeth causing further damage.
 - Rinse with water or a non-acidic mouthwash after vomiting.
 - Reduce the acidity of the oral environment (for example by avoiding acidic foods and drinks).
 - **Misuse of laxatives or diuretics** - advise that:
 - Laxatives and diuretics do not reduce calorie absorption and do not help with weight loss.
 - Laxative or diuretic use should be gradually reduced and stopped.
- **Risks of cardiac complications:**
 - Put an alert in the person's prescribing record to highlight the potential risk of adverse drug effects.
 - Adverse drug effects, such as prolonged QT interval and cardiac dysrhythmias are more likely in malnourished people and those with electrolyte abnormalities - review all drugs.
- **Contraception:**
 - Women who are sexually active should have contraception even if they are amenorrhoeic.
 - IUD/IUS or implant are the most effective contraception, but note implant may be technically difficult if limited subcutaneous tissue.
 - Anorexia nervosa is a significant risk for osteoporosis so progesterone injectable requires careful consideration and informed discussion.
- **Osteoporosis risk in anorexia nervosa** (investigation and treatment likely to need specialist advice):
 - Explain the main way of preventing and treating low bone density is reaching and maintaining normal body weight.
 - Consider a bone mineral density scan after one year of being underweight in children and young people and two years in adults, earlier if bone pain or fractures.
 - If osteoporosis present advise avoiding impact sport.

SUICIDE PREVENTION

Suicide touches us all in General Practice. It is a rare but devastating event, yet seeing patients with suicidal ideation is a daily occurrence. An excellent recent BJGP paper [BJGP2021;71;409](#) discussing safety plans, quotes that 80% of people who die by suicide contact primary care in the year before. Clearly, we have a key role to play in suicide prevention but managing risk in primary care is a common concern for us. *Safety plans have long been advised as a tool to help to reduce risk, and new research shows that they can be highly effective.*

► *Safety plans interventions for suicide prevention* [BJPsych 2021](#)

Safety plans have been recommended for a long time in guidelines on suicide prevention, but we have lacked evidence as to how effective they are.

- This new study was a meta-analysis, of safety planning-type interventions for patients at risk of suicide
- 6 studies, n=3,500 patients were included
- *The relative risk of suicidal behaviour amongst patients who received a safety plan was 0.57 compared to controls, with a NNT of 16*
- There was no significant effect found for suicidal ideation
- **Conclusion:** this is the first study to report a meta-analysis of safety plans for suicide prevention, and it shows a *43% reduction in suicidal behaviour in patients receiving safety plans*. These results support the use of safety plans in clinical guidelines on suicide prevention.

An excellent recent BJGP paper [BJGP2021;71;409](#) (*'How can we work together to keep you safe?'*) discusses safety plans more generally. During an acute suicidal crisis people often feel a sense of hopelessness and a 'claustrophobic sense of being trapped' and that things will never improve. Safety planning is a 'simple and brief' intervention whereby clinician and patient work collaboratively to identify:

- **Signs** such as behaviours, feelings or situations in the patient's life that can trigger a suicidal crisis
- **Protective factors** such as coping strategies or resources that the patient can use during a crisis to help them cope and to break the cycle of suicidal thinking

To co-create an individualised safety plan with patients, the paper identifies 5 key questions to ask our patient:

- *What are the signs e.g. stressful situations, or distressing thoughts or feelings that may trigger suicidal thoughts?*
- *What are some of the things you can do to distract yourself when having suicidal thoughts?*
- *Who would you turn to (e.g. a friend or relative) when feeling suicidal?*
- *Which professionals or agencies can you contact when feeling suicidal?*
- *What things (e.g. sharp objects and medication) can we remove or limit access to in order to keep you safe?*

It is important to keep the safety plan brief and simple, ideally using the patients own words so that it is personalised to them and to work with the patient to identify reasons to stay alive e.g. what keeps them going, and what can they look forward to in the future? **The paper concludes that 'safety planning is a quick, practical and effective intervention that can save lives'.**

The RCPsych has useful information and a simple safety plan template [Feeling Overwhelmed](#). Also recommended is [Staying Safe, My Safety Plan](#).

► *How can we assess and mitigate risk in an evidence-informed way?*

This difficult question was explored in an excellent paper '*Exploring thoughts of suicide*' [BMJ2017;356:j1128](#), which has a superb [infographic on assessing risk](#).

Suicide: Background Facts

[BMJ2017;356:j1128](#) & [BMJ2015;351:h4978](#) & [RCGP/RCPsych Guidelines](#)

- 5,000 people die every year in the UK by suicide
 - This is twice as many as die in car accidents
 - Only 25% are known to specialist mental health services, but *30% have seen their GP within a month of their death*
 - Suicide rates are 3 times higher in men than women
- After a fall in incidence, **suicide rates have started to rise again**
 - It is one of the top 3 causes of death in people aged under 44
 - The highest rates are in men aged 30 to 60
 - Rates are increasing in teenagers aged 15 to 19
- **Suicide risk is most commonly in people with mental illness** but varies substantially by socio-demographic (age, gender) and clinical features
 - Risk is highest within the first few months after diagnosis across all mental health disorders
- **Highest risks for suicide** are (odds ratios in brackets)
 - Previous self-harm (5)
 - Beware escalating self-harm behaviour
 - 4% of people who attend A and E with self-harm will kill themselves within 5 years
 - Previous mental illness (4)
 - Severe depression (2)
 - Misuse of drugs or alcohol (2 if also depressed)
 - Male sex (3)
- Protective factors that make suicide risk less likely include
 - Dependent children, religious belief and other family members who would be distressed
- **Combinations of risk factors are more important than individual characteristics**
- Accurate methods of predicting suicide remain elusive and the best approaches to reducing the risk of suicide remain unclear.
 - Although admission to the hospital is supposed to create a safe environment, the risk of suicide whilst an inpatient remains high. The risk is high also in the first few months after discharge.

► *How should GPs assess suicide risk?*

This paper [BMJ2017;356:j1128](#) explored the evidence which, sadly, is often of low quality. However, some conclusions can be made:

- 3.5% to 5% of all people report suicidal thoughts in any one year, and 0.5% to 0.7% will make a suicidal attempt
- **Asking about suicide does not increase the risk of self-harm.** We then need to explore ideas, intent and plans they may have in place.
- Assessment of risk needs to look at both **risk factors** and **protective factors**, from which we need to make a balanced overall assessment of risk

■ Risk factors

- A recent meta-analysis of risk factors for self-harm and suicide found only four predictors with strong evidence to support them (previous episodes of self-harm, male gender, being unemployed and having physical health problems). Additional risk factors identified from the UK inquiry into suicide also identifies active mental illness, living alone, being unmarried and drug and alcohol dependence as additional risk factors
- Some of these risk factors are static (i.e. cannot be changed, such as gender, demographics or past medical history) but some are dynamic and can be modified (e.g. untreated active mental illness). Identifying dynamic risk factors which may be amenable to change can help with a management plan. Combinations of risk factors are more important than individual characteristics
- **Protective factors** include family support, having children at home, strong religious faith and a sense of responsibility and problem-solving skills
- From this balance of risk and protective factors, we need to make an *individualised assessment of overall risk*
 - **Low risk:** manage in primary care with treatment of mental health problem
 - **Moderate risk:** careful primary care follow-up, treatment of mental health problem, consider mental health referral and signpost further support e.g. Samaritans
 - **High risk:** urgent mental health referral

► What do NICE say? [NICE NG222 Depression in Adults](#)

The recently updated NICE guideline on depression advises to assess self-harm risk and always ask directly about suicidal ideation and intent. If there is a risk of self-harm or suicide

- Assess whether the person has adequate social support and is aware of sources of help
- Arrange help appropriate to the level of risk. If considerable immediate risk to themselves or others refer urgently to specialist mental health services
- Advise the person to seek further help if the situation deteriorates
- Regarding antidepressants, do not withhold ADs if people feel suicidal but take toxicity into account when choosing antidepressants. TCAs (except lofepramine) & venlafaxine are associated with greatest risk.
 - People should be advised about the potential risk of increased agitation, anxiety and suicidal ideation in the initial stages of treatment and this needs to be appropriately 'safety netted'.
 - If a person with depression is considered at risk of suicide, do not withhold treatment but consider referral and increasing the level of support
 - Monitor for suicidal ideation in the early weeks of treatment and anyone at increased risk of suicide and young adults under the age of 25 should be reviewed at one week after starting.

KISS: Suicide Risk Assessment in Primary Care

Based on [BJGP2021;71;409](#) & [BMJ2017;356;j1128](#) & [BMJ2015;351:h4978](#)

'Suicide mitigation starts from the assumption that suicidal thoughts need to be taken seriously and met with empathy and understanding on every occasion'

General:

- *'Suicidal thoughts need to be taken seriously and met with empathy and understanding on every occasion'.*
- Establish rapport. Engender trust and hope wherever possible. Demonstrate acceptance of & compassion for the patient and their situation. Use open questions to explore, prior to closed questions about planning and intent.

Assess for mental illness and other risk factors:

- Check for symptoms of mental illness, psychiatric history and misuse of drugs and alcohol.
- Distressing psychotic phenomena e.g. command hallucinations, persecutory delusions are 'red flags' requiring urgent specialist assessment.
- Other risks include previous/escalating self-harm, male, unemployment, social isolation, job (health care professionals, farmers).

Assess (and document) current suicidal thoughts, plans and intent:

- Consider using **progressive questioning**. Exploring in sequence:
 - Hopelessness - *'nothing to live for', 'I'm a burden';* Ideas of self-harm; Ideas of suicide - *'have you been feeling life is not worth living'.. 'have you thought about ending your life?'*
 - Planning - (a plan is a significant risk - 56% probability of a suicide attempt in the next year) - *'have you made any plans to end your life'...'please tell me more about these plans'.*
 - Lethality and Access. Intent - *'would you carry out these plans'.*
- Risk of harm increases moving through the questions; beware escalating behaviour.

Establish risk (risk levels are fluid and have multiple determinants):

- **Low risk:** Hopelessness, ideas of self-harm, ideas of suicide, no planning
 - Manage in primary care with treatment of mental health problem
- **Moderate risk:** Ideas of suicide with planning
 - Consider careful primary care follow-up or mental health referral
- **High risk:** Urgent mental health referral

Management in primary care:

- Develop an **individualised SAFETY PLAN** - features should include:
 - *What are the signs that may trigger suicidal thoughts?*
 - *What are some of the things you can do to distract yourself when having suicidal thoughts?*
 - *Who would you turn to (e.g. a friend or relative) when feeling suicidal?*
 - *Which professionals or agencies can you contact when feeling suicidal?*
 - *What things can we remove or limit access to in order to keep you safe?*
- Click here for safety plan resources: [Printable safety plan](#) [Digital safety plan](#).

'End by showing compassion, engendering hope & summarising shared agreement of the plan'.

INSOMNIA

The Hot Topic for this most common of presentations is new evidence looking at the efficacy of cognitive behavioural therapy for insomnia (CBTi) and how this may be delivered across digital platforms as a self-help tool. *In May 2022 NICE approved Sleepio*, the self-help digital app based on CBTi, to treat insomnia and insomnia symptoms [NICE 2022, MTG70](#).

► *CBT for insomnia in primary care* [BJGP July 2019](#)

Cognitive behavioural therapy for insomnia (CBTi) consists of a number of sessions (typically 4 or 6) delivered 1:1 or in a group setting and focuses on managing anxieties related to insomnia and relaxation training with behavioural change such as sleep restriction. This involves reducing the amount of time that patients with insomnia spend in bed to improve their 'sleep efficiency'. This new systematic review of CBT for insomnia in primary care was published in summer 2019 [BJGP July 2019](#).

- This was a systematic review of 13 trials (n=1,500) in primary care settings, comparing CBTi to various controls (control groups typically consist of sleep hygiene advice)
- Medium to large therapeutic responses for sleep were seen, compared to controls over 4 to 6 sessions
 - patients fall asleep up to 30 minutes earlier than in controls
- Benefits were well maintained for follow-up periods, which ranged from 3 to 12 months
- Conclusion: *These results confirm the efficacy of CBTi for insomnia in primary care, and that CBTi should be the first-line treatment*

Face-to-face CBTi may be the current best treatment for insomnia but is limited by availability, so there is great interest in self-delivered digital CBTi across an interactive interface...

► *Digital CBT for insomnia* [JAMA Psychiatry 2019;76;21](#)

- A 12-month UK-based study. Patients self-reported symptoms of insomnia (n=1700) were recruited, and randomised to dCBT for insomnia (using the [Sleepio](#) app) with the control group being sleep hygiene education (SHE)
 - The dCBT programme is structured into six 20-minute sessions which people could complete within 12 weeks
 - Treatment content includes behavioural, cognitive and educational components
- Programmes were completed within 12 weeks of recruitment and there was a follow-up, up to 24 weeks
 - 50% of the dCBT group completed all 6 sessions of the programme
- Use of dCBT was associated with improvements in functional health and psychological well-being and large improvements in sleep-related quality of life
 - Symptoms of depression, anxiety and daytime sleepiness were all improved
- Conclusion: compared to sleep hygiene advice alone, dCBT improves both nighttime and daytime aspects of insomnia

Following this research Sleepio has been approved for use in the NHS by NICE [NICE 2022, MTG70](#)

- Sleepio is recommended as a cost-saving option for treating insomnia and insomnia symptoms in primary care for people who would otherwise be offered sleep hygiene or sleeping pills
- For people who may be at higher risk of other sleep disorder conditions (e.g. sleep apnoea, restless legs etc) a medical assessment should be done before referral to sleepio
- NICE state that face-to-face CBTi is current best practice, but that availability is limited so Sleepio increases patient access to CBTi and increases options for GPs when seeing patients with insomnia
- At a cost of £45 per person, Sleepio is cost-saving compared with usual treatment in primary care

KISS: Insomnia

Based on [NICE 2022, MTG70](#) & [BJGP2019;69:152](#) & [BJGP July 2019](#)

1 Assess to look for secondary causes

- External stimuli e.g. noise, light etc
- Psychological distress (depression, anxiety etc. is the cause in 50% of cases)
- Physical health problems e.g. pain, heartburn, restless legs, hot flushes, sleep apnoea (these are very common, contribute to 40% of cases and are under-recognised)
- Stimulants e.g. alcohol, caffeine, decongestants, SSRIs, beta-blockers, decongestants, diuretics, steroids, recreational drugs etc
- Give the patient a [sleep diary](#) at the end of the first consultation and arrange a follow-up.

2 Make the 'diagnosis' and treat the underlying cause

- Together with all the usual sleep hygiene advice ++ [NHS Choices sleep advice](#)
- For detailed and very useful self-help, including sleep hygiene advice see [click here](#)
- For an excellent and relatively cheap self-help book see [Coping with insomnia](#)

3 Primary insomnia (is the sole cause in only 12% of cases)

- Sleep hygiene advice++ [NHS Choices sleep advice](#)
- Relaxation techniques e.g. listening to music [Cochrane](#)
- CBT for insomnia (CBTi)
 - CBTi is increasingly available in primary care, either face-to-face in group therapy
 - digital CBTi using [www.sleepio.com](#) is approved by NICE in 2022 for use in primary care
 - If patients are at risk of a sleep disorder, such as sleep apnoea or restless legs, assess this first before referring to Sleepio
- Consider advising 'sleep restriction': calculate patients 'sleep efficiency' (how much time asleep/how much time in bed) and advise bed-time restriction (i.e. reducing the amount of sleepless time in bed)

4 Benzodiazepines & Z drugs

- Tolerance can develop after only 3 to 4 weeks and high risk of addiction as well as other adverse events such as cognitive impairment, daytime sleepiness, impaired sleep quality, decreased reaction time and increased risk of accidents and aggravation of depression
- If used, use only in severe case with intermittent use for 1 week or less and give [patient information](#)
- Advise patients of a high risk of tolerance & addiction & that the drugs do not tackle the underlying cause, for which they need CBTi
- For patients on long-term BZPs explain that discontinuation leads to improved cognitive and psychomotor functioning and regularly offer the opportunity for supported withdrawal
- Other hypnotics (e.g. sedating TCA antidepressants and antihistamines) also have a high side effect burden due to their anticholinergic properties)
- **Melatonin:** There is [evidence](#) that melatonin is modestly more effective than placebo

PANIC DISORDER

There has been a new systematic review and network meta-analysis of trials for panic disorder [BMJ2022;376:e066084](#) published. NICE updated their guideline on generalised anxiety disorder (GAD) and panic disorder in adults in 2020, [NICE 2020 CG113](#). Panic disorder is a discrete form of generalised anxiety disorder, however, there is clearly an overlap with other mental health disorders. As for GAD, the first line interventions are knowledge, self-help and psychological interventions.

Background, Panic Disorder [BMJ2022;376:e066084](#)

- Lifetime prevalence 1% to 5%
- Characterised by recurrent and unexpected panic attacks associated with several comorbid conditions such as anxiety, depression and cardiovascular disease and has a significant adverse effect on function
- Agoraphobia is a strong fear or anxiety provoked by real or anticipated exposure to a wide range of situations

► Drug treatment for panic disorder [BMJ2022;376:e066084](#)

This new systematic review of drug treatments for panic disorder with and without agoraphobia combined the outcomes of 87 studies including 13,000 patients and 12 different classes of drugs. However, the studies were mostly of low quality, with moderate to low certainty of evidence. Findings:

- SSRIs provide high rates of remission with low risk of adverse events for the treatment of panic disorder
- *Among SSRIs, sertraline and escitalopram were found to have the highest remission rates and low risk of adverse events*
- Beta-blockers alone were not effective however SSRIs combined with beta-blockers were associated with reduced anxiety symptoms, and 'top ranked' for improving anxiety and depression scores
- Other drugs associated with significantly higher remission rates than placebo included SNRIs, TCAs, benzodiazepines and MAOI
 - The authors propose that SNRIs whilst ranked lower than SSRIs for remission might recommend a second-line treatment
 - TCAs were superior to placebo for efficacy but worse for adverse events
 - Not surprisingly benzodiazepines were associated with a higher risk of adverse events
- **Conclusion:** This evidence supports the use of SSRIs as first line drug choice for the pharmacological management of panic disorder, with sertraline and escitalopram having the most favourable profile; if ineffective or not tolerated, the evidence supports a SNRI as second-line; betas blockers alone were ranked worst for remission, but SSRI plus beta-blocker were top-ranked for improving anxiety and depression scores

KISS: Panic Disorder

Based on [BMJ2022;376:e066084](#) and [NICE 2020 CG113](#)

1 Make the diagnosis

- Panic disorder is a separate anxiety disorder from GAD, characterised by recurring, unforeseen panic attacks followed by persistent worry about another attack or concern about its consequences
- Isolated 'panic attacks' do not necessarily constitute a panic disorder

2 Check for co-morbidity

- Depression & substance misuse are particularly common co-morbidities

3 Management

- **Step 1, recognition and diagnosis**
 - Identification, assessment, education and active monitoring; signpost self-help [NHS Panic Disorder](#)
- **Step 2, offer treatment in primary care**
 - Low-intensity psychological interventions (individual non-facilitated self-help or facilitated self-help based on CBT principles)
- **Step 3, consider referral** for psychological therapy (CBT) or offer drug therapy
 - **Drug treatment**, offer an antidepressant if the disorder is longstanding or if the patient has not benefited from or declined psychological help
- • ○ Offer an SSRI licensed for panic disorder
 - Assess regularly, but the full benefit may not be seen for 12 weeks
 - Doses at the upper end of the indicated range may be necessary
 - If beneficial, continue for a further 6 months, after which the dose can be tapered and the drug stopped
 - Recent evidence supports sertraline or escitalopram as first choice [BMJ2022;376:e066084](#)
 - If SSRI is unsuitable, or no improvement after 12 weeks, consider TCA imipramine or clomipramine (NICE) or SNRI ([BMJ2022;376:e066084](#)) as second-line drug
 - Benzodiazepines, sedating antihistamines or antipsychotics should **not** be prescribed for panic disorder
 - There is a lack of evidence to support the use of propranolol & it is therefore not recommended in guidelines; however recent evidence suggests that SSRI plus beta-blocker may give added benefit [BMJ2022;376:e066084](#)
- **Step 4**
 - Refer for specialist help, for highly specialist treatment such as complex drug and psychological interventions from multi-agency teams

BIPOLAR DISORDER

Bipolar is common (affects 2% of the population) and associated with high disability and risk of suicide, but it can easily be missed in primary care because people tend to present in the depressed phase. ***The suicide risk in bipolar is 20 times that of the general population.*** NICE updated their guideline in 2020 [NICE CG185 2020](#). Recent research [Br J Psychiatry 2019](#) suggests that in terms of drug management, guidelines are not being followed with 'concerning' data that the most common form of treatment for bipolar disorder in Scotland is antidepressant monotherapy and that lithium prescribing is in decline.

► ***NICE Bipolar guideline*** [NICE CG185 2020](#)

NICE updated their bipolar guideline in 2020, and they have amended recommendations in line with [MHRA guidance on use of valproate in women and girls](#).

- Bipolar disorder recognised as Type 1 (mania and depression) and the less severe Type 2 (hypomania and depression)
- The lifetime prevalence of Type 1 is 1%, and Type 2 is 0.4%
- It most commonly presents between the ages of 15 and 20 and has a strong genetic influence
- It is associated with significantly **increased risks of suicide and chronic physical illness**
- Around 2/3 of people with bipolar illness have a psychiatric co-morbidity, such as anxiety, impulse control disorders or substance misuse
- **The risk of recurrence after a mood episode is high**, (50% at one year, 70% at 4 years) and this has important implications for long-term management and relapse prevention
- Whilst drugs are important, the **evidence base for psychological treatments is strong** both in acute treatment and in preventing depression.
- The guideline emphasizes the **importance of physical lifestyle factors** (smoking, diet and exercise), which have been inadequately addressed in the past and has contributed to high premature death rates. These factors should be given high priority in management.

Learning Points and Quality Improvement Ideas: Bipolar Disorder

- Whilst drugs are important the evidence base for psychological therapies both for acute treatment and relapse prevention is high
- Physical lifestyle risk factor management is a key part of management
- Antipsychotics are first line treatment for mania, quetiapine for depression and lithium for relapse prevention
- Do not start SSRIs in the depressed phase, and stop them if people become hypomanic
- In suspected cases, consider using the [Mood Disorder Questionnaire](#)

KISS: Bipolar Disorder - Background

[NICE CG185 2020](#) [NICE CKS 2020](#)

- **Bipolar disorder** is characterised by episodes of both depressed mood and elated mood associated with increased activity (hypomania or mania).
- Bipolar disorder is recognised as Type 1 (mania and depression) and the less severe Type 2 (hypomania and depression).
- **A manic episode** is abnormally and persistently elevated, expansive, or irritable mood lasting at least 1 week, usually beginning abruptly and accompanied by at least three additional symptoms:
 - Increased energy or activity, restlessness, and a decreased need for sleep.
 - Pressure of speech or incomprehensible speech.
 - Flight of ideas or racing thoughts.
 - Distractibility, poor concentration.
 - Increased libido, disinhibition, and sexual indiscretions.
 - Extravagant or impractical plans (for example business investments, spending sprees).
 - Psychotic symptoms: delusions (usually grandiose) or hallucinations (usually voices).
- **A hypomanic episode** is similar to a manic episode but is not severe enough to cause impairment in functioning, need hospitalisation or have psychotic features.
- **A depressive episode** is a period of at least 2 weeks during which there is either depressed mood or loss of interest or pleasure in nearly all activities and is diagnosed in the usual way.
- The lifetime prevalence of Type 1 is 1%, and Type 2 is 0.4%.
- It most commonly presents between the ages of 15 and 20 and has a strong genetic influence; incidence 3.4% in 16-24 yr olds, lower in older age groups.
- Around 2/3 of people with bipolar illness have a psychiatric co-morbidity, such as anxiety, impulse control disorders or substance misuse.
- Suicide risk is very high and is the psychiatric condition with the highest suicide rate
- **The risk of recurrence after a mood episode is high**, (50% at one year, 70% at 4 years) and this has important implications for long-term management and relapse prevention.
- Whilst drugs are important, the **evidence base for psychological treatments is strong** both in acute treatment and in preventing depression.
- **Recent research** [Br J Psychiatry 2019](#) suggests that in terms of drug management, guidelines are not being followed with 'concerning' data that the most common form of treatment for bipolar disorder in Scotland is antidepressant monotherapy and that lithium prescribing is in decline.
- **Bipolar is easily missed in primary care** as patients often present in the depressed phase and are incorrectly diagnosed with unipolar depression. A study ([BJGP2016;66:e71](#)) suggested 7% of patients diagnosed with depression actually have bipolar.
- The NICE guideline emphasises the **importance of physical lifestyle factors** (smoking, diet and exercise), which have been inadequately addressed in the past and has contributed to high premature death rates. These factors should be given high priority in management.

KISS: Bipolar Disorder - Assessment and Management

[NICE Bipolar Disorder CG185 2020](#)

Diagnosis of suspected cases

- **Ask patients presenting with depression about previous episodes of over-activity or disinhibited** behaviour (especially in atypical cases, if there is a positive family history or in patients who do not respond to antidepressants).
 - Do you currently, or have you in the past experienced mood that was higher than normal?
 - At the same time did you have an increase in your energy levels so that you were much more active or did not need as much sleep?
 - Yes to either question should prompt further evaluation. A self-rating questionnaire the [Mood Disorder Questionnaire](#) has been developed to help identify patients.
 - **Refer all suspected cases for diagnostic confirmation**, urgently if they are a danger to themselves or others.
 - Do not start SSRIs in the depressed phase as they are ineffective, and may be harmful.

Management

- Monitor physical health at least annually, with a comprehensive check.
- **Drug treatment:**
 - Prescribing of mood stabilisers or antipsychotics should be specialist-led.
 - Do NOT start valproate in primary care to treat bipolar disorder.
 - Do NOT start lithium in primary care except under shared care agreements.
 - Antipsychotics are first-line for mania, but lithium is first-line to prevent relapse.
 - **Mania or hypomania:**
 - If a patient develops mania or hypomania and is taking an antidepressant, this should be stopped.
 - Antipsychotics should be offered first to treat mania or hypomania.
 - If the patient is taking lithium, check the level and increase the dose if needed.
 - **Depression:** bipolar depression should be treated with:
 - Quetiapine on its own, or fluoxetine combined with olanzapine.
 - Consider lamotrigine if no response to these.
 - If patients develop depression and are taking lithium, check the levels and increase the dose if they are low. If it is at the maximum, add in fluoxetine + olanzapine or quetiapine on its own.
 - **Relapse prevention:**
 - Lithium remains the most effective long-term treatment and is first line for long-term relapse prevention.
 - If ineffective, consider adding valproate (except in women of childbearing age).
 - If lithium poorly tolerated consider valproate or olanzapine instead, or quetiapine if it has been successfully used during a period of mania.
- **Psychological interventions:**
 - Adults should be offered high intensity psychological interventions (CBT or interpersonal therapy) & psychological interventions for relapse prevention.
- **Provide clear patient information:** [RCPsych Patient Info](#) & [NHS Patient information](#)

PERSONALITY DISORDER

Personality disorder (PD) is a controversial diagnosis. Yet, it is very visible to us in primary care as a common, burdensome and at times difficult to help long-term condition with many physical and psychiatric co-morbidities. Despite being so common, it is an area where GPs and primary care teams frequently feel challenged and have often felt unsupported by specialist services.

Personality disorders are a major Hot Topic. The WHO has recently re-classified [Personality Disorders in ICD-11](#) introducing major changes to how they are classified, including bidding farewell to Borderline PD and Emotionally Unstable PD. The NICE guideline [CG78 Borderline Personality Disorder](#) was updated in 2018 and is next due for review in 2021 and new [European Guidelines 2019](#) have published. There has also been a systematic review of the evidence by [BMJ Best Practice 2019](#) and a new systematic review from [Cochrane 2020](#) on psychological therapies for people with borderline PD. And new research [Personality and Mental Health 2019](#) has highlighted the greatly increased premature mortality of people with PD.

These publications help us to understand PD better and to answer the questions below. Crucially they remind us of *these essential learning points*:

- **People with PD have a greatly increased burden of physical disease** and shortened life expectancy of approximately 18 years compared to the general population
- **People with PD have a significantly increased risk of co-morbid and treatable mental health conditions**, such as severe depression, substance misuse and suicidal ideation
- **It is a myth that PD is not 'treatable'**, research shows improved outcomes with therapy and co-morbidities respond to treatment
- **People with PD have significant problems which affect the biological, psychological and social dimensions of their lives**; primary care is ideally placed to support them with a holistic, long-term approach based on continuity

► How are personality disorders defined? [ICD-11 classification for PD](#)

Personality disorders refer to long-lasting patterns of thinking about oneself and others which have a significant negative impact on one's ability to function in life, often resulting in extreme distress and social impairment. We all have aspects of our personality that can make life and relationships difficult at times, but it is at the extreme end of this spectrum when these character traits cause significant problems that are persistent and pervasive in life that it becomes personality disorder (PD).

Whilst different definitions of PD can seem confusing, there are common traits which people with PD have which we can recognise affecting **emotion** (especially emotional regulation), **cognition** (especially making appropriate decisions when under stress) and **behaviour** (especially impulsive and reckless behaviour, including self-harm, often in response to stress).

The new [ICD-11 classification for PD](#) in 2019 has introduced major changes to the definition of PD. It aims to simplify the diagnosis and classify the severity based on the impact that the PD has on functioning.

The definition focuses on the impairment of function which is classified as mild, moderate or severe based on a holistic evaluation of personality functioning and its emotional, cognitive and behavioural manifestations. Personality functioning may, of course, be impaired in various ways, so **different traits of PD** are defined to describe specific patterns which contribute to impaired personality functioning. This new definition replaces more narrow definitions of multiple different types of PD. Previously, there were 10 different types of PD (e.g. borderline or emotionally unstable, antisocial, narcissistic, avoidant, paranoid PD etc).

ICD-11 now greatly simplifies this definition, as a clinician can make a global diagnosis of PD and then categorise it as mild, moderate or severe depending on the impact. There are then six personality domain traits which may be prominent in some people with PD (see KISS summary below). Controversially **Borderline PD or Emotionally Unstable PD** is no longer a separate diagnosis but it remains as an optional qualifier for people with PD who exhibit a characteristic pattern of traits including emotional instability, impulsivity, fear of abandonment and repeated self-harm.

The **DSM-V** classification of personality disorders still retains 10 different types of PD, which are categorised into three separate clusters A ('odd, eccentric'; paranoid and schizoid PD), B ('emotional, erratic'; antisocial, borderline, histrionic and narcissistic PD) and C ('anxious, fearful'; avoidant, dependent and obsessive-compulsive PD).

KISS: Diagnostic requirements for Personality Disorders

Based on ICD-11 classification

■ General diagnostic requirements

- An enduring disturbance characterised by problems in the functioning of the self and/or interpersonal dysfunction (e.g. ability to develop and maintain relationships, to manage conflict and to understand others' perspectives)
- The disturbance has persisted for > 2 years
- The disturbance is manifest in patterns of **cognition**, **emotion** and with maladaptive **behaviours**
- The disturbance is manifest across a range of personal and social situations
- The patterns of the behaviour cannot be explained by primarily social or cultural factors
- The symptoms are not due to the direct effects of a medication or substance and are not better explained by another mental or behavioural disorder
- The disturbance is associated with significant distress or functional impairment

■ Aspects of personality functioning that contribute to the severity in PD

- **Emotional manifestations**
 - Range and appropriateness of emotional experience and expression, and the ability to recognise and acknowledge unwanted emotions
- **Cognitive manifestations**
 - Ability to make appropriate decisions in situations of uncertainty, especially under stress
- **Behavioural manifestations**
 - Flexibility in controlling impulse and modulating behaviour based on the situation and consideration of the consequences
 - Appropriateness of response to intense emotion and stress e.g. propensity to self-harm

■ **PD is then defined as mild, moderate or severe based on a holistic, individualised assessment**

- **Mild PD:** typical picture is someone in whom disturbances affect some areas of personality functioning but not others, creating some problems in relationships and social and occupational roles but is not associated with significant harm to self or others
- **Moderate PD:** disturbances affect multiple areas of personality functioning (e.g. ability to maintain relationships, or to control or modulate behaviour) causing marked problems in relationships and expected performance in expected roles; may be associated with harm to self or others and associated with the marked impairment in areas of functioning
- **Severe PD:** severe disturbances of functioning that seriously affect virtually all relationships, with severe compromise on the ability and willingness to perform expected social or occupational roles; specific manifestations of personality disturbance e.g. emotional instability are severe; is often associated with harm to self or others; is associated with significant impairment in all or nearly all areas of life including personal, family, social, education and occupational

KISS: Diagnostic requirements for Personality Disorders continued

Based on ICD-11 classification

• **Traits of Personality disorder**

▪ *There are 6 traits of PD, some of which may dominate in individuals with PD*

- **Negative affectivity:** a tendency to experience a broad range of negative emotions (e.g. anxiety, worry, anger, shame, depression, guilt, paranoia etc) with a frequency and intensity out of proportion to the situation
- **Detachment:** a tendency to keep socially and emotionally detached from others
- **Dissociality** (anti-social): a disregard for the rights and regards of others, resulting in self-centredness (including entitlement, grandiosity, narcissism) and lack of empathy including deceptive, manipulative, aggressive or exploitative of others e.g. may respond with anger or denigration of others if they are not granted admiration
- **Disinhibition:** a tendency to act rashly based on immediate external or internal (e.g. thoughts, sensations or emotions) without consideration of the consequences leading to impulsivity, irresponsibility and recklessness e.g. may lead to substance misuse, gambling or unsafe sex
- **Anankastia** (obsessive-compulsive): a narrow focus on one's rigid standard of perfection and right and wrong, resulting in behaviour which is rigid, highly perfectionist and controlling; often associated with emotional control and restraint

▪ **Borderline pattern qualifier**

- *Borderline pattern PD may be applied to individuals whose pattern of personality is characterised by a pervasive pattern of instability of interpersonal relationships, self-image and affect and marked impulsivity*
 - Patterns include emotional instability, recurrent episodes of self-harm, intense and unstable relationships, fear of abandonment, rejection hypersensitivity, lack of trust in others; impulsivity resulting in potentially self-damaging behaviours; chronic feelings of emptiness; inappropriate intense anger; transient dissociative like features or psychotic-like features (e.g. brief hallucinations or paranoia) in situations of high affective arousal; very low self-esteem, with an experience of self as profoundly different from others resulting in a painful sense of isolation and pervasive loneliness

► **So, should we abandon the diagnoses Borderline and Emotionally Unstable PD?**

This is controversial and psychiatrists do not agree. Borderline and Emotionally Unstable PD no longer exist as separate diagnoses in ICD-11 but they do in DSM-5. It is likely that many psychiatrists and mental health teams will continue to use the terms for some time yet.

Borderline PD dates back to a definition back in the 1930s of a category of patients who were in a 'borderline' between psychotic and neurotic groups and difficult to help. Despite being widely recognised it has remained perennially controversial as a diagnosis, hence the move a few years ago to emotionally unstable PD. A recent Lancet editorial, [Beyond the Borderline Lancet Psychiatry 2019](#) clarifies the disquiet that many health professionals feel with the label, stating '*Borderline personality disorder is not so much a diagnosis as a liminal state...the diagnostic criteria fall between objective descriptions worthy of medical manuals and value judgements based on received opinion*'. However, others argue [Lancet 2019](#) that the diagnostic label is useful as there is widespread understanding for patients and clinicians of what it means, and crucially it defines a specific category of patients who can then be helped with a specific formulation of evidence-based care as well as opening the door to better self-understanding and self-help.

What is clear however is that debate about terminology and definitions should not detract from making sure that people's problems are recognised, and that they receive good quality care to meet their unique needs.

► *How common is Personality disorder?*

Given how controversial the diagnosis is, it is no surprise that there are wide variations in the reported literature of the prevalence but it is typically quoted as being around 10% [BMJ Best Practice 2019](#). Because of all the problems created by PD, the prevalence in a primary care setting is likely to be much higher. An oft-quoted [study](#) found personality disorder in 24% of successive attendees in GP consultations, and 30% to 45% of psychiatric outpatients.

► *What causes Personality disorder?* [BMJ Best Practice 2019](#)

As every GP knows, there is a very strong link with early emotional trauma and abuse (leading to some people to prefer to use the term 'damaged personality'). However, there is also evidence for genetic factors contributing to PD with [research from twin studies](#) showing a significant genetic contribution.

Emerging evidence from [research](#) has shown that adverse childhood experiences such as sexual and physical abuse are closely related to enduring neurobiological changes in the brain causing dysfunction that can affect physical and mental health throughout life. *People who have suffered childhood abuse including maltreatment, neglect and sexual abuse are 400% more likely to develop PD.* Recent fascinating research [Front Psych 2019](#) suggests that such early childhood trauma may cause epigenetic changes (i.e. biochemical changes that affect how genes are expressed) in neurobiological function, which may then subsequently, be passed down to offspring heightening the risk of problems repeating through subsequent generations.

(NB If interested in the impact of abuse and trauma on the brain, and the long-term sequelae on cognitive, social and emotional functioning we would highly recommend the book [The Body Keeps the Score](#) by Bessel van der Kolk).

► *My patient has suffered sexual abuse and suffers from emotional instability. Does he/she have an emotionally unstable personality disorder or complex PTSD?*

There is recent and growing debate as to whether or not Borderline or Emotionally Unstable PD may have been over-diagnosed in people who have been sexually abused, and whether or not some of these patients may have instead **complex post-traumatic stress disorder**.

Complex PTSD (CPTSD) is also a new diagnosis in [ICD-11](#). *CPTSD is applicable to anyone who has experienced prolonged, repeated or multiple forms of trauma such as childhood abuse.* It is defined as including the three classic symptoms of PTSD (re-experiencing the trauma through flashbacks, avoidance of trauma reminders and a heightened sense of threat (hyper-vigilance and startle response). However, it also includes symptoms of 'self-organisation' including emotional dysregulation, interpersonal difficulties and low self-esteem which have significant overlap with PD, especially of the borderline or emotionally unstable type.

Patients with CPTSD will often experience emotional instability, relationship difficulties, dissociation and depersonalisation, feelings of shame and guilt, suicidal thinking and destructive or risky behaviour. There is also a strong link with long-term COMPLEX PTSD and physical comorbidity [BMJ 2019](#).

Recent research [Lancet Psychiatry 2019](#) shows that early life trauma exposure is associated with a 25% chance of lifetime PTSD and an associated broad range of complex psychiatric presentations, high risk and significant impairment in young people. The risk of PTSD was greatest after sexual assault, with 74% of young people experiencing sexual assault developing long-term PTSD. These symptoms of CPTSD reflect the impact that trauma has on emotional regulation, relationships and self-view.

The recognition of CPTSD as a diagnosis also raises a new debate about whether or not many people who have been victims of abuse and trauma may have been misdiagnosed as having PD when in fact they have CPTSD, and that they may be more helped by PTSD specific therapy such as trauma-focused CBT or EMDR (eye movement desensitisation and reprocessing). There is emerging [evidence](#) that such trauma-focused therapy is strongly beneficial for people with CPTSD.

See the PTSD chapter for more on this, and for patient information on CPTSD, see [NHS CPTSD](#) and [MIND CPTSD](#).

► *What is the impact of Personality disorder?*

Profound. People with PD compared to those without a PD have a high burden of physical and psychiatric co-morbidities, with health inequalities leading to poor health and social outcomes and greatly increased premature mortality.

Back in 2012, research [J Psycho Res 2012](#) looking at a cohort of people with PD who were using mental health services had a life expectancy at birth of 59 years for men and 63 for women, *18 years shorter than in the general population*. More recent research [Personality and Mental Health 2019](#) including patients with less severe PD shows that:

- **People with PD lose 12 to 18 years of life ($p < 0.0001$) compared to the general population**
 - There is a significant psychiatric co-morbidity in people with PD and significant other mental health conditions, such as chronic depression and substance misuse, which also affect health and longevity
- **Causes of premature death in people with PD were**
 - Cardiovascular 29%
 - Cancer 33%
 - Suicide 5%
 - Other causes 33%
- People with PD have a significantly increased risk of cardiovascular disease, that is not solely explained by socioeconomic status or lifestyle factors such as smoking or alcohol misuse
- People with PD have a high burden of psychiatric co-morbidity, especially chronic depression
- People with PD have a significantly **reduced quality of life**, with very high rates of educational under-achievement, unemployment and high rates of trauma (including sexual exploitation, rape and abuse), marginalisation, homelessness and forensic issues
- Aspects of PD can make patients harder to help, less likely to get good care and make problems harder to help

► *What is the suicide risk in Personality disorder?*

There is a strong association between suicide and self-harm. Suicide risk in people with PD is high. It has been estimated that 75% of people with borderline PD have had at least one suicide attempt and up to 10% complete suicide over 30 years of follow-up. Research from a large database in UK primary care [BMC Family Practice 2016](#) found that

- A diagnosis of PD is strongly linked with an increased risk of suicide
- There was a 20-fold increase in suicide risk for patients with personality disorder compared to controls with no recorded psychiatric disorder
- People with PD have a 4 times higher risk of suicide than people with other mental illness
- The risk was especially high in patients with Borderline PD and PD with comorbid alcohol misuse, with a 37 and 45-fold increased risk respectively compared to controls
- Significant increases in risk were seen across all ages and both genders with PD, but especially aged 16 to 39 years and women with PD had a greater risk of suicide than men
- **Conclusion:** *There is a large elevation in suicide risk among patients diagnosed with PD, especially of the borderline type and with comorbid alcohol use*

► **How does Personality Disorder present in primary care?** [BMJ Best Practice 2019](#)

It is most likely to present in primary care with co-morbid presentations of other mental or physical health conditions. People with PD are most likely to present with depression, anxiety, self-harm, substance or alcohol misuse or social problems. Presentations in primary care are likely to include:

- Depression and/or anxiety, which is likely to be chronic and unremitting and may not respond to treatment
- Problematic substance misuse and/or dependency
- Presence of multiple medically unexplained symptoms
- Signs of repeated self-harm (cutting)

► **How do we distinguish serious mental illness, such as bipolar disorder or psychosis, from Personality Disorder in primary care?** [BMJ Best Practice 2019](#)

This is crucial, as patients with bipolar disorder for example will of course also have great instability of mood and personality changes which can be similar to PD traits when they are ill. They may also present in primary care with depression that is unresponsive to standard treatment. *In general however in patients with personality disorder changes in mood are reactive, in response to interpersonal events or internal experiences, compared to bipolar disorder.* They are also less likely to have somatic symptoms such as insomnia or appetite loss. The [Mood Disorder Questionnaire](#) can help to point toward bipolar disorder.

Persistent dysphoria and depression can be a symptom of PD associated with a profound and persistent sense of hopelessness and worthlessness, and of course, can make a diagnosis of depression in this group more difficult. A [recent meta-analysis](#) shows that when patients have co-morbid depression and PD, patients generally respond less well to treatment of the depression than in patients without depression without PD. However, some patients do respond so diagnosis and treatment remain worthwhile.

Patients with PD may also report psychotic symptoms, such as hallucinations or paranoia. However these tend to be transient and in reaction to stress, and they do not have persistent hallucinations or delusions.

All mental health conditions, however, are more common in people with PD, in which case they should be seen as being co-morbid.

► **Is it helpful for patients having a diagnostic label of PD?**

This is controversial, but the simple answer is for some yes and for others no. Some patients dislike receiving a diagnosis of personality disorder, whilst for others, it is helpful [Lancet 2019](#). It is a balance between the potential positives of opening the door to greater self-understanding, self-help and support balanced against the stigma, social consequences and associated concerns and anxiety that a diagnosis may bring. Qualitative [research](#) shows that the positive and negative aspects of reaching a diagnosis of borderline PD are similar to those receiving a diagnosis of bipolar disorder. Clearly how the diagnosis is given is vital, but a positive message that it is treatable and that a diagnosis can help people to better self-understanding can be very empowering.

Every patient is unique and we need to make individualised judgements based on patient's needs and wishes. Clinicians may incorrectly assume what a patient's attitudes to diagnosis are, so clear patient-centred communication is needed.

► **How should we approach patients with PD in primary care?**

As the review in [BMJ Best Practice 2019](#) states, *'the establishment of a stable, supportive physician-patient relationship lies at the core of the approach to managing personality-disordered patients in a primary care setting'*. For patients with PD, *'the foundation of treatment is the physician-patient relationship'* and *'all patients will benefit from a consistent and stable relationship, characterised by clear communication and well-established boundaries'*.

An approach based on mutual respect, understanding and empathy which prioritises continuity and consistency is key for a successful outcome for the patient, the health professionals and the practice. Sadly research has shown that patients with a label of PD may be stigmatised by health professionals, who may make antagonistic judgements towards them and be less empathetic and proactive in their care.

► *How should patients be managed?*

Patients should ideally be referred for personality disorder-specific psychotherapy, which has been shown to effectively reduce symptoms of PD [BMJ Best Practice 2019](#). Although NHS provision is poor and waiting lists are long, patients should be encouraged to access this as outcomes are improved. Many different psychotherapeutic interventions have been studied including cognitive behavioural therapy (CBT), dialectical behaviour therapy (DBT) which is similar to CBT, transference-focused psychotherapy and mentalization-based therapy (MBT).

A recent systematic review [Cochrane 2020; CD012955](#) of 75 RCTs n=4,500 examining 16 different types of psychological intervention found that

- BPD tailored psychotherapy reduces borderline PD severity and suicidality and may reduce self-harm and depression compared to usual care
- Psychotherapy improved psychosocial functioning compared to usual care
- DBT may be better at reducing BPD severity, self-harm and improving psychosocial functioning
- MBT may be more effective at reducing self-harm, suicidality and depression
- **Conclusion:** *PD specific psychotherapy shows beneficial effects for primary outcomes compared to usual care, however, there was no unequivocal high-quality evidence to support one modality of intervention over another*

Mentalization-based therapy (MBT) has a growing evidence base in PD. Mentalization is learning to think about thinking. It aims to improve how we make sense of what is in our own and other's minds, and how our thoughts, feelings and beliefs link to our behaviours and actions especially when we are stressed or distressed. It is an integrative therapy which also brings together aspects of CBT and psychodynamic therapy.

It is particularly useful for people with PD who suffer from long-term relationship difficulties and who experience intense emotional distress and overwhelming feelings which may result in destructive behaviour including self-harm.

MBT is useful for *learning how to manage strong emotions*. A useful metaphor for people with PD is the emotional thermometer. In people with PD, in response to stress as the emotional temperature rises then understanding of context, consequences and empathy (i.e. the identification with an understanding of another's situation, feelings and motives) falls leading to a rapid loss of our pre-frontal control on our behaviour and actions which then may have destructive consequences.

► *What about medication in people with PD?*

People with PD are often on multiple psychotropic medications, despite [NICE CG78](#) concluding that there is no evidence that medication is helpful in people with borderline PD. The guideline states:

- Drugs should not be used for the treatment of borderline PD or for the individual symptoms associated with it e.g. repeated self-harm, emotional instability, risk-taking behaviour and transient psychotic symptoms
- Antipsychotic drugs should not be used in PD
- In a crisis situation, however, short-term use of a sedative *may* be 'cautiously' considered (for less than a week)

- However, co-morbid mental health conditions such as depression and anxiety are common and for these medications may be considered as part of an overall treatment plan according to their individual guidelines e.g. for depression, anxiety or PTSD (NB research suggests that patients with these mental health conditions that are co-morbid with PD may respond less well to treatment, however, some may respond so treatment should not be denied because someone has a PD)
- For patients with PD who also have major psychosis, dependence on alcohol or class A drugs or a severe eating disorder refer to specialist services

KISS: Management of People with Personality Disorders in Primary Care

Based on [NICE CG78](#); [ICD-11 classification for PD](#); [BMJ Best Practice 2019](#); [Cochrane 2020](#)

■ General principles

- *Establishment of a stable, supportive physician-patient relationship lies at the core of the approach to managing patients with personality disorder (PD) in primary care*
- Aim to maximise continuity and build a relationship built on trust
- Clear, honest and open communication is needed that is respectful of the patient's needs and wishes whilst respecting boundaries
- Engage with your patient with a positive and optimistic outlook; it is a myth that PD is not treatable, outcomes are improved with a broad range of psychosocial interventions
- Self-harm is not attention-seeking behaviour; it is a dysfunctional emotional reaction to stress
- Be aware that people with PD have:
 - A greatly increased risk of psychiatric comorbidities including depression, anxiety, PTSD etc
 - Greatly increased risk of self-harm and completed suicide
 - Greatly increased risk of physical co-morbidity, especially cardiovascular disease & cancer
 - Premature mortality of approximately 18 years compared to people without PD

■ Diagnosis

- Be aware of the strong link between personality disorder and childhood abuse and trauma, but also be aware that some patients with a history of abuse may have complex PTSD
- Be mindful that some patients find a diagnosis of PD useful as it opens the door to greater self-understanding, self-help and support whilst others will not due to the stigma, social consequences and anxiety that a long-term diagnosis may bring; explore this sensitively with patients and make individualised judgements based on patient preference; but do not be afraid of discussing the diagnosis in a positive and optimistic way
- The new ICD-11 diagnosis of PD is simpler and less restrictive: it classifies people as mild, moderate or severe according to effect on function and no longer includes multiple sub-types

■ Management

• Acute management

- Assess suicide and self-harm risk
- Suicide risk is much higher in borderline PD type and co-morbid alcohol misuse
- Refer to and utilise specialist mental health services wherever available
- Patients with PD may exhibit psychotic symptoms, such as hallucinations or paranoia; these tend to be transient and in reaction to stress and are not persistent hallucinations or delusions
- ***Drugs should not be used for the treatment of PD or for the individual symptoms associated with it*** e.g. emotional instability and transient psychotic symptoms
- Antipsychotic drugs should not be used in PD
- In a crisis, short-term use of a sedative *may* be 'cautiously' considered (for less than a week)
- Co-morbid mental health conditions such as depression and anxiety are common, and for these medications may be considered as part of an overall treatment plan

• Longterm management

- Continuity is key, ALL patients will benefit from a consistent and stable relationship characterised by clear communication and well-established boundaries
- Optimise physical health with a lower threshold for CVD primary prevention interventions, as per other major mental health conditions [Lester Cardiometabolic Framework](#)
- For patients with PD who also have major psychosis, dependence on alcohol or class A drugs or a severe eating disorder refer to specialist services
- Refer for specialised, PD focused psychotherapy (e.g. dialectical behaviour therapy DBT and mentalization-based therapy MBT) if available via specialist complex needs' services and be positive about these outcomes

KISS: NB Tips for working with people with Personality Disorder

With thanks to the [Training & Vocational Initiatives in Personality disorder from Oxford Health](#) and [Personality Disorder Positive Outcomes Programme](#)

- Be **professional** at all times, genuine and consistent in your approach and non-judgemental
- **Listen** actively, be interested in your patient and their life and make them feel 'heard'
 - Validate and acknowledge suffering, pain or distress
 - But don't try to be their friend and slip into a collusive relationship
- Keep **clear and consistent boundaries**, and make sure that colleagues and practice staff are aware to avoid 'splitting'
- **Don't try and 'fix it'**, but encourage your patient to take responsibility for themselves, to let them know that they have choices and to gently push them to find their own solutions
 - Try to empower your patient to find coping strategies which are helpful to them
 - Don't make suggestions but gently push your patient to find their own solutions
- Be **positive and optimistic** in your outlook, but patient and realistic; progress will be slow
- Be **empathetic but not sympathetic**, which can be unhelpful and lead to co-dependency
 - *Empathy is understanding* i.e. identifying oneself mentally with another person
 - *Sympathy is sharing* i.e. sharing feelings similar to that of the other person
 - Empathy, therefore, acknowledges suffering, and you understand how another person may feel as they do but there is a greater degree of emotional detachment and you do not share that feeling
 - See [Sympathy vs Empathy](#) and [Brene' Brown short video on empathy](#)
- Bring third parties **together** WITH your patient (no one likes things done 'behind their back')
- Be aware that **behaviour is a means of communication**, and that people with PD may have difficulty in articulating unmet needs which may be expressed through challenging behaviour
- Be self-aware of emotions that your patient invokes in you, and beware being tipped in one of two directions on the '**rescue-blame see-saw**'
 - Patients may evoke great feelings of sympathy in you and a desire to 'rescue' them and to 'fix' their problems, leading to co-dependence and a reduction in your patient's agency (ability to self-help)
 - Patients may evoke very negative feelings, including blame and anger, which will be destructive for you, your patient and your relationship
 - Be aware of your response to your patient, when you may be tipped in either direction and remain balanced and neutral in your response
- Understand that **self-harm is not attention-seeking**, but a dysfunctional reaction to stress; try to empower your patient to find alternative coping strategies and to focus on things which may help to build their self-esteem
- **In response to challenging behaviour** [Evaluation of the Angry Patient](#)
 - Actively listen, and try to identify the emotion that lies behind the words
 - If your patient 'blows up' with a rant they will not listen or hear you; wait for the rant to be over
 - Avoid getting defensive but acknowledge the emotion ('I can see why you would feel angry' etc) and respond appropriately
 - Keep cool and professional: maintain sincere eye contact (not aggressive staring!), use your patient's name, remain calm and consistent
 - Use de-escalation techniques i.e. listen, acknowledge the grievance and frustration, shift the focus onto how to solve the problem
 - Stick to boundaries and be consistent
 - If you or your staff are physically threatened or fear for your safety, call the police or security
- ***Make time to reflect on your own feelings and for confidential professional de-briefing***

The rescue-blame seesaw © 2019 Rob Schafer & Fiona Blyth based on

1. Responsibility Without Blame: Empathy and the Effective Treatment of Personality Disorder, Hanna Pickard, Philosophy, Psychiatry, and Psychology 18 (3):209-224 (2011)
2. Hamilton, L (2010). Boundary Seesaw Model: Good fences make for good neighbours. In Tennant, A. & Howells, K. (Eds.), Using Time, Not Doing Time: Practitioner Perspectives on Personality Disorder & Risk(p181-194). Chichester: Wiley-Blackwell.

NB Learning Points, Quality Improvement Ideas and further resources

- *People with PD have a greatly increased burden of physical disease*, and shortened life expectancy of approximately 18 years compared to the general population
- *People with PD have a significantly increased risk of co-morbid and treatable mental health conditions*, such as severe depression, substance misuse and suicidal ideation
- *It is a myth that PD is not 'treatable', research consistently shows improved outcomes with* therapy and co-morbidities respond to treatment
- *People with PD have significant problems which affect the biological, psychological and social dimensions of their lives; primary care is ideally placed* to support them with a holistic, long-term approach
- Self-harm in people with PD is not 'attention-seeking behaviour'; it is a dysfunctional reaction to stress
- The new ICD-11 diagnosis of PD is simpler and less restrictive: it classifies people as mild, moderate or severe according to effect on function and no longer includes multiple sub-types
- There is a strong link between personality disorder and childhood abuse and trauma, however, some of these patients may have complex PTSD
- Be mindful that some patients find a diagnosis of PD useful whilst others will not; explore this sensitively with patients, do not be afraid of discussing the diagnosis in a positive and optimistic way and make individualised judgements based on patient preference

Quality Improvement Ideas

- Undertake some whole **team-based PD training** via the [Knowledge and Understanding Framework](#), or contact your local complex needs service for guidance on training (many local training schemes for whole practices are being rolled out e.g. the [PD Positive Outcomes Programme](#) in the South East region of England)
- Audit of people with **PD and physical health**: are their physical health needs being met? Consider inviting them for annual physical health checks as we do for patients with bipolar and psychotic disorders
- **Medication reviews**: patients often have multiple psychotropic medications, despite none being licensed for personality disorder and NICE advising that they should not be used except to treat co-morbidities
- Follow-up of patients after deliberate self-harm to assess and mitigate suicide risk

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Further resources:

- [NHS Personality Disorder](#)
- [Rethink mental health](#)
- [MIND personality disorders](#)

POST TRAUMATIC STRESS DISORDER

Post traumatic stress disorder (PTSD) had a new NICE guideline in [2018 NICE NG116](#), an updated [NICE-CKS guideline in 2020](#) and a review of the evidence in [BMJ Best Practice in 2019](#).

The new [ICD-11](#) classification of disorders from the WHO includes a new diagnosis of [complex post traumatic stress disorder](#), which is recognised in the NICE guideline. Since the NICE guideline was published there has been growing interest in the role of EMDR (Eye Movement Desensitisation Reprocessing) therapy.

► *When should we consider PTSD as a diagnosis?*

Importantly PTSD should be considered as a possibility in patients repeatedly presenting with unexplained physical symptoms and in people with depression, anxiety and substance abuse. It most commonly presents with these co-morbidities and may present many months or years after the trauma.

The NICE guideline encourages us to ask people if they have experienced trauma and if so to then ask about the 3 core symptoms of PTSD of

- **Re-experience:** re-experiencing in the form of vivid intrusive memories, flashbacks or nightmares typically accompanied by a sense of fear, horror and strong physical sensations
- **Avoidance:** of thoughts, situations, places or people reminiscent of the events
- **Hypervigilance:** persistent perceptions of heightened threat e.g. enhanced startle response to unexpected noises

► *What is complex PTSD?*

Complex PTSD is a new diagnosis in [ICD-11](#) which is recognised in the [NICE guideline](#) and is a subset of PTSD.

- It is a disorder that may develop following exposure to a traumatic event or series of events of an extremely threatening or horrific nature, most commonly prolonged or repetitive events from which escape is impossible e.g. repeated childhood sexual or physical abuse, torture, slavery or prolonged domestic violence
- The three key diagnostic criteria for PTSD are met
- In addition, complex PTSD is characterised by severe and persistent
 - **Emotional dysregulation:** problems in regulating mood and emotions
 - **Negative self-concept:** negative beliefs about oneself as diminished, defeated or worthless and with feelings of guilt, failure or shame related to the trauma
 - **Interpersonal difficulties:** difficulties in sustaining relationships and in feeling close to others
 - These symptoms cause significant impairment in personal, family, social, educational and occupational domains of life

As complex PTSD includes persistent problems in regulation of mood and emotions, persistent negative self-perception and difficulties in sustaining relationships following a background of abuse there is clearly an overlap with **borderline or emotionally unstable personality disorder** (see personality disorder chapter). There is considerable debate in the [literature](#) about how complex PTSD and PD can be differentiated, and indeed whether many survivors of abuse may have been misdiagnosed with PD when in fact they have complex PTSD and benefit from specific trauma-focused therapies.

► *What is EMDR? And does it work?*

One of the most eye-catching recommendations of the [2018 NICE NG116](#) guideline was that as well as trauma-focused CBT that EMDR (Eye Movement Desensitisation & Reprocessing) therapy should be offered as an option for PTSD in adults if patients present with non-combat trauma.

So, what is EMDR? EMDR is a specific desensitisation technique for trauma. It is a cognitive psychotherapy which involves sequential exposure to the traumatic memory. It is delivered in a phased manner and includes psychoeducation about reactions to trauma, and managing distressing memories, situations and triggers. It has the crucial added ingredient of simultaneous bilateral eye movement guided by the therapist which helps the brain to reprocess the memory.

The idea is that following trauma, the memory of the event may become 'stuck' in the brain. Whenever the memory comes to mind, it can trigger a reaction much more intense than a usual memory e.g. vivid flashbacks. EMDR aims to 'unstick' the memory and to process it so that it is no longer so intense.

It does this by the person recalling the trauma with a therapist whilst they simultaneously move their eyes from side to side, or have alternative bilateral stimulation such as a sound in each ear or a tap on each hand alternately. The patient, therefore, has to recall the emotionally disturbing events, usually in brief sequential doses, whilst simultaneously focusing on a bilateral external stimulus. The bilateral, side-to-side stimulus stimulates the natural processing system in the brain so that the memory is reprocessed, desensitised and becomes less intense. The memory thus becomes less emotionally charged, in a process called 'adaptive information processing'.

The eye movements mimic what occurs naturally during **rapid eye movement REM sleep**. It is thought that during REM sleep, the rapid eye movements activate parts of the brain which help us to process information and memories. This helps our minds to naturally heal. However, with difficult trauma, this normal REM system is overwhelmed and we do not heal. EMDR thus helps the brain to process the trauma and unblock the memory, by processing the memory whilst mimicking the eye movements of REM sleep.

EMDR is not new. It was first described in the literature by [Shapiro in 1989](#). Since then evidence for its efficacy has accumulated. A meta-analysis of randomised controlled trials back in 2014 [Plos One 2014](#) found that EMDR significantly reduces the symptoms of PTSD, depression, anxiety and subjective distress in patients with PTSD. This finding has been replicated in other studies since, including a systematic review and [meta-analysis of studies in 2018](#) found that EMDR was more effective than CBT for PTSD. EMDR has also been shown to be effective in children and adolescents [J Child psychology 2017](#).

Typically EMDR sessions are 60 to 90 minutes and a course of therapy lasts 8 to 12 sessions. Patients often report improvement after only 2 or 3 sessions. It can also be used for phobias. EMDR is available on the NHS via IAPT services and can be delivered online see [EMDR UK](#).

See the useful short video '[What happens in an EMDR session](#)' from NHS South Yorkshire.

KISS: Post Traumatic Stress Disorder in Adults

Based on [NICE-CKS guideline in 2020](#) & [2018 NICE NG116](#) & [BMJ Best Practice in 2019](#)

► 3 key learning points

- *PTSD is common and treatable with trauma-focused therapies*
- *Patients generally present with co-morbid depression, anxiety, substance/alcohol misuse or chronic pain or unexplained physical symptoms*
- *Ask such patients about a history of trauma, and if they fulfil the criteria for PTSD explore the diagnosis with them and refer for specific trauma-focused therapy*

► Diagnosis

- **PTSD** may develop following exposure to an extremely threatening or horrific event or series of events
- It is characterised by ALL of the following
 - **Re-experience:** re-experiencing the trauma in the form of vivid intrusive memories, flashbacks or nightmares; typically accompanied by a sense of fear, horror and strong physical sensations
 - **Avoidance:** of thoughts, situations, places or people reminiscent of the events
 - **Hypervigilance:** persistent perceptions of heightened threat e.g. enhanced startle response
- These symptoms are persistent and cause functional impairment in relationships and life
- **Complex PTSD** may develop following exposure to a traumatic series of events, most commonly prolonged or repetitive events such as repeated sexual or physical abuse
- As well as the 3 core symptoms of PTSD, **complex PTSD** is characterised by severe and persistent
 - Problems in regulating mood and emotions
 - Negative beliefs about oneself with feelings of guilt, failure or shame related to the trauma
 - Difficulties in sustaining relationships and in feeling close to others
- *Be aware that patients with PTSD usually present with co-morbid depression, anxiety, alcohol or substance misuse, or with chronic pain or unexplained physical symptoms*
- Think about asking these patients if they have experienced traumatic events such as serious accidents, physical or sexual abuse, domestic abuse, work-related trauma, war and conflict or torture
- *Be aware that people with PTSD will be apprehensive, anxious or ashamed and may be reluctant to have treatment due to avoidance; they may believe it is untreatable and trust may be an issue*

► Management

- Explain PTSD and *reassure that PTSD is a treatable condition even if patients present very late*
- Assess the risk of self-harm
- Advise people to access and utilise peer support groups
- **Refer for individual trauma-focused cognitive behavioural therapy CBT** intervention
 - patients should receive 8 to 12 sessions of individual therapy
- Consider **eye movement desensitisation and reprocessing (EMDR) therapy** if patients present 1 to 3 months after the trauma, if they have a preference for EMDR
- Offer EMDR to adults with PTSD who present ≥ 3 months after the trauma
- Do not offer **drug treatments**, including benzodiazepines, to adults with PTSD
 - However, if patients have a preference for drug treatment consider venlafaxine or sertraline
 - Consider antipsychotics such as risperidone in addition to psychological therapies if disabling symptoms such as severe hyperarousal or psychotic symptoms that have not responded to other drug or psychological treatments
- For people presenting with **PTSD and depression**
 - Usually, treating the PTSD first because the depression will often improve with successful PTSD treatment
 - Treat the depression first if it is severe enough to make the psychological treatment of PTSD difficult, or if there is a risk of self-harm

BURNOUT

Burnout has of course been a Hot Topic for healthcare professionals for many years as long hours, increasing demand, a workforce shortage and a high administrative burden take their toll. A recent survey of 200 GPs [BJGP2019;69;684](#) reported alarmingly high levels of burnout and poor well-being amongst the respondents. Not surprisingly, recent evidence from a systematic review [Ann Int Med 2019](#) shows that burnout in health professionals is frequently associated with poor quality of care and major medical errors.

The COVID-19 pandemic is resulting in a significant increase in medical burnout. As well as the increased work pressure, and dealing with the impact of the pandemic on our patients and ourselves, in primary care the impact of social distancing and remote consulting dilutes many of the positive aspects of the job (human contact with patients and colleagues) and increases the negative aspects (such as living with risk and uncertainty).

But what do we actually mean by burnout, how is it defined, how does it differ from depression and how should we recognise it not just in ourselves and our colleagues but also in our tired and exhausted patients? Burnout is included in the [2019 ICD-11 International Classification of Diseases](#), and now has an updated and more detailed definition. *Given the significance of overwhelming fatigue in the definition, it is likely that we may not just be missing burnout as a diagnosis in ourselves and our colleagues but also in our patients.*

In the last year, a number of papers have been published relating to burnout in healthcare. But first, what do we actually mean by burnout?

► *What do we mean by Burnout?*

In [2019 Burnout was recognised by the WHO](#) as an 'occupational phenomenon', and it is included in the ICD-11 International Classification of Diseases. It is defined as a psychological syndrome resulting from chronic workplace stress, and NOT as a distinct disease or medical condition.

Burnout [WHO Definition 2019](#)

- Burnout is a psychological syndrome resulting from chronic workplace stress. It is characterised by three dimensions:
 - Feelings of energy depletion or exhaustion
 - Increased feelings of distance from one's job, or feelings of negativism or cynicism related to one's job
 - Reduced professional efficacy
- Burnout specifically applies to an occupational context, and should not be applied to describe experiences in other areas of life

Much of the research work done on burnout has been by Prof Christina Maslach. Writing in [World Psychiatry 2016](#) she defines burnout as a psychological syndrome emerging as a prolonged response to chronic interpersonal stress, resulting in the three key dimensions of:

- **Exhaustion and fatigue**
- **Feelings of cynicism and detachment from the job**
 - Leads to depersonalisation, irritability, loss of idealism, withdrawal and in healthcare workers compassion fatigue and loss of empathy
- **A sense of ineffectiveness and lack of accomplishment**
 - Reduced personal accomplishment, low productivity, creativity and capability, poor coping, negative self-evaluation

Crucially the significance of the definition is that it is placed in a social occupational context and that it involves the person's conception of both themselves and others. This is the crucial factor that differentiates burnout from depression: depression and burnout are independent of each other, with burnout being occupational-specific and depression more generalised and context-free. However, they can both predict the development of the other. Burnout can lead to depression, and depression increases the risk of burnout.

The best-validated research tool for measuring burnout is the [Maslach Burnout Inventory](#) (unfortunately not freely available) which assesses scores across the three domains of emotional exhaustion, depersonalisation and personal accomplishment. Whilst being wary of generalisation, there is some research to suggest that women and men may differ in their development of burnout in that men are more likely to develop cynicism first and inefficacy later and vice versa in women.

Burnout is often attributed to excessive workload, however, Maslach has identified other key areas of risk for burnout including a lack of choice or control over one's work, lack of recognition and reward, not being part of a supportive team, lack of fairness or respect and a lack of shared values or meaningful work. These were discussed in [BMJ2019;366:1477](#) and are identified as the six *key domains of risk for burnout*.

- **Sustainable workload**
 - Constant stress and work overload with no time to rest, recover and restore balance and loss of boundaries are highly predictive
- **Choice and control**
 - A lack of autonomy or control is a consistent predictor of burnout
- **Recognition and reward**
 - Not just financial but in terms of insufficient recognition and reward for your work; lack of such reward increases vulnerability to burnout and is strongly linked with feelings of inefficacy
- **Community**
 - Being part of a supportive team community is a strong protective factor, and relationships that are characterised by a lack of support and trust or have unresolved conflict will lead to burnout
- **Fairness**
 - A sense of injustice and being unfairly treated is strongly predictive of burnout; a sense of fairness, respect and justice in the workplace is highly protective
- **Clear values and meaningful work**
 - Feeling that the work that you do is far removed from what you think is useful to do (NB GP-related administrative and regulatory work springs to mind!) and what you should be doing is highly predictive

This is important as whilst sustainable workload is very important, it is not the only risk for developing burnout. Recent papers [BMJ2019;366:1477](#) stress that to effectively combat burnout, all of these domains of risk need to be addressed in the workplace.

► *Burnout amongst doctors*

Burnout amongst doctors was explored in a BMJ paper [BMJ 2017;358;j3360](#). There are a number of reasons why doctors are particularly prone to burnout, including:

- Individual factors e.g. perfectionism or denial and avoidance as coping strategies etc
- The 'medical culture' e.g. conspiracy of silence, ignore distress, long hours etc
- Organisational factors e.g. chaotic work environments, bureaucratic demands, poor leadership, lack of support etc

Most interventions for burnout have focussed on individual factors, such as increasing resilience, encouraging mindfulness and stress reduction techniques etc. There is evidence that these can be helpful, however, *systems level organisational change is more effective*.

A recent paper *burnout in healthcare: the case for organisational change* [BMJ2019;366:1477](#) stresses that burnout is an occupational problem, and it, therefore, needs system-based organisational change to address it. Interventions to tackle burnout are often focused on individuals rather than taking a systems-based approach to the problem. Too much focus on individual interventions to build resilience may divert attention away from changing toxic work environments, and the need to develop more healthy work environments. We need to broaden our approach to burnout, and there needs to be more focus on developing healthy workplaces based on addressing Maslach's key 6 areas of risk. Small changes in the workplace can be surprisingly effective.

► *Burnout in the Age of COVID-19*

The COVID-19 pandemic is resulting in a significant increase in medical burnout. As well as the increased work pressure, and dealing with the impact of the pandemic on our patients and ourselves, in primary care the impact of social distancing and remote consulting dilutes many of the positive aspects of the job (human contact with patients and colleagues) and increases the negative aspects (such as living with risk and uncertainty). See Neal's blog on this [Burning the Covid Candle](#).

► *What is parental burnout?*

Parental burnout is not a condition in DSM or ICD-11. But it is increasingly recognised as a phenomenon, and there is a growing [evidence base](#) that it is a '*serious condition that urgently requires more attention*'. The evidence was recently discussed in [New Scientist 2021](#).

- Parental burnout, like occupational burnout, is primarily an 'exhaustion disorder'
- **The three main symptoms are:**
 - Profound fatigue and emotional exhaustion
 - Becoming emotionally distanced from your children
 - Loss of pleasure and fulfilment in the parental role
- **Who is at risk of parental burnout?**
 - Perhaps not surprisingly, women are more at risk than men
 - Going to work is a protective factor, hence having to work at home and juggle homeschooling and childcare is a major risk
 - Personal factors such as perfectionism and parenting practices (e.g. trying to complete the home school curriculum, plus extra-curricular, and be the fun parent and cook organic meals etc)
 - Feeling a cultural pressure to make sure that your children develop to the maximum of their potential in every domain, including that they feel emotionally secure, competent and proud of themselves
 - Studies show often parents just try 'too hard' e.g. faking happiness
- **What has been the effect of the covid lockdowns on parental burnout?**
 - Interestingly, there is considerable variation
 - For a third of parents, feelings of parental burnout increased significantly during the lockdowns as they have to juggle work, homeschooling etc; however, for another third, being at home all the time actually improved their feelings of parental burnout; this is likely to represent the difference in personal factors and attitudes to parenting
- **What helps parental burnout?**
 - As ever recognising the problem and talking about it; group therapy, knowing that you are not alone, is particularly helpful
 - Being aware of your stress enhance factors and your emotional resources that work for you
 - For parents who recognise perfectionist traits, lower parenting expectations to 'good enough is good enough'
 - For some people individual approaches such as mindfulness can help; for others it makes it worse as it just 'yet another thing to do' in a hectic day
- **Am I suffering from parental burnout?**
 - Researchers have developed a test you can do online [Parental Burnout Test](#)
 - For more information from the team who have done most of this research, see [Parental Burnout](#)

NB Learning Points & Quality Improvement Ideas

- Burnout is now in ICD-11
- It is an occupational phenomenon, classified as a psychological syndrome characterised by 3 key features of
 - Feelings of energy depletion or exhaustion
 - Feelings of negativism or cynicism related to one's job
 - Reduced professional efficacy
- Burnout is independent from depression and occupationally specific. However, burnout leads to depression and depression increases the risk of burnout
- *Does your patient have burnout?* Given that we see a lot of patients with stress-related illness and fatigue, it is likely that occupational burnout may be an unrecognised cause of the presentation; consider exploring workplace-related stress in your patients presenting with fatigue and exhaustion
- *Do you or your colleagues have burnout?* Burnout is highly prevalent amongst doctors and healthcare professionals and is strongly linked with medical error. Examine the key features, and the 6 domains of risk, and look out for it in both yourself and your colleagues. We need to look after ourselves and look out for each other...
- Interventions to prevent burnout have very much focussed on individuals, helping them to build resilience and manage stress. Such interventions can be useful, however, there is increasing recognition that to prevent burnout systems-wide organisational change is needed to create healthier workplaces.
- Be aware, in our patients and ourselves, of the risks of [Parental Burnout](#), especially during lockdown
- **QI Idea:** look at Maslach's 6 domains of risk: what changes need to be introduced into your workplace to help to prevent burnout? Small changes in the workplace can be highly effective.
- See Zoe Norris' [GP Burnout](#) and Jessica Buchan's '[I faced burnout as a GP](#)'
- Burnout and COVID -19, see [Burning the Covid Candle](#)
- **Resources for GPs with Mental Health Concerns**
 - [NHS GP Health Service](#) (for GPs in England)
 - [Doctors Support Network](#)
 - [BMA Doctors Support Service](#)
 - [BMA Counselling & Doctors Advice Service](#) (Tel: 0330 123 1245)
 - [RCGP, GP wellbeing help and support resources](#)

OBSESSIVE COMPULSIVE DISORDER

When you see anxious or distressed patients, how often do you ask about intrusive thoughts? An excellent recent review paper [BMJ2019;367:16574](#) explored intrusive thoughts in the context of *perinatal obsessive-compulsive disorder*.

- Intrusive thoughts are distressing, unwanted thoughts, images, impulses or urges that come into our minds in an unbidden and unplanned way
- These thoughts may lead to compulsive behaviour and high levels of anxiety
- Compulsions may serve to alleviate a sense of threat from the intrusive thoughts and may include avoidance, reassurance seeking and suppressing thoughts
- People with OCD may have violent or distressing intrusive thoughts but do not act on these thoughts; the intrusive thoughts are ego-dystonic i.e. they do not fit with and are often repugnant to the person
- OCD is more common in pregnancy and the postnatal period, and distressing intrusive thoughts such as harming the baby are common
- Patients will not feel able to disclose these intrusive thoughts, and they need to be sensitively explored (e.g. *'sometimes people experience distressing thoughts or images of harm coming to their baby, deliberately or accidentally...do you have those?'*)
- Patients often attribute having these thoughts to be due to being a bad person; being given a clear diagnosis and explanation of OCD with appropriate offer of help can be a relief

KISS: Obsessive Compulsive Disorder

Based on [BMJ2019;367:16574](#) & [CKS 2018](#); [BMJ BestPractice 2017](#)

- ***OCD affects 1% of all adults & shame commonly prevents presentation***
 - Patients may present with physical sequelae e.g. dermatitis or with co-morbid mental health problems e.g. anxiety, depression or substance misuse
 - OCD is more common in pregnancy and the postnatal period (2% to 4% of women)
 - Obsessions and compulsions may be entirely cognitive and hidden
 - Intrusive, distressing and unwanted thoughts are a key feature
- ***Be alert to the diagnosis, and ask the screening questions***
 - *Do you wash or clean a lot? Do you check things a lot? Is there a thought that keeps bothering you that you would like to get rid of? Do your daily activities take a long time to finish? Are you concerned about putting things in a special order? Do these problems trouble you?*
 - *Enquire about intrusive and distressing thoughts*
- If diagnosis likely, the patient can do a validated assessment tool [click here](#)
- Assess for ***co-morbid mental health problems*** e.g. depression, substance misuse etc
 - assess suicidal and self-harm risk and any safeguarding issues
- **Management**
 - ***Up to 75% of people get better with treatment***
 - **Step One:** identification, education, support and self-help
 - Patient information from [MIND](#); Self-help recommended books [click here](#)
 - **Step Two** for mild to moderate OCD
 - Refer for individual or group CBT which is highly effective
 - **Step Three** for moderate or severe functional impairment
 - Refer for CBT
 - Consider SSRI: good quality evidence that sertraline, citalopram, fluoxetine and paroxetine are effective and are all first-line choices. May need higher doses and take up to 12 weeks to respond. When controlled CBT needed to reduce relapse rate on stopping.
 - If SSRIs are ineffective, contraindicated or not tolerated there is some evidence for clomipramine

GENERALISED ANXIETY DISORDER

NICE updated their guideline on generalised anxiety disorder (GAD) and panic disorder in adults in July 2019, [NICE 2019 CG113](#). See the KISS summary for GAD below and a separate chapter for panic disorder.

We are often reluctant to make a formal diagnosis of a 'disorder' in an anxious patient for fear of 'labelling' but the *guideline encourages a proactive approach to early diagnosis*. We are encouraged to identify and communicate the diagnosis of GAD as early as possible to help identify the disorder and start effective treatment as soon as possible.

GAD is common (lifetime prevalence of 6%), but it is particularly prevalent in primary care settings where it occurs among 7% to 8% of our patients [NEJM2015;373:2059](#). However, it often goes undiagnosed because it presents with physical symptoms (headaches, GI symptoms etc) and patients rarely report the symptom of worry. With a focus on physical symptoms, the underlying GAD is often missed. It is more common in women than men (7% vs 4%) and interesting that it is most common in the 45 to 59-year-old age group. It is no surprise that people with GAD consult GPs more often than matched controls, yet they are often not diagnosed. Psychiatric co-morbidity is very common: major depression (estimated in 30-60% of cases of GAD), social anxiety disorder (34%) and alcohol misuse (38%).

The NICE [NICE 2019 CG113](#) guideline *encourages us to consider the diagnosis of GAD in patients who have a chronic physical health problem, seek reassurance about somatic symptoms and repeatedly worry about different issues*.

► ***When, if ever, should we use benzodiazepines for anxiety?*** [BJGP2019;69:152](#)

The NICE guideline states we should not offer BZPs for the treatment of GAD except as a *short-term measure during a crisis*. This new review of the evidence [BJGP2019;69:152](#) gives further useful prescribing advice for benzodiazepine (BZPs) for anxiety. Tolerance to these drugs develops rapidly and the risk of addiction is high, as well as the other well-known adverse effects. They recommend:

- The most appropriate first-line treatments for anxiety are self-help strategies, counselling and CBT
- BZPs should only ever be prescribed in the lowest effective dose for short-term use (maximum of 2 to 4 weeks) for severe anxiety and panic usually in a crisis situation
- If they are used, patients would be warned that they are highly addictive and of the risks with driving, operating machinery etc
- BZPs are never appropriate for short-term use for milder anxiety

KISS: Generalised Anxiety Disorder

Based on [NICE 2019 CG113](#) & [NEJM2015;373;2059](#)

1 Make the diagnosis

- Use the GAD-2 screening questions in suspected cases including patients with anxiety related to somatic symptoms or chronic health problems
 - *During the past 4 weeks have you been bothered by feeling tense, anxious or worried most of the time?*
 - *Are you frequently tense, irritable and having trouble sleeping?*
- Answering yes prompts further assessment e.g. using the [GAD-7 questionnaire](#)
- Identify and communicate the diagnosis of GAD as early as possible to help identify the disorder and start effective treatment as soon as possible

2 Check for co-morbidity

- Depression & substance misuse are particularly common co-morbidities
- For people with GAD and a co-morbidities anxiety or depressive disorder, treat the primary dominant disorder first

3 Management

- **Step 1**
 - Identification, assessment, education and active monitoring; signpost [NHS self-help resources](#)
- **Step 2**
 - **Low-intensity psychological interventions** (individual non-facilitated self-help, supported self-help using written or electronic materials or psycho-educational groups based on CBT principles)
- **Step 3**
 - Offer higher intensity psychological treatment OR drug therapy
 - **Psychological**
 - CBT or applied relaxation therapy in 12 to 15 weekly sessions
 - **Drug treatment**
 - **SSRI** first-line, sertraline is the first-line choice
 - Warn re usual risks and interactions, and see adults under 30 after 1 week to assess suicidal thinking and self-harm risk
 - Review effectiveness every 2 to 4 weeks for the first 12 weeks, and thereafter every 12 weeks
 - If ineffective after 4 to 8 weeks, offer alternative SSRI or SNRI (duloxetine or venlafaxine)
 - Once effective drug. Continue for 12 months as relapse rates are high
 - **Pregabalin**, offer only if unable to tolerate SSRIs or SNRI; be aware of the risk of misuse & dependence
 - **Benzodiazepines**: do NOT use other than very short-term crisis measure. Do NOT use antipsychotics
 - **Propranolol**, although widely used in practice it lacks an evidence base [J Psychopharm 2016;30;128](#) and is not recommended in guidelines
- **Step 4**
 - Refer for specialist help, for highly specialist treatment such as complex drug and psychological interventions from multi-agency teams

MENTAL ILLNESS IN CHILDREN & YOUNG PEOPLE

Mental illness in children and young people is a major Hot Topic. We seem to have an 'epidemic' of mental health problems, there are increasing concerns over social media use and influence following recent high-profile tragedies and child and adolescent mental health services (CAMHS) are hugely overstretched. Against this background, in the summer of 2019, [NICE produced a new guideline, NG134](#). We have also had a recent editorial on the mental health of young people and especially self-harm [BJGP2019; 69;18](#) and another on the under-provision of mental health services for children and young people [BJGP 2019;69;112](#) and a recent editorial on depression and anxiety in young people and adolescents [BJGP2020;70;56](#).

► *How big is the problem?*

A large survey of data from England has published in November 2018 [NHS Mental Health of Children & Young People](#). It created huge debate, including a response from the RCGP Nov 2018 calling for better access to mental health services for young people & more support for GPs. The finding that 1 in 4 women aged 17 to 19 has a significant emotional or mental health problem is particularly striking.

- 13% (one in eight) 5 to 19-year-olds have at least one significant mental health disorder, rising to 17% of 17 to 19-year-olds, with girls (24%) significantly more likely than boys (10%) to have a mental health disorder.
- The prevalence has slightly increased over the last 10 years.
- There is a **strong link with self-esteem**:
 - High self-esteem is 5 times more common (25%) in those without a mental health disorder, compared to those that do (5%).
 - Boys (26%) are more likely to have high self-esteem than girls(19%).
- Children and adolescents with a mental health disorder are **twice as likely to have been bullied** in the last year (59%) as those without a disorder (32%).
- **Self-harm**: 26% of those with a mental health disorder report self-harm rising to 50% in girls aged 17-19.
- **Social media**:
 - Children and young people with a mental health disorder have heavier social media use (30% of those with a disorder were on social media for > 4 hours a day, compared to 12% for daily users without a disorder).
 - Girls with a mental health disorder were more likely to compare themselves to others on social media, and boys and girls with a mental health disorder were much more likely to be a victim of cyberbullying compared to those without a disorder (42% vs 18%).

As the editorial [BJGP 2019;69;112](#) stresses, the data from this report highlights how critical the under-provision of mental health services for this group is. *Only 25% of children or young people referred to CAMHS are actually seen and* for those that are seen waiting times are typically up to 10 months.

► *Self-harm in young people, a challenge for general practice* [BJGP2019; 69;18](#) & [BJGP2017;67;542](#)

Every GP is aware of the current 'epidemic' of deliberate self-harm (SH) in teenagers and young people. It has been the subject of some important **original research** from primary care [BMJ2017;359;j4351](#) and three recent **BJGP editorials** [BJGP2019; 69;18](#), [BJGP2017;67;542](#) and [BJGP2019;69;224](#). A recent editorial has also stressed the need for better suicide prevention strategies for young people [BJGP2020;70;104](#), with the age range from 15 to 19 identified as a time of increased risk.

These editorials summarised the scale of the challenge we face:

- SH is complex, multifaceted and multifactorial and is a major risk factor for suicide and mortality in young people; SH behaviour evolves over time thus all self-harm needs to be taken seriously.
- SH is increasingly common, with prevalence rates ranging from 10% to 68% in the community. Of those who self-harm, only 12% seek professional help.

- There is evidence of the inverse care law, with incidence of self-harm being highest and the rates of referral the lowest in the most deprived communities; self-harm by poisoning shows a particularly strong association the increasing rates of socioeconomic deprivation [BJGP2019; 69;18](#).
- Although frequently low-risk and transient, the behaviour can persist and become life-threatening.
- Young people will rarely present to primary care citing the SH as the reason why they are attending.
- There are ***different drivers of self-harm*** in young people, all of which may or may not be associated with clinical anxiety and depression, broadly characterised as:
 - SH to cope with developmental stressors, such as bullying or exam pressure.
 - SH to cope with overwhelming distress.
 - SH to end dissociation and to 'feel alive'.
 - SH with suicidal intent.
- The literature offers ***some consensus as to how to talk to young people*** about SH - see KISS below.

► ***Reducing risk following self-harm*** [BJGP2019; 69;18](#)

This editorial focuses on the need to reduce suicidal risk following self-harm with careful prescribing. Research has demonstrated that analgesics and antidepressants are the most commonly used drugs for overdoses. NICE guidance on self-harm [NICE CG133](#) dating back to 2011 is unequivocal in its recommendation to NOT prescribe tricyclic antidepressants after episodes of self-harm because of their greater level of risk. Despite this recent studies in primary care cohorts have demonstrated a high degree of TCA prescribing after episodes of self-harm, 6% of adolescents aged 10 to 19, 10% of adults to age 65 and 12% of adults aged over 65. The paper concludes that patients after an episode of self-harm should NOT be prescribed tricyclics, and recommends an alert on the primary care record about the avoidance of TCAs in patients who have been previously coded as having an episode of self-harm.

► ***The incidence, clinical management & mortality risk following self-harm in children and adolescents*** [BMJ2017;359;j4351](#)

Self-harm in children and adolescents is a major public health concern in many countries. Self-harm correlates highly with depression and anxiety and is associated with an increased risk of mental illness in adulthood. It is also the **strongest risk factor for suicide**, with suicide being the second commonest cause of death under the age of 25 (after road traffic accidents). In the UK, suicide rates among adolescents aged 15 to 19 have increased. The [UK National Suicide Prevention Strategy 2017](#) has widened its scope by aiming to reduce self-harm rates as a common precursor to suicide. **See also the chapter on suicide prevention and safety plans.**

This important cohort study from primary care looked at 17,000 patients aged 10 to 19 who harmed themselves between 2011 and 2014 (NB the study therefore just looked at people who had been coded on their primary care record, thus will just represent the 'tip of the iceberg' of self-harm in the community).

Main results:

- The annual incidence of self-harm increased in girls compared to boys, with a sharp 68% rise in girls aged 13 to 16. Considerably higher incidence rates were observed in girls (37.4 per 10,000) compared to boys (12.3 per 10,000).
- 84% of coded cases were overdoses, 12% cutting episodes and the remaining various methods such as scalding, hanging, suffocation and jumping.
- Referrals to mental health services were less likely in the most socially deprived areas, even though the incidence of self-harm was significantly greater in these areas (suggesting Tudor Hart's inverse care law).
- Children and adolescents who self-harm were 9 times more likely to die during follow-up due to suicide or fatal drug poisoning.
- **Conclusion:** self-harm is common, the incidence is rising, it is often repeated and is strongly associated with suicide.

► *What have NICE recommended?* [NICE 2019, NG134 Depression in children & young people](#)

The headline recommendation, as discussed in [BMJ2019;364:1364](#), is to offer digital CBT as a first-line treatment to enable more young people to get faster access to psychological help. This should be the first-line treatment for mild depression and can be accessed via mobile phones, tablets or computers. NICE say they have based their recommendations for psychological therapies on effectiveness, availability and cost. However, a range of options other than digital CBT should be available and choice of treatment should be based on clinical need and patient/parent preference. As well as digital CBT there is also evidence that group therapy is effective, and this should be an alternative first-line treatment. See the KISS below for a summary.

The very important practical point of 'how' the digital CBT is to be delivered is not made clear, especially as some of these programmes need to be supported.

Presumably, this will be left to local arrangements but there are a number of these programmes that young people and their families can access themselves, see the resources box below.

Mental Health Resources for Children and Young People

- For children, young people and their families [Happy Maps](#), great information and a local directory for your area for help and support (developed by Dr Jasmin Krischer, a GP in Bristol, and funded by NHS England).
- Support for young people and parents [Young Minds](#) for children & adolescents up to age of 19 [Childline](#).
- For people aged 16 and over [SANE](#), [MIND](#).
- Free online emotional support, including online counselling [Kooth](#).
- Online digital CBT (NB there is a charge for some of these): Living life to the full for young people [LLTTF-Young People](#) and [Mood Gym](#).
- Crisis and self-harm: free 24/7 text support service [Young Minds Crisis Messenger](#).
- Self-harm help & avoidance [Calm Harm App](#).
- [Self-harm resources](#) from RCPsych.
- [Feeling overwhelmed](#) information from RCPsych.
- [Think Ninja](#) - mental health support app for 10-18 year-olds.
- [Princes Trust](#) - excellent signposting information for mental health issues for young people.
- [CHUMS](#) - mental health support for children and young people.
- [Book recommendation for children with anxiety](#) (very good for parents too!) - 'What to do when you worry too much. A kid's guide to overcoming anxiety.' by Dawn Huebner, published by the American Psychological Association.

KISS: Mental Illness in Children and Young People - Management

[BJGP2019; 69;18](#) [BJGP2017;67;542](#) [NICE NG 134, 2019](#)

Addressing self-harm:

- It is important that young people feel they have been listened to and that there is a recognition of self-harm as an expression of personal distress.
- We need to take disclosure of self-harm seriously, respond empathetically with compassion and consideration to build rapport and to help the person feel listened to.
- Non-judgemental listening and effective communication backed up with signposting and help from outside organisations and ongoing support are needed.
- Young people report that self-help resources are useful but we lack robust evidence of effectiveness.
- Having a 'youth-friendly' practice is a vital component of care.
- The editorial [BJGP2019; 69;18](#) recognises the challenges that we face in primary care, but emphasises the 'exceptional potential' of the GP consultation to have a positive impact, and that it is a vital 'opportunity not to be missed' to prevent escalating self-harm and to mitigate suicide risk.
- **See also the chapter on suicide prevention and considering use of safety plans.**

Managing depression:

- Consider the social, educational and family context.
- Always ask the patient and their parent(s) or carer directly about:
 - Drug & alcohol use, the experience of being bullied or abused, self-harm, ideas about suicide.
 - Give the child the opportunity to talk in private about these issues.
- Special attention should be paid to issues of confidentiality, consent (including Gillick competence), child protection and the use of the Children Act & Mental Health Act.
- **Referral criteria:**
 - Active suicidal ideas or plans, or high-risk recurrent self-harm.
 - Depression with multiple risk factors.
 - Mild depression that has not responded to interventions after 1-2 months.
 - Moderate, severe or recurrent depression.
 - Self-neglect, including a significant reduction in eating.
- **Mild depression:**
 - Watchful waiting.
 - Digital CBT or group therapy (CBT, interpersonal therapy or mindfulness), according to patient/parent preference.
 - If needs are not met after 2 to 3 months, refer for individual CBT or family therapy.
- **Moderate to severe depression:**
 - 5 to 11-year-olds: family-based psychotherapy or individual CBT +/- fluoxetine.
 - 12 to 18-year-olds: individual CBT +/- fluoxetine.
 - Combined psychological therapy & fluoxetine (*fluoxetine should only be prescribed after assessment by a child & adolescent psychiatrist and on specialist advice*)
 - Do NOT offer antidepressants to children or young people except in combination with psychological therapy; specific arrangements must be made for careful monitoring e.g. weekly contact for the first 4 weeks.
 - If needs not met, psychosocial or brief psycho-dynamic intervention & family therapy +/- fluoxetine.
 - If depression unresponsive, recurrent or psychotic for intensive psychological therapy +/- fluoxetine (or other SSRI such as citalopram or sertraline if fluoxetine is ineffective) +/- augmentation with an antipsychotic.

ADHD IN ADULTS

Attention deficit hyperactivity disorder has long been a controversial diagnosis in primary care. A recent BJGP review [BJGP2020;70;467](#) highlights how it can commonly present in primary care with underlying co-morbidities such as depression, anxiety, mood instability and substance misuse. The paper concludes that ADHD in adults in primary care is common, impairing and often misdiagnosed and yet often highly responsive to treatment.

KISS: ADHD in Adults Based on [BJGP 2020;70;467](#) and [NICE NG87 2019](#)

► Key learning points

- ADHD in adults is common, affecting 2.5% of adults in the UK
- It commonly presents to primary care with the consequences e.g. depression and/or anxiety, mood instability, substance and alcohol misuse, problems related to impulsive behaviour, social problems including forensic and social, educational and professional underachievement
- Although there is great under-provision of diagnostic and treatment services in the NHS, treatment is highly effective both in reducing symptoms and improving function

► Clinical features

- ADHD is a highly genetic condition and has a significant crossover with other neurodevelopment conditions including learning disability, autism and dyslexia
- *It is defined as a persistent and pervasive pattern of > 6 months of inattention and/or hyperactivity-impulsivity that has a direct negative impact on academic, occupational or social functioning (ICD-11)*
- In adults, hyperactivity is less marked (although fidgeting is common) and generally manifests as an internal restlessness and inability to switch off
- Attention is dysregulated rather than deficient e.g. patients can focus on topics of interest to them, sometimes with hyper-focus, but this is typically not sustained
- Common features include procrastination and failure to complete tasks
- Patients are left feeling unreliable and disorganised; they underachieve and have low self-esteem; they often experience mood instability, anger issues and relationship problems as well as underachieving in educational and occupational environments
- There is a very high frequency (80%) of comorbid mental health problems, including anxiety, depression, substance or alcohol misuse, bipolar disorder or emotional instability
- *Clinicians should be alert to the possibility of adult ADHD in patients with treatment-resistant depression or anxiety, 'rapid cycling' bipolar disorder or anyone with ≥ 3 psychiatric diagnoses*

► Screening for ADHD and diagnosis

- The WHO have a free screening tool for patients, the [Adult Self Report Scale ASRS](#)
- In a general population, it is highly specific 98%, with a sensitivity of 70%
- *Adults presenting with symptoms of ADHD in primary care should be referred for specialist assessment where there is evidence of typical manifestations that began during childhood and have persisted through life and resulted in significant impairment* [NICE 2018](#)
- A diagnosis of ADHD should only be made by a specialist

► Treatment for ADHD

- ADHD medications are 'highly effective, well-tolerated and generally safe' [BJGP 2020](#)
- NICE recommends both methylphenidate and lisdexamfetamine as first line treatments
- There is evidence [Lancet 2018](#) that amphetamine-based drugs such as lisdexamfetamine have superior tolerability and efficacy in adults and methylphenidate in children and adolescents
- NICE advise 6 monthly monitoring of BP, pulse and weight plus an annual review
- Non-pharmacological strategies e.g. psychology, education and lifestyle advice should also be used

HEALTH ANXIETY

Health anxiety frequently appears in the consulting room, yet rarely in the journals. So, a recent BMJ '10 minute consultation' review [BMJ2019;364:1774](#) and an editorial on the topic is welcome [BMJ2016;353:i2250](#). Health Anxiety is common, persistent and a significant generator of long-term morbidity and sick leave in primary care yet it is very treatable with cognitive behavioural therapy (CBT). However, it often goes unrecognised and is inappropriately managed by reassurance and investigations.

► *How do we define Health Anxiety?*

The term itself is relatively new. The new [DSM-5 criteria](#) redefines the stigmatizing 'hypochondriasis' and replaced it with 'illness anxiety disorder' and 'somatic symptom disorder'. But, these are narrowly defined. The preferred clinical diagnosis of Health Anxiety is defined by cognitive and emotional symptoms that allow it to co-exist with other diseases. Indeed, Health Anxiety is often found in conjunction with other disorders including physical ones.

► *How common is Health Anxiety in primary care?*

The BMJ editorial states that Health Anxiety 'is reaching epidemic proportions', with 3.4% of people in the community meeting the criteria, up to 9.5% in primary care and up to 20% in secondary clinics. *The incidence is thought to be rapidly rising, partly due to the increase in cyberchondria* (the excessive use of health internet sites to fuel health anxiety & spread pathogenic ideas). For people with Health Anxiety, the internet is a 'menace' as they pay selective attention to the most serious explanation of symptoms. This has contributed to an increasing 'pathologisation' of society

► *What treatments are effective?*

Several psychological treatments are now of proven effectiveness in Health Anxiety. These include:

- Traditional CBT
 - research [Lancet2014;383:219](#) shows that 5 to 10 sessions of group CBT produces sustained symptomatic benefit at 2 years compared to standard care
- Internet-based CBT and guided self-help
- Group based mindfulness
- Acceptance and commitment therapy (for a simple explanation of this form of therapy, see [ACT](#))
- **Conclusion:** the paper [BMJ2016;353:i2250](#) concludes that physicians in primary care need to be more aware of this important diagnosis, to make the diagnosis using these simple criteria and to offer these established, effective and evidence-based treatments.

CBT can also reduce frequent attendance in primary care. Research [BJGP2016;66:e729](#) shows that of long-term frequent attenders in primary care (more than 30 face-to-face consultations with GP or nurse over 2 years, excluding scheduled appointments for routine monitoring e.g. chronic disease checks) who took up an offer of CBT, 50% of these halved their consultation rate over the next year and reported improved quality of life and better mental health scores.

NB Learning Points: Health Anxiety

- Up to 10% of our patients in primary care have Health Anxiety
- Health anxiety can, and often does, co-exist with physical health problems
- The diagnosis is well accepted by patients if it is respectfully explained. Reassurance and investigation do not work. There is a range of effective, evidence-based psychological therapies which patients should be offered.

KISS: Health Anxiety

[BMJ2019;364:1774](#) & [BMJ2016;353:i2250](#)

Background:

- **What is health anxiety?**
 - The term itself is relatively new. The new DSM-5 criteria redefines the stigmatizing 'hypochondriasis' and replaced it with 'illness anxiety disorder' and 'somatic symptom disorder'.
 - But, these diagnoses are narrowly defined. The preferred clinical diagnosis of Health Anxiety is defined by cognitive and emotional symptoms that allow it to co-exist with other diseases.
 - Health Anxiety is often found in conjunction with other disorders including physical ones.
- How common is Health Anxiety in primary care?
- The BMJ editorial states that Health Anxiety 'is reaching epidemic proportions', with 3.4% of people in the community meeting the criteria, up to 9.5% in primary care and up to 20% in secondary clinics.
- The **incidence is thought to be rapidly rising**, partly due to the increase in [cyberchondria](#) (the excessive use of health internet sites to fuel health anxiety & spread pathogenic ideas). For people with Health Anxiety, the internet is a 'menace' as they pay selective attention to the most serious explanation of symptoms. This has contributed to an increasing 'pathologisation' of society.

Assessment and management:

- Consider Health Anxiety as a diagnosis in patients reporting excessive worry about their health which is causing distress, suffering and functional impairment.
- **Avoid routinely reassuring the patient that all is well** and acknowledge instead that the patient is worried and suggest seeking ways to address this.
- Explore possible co-morbid mental health problems such as depression, generalised anxiety disorder, OCD, panic disorder and medically unexplained symptoms.
- **Suggested strategies:**
 - Avoid repeated reassurance: *'reassurance is like an addictive drug, providing immediate relief but wearing off quickly'*.
 - **Validate that the symptoms** are real and genuine but **re-align the focus on the anxiety** as the most important symptom and how distressing this can be.
 - Acknowledge the anxiety: *'it is clear that you are very worried about your health, we have ways of helping this and this will not affect other treatment you are receiving'*.
 - Provide alternative explanations for somatic symptoms e.g. that a wide range of physical symptoms can be symptoms of anxiety rather than the physical disease: misinterpretation of symptoms and also of health information is a key feature of health anxiety.
- Offer referral for CBT focused on health anxiety, which is highly effective:
 - 5 to 10 sessions of a group or individual CBT produce sustained benefits at 2 years and halve GP consultation rates.
- **Drug treatment?**
 - Antidepressants - consider if significant co-morbid depression, GAD or OCD.
 - They may be considered for debilitating health anxiety for patients who are interested in treatment but decline talking therapies; there is a weak and limited evidence base, but fluoxetine has been shown to be more effective than placebo in reducing health anxiety.

Further resources:

- Self-help book [Overcoming Health Anxiety](#)
- NHS Choices [Health Anxiety information](#)
- Excellent [self-help leaflet & guide](#) from NHS Northumberland

PSYCHOSIS AND SCHIZOPHRENIA

Schizophrenia has been contested as a diagnosis in recent years, for example, see the fascinating paper '*Schizophrenia doesn't exist*' [BMJ2016](#). However, as far as the WHO are concerned in 2021 it does exist. *It is still very much in ICD-11 as a primary psychotic disorder* [ICD-11 6A20 Schizophrenia](#). It is defined as being characterised by disturbances in multiple mental modalities including thinking (e.g. delusions, disorganisation in the form of thought), perception (hallucinations), self-experience (e.g. the experience that one's feelings or thoughts are controlled by an external force), cognition (e.g. impaired attention and memory), volition (loss of motivation), affect (e.g. blunted emotional expression) and behaviour (e.g. behaviour that appears bizarre, purposeless or inappropriate). These symptoms need to have persisted for at least a month and are not a manifestation of another condition or caused by the effects of substances on the CNS. There is increasing recognition however that transient psychotic symptoms are common, including in response to stress, and that a more positive approach needs to be taken with schizophrenia towards long-term recovery.

KISS: Psychosis and Schizophrenia - Background

[CKS 2021](#) [NICE CG 178, 2014](#)

Background:

- The exact cause is unknown but there is strong a genetic element (>7x increased risk of psychotic disorder in people with a parent with schizophrenia); other risk factors include psychosocial factors (e.g. stressful life events, childhood adversity etc), drug misuse (esp. cannabis) and steroid use.
- The prevalence of psychosis in England is about 0.5% with schizophrenia remaining the most common cause of major psychosis.
- 4/5 people will show some response to treatment after the first episode.
- 1/5 will have no further psychotic episodes.
- However, ~15% will experience persistent symptoms unresponsive to treatment at 2 years.

Remind me what psychotic symptoms are:

Psychotic symptoms may be 'positive' or 'negative':

- **Positive symptoms** include:
 - Hallucinations (perceptions in the absence of stimulus). The most common are auditory hallucinations, in which voices are heard and may provide a running commentary on actions, argue with the person, command, or echo the person's thoughts.
 - Hallucinations which are visual, smell, taste, or tactile in nature occur less commonly.
 - Delusions (fixed or falsely-held beliefs) can include:
 - Delusions of reference (the belief that ordinary events, objects, or the behaviour of others have a meaning specifically for the person, for example, that people on the radio are talking to, or about, them).
 - Delusions of control (the belief that the person's thoughts, feelings, or behaviour are being controlled by others. They include thought insertion, thought withdrawal and thought broadcasting).
 - Delusions of persecution (the belief that other people are plotting against the person).
 - Disorganised behaviour, speech, and/or thoughts (thought disturbance).
- **Negative symptoms** include emotional blunting, reduced speech, loss of motivation, self-neglect, social withdrawal.

KISS: Psychosis and Schizophrenia - Assessment and Management [CKS 2021](#) [NICE CG 178, 2014](#)

Assessment:

- **Psychosis may be preceded by a prodromal period** that can last from a few days to around 18 months, which offers a critical treatment window to delay or prevent transition to frank psychosis. Symptoms include:
 - Transient, low-intensity psychotic symptoms — intermittent, self-limiting episodes, typically lasting less than a week. May include hallucinations/unusual perceptual experiences, unusual thoughts and may manifest as unusual or uncharacteristic behaviour.
 - Reduced interest in daily activities — may manifest as poor personal hygiene and/or reduced performance at school or work.
 - Problems with mood, sleep, memory, concentration, communication, affect, and motivation.
 - Anxiety, irritability, or depressive features.
 - Incoherent or illogical speech — suggestive of thought disturbance.
- **We should routinely ask about:**
 - **Substance misuse** in psychosis, and refer all patients with suspected psychosis with known or suspected substance misuse (dual diagnosis rates are high, ~50% in patients with psychosis).
 - **PTSD** as patients with psychosis have often suffered significant previous trauma.

Management:

- Undertake a **risk assessment to self and others:**
 - Should include risk of self-harm or suicide, misuse of drugs/alcohol, risk of unintentional harm (caused by disorganised behaviour etc.)
 - Risk to others including family and the public - confrontational behaviour, history of violence, carrying weapons etc.
- If considered high risk arrange for same-day specialist mental health assessment.
- If not considered high risk refer urgently to early intervention in psychosis (EIP) team, or if unavailable to a crisis resolution/home treatment service.
- If presenting in the prodromal phase refer urgently to EIP or specialist mental health service.
- **Do NOT start antipsychotics in the community** (unless under the guidance of psychiatrist).
- Secondary Care/EIP management is likely to include CBT (to all patients, including in the acute phase), an antipsychotic medication.
- The secondary care team should maintain responsibility for monitoring physical health and the effects of antipsychotic medication for at least the first 12 months or until the person's condition has stabilised; thereafter, monitoring may be transferred to primary care under shared care arrangements.
- Once stable **GPs should carry out a physical and mental health review at least annually**, which should include:
 - Weight and waist circumference, pulse and BP.
 - **Blood tests** - FBC, U&E, LFT, Fasting blood glucose, HbA1c, lipid profile and prolactin levels (annual prolactin levels not needed for aripiprazole, clozapine, quetiapine, olanzapine).
 - **Carry out ECG** if taking haloperidol, pimozide or sertindole, or if they have had a previous ECG abnormality, or if they have an additional risk factor for QT prolongation (e.g. taking another medication that can increase the QT interval such as erythromycin, co-trimoxazole, or pregabalin).
 - A QT interval of >450 ms may be cause for concern; QT interval > 500 ms should prompt specialist advice.
 - Assessment of any movement disorders; reviewing adherence and side effects to treatment.
 - Screening for psychotic symptoms and co-existing mental health symptoms.
 - Assessment of nutritional status, diet and level of physical activity.
 - Ask about alcohol intake and substance misuse.
- At ALL stages offer help to stop smoking using nicotine replacement if appropriate.



GOUT

Gout is the most common inflammatory arthritis, increasing in prevalence and responsible for significant long-term disability. In *June 2022 NICE published a new guideline* [NICE NG219, 2022](#) based on a new review of the evidence, however as discussed in a recent DTB editorial [Drug and Therapeutics Bulletin 2022;60:130](#) and a follow-up [NB Blog](#) this guideline still has a number of gaps, and there is still a lack of evidence and consensus in a number of areas of gout management.

► Gout, [NICE NG219, 2022](#)

See below for a full KISS of the guideline. There are some noticeable differences from other guidelines in recent years such as [BSR Gout guideline 2017](#). Notably, NICE now recommend based on recent evidence:

- Not *all* patients need to be offered urate-lowering therapy (ULT) after an acute attack of gout.
- When ULT is offered, we should treat serum uric acid (SUA) to a target but this target is more relaxed < 360 µmol/l than previous guidelines which have recommended a < 300 µmol/l target.
- Assessment as to whether or not the patient has chronic kidney disease (CKD) is important, as if they have GFR stage 3 to 5 they should be all be offered ULT:
 - Treatment of gout in people with CKD can be challenging, especially in CK GFR stages 4 and 5. The guideline recommends rheumatology referral if patients have CKD stages 3b to 5.
- It is not considered essential to have 3 months of NSAID or colchicine to prevent flares when starting long-term urate-lowering therapy:
 - Although this has been standard advice for decades when starting ULT, NICE note a total lack of evidence for NSAIDs in this context; there is evidence for colchicine, but of course, it comes at a cost of potential GI side effects; we should discuss the pros and cons with the patient, and if the patient opts to take it we should offer colchicine first, with low dose NSAIDs and steroids as a second-line option if it is not tolerated.
- The guideline is in agreement with previous guidelines, that it is crucial that patients understand that this is a lifelong condition and to have information to hand as to how to reduce the risk of flares. However, the guideline states that there is insufficient evidence that any specific diet prevents flares or lowers SUA. We should advise patients that excess body weight or obesity or excessive alcohol consumption may exacerbate gout flares and symptoms, and otherwise just advise a usual healthy diet.

But, there are some big 'buts' to this guideline as highlighted in the [DTB editorial](#). The NICE guideline gives a more pragmatic framework to gout management although it's important to **highlight a number of the uncertainties** that we still have particularly when it comes to urate-lowering therapy (ULT) and that **large parts of the recommendations are based on expert opinion rather than good evidence**:

- Overall, the quality of the evidence for ULT is low and very little of it is long term and there is little evidence on identifying who should be offered ULT for long-term management of gout.
- NICE stick to a 'treat-to-target' strategy if ULT is used, however, there is no evidence identifying what level should be aimed for, and the committee agreed the target level by consensus.
- It's also worth noting that using a 'treat-to-target' strategy is by no means universally accepted with the American College of Physicians recommending a 'treat to avoid symptoms' strategy.
- In addition, NICE suggests that increasing ULT treatment from 40% of patients following a flare to 96% is cost-effective and will lead to savings. However, there are actually very few cost-effectiveness studies on which to base these recommendations and NICE's modelling did not include clinician time in the management of ULT (nice to know our time is appreciated...)

It may come as a surprise to many of us that for such a common condition we still have a very limited evidence base on which to rely for management, meaning we should use a larger slice of clinical judgement and share these uncertainties when it comes to discussions on ULT with our patients.

KISS: Gout

Based on [NICE NG219, 2022](#) & [BSR 2017](#)

Diagnosis

- **Diagnosis of acute gout remains clinical** in people with rapid onset of severe pain, often overnight, together with redness and swelling, commonly in one or both first MTPs or tophi, but may appear in other joints e.g. midfoot ('bootlace' area), ankle, knee, hand, wrist, elbow.
- Assess for the possibility of septic arthritis and other inflammatory arthritis:
 - if suspect septic arthritis, immediate referral
 - if diagnosis is uncertain, consider referral to secondary care for joint aspiration +/- imaging.
- Measure serum urate level (SUA):
 - If typical clinical presentation and SUA is $> 360 \mu\text{mol/l}$ (6mg/dl) this confirms diagnosis.
 - If SUA is < 360 and was done during acute flare, repeat > 2 weeks after the flare has settled.

Management of acute gout flares:

- Offer either a **NSAID, colchicine or a short course of prednisolone for first-line treatment** based on the patient's co-morbidities; consider adding a PPI if using an NSAID/oral corticosteroids. NICE do not give guidance on dosing but BSR recommend:
 - Colchicine 500mcg BD-QDS (noting that lower doses often as effective with far fewer side effects).
 - Or if oral steroids used prednisolone 35mg daily for 5 days.
- If NSAIDs or colchicine are C/I or not tolerated, consider an IA or IM injection of corticosteroid.
- Advise people that **ice packs** may help alleviate pain.
- Consider a follow-up appointment to measure SUA (at least 2w after acute attack settled), assess co-morbidities inc. CKD and CVD, review medications and to provide lifestyle/dietary advice.

Long-term management:

- Offer urate-lowering therapy (ULT) to people with gout who have:
 - Multiple or troublesome flares
 - CKD stages 3 to 5
 - Diuretic therapy
 - Tophi or chronic gouty arthritis
- Discuss the option of ULT using a treat-to-target strategy for people who are not in these groups.
- Start ULT at least 2 to 4 weeks *after* the acute flare has settled.
- **Target SUA level (note uncertainties discussed above):**
 - Start with a low dose of ULT and monitor SUA levels monthly until target $< 360 \mu\text{mol/l}$ is reached.
 - Consider a lower target level of SUA $< 300 \mu\text{mol/l}$ if they have ongoing flares despite SUA < 360 , or have tophi or chronic gouty arthritis.
- **Urate lowering therapy:**
 - Offer allopurinol or febuxostat as first-line treatment, taking into account co-morbidities and preferences; if major CVD offer allopurinol and not febuxostat.
- Discuss the pros and cons of taking medication to prevent gout flares when starting or titrating ULT, and if people choose to have treatment offer colchicine during the start and titration of ULT. Consider low-dose NSAID or oral corticosteroid if colchicine is contraindicated or not tolerated.
- Ensure people understand that ULT needs to be continued after SUA is reached and is considered a lifelong therapy.
- Consider annual monitoring of SUA.
- Refer to a rheumatologist if the diagnosis is uncertain, treatment is contraindicated, ineffective or not tolerated; they have CKD stage 3b to 5; they have had an organ transplant.
- Patient information: [GOUT, NHS patient information](#) & [Versus Arthritis](#)

OSTEOARTHRITIS

Some interesting new research on the **efficacy of steroid and LA injections** for hip arthritis has been published in the BMJ [2022;377:e068446](#), and this has led to an accompanying editorial suggesting we change the way we think about osteoarthritis moving away from the concept of 'wear and tear' arthritis towards a model of **'tear, flare and repair'** [BMJ2022;377:o1028](#). NICE have also published a new guideline on osteoarthritis in October 2022 ([NG226](#)) which introduces significant changes to management. At the same time the NICE guidance was published we have had important pieces of research which support two of the NICE recommendations, namely that **XRays in early suspected OA are of little benefit** ([BJGP 2022; 72 \(723\)](#)), and that **overweight and obesity should not be a preclusion to knee joint replacements** ([BMJ 2022;379:o2473](#)). One treatment that is not mentioned by NICE is antidepressants, but a recent Cochrane review [CD012157](#) suggests there may be a group of people who benefit from duloxetine.

► **The hip injection trial, and the 'tear, flare and repair' model for OA**

The HIT Hip Injection Trial [2022;377:e068446](#) was an excellently designed trial with positive results. This is extremely welcome in that therapeutic options for OA have always been limited, and following the latest NICE guideline (see below!) are now even more restricted!

- 200 adults aged over 40 with at least moderate pain from OA hip were randomised to receive either a single ultrasound-guided intra-articular hip injection of steroid (triamcinolone 40mg) plus local anaesthetic (lidocaine), OR USS guided injection of lidocaine alone OR best current treatment i.e. advice and education. Participants in the two injection arms were blinded to which treatment they received.
- Greater improvement in pain and function over 6 months was reported with the two injection groups:
 - Patients in the combined injection group had pain scores reduced by 50% at 2 weeks, with a mean improvement of 25% over 6 months
 - The NNT at 2 months were just 2 for improved sleep, 3 for feeling better and 3 for no limits to physical activity
 - At 6 months pain relief was not sustained but functional improvements persisted
- There was no difference in efficacy between steroid + LA injections and LA alone, unless patients had USS confirmed synovitis or effusion where a clinically significant benefit of the added steroid was seen
- **Conclusion:** *for people with mild to moderate OA hip, ultrasound-guided injection of steroid + LA can give rapid improvement in pain and function which lasts for several months*

Given the very long waits for hip replacements, we have to hope that commissioners make this a service widely available for GPs to refer patients to. Discussing the results in an accompanying editorial discussing **'tear, flare and repair'** [BMJ2022;377:o1028](#) the authors point out

- Given the lack of effective non-surgical therapeutic options in OA, the findings of the HIT trial are clinically important for patients with mild to moderate OA hip
- The trial supports the premise that steroids work better for patients with baseline synovitis and effusion, justifying an anti-inflammatory approach
- This supports the model of 'tear, flare and repair' model of OA
- The previous view of OA as 'wear and tear' should be abandoned; it was negatively framed, reinforcing expectations of inexorable decline and loss of quality of life and is inconsistent with evidence and advice on physical activity
- They conclude that patients and physicians should 'warmly welcome' this treatment option, bearing in mind the usual caveats about infection risk

► **Antidepressants for OA?** [Cochrane 2022, Issue 10. Art. No.: CD012157](#)

As we will discuss below there are very few pharmacological interventions that have been shown to be effective for OA, with the new NICE guideline reflecting that. Given it is a chronic pain condition it's an important question as to whether antidepressants might improve OA-related pain and disability.

We have [RCT evidence](#) nortriptyline is ineffective for OA knee, but a recent Cochrane review (which included that RCT on nortriptyline) has looked at a wide range of antidepressant drugs with somewhat mixed results:

- This review included 9 trials comparing antidepressant to placebo or comparator drug, 2122 patients with hip or knee OA (most just knee OA), most participants were female, with mean ages of 54.5 to 65.9 years; trial durations were 8 to 16 weeks.
- The vast majority of trials (6) looked at duloxetine with a single trial looking at each of nortriptyline, fluvoxamine and milnacipran.
- There was high-certainty evidence that antidepressants resulted in a clinically unimportant improvement in pain compared to placebo with a mean difference of -0.59 on a 0-10 scale (95% CI = -0.88 to -0.31).
- However, when **clinical response was defined as >50% improvement in pain over 24 hours there was high certainty evidence of a significant benefit in the antidepressant group** (RR 1.55, 95% CI 1.32 - 1.82); this corresponded to an absolute improvement of 16% more responders getting a >50% pain improvement and a number needed to treat of 6.
- There were clinically insignificant improvements in function and quality of life.
- There were **significantly more adverse events in the antidepressant group** (minor only, mostly dry mouth, constipation, nausea) with RR 1.27, 95% CI 1.15 - 1.41 and a NNH of 7, with higher withdrawal rates (double) in the antidepressant group.
- **Conclusion:** antidepressants (mainly duloxetine) cause a small, clinically insignificant, improvement to mean pain and function levels for people with OA (predominantly knee OA); however, **there is a small subgroup of people who may derive much more significant improvement in pain levels at the cost of higher side effects.**

The authors conclude, not unsurprisingly, that given there seems to be a small subgroup that gain much more benefit from antidepressants, these results suggest that *'pain is multifactorial in osteoarthritis and the underlying mechanisms may vary between individuals, it is possible that only individuals with a specific underlying pain mechanisms will be responsive to this therapy'*. It is likely that the pain (especially in knee OA) is likely to be a combination of direct damage (nociceptive), neuropathic and nociplastic (caused by a disturbance in peripheral and/or central nervous system pain processing). In practice of course it will be very difficult to ascertain who may or may not be in that group that might get a decent improvement in pain levels, but these results suggest a trial of duloxetine is not unreasonable, with the caveat that you're as likely to get side effects from the treatment (NNH of 7) as you are benefit (NNT of 6 for 50% improvement in pain).

► **What about supplements, glucosamine, chondroitin etc?** [Drug and Therapeutics Bulletin 2022;60:4](#)

The NICE guideline (discussed below) continues to actively recommend that glucosamine should not be routinely offered for OA, and this is supported by recent evidence reviewed in the DTB, which looked at glucosamine as well as number of other pharmacological options that have been touted to reduce OA progression and possibly improve pain:

- This systematic review and network meta-analysis of 28 studies and almost 12,000 patients, evaluated long-term efficacy of 12 classes of drug in terms of OA progression and pain.
- The **primary structural outcome was joint space (on X-ray) or cartilage volume (on MRI)**, as well as pain.
- Most studies (23) focused on knee OA with the remainder focusing on hip OA.
- **Drug classes** included glucosamine, chondroitin, bisphosphonates, strontium, vitamins E and D, doxycycline plus a few other less well-known ones including MMP inhibitors.
- Small improvements were seen in structural outcomes for chondroitin, doxycycline and strontium, but they had **no effect on need for joint replacement**.
- Small improvements were seen in pain scores and joint function for glucosamine and Vitamin D.
- However, **none of the small improvements were clinically meaningful and the authors conclude that none of the drugs studied has long-term disease-modifying benefits.**

This evidence very much backs up the NICE guidance below that the focus of OA treatment (at least before joint replacement) should be non-pharmacological.

► [NICE guideline on osteoarthritis 2022](#) [NICE NG226 Oct 2022](#)

No surprises here, but there are significant changes in that paracetamol and weak opioids are not recommended (how many patients do we have with OA on endless repeats of co-codamol??). The guideline emphasises lifestyle changes including weight loss and tailored exercises such as muscle strengthening and aerobic exercise. Referral for supervised exercise is recommended as it increases adherence and provides additional social support. Importantly, NICE stress the evidence that this improves not just outcomes related to the OA but also improves the overall quality of life. The diagnosis of OA can be made clinically in patients aged over 45 with activity-related joint pain who do not have morning joint-related stiffness or morning stiffness that lasts more than 30 minutes. As far as drugs are concerned, topical and oral NSAIDs remain as treatment options, but importantly NICE do NOT routinely recommend paracetamol or opioids.

When it comes to referral for consideration for joint replacement, this should be based purely on a holistic clinical assessment. Importantly the guideline is explicit that people should NOT be excluded from referral on the basis of being overweight or having obesity (important as many CCGs do 'bounce back' referrals in patients with obesity).

As far as the lack of drug treatment options go, high-quality evidence that paracetamol is ineffective in OA has been gathering for a few years [Cochrane 2019](#), and it is interesting to see that it is now enshrined in these new guidelines. Similar lack of efficacy has been demonstrated recently for tramadol [Cochrane 2019](#) and also low dose TCA [BJGP 2021; 71 \(708\): e538-e546](#).

► [What about exercise?](#)

- The quantitative data was inevitably of lowish quality, as performing high-quality trials with exercise interventions is difficult, and there are many modalities available; however, there was enough data to suggest **exercise interventions do improve pain, physical function and depression**.
- The qualitative data was of good quality and had some very useful points for us in primary care:
 - Patients wanted better information and advice about the **safety and value of exercise in OA**.
 - They want exercise recommendations tailored to their individual needs.
 - They suggest **inappropriate health beliefs** (e.g. that more exercise will cause worsening joint damage) **need to be challenged**.

► [What about X-rays for early hip OA?](#) [BJGP 2022; 72 \(723\)](#) [NB Blog 1.12.2022](#)

One of the previous, and now ongoing, recommendations from NICE is that OA diagnosis can be made clinically without X-ray if typical symptoms and signs are present. This was predominantly based on data looking at knee OA, but the recent [CHECK cohort trial](#) supports this strategy in early suspected OA of the hip also. We have reviewed this in a [NB blog](#) as well. What did it find?

- This was a Dutch cross-sectional cohort study that recruited just over 1000 participants aged 45-65 who had had hip and/or knee pain and hadn't been to their GP in the 6 months prior for this problem.
- They all had hip X-rays which looked for **evidence of radiographic OA (ROA)** of the hip which was either 'definite' or 'early-stage' based on radiographic criteria, and the **association between XR findings and pain was assessed**.
- There was a modest association between hip pain and hip ROA - compared with pain-free hips the odds ratios for painful hips was ~ 1.5 for both definite and early-stage ROA.
- However, both the **sensitivity and specificity of hip ROA for symptomatic clinical OA was poor**:
 - Prevalence of definite ROA was 13.3% in painful hips but was still 9.5% in pain-free hips
 - Prevalence of early-stage ROA was 41.2% in painful hips but was still 31.4% in pain-free hips
 - When sensitivity analyses were conducted and clinical OA criteria were applied to the cohort, X-rays would miss a significant number of people with clinical OA that wasn't seen on X-ray.
- **Conclusion: Xray provides little assistance to help identify people with hip OA in the early stages.** The authors suggest GPs should start conservative treatment for OA based on clinical criteria without X-ray confirmation.

It's important to note that this study (and nor does NICE say) we should never do X-rays. There will obviously be a time with further presentations down the line, when an X-ray will be needed, especially if onward referral to MSK clinics or secondary care is needed. The other point to highlight is that **NICE does recommend considering X-rays if there are atypical features or alternative diagnoses are being considered.**

► ***What about referral for knee joint replacement for people with obesity?*** [BMJ 2022;379:o2473](#)

Another of the (not new) recommendations from NICE is '*Do not exclude people with osteoarthritis from referral for joint replacement because of...overweight or obesity, based on measurements such as body mass index (BMI)*'. This has been widely bypassed in many areas - it is essentially rationing under the pretext that people with overweight and obesity may have poorer outcomes after joint replacement, although any evidence to support this is based on small studies which often looked single outcomes.

New evidence, initially published in [PLoS Med 2021;18:7](#) and then recently reviewed as part of a [NIHR alert in the BMJ](#), both **supports the NICE stance on this and further questions the BMI limitations imposed in many areas at least with respect to knee replacements.** What did this evidence show?

- This study looked at **just shy of half a million people on the UK National Joint Registry** who had had **total knee replacements**.
- The outcomes reviewed were those requiring repeat surgery within 10 years, death within 90 days of surgery and knee pain and function 6 months after surgery.
- 10-year repeat surgery rates were slightly higher in those with obesity class 1 and 2 vs those with normal BMI, but still well below the 'benchmark' 5% level generally considered to be acceptable.
- Death rates were low, but trended towards lower death rates in those with overweight or obesity.
- Pain and function at 6 months were slightly poorer in those with overweight or obesity but these differences were clinically insignificant.
- **Conclusion: BMI alone is not an appropriate way to decide who is suitable for total knee replacement.**

This study strongly supports the NICE stance that **BMI alone should not preclude us from referring people for consideration of joint replacement for OA.** Yes, we should support people who are overweight to lose weight both pre and post-op as it may help their overall health, and possibly may reduce joint revision rates 20-30 years down the line, but we should not use BMI as a binary decision to preclude some people for surgery.

KISS: Osteoarthritis**[NICE NG226 2022](#)****Diagnosis:**

- **Can be made clinically and without investigation if:**
 - **The person is >45 AND** has activity-related joint pain AND have no, or <30 minutes, of early morning joint stiffness.
 - Do not routinely use imaging to diagnose OA unless there are atypical features suggesting an alternative diagnosis.

Management:

- *Advise patients that the core treatments for OA are weight loss (if appropriate) and therapeutic exercise, along with information and support.*
- **Good quality information:** Excellent information from **Versus Arthritis** - [click here](#)
 - NICE highlight the importance of **challenging common misconceptions** about the disease as discussed in the [qualitative section of a 2018 Cochrane review](#) e.g. that exercise will cause more joint damage where as we should highlight the safety and value of exercise.
- **Exercise therapy:**
 - Offer tailored therapeutic exercise to ALL patients with OA, which should include local muscle strengthening and general aerobic fitness.
 - Patients need reassurance that although regular and consistent exercise may initially cause discomfort, it will be beneficial to their joints and long-term adherence increases its benefits.
 - Consider combining therapeutic exercise with education programmes.
- For people with OA with **excess weight or obesity**, advise them that weight loss will reduce pain, and improve function and quality of life:
 - Any weight loss will be beneficial but losing 10% is a target to aim for if this is achievable.
- **Manual therapy?** (e.g. manipulation, mobilisation, soft tissue techniques). Not enough evidence to support its use alone, but can be considered alongside exercise therapy for hip and knee OA.
- **Do NOT offer** acupuncture, electrotherapy (e.g. TENS, ultrasound therapy).
- **Do NOT offer** devices such as insoles, braces, tapes, splints etc unless joint instability or abnormal biomechanics loading.
- **Pharmacological treatments:**
 - If needed, use alongside to support therapeutic exercise in the lowest dose for the shortest time.
 - Offer a **topical NSAID** for OA knee, and consider for other joints.
 - If topical NSAID is ineffective, **consider an oral NSAID** taking into account risk factors and co-morbidity and consider adding a PPI.
 - **Do not routinely offer paracetamol or weak opioids**, unless for short-term pain relief or all other treatments are contraindicated, not tolerated or ineffective.
 - **Do NOT offer** glucosamine or strong opioids, or IA hyaluronic injections.
 - Consider intra-articular steroid injections when other treatments are ineffective or unsuitable, but make people aware this only gives short-term relief (see joint injections chapter for more detail).
- **Referral for joint replacement:**
 - Consider referring people with hip, knee or shoulder OA for joint replacement if their joint symptoms are substantially impacting their quality of life and non-surgical management is ineffective or unsuitable.
 - Base referral decisions on global clinical assessment and not on symptom/severity scores.
 - Do NOT exclude people with OA referral for joint replacement because of age, smoking, comorbidities or overweight or obesity, but make them aware of increased surgical risks.
 - Arthroscopy and debridement should *not* be offered.

SHOULDER PAIN

There has been little in the way of new research or evidence regarding shoulder problems in the recent past but we've had some usual evidence and updates over the past few years. There has been an excellent new BMJ clinical review on **frozen shoulder** [BMJ2022;377:e068547](#) and a useful updated [primary care guideline 2020](#) from the British Elbow and Shoulder Society (BESS) in conjunction with the British Orthopaedic Association (BOA), and most recently some important evidence [BMJ 2022;378:o1776](#) on the management of the commonest shoulder problem we see in primary care, rotator cuff related shoulder pain...

► **Physiotherapy +/- CSI for RCRSP: GRASP trial** [Lancet 2021; 398: 416–28](#) [BMJ 2022;378:o1776](#)

First terminology - what on earth is rotator cuff-related shoulder pain (RCRSP)?! This condition has been called many things over the years - subacromial bursitis, shoulder tendinitis, supraspinatus tendinopathy, impingement etc. It is now thought the problem is primarily due to tendinopathy and degeneration of the rotator cuff/tendons, but is probably a combination of lots of different pathologies with a broadly unifying presentation and management. The preferred terms currently seem to be subacromial shoulder pain or rotator cuff-related shoulder pain (RCRSP).

As discussed in our recent [NB blog](#) **RCRSP is the most common shoulder problem we see, accounting for 70% of all shoulder presentations**, yet for such a common condition the evidence base on how to best manage it is limited. Many of us will have used corticosteroid injections (CSI) as a mainstay of treatment for years, but as discussed in the Lancet paper, **CSI probably only gives short-term benefit at best**, and what evidence we have is based on small trials with no data beyond 6 months. So **what about physiotherapy?** Again the evidence is limited. The latest [Cochrane review 2016](#) cited a lack of any convincing evidence for manual therapies, with the more recent Lancet paper confirming most studies to be small, short-term (<6 months) with considerable heterogeneity. So this well-conducted RCT (GRASP trial) is welcome.

- This is the largest RCT to date on rotator cuff shoulder disorders and was a multi-centre trial based in the UK and funded by the NIHR.
- It randomised >700 patients to one of 4 arms - one-off physio appointment with self-guided exercises at home (with support materials - booklet and videos) or up to 6 progressive face-to-face physio sessions, each with or without CSI (using a standard injection dose ≤40mg triamcinolone or methylprednisolone).
- Participants reviewed at 8 weeks, 6 months and 12 months with the primary outcome being the Shoulder Pain and Disability Index (SPADI).
- There was **no difference in SPADI scores between the progressive physio group and the one-off physio group at 12 months** (adjusted mean difference -0.66, 95% CI = -4.52 to +3.20) or at any point during the trial.
- There was also **no difference between CSI vs no CSI at 12 months**; there was a short-term benefit of CSI at 8 weeks but this had disappeared by 6 months.

This study has important implications for us in primary care. Many PCNs will now have physios or MSK practitioners in their teams and this **study would support using them for an initial point-of-contact session for rotator cuff shoulder disorders followed by guided self-help exercises at home**. Indeed the conclusion of the GRASP trial was that *'a single face-to-face session with a physiotherapist is likely to be more cost-effective and is not significantly different in terms of clinical outcomes when compared with a comprehensive physiotherapy intervention of up to six face-to-face sessions'*. If you don't have direct initial access to physiotherapy, it would also seem reasonable to make an initial diagnosis in primary care and direct patients to information about the condition and progressive home exercises for 6-12 weeks, which can be accessed for free on the excellent BESS (British Elbow and Shoulder Society) website -see KISS below.

As for CSI this study confirms the **'direction of travel'** that **CSI can be considered an adjunct to exercise therapy rather than a treatment in its own right**, and that it may be most beneficial for those with more severe symptoms (e.g. if unable to effectively do the exercise regime). The other important point on CSI highlighted in the BESS primary care shoulder guideline is that **no more than 2 injections are recommended as there is emerging evidence that repeated injections may cause tendon damage**.

► Frozen shoulder [BMJ2022;377:e068547](https://doi.org/10.1136/bmj-2022-068547)

This was a comprehensive review of the evidence. It does not tell us anything new as sadly there is not much new to be said about this common, troublesome and difficult-to-treat clinical problem. Important revision points from the paper are:

- It is most commonly a condition of the 5th decade, peak incidence between the age of 40 and 60
 - Patients with diabetes are at higher risk of frozen shoulder (up to 20% lifetime prevalence) and are more likely to have bilateral symptoms
- **Diagnosis** is clinical
 - History: gradual onset, pain at night, often anterolateral pain at onset
 - Exam: painful movement restriction, passive ext rotation < 30 degrees, passive elevation < 100 degrees
 - Investigations: X Rays are recommended to check for glenohumeral arthritis, and USS and MRI may be clinically indicated
- **Recovery** times are very variable but can be years and some patients are left with persistent residual pain and discomfort
 - Most people make a full recovery although recovery time is commonly between one to three years
 - Patients go through distinct phases of freezing/proliferation, frozen/adhesive stage and thawing/resolution
 - The good news is that recurrence is rare
- **Treatment:** physiotherapy supplemented if needed by intra-articular steroid injections remains the mainstay of treatment
 - The UK FROST trial compared early physio and steroid injections with various secondary care options including manipulation under anaesthetic and arthroscopic surgery and found no improvement in outcomes
 - There is no difference in efficacy between ultrasound-guided and 'blind' intra-articular injections
- **Referral** to secondary care is recommended if cases are atypical, there is marked functional limitation or persistence of pain despite primary care intervention after 3 months

NB Learning Points: Shoulder Pain

- History and examination is generally all that is needed for assessment of shoulder pain, but our first questions should be: *are there any red flags and could this be referred pain?*
- For a *great video on shoulder examination* see [Versus Arthritis](#)
- Plain X-rays are indicated if a malignancy is a possibility, if GHJ disorders are suspected, or if there is a failure of conservative treatment but **ultrasound or MRI should NOT be used in primary care** unless part of a specific local treatment pathway.
- **Exercise therapy or physiotherapy** are the cornerstones of managing most shoulder problems, but we need to warn patients this takes time, but most people will get improvement by 6-12 weeks.
- **Corticosteroid injections (CSI)** do have a role in shoulder problems, in some situations, but they should be seen as a treatment *in addition to* exercise therapy/physio, not instead of it.
- **Rotator Cuff Related Shoulder Pain (RCRSP)** is now the preferred term for rotator cuff, impingement and related tendinopathy conditions:
 - Recent high-quality RCT evidence suggests a one-off physio session with guided self-help is as effective as progressive face-to-face physio sessions for RCRSP.
 - This trial also confirmed that **CSI for RCRSP does not improve outcomes at 12 months** - there are benefits in the short term at 8 weeks, with a trend to suggest greater improvement in those with more severe symptoms.
 - BESS guidance advises a **maximum of two CSI for RCRSP** due to the risks of tendon damage if repeated injections are done.
- For more detailed information & self-help exercise leaflets on shoulder conditions - [click here](#)

KISS: Shoulder Pain

[BMJ2022;377:e068547](#) and [BESS/BOA Shoulder Pain Guideline for Primary Care 2020](#)

Rule out Red Flags

- Joint Infection - red skin, fever, systemically unwell = same-day emergency referral.
- Unreduced traumatic shoulder dislocation = same-day emergency referral.
- Malignancy - mass, swelling, unexplained deformity, night pain, previous cancer, consider referred pain from apical lung cancer = 2-week cancer referral.
- Acute rotator cuff tear (can be 'easily missed' [BMJ 2017;359:j5366](#)) = urgent referral, ideally next available outpatient clinic: Trauma, pain, weakness; unable to abduct arm above shoulder level, especially if not limited by pain. Be alert to acute tears after shoulder dislocations, and do not be reassured by patients discharged from ED with 'normal' X-ray.
- Systemic/widespread symptoms e.g. PMR, inflammatory arthritis.

General principles of managing shoulder pain (NB quick video reminder on shoulder exam [click here](#)):

- Management should be personalised, and key aspects to consider are the duration of symptoms, patient expectation, hand dominance, occupation, and level of activity/sports.
- **Is it shoulder pathology** or is it pain from the neck or referred pain from elsewhere?
 - Commonest non-shoulder cause of pain is the neck - see different pain distribution for neck/shoulder ([click here](#)) - move neck then shoulder to see which part reproduces the pain.
 - Consider also referred pain from lung apex (?malignancy) and diaphragm (shoulder tip pain).
- **Physiotherapy/exercise therapy** is a key component of most shoulder problems, but warn patients that improvement will take 6-12 weeks. If physio access is limited see links below for home exercises.
- **Steroid injections:**
 - Can be considered in some situations (see below) but should be done in addition to (not instead of) physio/exercise therapy.
 - Do NOT need to be image-guided in primary care.
- **Imaging** - X-ray can be considered (see below) but ultrasound or MRI should NOT be used in primary care unless part of a specific local treatment pathway.

Is there a history of instability? Does the shoulder partly or completely come out of the joint?

- If traumatic refer urgently to orthopaedics (probably fracture clinic) - younger patients at risk of labral injury and high risk of recurrence, older patients at risk of acute cuff tear.
- If atraumatic (typically young patients aged 10-35) = refer physio; warn patients can take 6 months to resolve; consider early referral to shoulder specialist if frequent ED attendance, persistent displacement, aged <18 or regular absence from school or work.

Suspected AC joint pathology (typically aged >30):

- Pain and tenderness localised to AC joint; high arc pain; +ve cross arm ('Scarf') test.
- **Management:**
 - Rest, NSAIDS, analgesia, modify activity (e.g. avoid cross-body shoulder adduction, avoid lifting).
 - Consider physio and consider ACJ steroid injection if conservative measures not working.
 - Xray and consider secondary care referral if not improving despite the above measures.

KISS: Shoulder Pain continued...

Glenohumeral joint (GHJ) disorders include frozen shoulder and GHJ OA and present with globally restricted painful range of movement; the key feature to differentiate from other shoulder disorders is reduced *passive* external rotation. Xray should be considered to help differentiate between frozen shoulder and OA, as well as to rule out other causes e.g. avascular necrosis.

Frozen shoulder:

- **Background:**
 - Generally presents in 40 to 60-year-olds; more common in those with diabetes or CVD.
 - 3 phases: painful phase, stiff phase, resolution phase.
 - Generally settles spontaneously over 18-24 months but can be persistent for several years in some.
- **Presentation:**
 - Gradual onset pain in the shoulder/deltoid region and stiffness.
 - Progresses to a globally restricted painful ROM in all planes, disproportionately affecting passive external rotation.
- **Management:**
 - Analgesics, NSAIDs (if appropriate, consider PPI cover).
 - Exercise therapy for 6-12 weeks - physio if available but home exercises are also recommended (PIL/exercises from OUH [click here](#)).
 - IA steroid injection can be considered - ideally in the painful/early phase.
 - Refer if no improvement by 3 months (sooner if severe symptoms or unable to manage treatments).

GHJ OA:

- **Background:**
 - Generally presents in > 60-year-olds.
 - Primary OA is relatively rare as non-weight-bearing joint.
 - Look for signs of OA in other joints which may increase the likelihood of GHJ OA.
- Presents in a similar fashion to frozen shoulder hence need to consider Xray to help differentiate.
- **Management:**
 - Analgesics, NSAIDs (if appropriate, consider PPI cover).
 - Exercise therapy - consider physio or home exercises ([click here](#)).
 - IA steroid injection can be considered for short-term relief of OA flare.
 - Refer if failure of conservative treatments or patient wishes to consider shoulder replacement but there are very limited other surgical options.

Subacromial shoulder pain AKA 'Impingement' or rotator cuff-related shoulder pain (RCRSP):

- Covers a wide range of underlying pathologies generally caused by rotator cuff tendinopathies or partial/degenerative cuff tears; the commonest cause of shoulder pain in primary care.
- **Presentation:**
 - Shoulder/deltoid pain, is generally worse in overhead positions and painful to lie on.
 - Active movements are generally more painful than passive; preserved passive external rotation.
 - Painful arc of abduction and pain on abduction with thumb down/worse against resistance (Jobe's test/empty can test).
- **Management:**
 - Exercise therapy and physio should be the cornerstone of treatment ([click here](#) for home exercises if physio access is limited).
 - IA steroid injections can be considered to aid physio/exercise but NO MORE than 2 (evidence suggest repeated injections increase the risk of tendon damage).
 - Refer (and consider Xray) if no better by 6-12 weeks.

OSTEOPOROSIS

Current guidance for managing osteoporosis and preventing fragility fractures remains a minefield with many conflicting messages. *In 2019 we had a useful update* from the [National Osteoporosis Guideline Group \(NOGG\)](#). Whilst there are increasing concerns surrounding the adverse effects of osteoporosis treatments and the danger of 'overdiagnosis' (see [excellent article BMJ May 2015](#)), it is clear that the morbidity and mortality associated with osteoporosis are substantial. Although the use of bisphosphonates is debated (the '*marmite of bone protection - you either love it or hate it*' as discussed in a recent [BMJ blog](#)) a systematic review from 2016 [Health Tech Assess 2016;20;1](#) gave us high-quality evidence that bisphosphonates do reduce fragility fractures, however, the benefit to risk ratio is very debatable in the lowest risk patients (in whom paradoxically the majority of the fractures will occur). The other pressing and unanswered question is how long bisphosphonates should be used for, as the majority of the data looks at outcomes for ≤ 3 years of treatment. **Is there any evidence for a 'drug holiday'?** A recent systematic review ([Ann Intern Med. 2019;171\(1\)](#)) attempted to answer these questions and a subsequent DTB review in 2021 gave useful pragmatic guidance ([Drug and Therapeutics Bulletin 2021;59:35-36](#)). We have had an excellent DTB review ([DTB August 2020](#)) questioning the routine use of calcium (in particular) as well as Vitamin D in osteoporosis, as well as an important safety warning on Denosumab from the [MHRA August 2020](#) highlighting the risks of rebound fractures if treatment is stopped or delayed. And finally, we had a useful [DTB review Nov 2021](#) on steroid-induced osteoporosis and fracture risk with inhaled and oral steroid use in asthma.

► *Osteoporosis risk with oral and inhaled corticosteroids in asthma* [DTB Nov 2021](#)

We have pretty clear guidance from NOGG (see below) on managing people at risk of steroid-induced osteoporosis if they are on regular oral daily doses. But **what about the large number of people on inhaled corticosteroids (ICS) or who have intermittent repeat courses of oral corticosteroid (OCS)? What is their osteoporosis and fracture risk?**

- 70,000 people with asthma were identified through the UK CPRD, most women (~75%), average age 70.
- People with osteoporosis or osteoporotic fractures (vertebral, hip, forearm/wrist, humeral) were matched to controls.
- There was inevitably a dose-response for numbers of prescriptions and cumulative dose of both ICS and OCS for osteoporosis, with the risks highest with OCS.
- However, probably more relevant were the **fracture risks**:
 - Cumulative dose of OCS (prednisolone equivalent) 1001-2500 mg was linked with an increased risk of fracture OR = 1.54 (95% CI, 1.10 to 2.14).
 - Cumulative dose of OCS >2500 mg increased the risk further still OR = 1.99 (95% CI, 1.30 to 3.04).
 - For ICS, there was an increased fracture risk for those who received a cumulative dose of >120 mg/year, OR = 1.20 (95% CI, 1.08 to 1.42).

This data confirms an **increased fracture risk for people with asthma on regular high-dose ICS and especially for those needing regular courses of oral steroids**. A yearly cumulative dose of >2500mg of oral prednisolone is massive and likely to be rare, but we may well have a number of patients that fit in the lower **OCS risk group of >1000mg/year who still have a >50% increased relative risk of osteoporotic fracture**. 1000mg would be the equivalent to 5 x 40mg 5 day courses, or only 3 x 50mg 7 day courses. Although the risks are lower with ICS, there was still a **20% increased relative risk of fracture for those on >120mg/yr**, and by way of example Fostair® 100/6, 2 puffs BD regularly for 1 year = >140mg/yr, putting many people in this risk bracket. And one final thought - although this data was specific to people with asthma, we will **see many patients with COPD who will reach the OCS thresholds above** with repeat courses for exacerbations, many of whom will have higher baseline risk anyway. Currently, the issue of when to consider bone protection for repeat short courses of oral steroids is not adequately addressed in the national guidance, but [guidance from some localities](#) recommends we **consider formal osteoporosis risk assessment/treatment for anyone whose cumulative (prednisolone equivalent) yearly dose exceeds 1500mg**.



► ***Denosumab: Risk of vertebral fractures after stopping or delaying treatment*** [MHRA Aug 2020](#)

Osteoporosis treatments are not without risks. The bisphosphonates are associated with a few rare, but serious, side effects including atypical femoral fractures and osteonecrosis of the jaw. A few years back [Strontium Ranelate was discontinued](#) due to increasing concerns about cardiovascular risks, and more recently the risks of Denosumab have started to become apparent. Denosumab is also associated with increased risks of osteonecrosis of the jaw and external auditory canal. But probably of more concern has been the increasing recognition that **if treatment is abruptly stopped, or delayed, there appears to be a risk of rebound multiple vertebral fractures**. The [NICE rapid COVID guideline on rheumatological conditions](#) originally published in April 2020 confirmed that injections of Denosumab should not be delayed during the pandemic, and subsequently, we have strengthened advice from the MHRA on this issue.

- UK data suggests an increased risk of vertebral fractures, including multiple fractures, in patients after stopping or delaying treatment with Denosumab.
- In some cases, the fractures were described as 'life-changing'.
- Incidents were reported within 18 months and in some cases 9 months, of cessation.
- Patients with previous vertebral fractures may be at the highest risk.
- The MHRA advise the benefit/risk with Denosumab need to be carefully evaluated before initiating treatment, especially in those at increased risk of vertebral fractures.
- **Patients should not stop Denosumab without specialist review** and they note '*the optimal duration of denosumab treatment for osteoporosis has not been established; re-evaluate the need for continued treatment periodically based on the expected benefits and potential risks of denosumab on an individual patient basis, particularly after 5 or more years of use*'.
- **Good patient information on Denosumab from the Royal Osteoporosis Society** - [click here](#).

All of which does make you wonder why you would initiate this treatment in the first place? Surely all people in whom you are considering treatment will be at increased risk of vertebral fracture? Once on treatment how and when do you stop, given optimal treatment duration has not been established? Do you reduce dosing frequency slowly or switch to an alternative treatment? These are all questions to which we do not have answers. The [NOGG guideline](#) from 2017 (updated 2019) noted the concern that rapid bone loss has been reported after cessation of Denosumab and that switching to a bisphosphonate when stopping Denosumab should be considered (in which case why not be on the bisphosphonate in the first place?). Clearly, a difficult area to negotiate, we will need expert advice and guidance for our rheumatology colleagues on where this drug now sits in the osteoporosis treatment armoury, and importantly for any of our patients who are already on these injections, **we need to make sure their doses are not delayed and get specialist advice before considering stopping or switching treatment**.

► ***Osteoporosis: evidence for Vitamin D and calcium in older people*** [DTB August 2020](#)

This was a really interesting article questioning the use of both calcium and vitamin D routinely to either reduce fracture risk or as a supplement in osteoporosis treatment. As this review points out, calcium and vitamin D are often lumped together when talking about treatment, but they are 2 distinct entities and should be looked at separately.

Does calcium or vitamin D improve BMD? The authors quote evidence that calcium supplementation increases bone mineral density (BMD) by about 1% in the first year of use, but there is no further increase beyond the first year. Vitamin D supplements generally do not affect BMD, although in those with significant deficiency (<30 mol/L) there does seem to be an increase of 1-2% per year in BMD, but those with vitamin D levels above this show no response.

Does calcium or vitamin D reduce fracture risk? The meta-analysis evidence on this is conflicting, but most of us will be aware of evidence that suggests that calcium, with or without vitamin D, reduces fracture risk in frail nursing home residents. Interestingly reviewing this in more detail shows that one study dominates this data - the residents in the Chapuy study had very severe Vitamin D deficiency (average ~14 mol/L) so the reduction in fractures is more likely treatment of underlying osteomalacia rather than of osteoporosis. If this study is removed from meta-analysis and studies with community-dwelling residents are used, there is no effect on fracture risk with calcium supplementation. Trials of vitamin D supplements are also negative unless subsets of those with Vitamin D <30 are used, consistent with the BMD evidence above.

What about calcium and vitamin D with osteoporosis treatment? Most of the pivotal trials of osteoporosis medication were carried out with co-administration of calcium and vitamin D, so it had been assumed for some time these drugs were a required part of therapy, but what evidence we have when these supplements are not added suggests the bisphosphonates and raloxifene are as effective without supplements. The exception to this is denosumab in the context of renal failure (eGFR <35 ml/min) which can cause a hypocalcaemic state so calcium supplements are recommended. This review also reminds us that the potent antiresorptive drugs zoledronate and denosumab can cause hypocalcaemia in the context of low vitamin D, so vitamin D deficiency needs correcting before these drugs are given.

What are the downsides of calcium and vitamin D? This is probably the crucial part because if calcium or vitamin D are safe why not just add them in case deficiency is present? Well, there are significant potential downsides, mainly with calcium. Calcium supplements cause GI side effects in >40% and double hospital admission risk for this cause. They increase the risk of renal stones significantly (by 17% in the WHI study) and there is also some evidence of increased MI risk, although the data is conflicting on this.

So where does that leave us? Well, the authors give a very useful summary of recommendations for the use of calcium or vitamin D supplements in older people:

Population	Calcium	Vitamin D
Healthy, community-dwelling	Not recommended	Only in those at high risk of deficiency (low sunlight exposure, seldom outdoors, veiling, dark skin, high latitude*)
Frail, community-dwelling	Not recommended	Recommended
Institutionalised	Not recommended	Recommended
Osteoporosis +/- fractures	► Not required as co-therapy for bisphosphonates ► Use with denosumab with renal impairment	► Important to rule out vitamin D deficiency before using zoledronate or denosumab ► Recommended in those at high risk of deficiency (see above) or living in institutions *e.g. north Scotland almost 50% older women have vitamin D <30 at end of winter vs 10-15% in south England

► **Long-term drug therapy for osteoporosis and 'drug holidays'** [Ann Intern Med. 2019;171\(1\)](#)

This systematic review attempted to assess the optimal treatment duration of bisphosphonates as well as the effects of discontinuation of therapy ('drug holidays'). What did it find?

- 48 trials reviewing men or post-menopausal women being investigated or treated for fracture prevention.
- The **vast majority were healthy postmenopausal women** (84-100%), mean age 72, with most having osteoporosis (as defined by bone mineral density - BMD) or vertebral fracture history, and some with osteopenia (as defined by BMD); long term drug therapy was defined as treatment >3 years.
- **Results:**
 - In women with **osteoporosis 4 years alendronate reduced clinic fractures** vs placebo (HR 0.64, CI 0.50-0.82) as well as radiographic vertebral fractures.
 - In women with **osteopenia or osteoporosis 6 years zoledronate reduced clinical fractures** vs placebo (HR 0.73, CI 0.60-0.90) including both non-vertebral fractures and clinical vertebral fractures.
 - The **observational data suggests an increase in both AFF and ONJ**, with an absolute increased risk of AFF of ~0.20%, and this risk of AFF did seem to be higher with longer duration of bisphosphonate use (>5 years).
 - There was very limited data on drug holidays but what information there was suggested a possible (but by no means conclusive) reduction in some vertebral fracture outcomes, but no reduction in non-vertebral fractures, in those on longer-term treatment beyond 3-5 years.
 - The authors estimate that for every 1000 women with osteoporosis treated with alendronate for 4 years, or zoledronate for 6 years, approx 50-70 will avoid a clinical fracture at the risk of 2 having some form of atypical femoral fracture.

This study rather neatly confirms the state of play with bisphosphonate treatment for reducing fracture risk - bisphosphonates do reduce fracture risk in the short term, but the data is extremely heterogeneous. On top of this, we have little or no convincing data to suggest any significant benefits of treatment beyond 3-5 years. In summary for a treatment we have had for many years now, there are still huge gaps in the evidence base and the authors of this conclude that *'To guide future decisions about osteoporosis treatment, considerable new research is warranted'*. **It is important to note that this study was reviewing bisphosphonates and did NOT include denosumab** - cessation of denosumab can cause rebound reductions in BMD and as reviewed in the MHRA above this is now associated with increased fracture risk, especially vertebral fractures.

► **Bisphosphonate holidays** [Drug and Therapeutics Bulletin 2021;59:35-36](#) [NB blog June 2021](#)

The idea behind bisphosphonate holidays is that these drugs have long half-lives, so the benefits of these drugs wear off slowly (over months-years), depending on the potency of the drug and how strongly it binds to bone e.g. risedronate and ibandronate wear off more quickly vs alendronate, with zoledronate the most durable. In contrast, the rate of AFF associated with bisphosphonate use goes up exponentially the longer you are on the drug, but drops quickly on stopping. Although rare, AFFs are nasty - surgically they are difficult to treat and have prolonged healing times. Hence a bisphosphonate holiday can quickly reduce AFF risk while maintaining the more durable benefits of the drugs on osteoporotic fracture.

This DTB review **suggests that after ~5 years of bisphosphonate use we can consider a 6-month break for risedronate and a 12-24 month break for alendronate**, with holidays being more beneficial for those at low/moderate osteoporotic fracture risk, especially women of Asian origin (whose AFF risks are higher with lower osteoporotic fracture risks). The review reminds us that this is usually a drug 'holiday', not a drug 'retirement', with those people at higher osteoporotic fracture risk likely to need longer-term treatment, albeit with intermittent breaks. However, it's worth a reminder that NICE sensibly tells us that in the context *of multimorbidity*, those taking a bisphosphonate for osteoporosis for at least 3 years, there is no consistent evidence of further benefit from continuing the bisphosphonate for another 3 years or harm from stopping the bisphosphonate after 3 years of treatment. NICE recommend we discuss stopping the bisphosphonate after 3 years considering patient choice, fracture risk & life expectancy. [NICE NG56, 2016](#)

KISS: Osteoporosis and the prevention of fragility fractures

NOGG 2019

Assessment of fracture risk

- Bone mineral density (BMD) testing alone should not be used for population screening due to low sensitivity (most fragility fractures occur in women who do not have osteoporosis)
- Fracture risk should be assessed in **postmenopausal women and men aged ≥ 50 with any of the following major risk factors** where assessment would influence management:
 - Low BMI ($\leq 19 \text{ kg/m}^2$)
 - Previous fragility fracture including vertebral fracture
 - Parental history of hip fracture
 - Current glucocorticoid treatment (any dose, by mouth ≥ 3 months)
 - Current smoking or Alcohol intake ≥ 3 units/day
 - Secondary causes of osteoporosis (including rheumatoid arthritis, type 1 diabetes, osteogenesis imperfecta, untreated hyperthyroidism, hypogonadism/premature menopause <45 years old, chronic malabsorption or malnutrition, chronic liver disease)
- NOGG then give a massive list of **other potential risk factors**, which are associated with low BMD or fractures or both, which **may indicate the need for risk assessment**, including thoracic kyphosis, height loss $>4\text{cm}$, type 2 diabetes, falls, a large variety of inflammatory/endocrine/haematological/muscle/respiratory/neurological diseases, and drugs (including antipsychotics, anticonvulsants, PPIs, aromatase inhibitors, androgen deprivation therapy)
- **Assess fracture risk** using the [FRAX tool](#) (those ≥ 70 with fragility fractures can be considered for treatment without the need for further assessment):
 - Those in the **low-risk** category can be reassured
 - Those in the **high-risk** category can be considered for treatment
 - Those in the **intermediate** category should have BMD assessment and their fracture risk reassessed in the FRAX calculator to guide treatment

Diagnosis/investigation of osteoporosis

- Osteoporosis is defined as a BMD T-score ≤ -2.5 , based on DEXA scanning
- The **preferred site for diagnosis is the femoral neck**
- DEXA scanning at the spine is not suitable for diagnosis of osteoporosis in older people, due to the high prevalence of degenerative changes which can cause an artefactual increase in BMD but can be used as an alternative if hip measurement not possible due to technical reasons or for younger postmenopausal women
- If neither hip nor spine measurements are possible, BMD at the distal radius can be considered
- Other techniques, including ultrasound, should NOT be used (often used in private screening)
- **Investigations** for those with osteoporosis:
 - FBC, ESR, CRP, U&E, bone profile, LFTs, thyroid function for all
 - Consider other investigations based on a clinical picture (e.g. XRay lumbar/thoracic spine to assess for vertebral fractures, Vitamin D, myeloma screen, hormone profiles)

KISS: Osteoporosis and the prevention of fragility fractures**NOGG 2019****Lifestyle & dietary measures**

- Stop smoking and reduce alcohol intake to ≤ 2 units/day
- Ensure adequate calcium and vitamin D intake: simple [dietary calcium calculator](#)
 - Calcium intake should be 700-1200 mg/day, **ideally through dietary intake**, but if below this give supplements
 - In postmenopausal women and older men at increased fracture risk *advise a daily dose of Vitamin D* colecalciferol 800IU/day
- Regular weight-bearing [exercise advice & osteoporosis](#)
- Careful assessment of falls risk, including recent falls history, and appropriate intervention

Pharmacological interventions in postmenopausal women and men

- Consider treatment if aged ≥ 70 with previous fragility fracture (no need for FRAX assessment in this group) or if above the FRAX treatment threshold
- **First-line treatment:** Alendronate or risedronate in the majority of cases
- **Second line** (if intolerant of oral bisphosphonates or contraindicated):
 - IV bisphosphonate (e.g. zoledronate yearly) or denosumab
- **Third line:**
 - Raloxifene (note really only evidence to support use for vertebral fractures)
 - HRT (generally restricted to younger postmenopausal women at high risk of fracture who also have menopausal symptoms)
- Teriparatide (very expensive!) restricted to those at very high risk of vertebral fractures
- Note **strontium** is NOT listed as a treatment option and has now [been withdrawn](#)

Glucocorticoid-induced osteoporosis

- Consider bone protection if on steroids for **women aged ≥ 70 , or with previous fragility fracture, or if taking ≥ 7.5 mg/day prednisolone** (or equivalent)
- Other individuals on glucocorticoids should have [FRAX assessment](#) (FRAX gives you 2 points on the graph for prednisolone doses 2.5-7.5mg/day or ≥ 7.5 mg/day)
- If on high doses (e.g. pred ≥ 15 mg/day) upward adjustment of fracture risk probably needed
- **Alendronate and risedronate** are first-line treatment options

Osteoporosis in men

- Treatments have been less extensively evaluated in men but similar options and thresholds are recommended as for postmenopausal women. Secondary causes of osteoporosis are more common in men so need 'thorough investigation'

[Click here for excellent patient information from Versus Arthritis](#) (previously Arthritis Research UK)

KISS: Osteoporosis and the prevention of fragility fractures [NOGG March 2017](#)

Risks and contraindications for treatment

- Oral and IV bisphosphonates and denosumab are **contraindicated in hypocalcaemia**:
 - Risk of hypocalcaemia with denosumab, needs checking before each dose, see BNF
- Oral bisphosphonates are contraindicated in those with abnormalities of the oesophagus that delay emptying e.g. achalasia or stricture
- All bisphosphonates have rare but serious adverse effects: (MHRA Dec [2014](#) & Dec [2015](#)):
 - **Oesophagitis** - PPIs not helpful as due to local irritant effect so switch treatment
 - **Atypical femoral fractures** - advise patients to report unexplained hip/groin pain
 - **AF** however, no difference in cardiac death or VTE has been noted
 - **Osteonecrosis of the jaw and external auditory canal (EAC)** - advise patients to have a dental check asap and to report any unexplained ear pain or discharge
- **Denosumab** has also been shown to increase the risk of osteonecrosis of the jaw and EAC

Treatment duration and reassessment

- Review treatment after 3-5 years (3 years for zoledronate, 5 years for other treatments)
- If ≥ 75 , previous hip or vertebral fracture, ongoing steroid use (prednisolone ≥ 7.5 mg daily) or fracture occurs on treatment, **continue treatment without reassessment**
- All others should have **reassessment with FRAX**:
 - If above NOGG threshold or hip T-score ≤ -2.5 continue treatment
 - If below NOGG threshold and hip T-score > -2.5 stop treatment & re-assess 1.5-3 yrs
- If **denosumab is stopped** we should consider switching to an alternative therapy (e.g. bisphosphonate) due to the risk of rebound BMD loss
- There is **no evidence base to guide treatment beyond 10 years**
- NICE advise that we consider stopping bisphosphonates after 3 years for patients with multimorbidity

NB Learning Points & Quality Improvement Ideas: Osteoporosis

- Recent data reviewed in the DTB has highlighted the increased risk of osteoporotic fractures for people with asthma on high-dose ICS, or receiving regular repeat short courses of oral steroids, with a cumulative (prednisolone equivalent) dose of > 1000 mg increasing fracture risk by $> 50\%$.
- A recent DTB article has highlighted that calcium supplements are not indicated in most people, even if on osteoporosis treatment, and increase the risk of GI upset and renal stones.
- There are increasing concerns about rebound bone loss and multiple vertebral fractures if Denosumab treatment is stopped or delayed, especially in people at high vertebral fracture risk. The [MHRA now advise Denosumab should not be stopped without specialist review](#).
- A recent DTB review suggests that after ~ 5 years of bisphosphonate use we can consider a 6-month break for risedronate and a 12-24 month break for alendronate, especially for those at lower fracture risk - this can reduce the risk of atypical femoral fractures while maintaining the benefits of bisphosphonates, which are more durable due to their long half-life.
- NICE advise considering stopping bisphosphonates after 3 years in patients with multimorbidity.
- ***Our role is to help patients make informed decisions.*** For a [great decision aid click here](#)
- **Quality Improvement Ideas:**
 - Review those with multimorbidity on bisphosphonates for > 3 years - is treatment still indicated?
 - Audit patients receiving long-term steroids, are they taking bone protection?
 - Audit those on calcium supplements - are they really needed? Use the table above as a guide.

GIANT CELL ARTERITIS

Giant Cell Arteritis (GCA) presents that classic conundrum we face with many conditions we see in primary care - a condition that generally presents in a non-specific way and which is rare, yet has a relatively small window of opportunity to make the diagnosis and start treatment before serious complications can occur. Hot off the press the British Society of Rheumatology ([BSR guideline on GCA was updated in March 2020](#)), and we have a very useful review article from the BMJ published in May 2019 ([BMJ 2019;365:l1964](#)) which reviews the diagnosis and management of this condition gives some very useful guidance for us in primary care and gives some encouragement for a novel steroid-sparing treatment for those with refractory or remitting disease in the form of tocilizumab. Recently we have had an interesting systematic review and meta-analysis reviewing the clinical features of GCA, and those which increase and decrease, the likelihood of GCA being present ([JAMA Intern Med. 2020;180\(10\)](#)).

► *Remind me what giant cell arteritis is and who gets it...*

GCA is an inflammatory disease affecting medium and large blood vessels, classically the extracranial branches of the external carotid. It is a rare disease affecting 2.2 per 10,000 patient-years in the UK, with an incidence of 7-29/100,000 in Europe. It affects older adults, being very rare in the < 50s, with a mean onset aged 70, the highest incidence being in the 70-79 year old group and is more common in women. Consequences can be serious if the diagnosis is delayed including visual loss, stroke and aortic aneurysm. If untreated up to 50% of patients with GCA can sustain unilateral visual loss within days to weeks of symptoms developing. Visual loss nearly always occurs before steroid initiation (or shortly after initiation) highlighting the importance of immediate treatment if GCA is suspected. Large vessel vasculitis is common and is thought to account for the systemic symptoms (fatigue, fever, weight loss); up to 83% of people have some degree of aortitis which can lead to vascular stenosis, aneurysm, dilation, dissection or rupture.

► *Assessing suspected GCA*

For a summary of the symptoms and signs associated with GCA see KISS below.

Differential diagnoses - quite wide, but includes:

- **Other headache types** (cluster, migraine, tension) - tend to be recurrent, visual aura resolves within minutes (c.f. persistent and progressive visual symptoms in GCA), nausea, photosensitivity.
- **Trigeminal neuralgia** - severe neuropathic shooting pain over the side of the face, lasts seconds/minutes.
- **Ophthalmic shingles** - diagnosis will be clear when rash presents but may have neuropathic pain before the rash is apparent.
- **TMJ dysfunction** - localised pain on chewing/speaking/yawning - more acute onset than jaw claudication.
- **Myeloma** - consider if systemic symptoms and ESR elevated out of proportion to CRP.

Investigations:

- **Bloods (need to be taken before first dose of steroid given)** - FBC, U&E, LFT, Ca, ALP, ESR, CRP.
 - ESR/CRP rarely normal - only 4% of cases have normal ESR and in >80% cases ESR >50.
 - Other typical findings include raised ALP and thrombocytosis.
- CXR.
- Urinalysis.

But of the features associated with GCA, which can best help identify this difficult diagnosis? This was the question asked by a recent systematic review and meta-analysis ([JAMA Intern Med. 2020;180\(10\)](#)). This review concludes that **no one single symptom, sign or blood test can adequately rule in, or rule out GCA**, however, there were features that were strongly associated with the presence or absence of GCA:

- **Features most strongly associated with GCA** (i.e. good positive predictive value):
 - Temporal artery features - tenderness, loss of pulsation or thickening.
 - Jaw claudication.
 - Limb claudication - this is interesting as it is rarely mentioned in the 'usual' features of GCA, but given it is a systemic disease, not surprising.
 - ESR >100, Platelets >400.
- **Features most strongly associated with absence of GCA** (i.e. good negative predictive value):
 - Age <70.
 - ESR <40.

So ultimately this useful systematic review supports the current guidance in terms of what symptoms, signs and blood tests help support, or refute, the diagnosis, with the **interesting addition that limb claudication is also strongly predictive of GCA**. However, it also confirms that to make the diagnosis we need to have a high index of clinical suspicion, that single symptoms/signs/blood tests cannot adequately rule in or out the diagnosis, and that confirmation of the diagnosis needs a prompt referral for further investigations.

► **Management**

- **All suspected cases need urgent secondary care review:**
 - The BSR recommend suspected GCA is seen as a 'medical emergency'.
 - If there is any visual involvement refer immediately (same day) for an ophthalmology review.
 - If no visual involvement refer for urgent rheumatology review - BSR recommend that ideally all patients should be seen within 24 hours, but within 3 days 'in all cases'.
 - If atypical symptoms discuss with a local specialist.
- **Secondary care investigations:**
 - Temporal artery biopsy used to be the gold standard, however increasing evidence for imaging (particularly the use of ultrasound of temporal arteries) suggests biopsy is not required in all cases.
 - The updated BSR guidance suggests that ultrasound is probably more sensitive, but less specific, for GCA and thus recommend **ultrasound can be used to rule out low-risk cases of GCA, and rule in high-risk cases** without the need for biopsies; those with intermediate risk of GCA (or those low risk in whom GCA cannot be ruled out with ultrasound) should go on to have a biopsy.
- **Steroids:**
 - **Should be started immediately in primary care in all suspected cases** - do not delay starting steroids while awaiting secondary care review. If GCA strongly suspected the first dose of steroids can be given whilst awaiting lab results as the risks of giving short-term steroids to patients who turn out to have an alternative diagnosis are 'acceptably low'.
 - Steroid regimes vary and evidence is inconsistent.
 - Those with ischaemic visual symptoms are recommended IV methylprednisolone before starting oral steroids unless there will be a delay to the IV treatment.
 - **BSR recommend 40-60mg prednisolone daily** for GCA without ischaemic visual manifestations:
 - Symptoms should reduce substantially within 1 week - if not consider alternative diagnoses (e.g. malignancy).
 - The following is a suggested tapering regime from BSR, but we should be guided by secondary care:
 - If good response continue a high dose of prednisolone 40-60mg until GCA symptoms and acute phase markers resolve.
 - Then taper by 10mg every 2 weeks until on 20mg. If still in remission:
 - Then taper by 2.5mg every 2-4 weeks till on 10mg. If still in remission:
 - Then taper by 1mg every 1-2 months.
 - **Note live vaccinations** are contraindicated in those on high-dose steroids (>20mg prednisolone daily for ≥ 2 weeks).

■ Other treatment:

- ?Aspirin - NO. The **updated BSR guidance advises against routine use of aspirin.**
- Consider PPI cover, especially for those on higher steroid doses.
- Consider osteoporosis risk and consider DEXA scanning and bone protection:
 - **NOGG** recommend those >70 years old and on prednisolone ≥ 7.5 mg daily should be commenced on bone protection (alendronate or risedronate).
 - Others should be considered for DEXA scanning to risk assess.
- **Exercise!** Although exercise programmes have not been formally evaluated in GCA the BSR highlight the importance of exercise to mitigate risks of deconditioning from the inflammatory response, and complications from steroid use (weight gain, diabetes risk, bone fragility).

■ Monitoring: See KISS below

► *What happens if the patient has a flare-up?*

- **Relapses are common**, with observational data suggesting >40% suffer at least 1 relapse, and **further management should be discussed with/directed by specialists.**
- A 'strong inflammatory response' (defined as ≥ 3 of fever/weight loss/ESR ≥ 85 /Hb <110) is associated with a high relapse rate and need for a prolonged treatment course.
- Consider relapse if new-onset ischaemic complications, PMR symptoms or recurrence of the GCA headache; inflammatory markers can be normal at this point.
- If uncomplicated GCA symptoms return (e.g. headache without ischaemic manifestations) BSR recommend going back to the dose of steroids that was previously effective.
- If new jaw or tongue claudication develops 40-60mg prednisolone is recommended.
- If new visual symptoms develop 60mg prednisolone or IV methylprednisolone recommended.
- BSR recommend considering methotrexate as a steroid-sparing agent if a relapse is experienced, or if jaw/tongue claudication or GCA-related aorta/proximal vessel inflammation return.
- NICE ([NICE TA158](#)) have recently agreed to use of **tocilizumab** (a humanised monoclonal antibody against the IL6 receptor) as a steroid-sparing agent and is **strongly recommended in the updated BSR guidance** for those at high risk of steroid toxicity or who relapse.

NB Learning Points & Quality Improvement Ideas: Giant Cell Arteritis (GCA)

- GCA classically presents with headache, but **be aware up to 25% of patients may present without a headache.**
- A recent systematic review has concluded that **temporal artery tenderness/thickening/lack of pulsation, as well as jaw and (interestingly) limb claudication, are all strongly predictive features of GCA.**
- Non-specific systemic symptoms are common - fatigue, weight loss, anorexia, sweats, and fevers.
- Patients with jaw claudication are at high risk of ischaemic complications.
- **ESR is normal in only 4% of cases** and in >80% of cases, ESR is >50.
- Any patients with visual involvement need an immediate same-day referral to ophthalmology.
- Patients without visual symptoms should be started immediately on oral prednisolone 40-60mg daily in primary care and referred urgently to rheumatology to be seen within 3 days.
- Aspirin is now NOT routinely recommended but consider PPI cover, especially on high steroid doses.
- **The majority of patients will need bone protection** with a bisphosphonate (alendronate or risedronate) including those >70 years old on prednisolone ≥ 7.5 mg daily and all others need a fracture risk assessment.
- **Quality Improvement Ideas:**
 - Review your patients with GCA still on treatment - are they having appropriate reviews including bloods (e.g. FBC, U&E, ESR, CRP, glucose)?
 - Review your patients with GCA on prednisolone - have they had their fracture risk assessed and are they on bone protection if appropriate?

KISS: Giant Cell Arteritis (GCA)

[BMJ 2019;365:l1964](#) and [BSR 2020](#)

► **Assessment**

- Typical **symptoms** include:
 - **Headache**: present in ~75% of cases; usually a dull severe headache in the temporal region, although can present with parietal or occipital pain; usually insidious in onset.
 - **Visual symptoms** - if present is typically partial or complete loss of vision or diplopia.
 - **Scalp tenderness** - is there tenderness when brushing hair?
 - **Jaw/tongue claudication**: present in 34% of cases (mostly in jaw); pain in the jaw/tongue on chewing - usually happens after minutes of chewing in the masseter muscles, due to ischaemia.
 - **Systemic symptoms** are common - fatigue, weight loss, anorexia, sweats, fevers.
 - **PMR symptoms** - may co-exist; proximal muscle pain and early morning stiffness.
- Be aware of the important differential of malignancy, especially **myeloma** - consider if systemic features and ESR elevated out of proportion to CRP.
- **Examination**:
 - **Temporal arteries** - palpate for tenderness, hard lumps and lack of pulsation.
 - **Vascular exam** - check for carotid bruits and differential BP in both arms.
 - **Eyes**: Check visual acuity and fields; fundoscopy - look for cotton wool spots, oedema, pale discs, and haemorrhages.
- **Investigations**:
 - **Bloods** - FBC, U&E, LFT, Ca, ESR, CRP; ESR/CRP rarely normal - **only 4% of cases have normal ESR** and in >80% cases ESR >50; other typical findings inc. ↑ ALP and thrombocytosis.
 - CXR and urine dip.

► **Management**

- **All suspected cases need urgent secondary care review** for further investigation (temporal artery biopsy used to be the gold standard but increasing evidence now supports temporal artery ultrasound to rule out low-risk cases, and rule in high-risk cases, thus reducing the need for biopsy):
 - If there is any **visual involvement refer immediately (same day) for an ophthalmology review**.
 - If no visual involvement refer for urgent rheumatology for review within 3 days.
- **Start steroids in all suspected cases in primary care immediately** and do not wait for an outpatient review, as irreversible visual loss can start within days of symptoms developing:
 - **Prednisolone 40-60mg daily**.
- Do NOT start aspirin, but consider PPI cover, especially when on higher steroid doses.

► **Monitoring/ongoing management**

- **Follow-up** recommended 'by a clinician with appropriate expertise' every 2-8 weeks for 6 months, every 3 months for second 6 months, then every 3-6 months ongoing.
- At each review check for worsening/new symptoms, and consider steroid-induced complications e.g. diabetes, osteoporosis; Check FBC, ESR +/- CRP as a minimum.
- Start **bone protection** (alendronate or risedronate) if >70 years old and on prednisolone ≥7.5mg daily; all others should be considered for DEXA scan.
- **Relapses are common** (>40%) and further management should be discussed with a specialist:
 - BSR recommends returning to previous effective steroid dose if headache returns, 40-60mg prednisolone if jaw/tongue claudication and 60mg prednisolone or IV Methylpred if eye symptoms.
 - Methotrexate recommended for relapses.
 - NICE have recently agreed use of tocilizumab for those with refractory or relapsing disease.
- **Patient information**: [PMR/GCA UK](#) and [versusarthritis](#) both have good information

POLYMYALGIA RHEUMATICA

Myalgic symptoms are such a common presentation in primary care in the older adult, and polymyalgia rheumatica (PMR), is going to be a differential in many cases. PMR itself is pretty common; it is the most common inflammatory rheumatic condition in older people, with an estimated incidence in the >40s of ~1%, and the vast majority of these cases will be diagnosed and managed solely in primary care. *However, there remain questions and challenges regarding both the diagnosis and management of PMR in General Practice, as discussed in a recent study in the BJGP* ([Br J Gen Pract 2018; 68 \(676\)](#)). The previous 'go-to' guidelines from the British Society of Rheumatology date back to 2009 (an update was due in 2018 but has not materialised yet), so it was timely that CKS have recently reviewed the evidence and updated their guideline ([CKS PMR January 2019](#)), although there is little in the way of new primary research on PMR.

Why is the diagnosis and management of PMR such a challenge? [Br J Gen Pract 2018; 68 \(676\)](#)

While the quantitative element to this study needs to be interpreted in light of the drawbacks of the study design (postal questionnaire of 5000 randomly selected GPs, of which ~25% responded), the qualitative review gives a very good insight into the difficulties we face in General Practice, in terms of both diagnosis and management of PMR. Although PMR is relatively common, the presentation can be highly variable, non-specific (particularly in the early stages), atypical and there is a wide differential diagnosis. But it is important to get the diagnosis and management right as the impact of the disease has significant implications for quality of life, and its subsequent treatment has implications on morbidity and mortality from corticosteroid adverse effects (most patients need oral corticosteroids for 1-2 years, of whom ~50% will develop adverse effects). Although this study suggests that we are generally good at assessing PMR in terms of clinical features and basic investigations with ESR +/- CRP in General Practice, the study highlighted two particular areas of uncertainty and gives insight into learning points for us in primary care:

- **Consider and investigate alternative diagnoses before starting corticosteroids:**
 - The study highlighted a low awareness of other diagnoses that can mimic PMR and further investigations that should be considered, particularly regarding malignancy
 - Only 45% of responders had tried to exclude malignancy as a cause for symptoms, with only 10-20% investigating for myeloma or lung cancer (both common differentials)
- **Starting dose of corticosteroids:**
 - Only ~55% of responders would initiate prednisolone at the current UK recommended dose of 15-20mg daily, with 40% of responders initiating prednisolone doses of >30mg
 - Although there is some discrepancy on recommended initial doses of corticosteroid for PMR, there is almost universal consensus that >30mg prednisolone is inappropriately high
 - Starting doses of prednisolone ≤10mg is associated with higher relapse rate and doses of ≥30mg can treat (or mask) other conditions (e.g. other inflammatory conditions or cancer), thus giving a 'false-positive' with the trial of corticosteroid

NB Learning Points: PMR

- Consider/rule out other diagnoses that can mimic PMR and investigate before starting steroids (see KISS below)
- Starting dose of prednisolone 15mg OD; consider increasing to 20mg if partial response but do not increase the dose any further - higher doses can mask/treat other conditions that mimic PMR (e.g. other inflammatory conditions, some cancers) giving a false positive response
- The response to 15-20mg prednisolone is usually rapid and dramatic with many symptoms resolving within 24-72 hours; if not re-consider the diagnosis

CISS: Polymyalgia Rheumatica (PMR)

[CKS January 2019](#)

Diagnosis/Assessment:

- Suspect PMR if **>50 years old** (although most >60) **with ≥ 2 weeks of core symptoms**:
 - Bilateral shoulder and/or pelvic girdle pain & stiffness AND
 - Stiffness lasting >45 minutes after waking or periods of rest
- Additional systemic symptoms may be present (in 40-50%) - low-grade fever, fatigue, weight loss, anorexia, depression
- **Exclude/consider other conditions that can 'mimic' or be associated with PMR**:
 - Giant cell arteritis - present in 15-20% of people with PMR
 - Infections - consider viral, osteomyelitis, TB, infective endocarditis
 - Cancer - myeloma, leukaemia, lymphoma, lung carcinoma
 - Endocrine disease - thyroid or parathyroid disease
 - Other inflammatory diseases - RA, SLE
 - Degenerative conditions - OA or bilateral impingement/frozen shoulder
 - Others - myositis/myalgia from statins, Vit D deficiency, fibromyalgia, CFS
- **Investigations to be done before starting corticosteroids**:
 - **In all cases** - FBC, ESR/CRP, U&E, LFT, Ca, CK, TSH, protein electrophoresis, rheumatoid factor, dip urine
 - Consider (depending on clinical features) - ANA, anti-CCP antibodies, CXR, urine BJP
- If PMR is the most likely diagnosis give a **trial of treatment**:
 - Prednisolone 15mg daily and review at 1 week - expect ≥ 70% improvement in symptoms within 1 week (typically many symptoms resolve within 24-72 hours)
 - If lesser response, consider increasing dose to 20mg, but if response still <70% refer
- **PMR diagnosis can be confirmed** if core symptoms present, other 'mimic' conditions excluded and there is a typical response to oral corticosteroids; ESR/CRP are typically raised but a diagnosis can be made if normal (but these patients need referring, see below)
- **Refer** if atypical features of PMR/concern about alternative diagnosis:
 - Red flags e.g. weight loss, night pain, neurological features
 - Younger than 60 years old or chronic onset of symptoms
 - Normal inflammatory markers or ESR >100 +/- very high CRP
 - <70% symptom response to 15-20mg prednisolone daily

Ongoing management:

- Flexible approach, individually tailored, **most will need 1-2 years of treatment**; refer if unable to reduce doses at reasonable intervals, or on steroids > 2 years
- Suggested **prednisolone regimen** (remember to give blue steroid card/warn of risks):
 - 15mg OD 3 weeks, then 12.5 mg OD 3 weeks, then 10mg for 4-6 weeks
 - Thereafter reduce by 1mg every 4-8 weeks
- **Bone protection** ([NOGG guidance 2017](#)):
 - Consider bone protection if >70 and on prednisolone ≥7.5mg daily
 - For all others assess risk with FRAX ([click here](#))
- **Monitoring** - Review 1 week after dose changes and at least every 3 months in 1st year, or urgently if they develop symptoms of GCA
- **Patient information** - [NHS Patient information sheet](#) and [Versus Arthritis](#)

JOINT AND SOFT TISSUE INJECTIONS

The advent of the COVID-19 pandemic has brought into sharp focus the use of corticosteroid intra-articular (IA) and soft tissue injections for musculoskeletal (MSK) conditions. Not that long ago they were the standard treatment for many (if not all) musculoskeletal conditions, whether inflammatory or not. However, over time we are getting **more evidence questioning the use of steroid injections in a number of non-inflammatory MSK conditions**. What evidence we have supporting their use is generally short-term only, and in some conditions (e.g. tennis elbow) the evidence suggests poorer outcomes in those who have had steroid injections.

Just before the pandemic hit the UK we had an excellent BMJ review in the uncertainties section ([BMJ 2020;368:l6923](#)), questioning the use of corticosteroid injections in knee OA ([read my blog here](#) for a review of this), and since then we have had a further RCT from the [Lancet April 2020](#) suggesting steroid injections for knee OA are inferior to physical therapies alone. Due to concerns that steroids may be linked with poorer outcomes in COVID-19 a guideline, supported by a wide range of professional bodies including the British Society for Rheumatology (BSR), the British Orthopaedic Association (BOA) and the RCGP, was [published and updated in June 2020](#). This **guideline recommends more judicious use of steroid injections in non-inflammatory MSK conditions** and is a good opportunity for us to consider more broadly a shift in attitude not to reach for the needle so freely and consider conservative measures, including physical therapies and exercise, more often. Their overriding principle is to *'only give a steroid injection if a patient has significant disease activity and/or intrusive and persisting symptoms, and there are no appropriate alternatives.'*

► [So what did this guideline say? BSR/BOA June 2020](#)

Background: The guideline came about due to **concerns about increased risks in people with COVID-19 who have had/are on steroids**. At the far end of the spectrum, those on oral prednisolone 20mg daily for ≥ 4 weeks are in the very high clinical risk (shielding) group, as are those on prednisolone ≥ 5 mg daily for ≥ 4 weeks if on concurrent immunosuppressive medication. Apart from the one exception of patients with severe COVID-19 on ventilators, where we now have evidence for improved outcomes with dexamethasone, most other evidence suggests potential harms with steroids. Evidence from previous pandemics suggested slower viral clearance with MERS-CoV and steroids, and potential harms with SARS and a recent study from China has also suggested delayed viral clearance of COVID-19 with corticosteroids.

Recommendations:

- **Inflammatory joints** - consider IA injection if active synovitis +/- effusion, and consider lowest clinically effective dose.
- **Non-inflammatory joints/soft tissue injections** (e.g. OA, shoulder pain, GTPS, CTS, trigger finger):
 - Simple analgesia, activity modification, splinting where appropriate and exercise as first-line and in most patients.
 - Only consider a steroid injection if a patient has high levels of pain/disability, has failed 1st line measures and ongoing symptoms will have a significant negative effect on health and wellbeing.
 - If injection considered choose minimum appropriate dose and where possible avoid simultaneous multiple-site injections.
 - Appropriate guidance on activity modification and exercise important after injections.
 - Exercise great caution in those already at higher baseline COVID-19 risk e.g. those > 70 , those with IHD, diabetes or chronic respiratory illness, and explain the risks carefully.

The guidance accepts that individual shared decision-making is important. For some patients, and in certain conditions, corticosteroid injections may give long-lasting effects or resolution of symptoms (e.g. trigger finger) and in others, they may avoid or delay the need for surgery, which may be of substantial benefit during a pandemic. However there is a risk, with a long incubation period of up to 14 days, that corticosteroid injections may be given to a 'well' patient who then goes on to develop COVID-19 with potential for a worse outcome with the virus, especially in the at-risk groups discussed above.

► [Intra-articular corticosteroid injections for knee OA](#) [BMJ 2020;368:l6923](#)

For a procedure that has been carried out for many years, it is somewhat surprising how little evidence there is to support this procedure, and as a very good recent review in the BMJ uncertainties section pointed out ([BMJ 2020;368:l6923](#)), there is now evidence of possible harms. The key evidence discussed in this review:

- The most comprehensive review on IA injections in knee OA comes from the [2015 Cochrane review](#). It concluded that there was **low-quality evidence of improvement in pain and function up to 6 weeks** but what benefits there were had all but disappeared by 3 months. This study was unable to comment on longer-term harms as the studies were so short-lived.
- A high-quality RCT in 2017 ([JAMA. 2017;317\(19\)](#)) found no clinical difference in pain with IA steroid injections vs saline every 3 months for 2 years, at any point in the 3 monthly follow-up period. Worryingly this study showed a small but statistically significant **deterioration in cartilage depth on imaging** in the steroid injection group - this is obviously only a surrogate marker and does not necessarily correlate to a deterioration in pain or function but is a concern nonetheless.
- A [meta-analysis in 2016](#) of individual patient data (which included both hip and knee OA) suggests that patients with more severe pain at baseline may derive more benefit from IA injections than those with less severe pain at baseline, but only in the short term (up to 4 weeks).
- We also now have [cohort study data from 2018](#) suggesting an association between IA injections and increased pain and stiffness in the longer term at ~2 years and [cohort study data 2019](#) suggesting **repeated IA injections may be linked to higher risk of disease progression and need for total knee replacement** - this data is not as robust as the RCT data, with increased risks of confounders, but again flags some concerns.

Most recently we have RCT evidence [NEJM April 2020](#) that **IA steroid injections are inferior in terms of pain and disability at 1 year compared to physical therapies**:

- This study randomised 156 patients to either IA injections (up to a maximum of 3) or physical therapies (face-to-face sessions with a therapist for passive movement exercises then a home exercise programme).
- Over the 12-month trial period the IA injection group had an average of 2.6 injections over the year and the physical therapies group had almost 12 sessions over the year.
- Interestingly this study suggests that physical therapies *were* as effective as IA injections at 4 and 8 weeks (previous data/assumption has been steroid injections give a quicker 'fix').
- By 6 months the physical therapies group were faring better and whereas any benefits from IA injections had pretty much finished by 8 weeks (as previous data suggests) the physical therapies group continued to improve throughout the year.

Conclusions: So where does this leave us regarding steroid injections for knee OA? Well, there is low-quality evidence that IA corticosteroid injections may give short-term improvement (<3 months) and there is some evidence that those with higher baseline pain scores may gain greater benefit. But there is **no evidence for longer-term benefits beyond 3 months**, with some evidence now suggesting harm in the longer term with possible increased pain and disability and an association with increased risk of progression to total knee replacement if used repeatedly. The **conclusion from the BMJ article is that moderate level low impact exercise should be the mainstay of treatment for people with mild/moderate OA knee** +/- referral to physio for a structured exercise programme. This is supported by the accompanying [editorial to the NEJM study](#) which concludes that this evidence suggests *'injections should not be used first, nor should they be used in place of a physical therapy program that includes exercise to manage symptoms of osteoarthritis of the knee'*. As the editorial points out there is not a suggestion that IA steroids should never be used, more that they should be used more selectively (e.g. significant flare of pain) rather than routinely. One of the caveats was that these patients had almost 12 sessions of physio - possibly not realistic in the NHS where one of the major barriers has been poor access to physiotherapy, but let's hope this may change with more widespread introduction of MSK practitioners in primary care.

NB Learning Points & Quality Improvement Ideas: Joint and soft tissue injections

- Guidance from a joint working group including the BSR, BOA and RCGP has highlighted concerns over steroid injections and poor outcomes with COVID-19.
- This guidance recommends that simple analgesia, activity modification, splinting where appropriate and exercise should be first-line treatments in most patients.
- Steroid injections should only be considered if these treatments have failed and there are ongoing significant levels of pain and disability and we need to be cautious using injections in those already at risk of poorer COVID-19 outcomes (those >70, those with IHD, diabetes etc.).
- There will however be situations where injections may give long-lasting benefits or even resolution of symptoms e.g. trigger finger, thus either delaying or removing the need for surgery.
- There is still significant uncertainty about the benefits of steroid injections in knee OA - current evidence suggests only short-term benefits <3 months with observational data suggesting an association with worse levels of pain/function in the longer term in those given steroid injections.
- We have RCT evidence that steroid injections are inferior to physical therapies for knee OA at 1 yr.
- **QI Ideas:**
 - Strongly promote physical therapies for MSK conditions and if physio access is poor, consider:
 - Excellent patient resources for home exercises from [VersusArthritis](#) and [BESS](#) (shoulder/elbow).

RHEUMATOID ARTHRITIS

Inflammatory arthritis has been a Hot Topic recently, with NICE producing updated guidance on spondyloarthritis in 2017, the British Society of Rheumatology giving updated guidance on DMARDs in 2017 and now *NICE updating its guideline on rheumatoid arthritis (RA) in July 2018* ([NICE NG100 July 2018](#)). It is not unreasonable to argue that there have been more advances in the management of inflammatory arthritis in the past 15 years than any other area of medicine. It used to be a disease with inevitable crippling consequences but is now a disease where patients can realistically hope for a very good quality of life.

Alongside the NICE guidance, we will also explore the other area of RA management of importance for us in primary care, namely the *cardiovascular risk associated with RA*. CVD risk accounts for 40% of all deaths in RA and a fascinating review article in the *BMJ April 2018* ([BMJ 2018;361:k1036](#)) explores the emerging evidence that reducing disease activity in RA may also reduce the excess cardiovascular disease burden.

But first, we thought it would be useful to revise *when to suspect RA*, which is not covered in much detail in the guideline:

KISS: When should I suspect Rheumatoid Arthritis?

[CKS 2019](#)

- RA typically causes *symmetrical joint synovitis of the small joints* of the hands and feet, although any synovial joint may be affected
- *Clinical features of synovitis* include:
 - **Pain** - usually worse at rest or during periods of inactivity
 - **Swelling** - around the joint (not bone swelling) with a 'boggy' feel on palpation
 - **Stiffness** - early morning stiffness usually lasts >1 hour
- Persistent synovitis and *poorer prognosis* are more likely when:
 - A greater number of joints are affected
 - The person cannot make a fist or flex fingers - helps differentiate from other causes of synovitis
 - There is a positive MCPJ squeeze test
 - The proximal IPJs and MCPJs are affected and there is a symmetry of joints affected
 - There is no other likely cause of inflammatory arthritis e.g. psoriasis
- In addition to joint synovitis, RA may present with *extra-articular features*:
 - Rheumatoid nodules: hard, firm swellings over extensor surfaces occur in a third of people with RA
 - Extra-articular features e.g. vasculitis features, or involvement of other body systems (e.g. eye, lungs, and heart)
 - Systemic symptoms of malaise, fatigue, fever, sweats, and weight loss
 - Family history of RA (although rates in monozygotic twins are only 15%)
- *The presentation of RA is variable*; most people have an insidious onset, but others can have a rapid or relapsing and remitting course

Management of rheumatoid arthritis in adults [NICE NG100 July 2018](#)

One of our key roles as GPs is to assess and refer suspected cases of RA urgently to a rheumatologist. Early treatment with disease-modifying anti-rheumatic drugs (DMARDs) improves functional outcomes, and as we will discuss later it may reduce cardiovascular disease.

► **When should we refer?**

- Refer from primary care any adult with suspected persistent synovitis of undetermined cause
- Refer urgently (even if ESR/CRP, rheumatoid factor and anti-CCP antibodies are normal) anyone with persistent synovitis if any of the following apply:
 - Small joints of hands or feet are affected
 - More than one joint is affected
 - There has been a delay of ≥ 3 months between symptom onset and seeking advice

► **But what about investigations?**

We should consider doing the following investigations in suspected RA, but doing these should not delay referral to a specialist (as above, even if these tests are normal we should still refer urgently):

- **Bloods:** Rheumatoid factor and anti-CCP antibodies
- **XRays** hands and feet

► **Treatment**

Although specialists will take over care at this point it is worth being aware of what the next steps will be for our patients if a diagnosis of RA is confirmed:

- 'Treat to target': NICE recommend that **RA should be treated with the aim of achieving remission**, or at least low disease activity if remission cannot be achieved
- **First-line drugs** will be conventional DMARDs (methotrexate, leflunomide or sulphasalazine) started as soon as possible, ideally within 3 months of symptom onset
 - Initially as monotherapy in escalating dose as tolerated, and then in combinations, if remission is not achieved
 - Hydroxychloroquine may be an alternative to conventional DMARDs for mild or palindromic arthritis (attacks of inflammatory arthritis similar to RA but between attacks joints return to normal)
- **Bridging corticosteroids** (i.e. short-term steroids to manage flares with recent-onset or established RA and to help achieve early remission) may be considered alongside DMARDs either oral, IM or intra-articular (NB we should NOT start corticosteroids before referral)
- If an inadequate response to conventional DMARDs **biologic agents** will be considered
- **Long-term glucocorticoid treatment** should only be considered as a last resort if all other options have been explored and been not effective or not tolerated

► **Symptom control**

NSAIDs should be considered for symptom control, but we need to be aware of the risks, and in particular with RA patients the increased CVD risk. If we use NSAIDs use at the lowest effective dose for the shortest time possible (i.e. step down as soon as possible) and co-prescribe a PPI. If patients also need to be on aspirin (e.g. with CVD) consider another analgesia before adding an NSAID.

► Non-pharmacological management

As with all musculoskeletal conditions, non-pharmacological management and exercise are absolutely crucial to improving outcomes and quality of life, and positive messages on the importance of exercise from us in primary care should not be underestimated. Adults with RA should have access to a multi-disciplinary team, specialist physiotherapy (hand exercise programmes should be considered), occupational therapy and podiatry.

► Diet and complementary therapies?

- Advise patients there is no strong evidence that diet alterations will improve RA symptoms
- Advise patients that although complementary therapies may provide short-term benefit, there is no evidence for long-term efficacy; complementary therapies should not replace conventional treatments

► Monitoring

Although much of the management of RA will be initiated in secondary care, we have an increasingly important role in primary care in the monitoring of risk factors associated with RA, as well as the monitoring of DMARDs and biologic agents as part of shared care guidelines.

- Monitoring of DMARDs and biologic drugs (see our separate chapters on DMARDs & Biologics in the Prescribing Hot Topics section, with KISS summaries)
- An annual review should include:
 - Review of CVD risk factors (use [QRISK3](#))
 - Osteoporosis risk (use either [FRAX](#) or [QFracture](#))
 - Review on quality of life and mental health

► Patient information

There are many options for rheumatoid arthritis patient information but the **Arthritis Research UK** website has an absolute wealth of information, everything for background information about the disease, specific information about the different drugs, as well as exercise, information [click here](#)

Increased cardiovascular risk in rheumatoid arthritis [BMJ 2018;361:k1036](#)

We have known for a long time that patients with RA have an increased risk of cardiovascular disease (CVD) and we are now accustomed to using [QRISK3](#) with RA as a risk factor. ***CVD accounts for the largest proportion of excess mortality in RA, about 40% of all deaths in RA.*** In two large meta-analyses, RA was associated with a 48% increased CVD event rate and a 50% increase in CVD mortality, compared to the general population.

► Role of the immune system in CVD

Currently, our focus has been on treating traditional risk factors of CVD in patients with RA to try to reduce CVD risk, but growing evidence suggests a central role of the immune system in CVD because the pro-inflammatory cytokines implicated in RA also enhance atherogenesis. Evidence is now emerging that higher disease activity in RA is associated with worse CVD outcomes, with disease-related inflammation contributing to plaque development and vulnerability to rupture.

► Could reduced disease activity in RA improve CVD outcomes?

This has ultimately led to the suggestion that reducing disease activity and inflammation in RA may improve CVD outcomes. ***There is increasing evidence that DMARDs do indeed improve CVD outcomes***, but it is also important to remember about drugs that increase CVD risks such as NSAIDs and corticosteroids.

CVD outcomes and drugs used in RA:

Drug	CVD outcomes in RA patients	CVD properties
NSAIDS	INCREASED CV risk Meta-analysis evidence: All CV events - RR 1.18	
Glucocorticoids	INCREASED CV risk Meta-analysis evidence: All CV events - RR 1.47 MI - RR 1.41 CVA - RR 1.57 CHF - RR 1.42	Pro-atherogenic side effect profile Risk dose and time-dependent
Methotrexate	REDUCES CV risk Meta-analysis evidence: All CV events - RR 0.72 MI - RR 0.81	Promotes cholesterol efflux Improves endothelial function Reduces oxidative stress
Combination conventional DMARDs	Uncertain Limited evidence of CV risk	Improves lipid concentration and function
TNF inhibitors (E.g. infliximab, etanercept, golimumab)	REDUCES CV risk Meta-analysis evidence: All CV events - RR 0.70 MI - RR 0.59 CVA - RR 0.57 Possible increased CHF risk	Improves surrogate markers of CVD Improves lipid indices Reduces oxidative stress
Non-TNF inhibitor DMARDs (e.g. tocilizumab, rituximab)	Uncertain Limited evidence of CV risk	Improved lipid parameters Improved oxidative stress and endothelial function

NB Learning Points and Quality Improvement Ideas: Rheumatoid Arthritis (RA)

- **Refer urgently all cases of suspected RA** based on clinical assessment; consider investigations but these should not delay referral and normal results do not rule out RA
- Patients with RA have a significantly increased CVD risk; **treat CV risk factors aggressively**
- **Quality Improvement Ideas** - review your patients with RA:
 - Have they had a CVD risk assessment in the last year with QRISK3?
 - Review NSAID use, especially those on long-term NSAIDs - can this be reduced?

NISS: Rheumatoid Arthritis (RA)

[NICE NG100 July 2018](#) [BMJ 2018;361:k1036](#) [Br J Gen Pract 2018; 68 \(669\)](#)

Clinical features:

- **Synovitis** - Pain, swelling, heat in affected joints; pain usually worse at rest or during periods of inactivity; swelling is around the joint (not bone swelling) giving a 'boggy' feel on palpation
- Usually **symmetrical synovitis of small joints of hands/feet**, but any joint may be affected
- **Stiffness in the morning** that usually lasts more than 60 minutes
- Positive metacarpophalangeal squeeze test

Referral:

- **Refer urgently** (even if ESR/CRP, rheumatoid factor and anti-CCP antibodies are normal) anyone with persistent synovitis if **any of the following apply**:
 - Small joints of hands or feet are affected
 - More than one joint is affected
 - Delay of ≥ 3 months between symptom onset and seeking medical advice
- **Investigations** (consider in primary care but **doing these should not delay referral**):
 - **Bloods** - to include ESR/CRP, Rheumatoid factor, anti-CCP antibodies
 - **XRays** - hands and feet

Treatment (all initiated and initially monitored in secondary care):

- The aim of treatment is to start DMARDs as soon as possible, ideally within 3 months of symptom onset, with an **aim of inducing disease remission**; this improves functional outcomes and there is increasing evidence it improves CVD outcomes as well
- Initial treatment with conventional DMARDs (methotrexate, leflunomide or sulphasalazine), initially as monotherapy and in combination if remission not achieved
- Biologic agents will be considered if inadequate response to conventional DMARDs
- Bridging corticosteroids may be considered for short-term control of flares

General management in primary care:

- **Symptom control**: consider NSAIDs with PPI cover but use at the lowest effective dose for the shortest possible time & try to avoid if patients need low-dose aspirin for CVD
- Do not forget the **importance of exercise**, particularly hand exercises and physio
- **Monitoring DMARDs** see the separate chapter on DMARDs & Biologics in Prescribing Hot Topics section
- **Annual review**: CVD risk factors (use [QRISK3](#)) - **treat aggressively** as RA patients have significantly increased CVD risk; osteoporosis risk (use either [FRAX](#) or [QFracture](#)); review on quality of life and mental health

Patient information - excellent resources from Versus Arthritis - [click here](#)

BACK PAIN & SCIATICA

NICE in late 2016 produced a controversial new guideline on low back pain and sciatica [NICE 2016, NG59](#). It was based on a thorough evaluation of the evidence...which reveals that most things don't work! Consistent with this we have evidence from a high-quality trial [NEJM2017;376:1111](#) that shows pregabalin is no better than placebo for acute and chronic sciatica, and gabapentinoids are no better than placebo in chronic low back pain [Plos August 2017](#). A recent BMJ therapeutics review ([BMJ 2020;369:m1315](#)) confirmed that there are **no treatment benefits of gabapentinoids over placebo for low back pain or sciatica**. Another area of controversy and uncertainty is the **use of muscle relaxants for acute low back pain**, and we have further evidence in the form of a recent systematic review and meta-analysis in the BMJ ([BMJ 2021;374:n1446](#)):

► ***Efficacy, acceptability and safety of muscle relaxants for low back pain*** [BMJ 2021;374:n1446](#)

This is one of those classic systematic reviews that tells you something...yet nothing at the same time. Spasm, particularly with acute low back pain, is a common problem and one we are faced with routinely in General Practice. **Do we prescribe a muscle relaxant?** The evidence base is very uncertain, which is reflected in the divergence of international guidelines with some advising muscle relaxants, some not, and some do not make any recommendation at all. As discussed below NICE have updated their guidance to recommend benzos are not used for sciatica, but leave the question open as to whether benzos (or other muscle relaxants) can be considered for acute low back pain and spasm, whereas the CKS guideline still recommends low dose diazepam for 5 days as an option for muscle spasm. Due to the known risks of benzos **some guidelines (e.g. US) recommend the use of non-benzodiazepine muscle relaxants for spasm. But are these any good?**

This systematic review and meta-analysis **reviewed the data on non-benzodiazepine muscle relaxants in low back pain**, and it nicely sums up the state of the evidence on this area - a bit of a mess. They identified 31 trials covering 6505 patients and the quality of the evidence was low or very low throughout. Summary:

- For acute low back pain (<2 weeks) there was very low certainty evidence that non-benzo antispasmodics reduced pain, but this improvement was considered to be clinically insignificant, and there was no reduction in disability.
- What little evidence there was for subacute and chronic back pain showed no improvements in pain or disability levels for non-benzo antispasmodics.
- There was low-quality evidence that non-benzo antispasmodics may increase the risks of adverse events (non-serious e.g. dizziness, nausea, drowsiness, headache).

So this is **poor quality evidence, very heterogenous and really cannot guide whether or not non-benzodiazepine muscle relaxants can help spasm with back pain**. Added to this is the fact that most of the trials identified looked at drugs that are not available in the UK (e.g. carisoprodol), with only 1 trial reviewed that looked at baclofen (the most likely one we would use in the UK if going down the non-benzo route), which showed no improvement in pain at 2 weeks. Yes, you got it, further good-quality trials are needed to address this question...if it is indeed a question we needed answering.

► ***Update to NICE guideline NG59, Dec 2020***

The NICE guideline can be welcomed as an honest and holistic guideline that encourages a biopsychosocial approach to management and is explicit about the limitations of medication and treatments. ***It focuses on identifying patients at high risk of chronicity early and an active, rehabilitative approach to management.*** All of this is to be lauded but the flip side is that as a consequence the guideline leaves us and our patients short of options, and with limited NHS access to many of the recommended interventions it will inevitably clash with routine clinical care and patient expectations. **NICE have recently ([NG59, December 2020](#)) updated their guideline** to reflect this new evidence, and to address the concerns over the harms of many commonly prescribed medications for back pain and sciatica (gabapentinoids, opiates and benzodiazepines). **The key changes are:**

- Do NOT offer gabapentinoids, other antiepileptics or benzodiazepines for sciatica.
- Do NOT offer opioids for chronic sciatica.
- Take a shared decision-making approach to consider withdrawal of opioids, gabapentinoids and benzodiazepines.
- Be aware of the risks of harm and limited evidence of the benefits of NSAIDs in sciatica.

So with the increasing evidence of harm, or lack of effectiveness, of many drugs and surgical treatments for low back pain, **NICE and most other international guidelines have shifted focus to non-pharmacological management strategies.** Although they are clearly safer than drug or surgical options, is there any evidence to support *those* treatments either? This was the subject of an excellent review article in the [Drugs and Therapeutics Bulletin July 2019](#).

► *Non-pharmacological treatment of low back pain in primary care* [DTB July 2019](#)

As with all DTB reviews, this was brutally honest and highlights the lack of good quality evidence for many of the non-pharmacological treatments for low back pain (LBP). Whilst it is a priority to move away from harmful and expensive treatments such as surgery, the DTB quite rightly point out that over-exaggerating the benefits of non-pharmacological treatments also risks patients getting sucked into healthcare dependency and unrealistic expectations. What evidence there is suggests small short-term benefits only, and the DTB argue that *'creating a semblance of an effective treatment (where there is none)...offers false hope and a barrier to truly engaging with a self-care approach to LBP.'*

First-line care - Good evidence for effectiveness:

- **Reassurance and self-care advice:**
 - It is telling that the **ONLY first-line non-pharmacological 'treatment' the DTB could recommend is reassurance and self-care.** There is strong evidence that patient education (as short as 5 minutes) can provide long-term reassurance, reduce pain-related distress and reduce healthcare utilisation for acute and sub-acute LBP. Some **useful points for us to discuss with our patients:**
 - Symptoms of LBP improve or fluctuate spontaneously over time, regardless of treatment.
 - Avoid bed rest, remain active and continue with usual activities including work, **despite pain.**
 - **Be honest with prognosis** - many episodes will recover substantially within 6 weeks, and 33% will recover at 3 months, but LBP often recurs and 65% still report some pain at 12 months.
 - Explain imaging is not needed - 'abnormal' scan findings (e.g. disc degeneration) are common in people with and without pain and imaging does not lead to better outcomes.
 - De-bunk any myths: There is a cure for LBP; Healthy means never experiencing pain; Pain is an accurate indicator of tissue damage; The spine is vulnerable and slow to heal.

Second-line care: treatments to consider - Safe interventions, but the true effectiveness is uncertain:

- **Exercise therapy:**
 - Encouraging exercise in any form is likely to be beneficial with good, if not conclusive evidence, for all forms of exercise - similar effects are seen across most options e.g. walking, yoga, and pilates.
 - Similar effects are seen with one-on-one or group exercise programmes.
 - There is some evidence exercise reduces the risk of further episodes of LBP.
- **Psychological therapies:**
 - There is some Cochrane evidence to support psychological therapies for chronic low back pain.
 - Further Cochrane evidence suggests multidisciplinary treatment can lead to modest improvements in pain, disability and work status.
- **Spinal manipulation and massage:**
 - The evidence for both spinal manipulation therapy (SMT) and massage is mixed.
 - A recent systematic review for SMT ([BMJ 2019;364:l689](#)) concluded it compared favourably to other recommended treatments for chronic LBP, but that it did not improve pain levels at 6 or 12 months compared to sham SMT (albeit this evidence was of poor quality).
 - A [Cochrane review from 2015](#) concluded there was no convincing evidence for massage therapy.
 - This evidence probably supports the NICE position that SMT and massage should only be offered in conjunction with an exercise programme.

■ Acupuncture:

- There is limited low-quality data showing the short-term benefits of acupuncture, but when it is tested in high-quality trials with a sham control it fails to show an effect.
- NICE currently do not recommend acupuncture for LBP.

■ Heat:

- NICE do not mention heat therapy, presumably due to limited evidence, but a [2006 Cochrane review](#) suggested short-term pain improvements with heat wraps.

Treatments to avoid

- The DTB concur with NICE recommendations that electrotherapies (e.g. TENS, ultrasound), traction and orthotics should not be offered.

What about risk stratification?

NICE recommend risk stratifying patients at the first encounter based on prognostic risk factors using the STarT Back tool, based on UK evidence that matching treatments to prognosis led to modest improvements in disability and was cost-effective. However, a recent US trial has failed to replicate this, so the DTB feel the effectiveness of this strategy cannot be confirmed yet.

Conclusions

The DTB review concludes that we need to recognise that no treatment, whether non-pharmacological or otherwise, represents a panacea for LBP, with current interventions giving small and short-lived improvements, while many do not work at all. Whilst we need to divert people away from expensive and/or harmful treatments, **we must not exaggerate the benefits of non-pharmacological treatments either as this will fuel healthcare dependency and may be a barrier to true self-management**. There are two themes that come out of this review in how we need to manage LBP, namely honesty and empowerment:

- **Honesty about the prognosis in LBP** - although many will be symptom-free at 3 months, the majority will have ongoing or recurrent episodes of pain at 1 year.
- Honesty about the lack of a cure for LBP and **not to over-exaggerate the benefits of any treatments**.
- **Empowering people in self-management** and supporting them in their efforts to live well with LBP with a recognition that advice and reassurance is doing 'something' and doesn't represent 'no treatment' - this will include validating their experiences, reassuring them and discussing the course of LBP.
- These principles are sound and evidence-based but as the DTB intimates this is all well and good in theory, but translating it into clinical practice, as we all know too well, can be very challenging...

KISS: Low Back Pain and Sciatica in Over 16s

NICE NG59 updated Dec 2020, CKS 2020 and DTB 2019

1 Exclude 'red flags' and secondary causes, such as:

- **Cancer** (see also NICE guidelines [metastatic spinal cord compression](#) as well as our cancer section and MSCC chapter):
 - Consider MSCC if history of cancer (esp. prostate, breast, lung, myeloma, but be aware in 10-20% of MSCC cases it is the presenting feature of cancer) and any of the following:
 - Severe or progressive lumbar pain, especially if nocturnal pain prevents sleep
 - Localised spinal tenderness
 - Pain aggravated by straining, coughing or sneezing
 - Radicular pain (e.g. down front or backs of thighs)
 - Neurological symptoms (e.g. limb weakness, difficulty walking, sensory loss, bladder or bowel dysfunction)
 - Consider myeloma if ≥ 60 with persistent back pain
 - Consider pancreatic cancer if ≥ 60 with weight loss
- **Infection** (e.g. discitis, vertebral osteomyelitis, or spinal epidural abscess). Red flags include:
 - Fever
 - Tuberculosis
 - Diabetes
 - History of IV drug use
 - Immunocompromised, including HIV and use of immunosuppressants
- **Trauma/spinal fracture** (see also NICE guideline on [spinal injury](#)) - red flags include:
 - Sudden onset severe central spinal pain which is relieved by lying down
 - Major trauma (e.g. road traffic accident or fall from a height $>3\text{m}$)
 - Minor trauma or strenuous lifting in those with or at risk of osteoporosis
 - Structural deformity of the spine (e.g. step from one vertebra to an adjacent)
 - Point tenderness over a vertebral body
- **Spondyloarthritis:** (see also NICE guideline [Spondyloarthritis](#) and Spondyloarthritis chapter)
 - Refer if < 45 years old **and 4 or more of:**
 - Low back pain <35 years old
 - Waking in the second half of the night
 - Buttock pain
 - Pain improves with movement or within 48hrs of NSAID
 - 1st degree relative with spondyloarthritis
 - Current or past arthritis or psoriasis or enthesitis
- **Cauda equina** - red flags include:
 - Severe or progressive bilateral neurological deficit of the legs e.g. major motor weakness with knee extension, ankle eversion, or foot dorsiflexion
 - Recent-onset urinary retention and/or urinary incontinence
 - Recent-onset faecal incontinence
 - Perianal or perineal sensory loss (saddle anaesthesia or paraesthesia)
 - Unexpected laxity of the anal sphincter

2 *Once 'red flags' are excluded:*

- **Make a positive clinical diagnosis** of low back pain and sciatica. Imaging is not routinely indicated in primary care unless it is likely to change management
- **Assess risk of chronicity** to guide management:
 - Consider using a risk stratification tool such as the [STarT Back tool](#) at first point of contact. This research-validated simple questionnaire assesses the risk of poor outcomes and persistent disabling symptoms. It incorporates physical & psychosocial risk factors ('yellow flags'), which then indicates risk (low, medium or high risk) to allow patients to be identified who will benefit most from more intensive intervention
 - Those who are likely to improve quickly and have a good outcome can be offered simpler and less intensive support e.g. reassurance, advice to keep active, and guidance on self-management. [Click here](#) for information/exercises on NHS choices and [here](#) for information and exercise advice from Versus Arthritis
 - Those with a higher risk of a poor outcome should be offered more complex and intensive support e.g. exercise programmes +/- manual therapy or psychological therapies

3 *Management options*

• **Non-pharmacological Interventions**

- Tailored self-management advice & information at all steps and encouragement to **continue with normal activities** - there is strong supportive evidence for reassurance and advice on self-management and the DTB argues this needs to be viewed as a 'treatment' in its own right
- **Promote & facilitate return to work** or normal activities
- NICE recommend considering a **group exercise programme** within the NHS e.g. biomechanical, aerobic, mind-body or combined approach, but there is evidence to support most forms of exercise
- Consider **manual therapies** (e.g. spinal manipulation, mobilisation or massage) and **psychological therapies** (e.g. CBT) only alongside exercise therapy
- Consider **multi-modal physical & psychological programmes** for those with persistent LBP when previous treatments not effective or significant psychological obstacles to recovery (i.e. those with 'yellow flags')
- There is some [limited evidence](#) **heat therapy** may reduce short-term pain and disability
- **Do not** offer orthotics, acupuncture or electrotherapies (e.g. TENS, ultrasound)

• **Pharmacological Interventions**

- Consider **oral NSAIDs** at lowest effective dose for the shortest duration of time considering adverse effects, but be aware **limited evidence for use in sciatica**; think about gastroprotection
- Consider **weak opioids** (e.g. codeine) +/- paracetamol for **acute** low back pain only if an NSAID is contraindicated, not tolerated or ineffective
- For **muscle spasm in low back pain**, consider a short course of a benzodiazepine, e.g. diazepam 2 mg tds for up to 5 days (NB this recommendation is from [CKS](#) and NOT the NICE guideline)
- For **sciatica**, NICE have updated their guideline based on strong evidence that we **should not** offer gabapentinoids, other antiepileptics or benzodiazepines.
- **Do not** offer paracetamol alone for managing low back pain (NB There is high-quality evidence [Cochrane 2016; CD012230](#) that it is no better than placebo in back pain)
- **Do not** offer opioids for managing **chronic** low back pain or sciatica
- **Do not** offer SSRIs, SNRIs, TCAs or anticonvulsants for managing low back pain - note evidence/reviews ([NEJM2017;376:1111](#) [Plos August 2017](#) [BMJ 2020;369:m1315](#)) confirm gabapentinoids are not effective for low back pain or sciatica

3 *Management options - Cont.*

■ Invasive Treatments for Low Back Pain & Sciatica

- Chronic low back pain
 - Consider referral for radiofrequency denervation
- Acute & severe sciatica
 - Consider referral for epidural injections of LA & steroid
 - Surgery. Consider referral for spinal decompression in those with sciatica where non-surgical treatments ineffective & radiological findings consistent with sciatic symptoms; do not allow a person's BMI, smoking status or psychological distress to influence the decision to refer
- Do **NOT** offer spinal (facet joint) injections, spinal fusion or disc replacement

NB Learning Points & Quality Improvement Ideas: Low Back Pain and Sciatica

- There is increasing evidence that **many drug and surgical interventions for low back pain and sciatica are either not effective or harmful**:
 - Paracetamol alone is ineffective for low back pain.
 - The gabapentinoids (and other antiepileptics) are ineffective for chronic low back pain and sciatica.
 - Opioids should only be considered (carefully) in acute low back pain and sciatica and not when chronic.
 - Spinal (facet joint) injections are not effective in low back pain or sciatica.
- A recent systematic review and meta-analysis has reviewed the **evidence for non-benzodiazepine muscle relaxants in back pain** - the evidence was of very poor quality and extremely heterogenous and showed (at best) a small improvement in pain levels for acute back pain at 2 weeks, but this was probably clinically not significant and disability levels were not improved.
- As a result, NICE have now shifted focus towards assessing patients with the STarT Back tool and matching non-pharmacological treatments depending on their risk of developing chronic back pain.
- However, a recent DTB review concludes that the evidence base for non-pharmacological treatments is also weak and warns that **over-exaggerating the benefits of non-pharmacological treatments runs the risk of causing ongoing healthcare dependency and a barrier to true self-management**.
- **Reassurance and advice on self-care** should be viewed as a first-line management strategy in all cases - it is the only well evidence-based non-pharmacological treatment and should be viewed as a 'treatment' in its own right.
- The DTB argues we need to be **honest about the prognosis** and lack of evidence for treatment with low back pain, **empower patients in self-management strategies** and support them to live well with their pain.

SPONDYLOARTHRITIS

Spondyloarthritis refers to a **group of inflammatory arthritic conditions** affecting both peripheral and axial joints with some shared features, including extra-articular manifestations. Also known as seronegative arthritis, this group of conditions includes ankylosing spondylitis (AS), psoriatic arthritis, reactive arthritis and enteropathic arthritis. The spondyloarthritis are distinct from rheumatoid arthritis, but due to the improvement in treatment options with the biological DMARDs, it is just as **important to recognise and manage this group early to improve outcomes**.

Picking up inflammatory back pain in primary care is difficult. We see large volumes of people with back pain, but inflammatory back pain is relatively rare. So it was timely that shortly after the NICE guideline on back pain (November 2016) was released, NICE also produced a very comprehensive guideline on spondyloarthritis in February 2017 - [Spondyloarthritis in over 16s: diagnosis and management](#), which gives us **good advice on whom we should refer**. The context of the guideline is clear - 'healthcare professionals in a non-specialist setting do not always recognise the signs and symptoms of spondyloarthritis, particularly spinal symptoms, which may be mistakenly attributed to other causes of low back pain.....This guideline seeks to raise awareness of the features of spondyloarthritis'. Currently, there is thought to be an [average of 8-10 years between symptom onset and diagnosis for ankylosing spondylitis](#). So our challenge is to pick these patients up earlier and get them on early biological treatment which can significantly reduce disability in the long term.

► Suspecting spondyloarthritis (SpA) [NICE Feb 2017 NG 65](#)

- Spondyloarthritis can have diverse symptoms and be difficult to identify. Features include:
 - **Musculoskeletal symptoms** e.g. inflammatory back pain (prolonged early morning stiffness that improves with moderate activity), enthesitis and dactylitis
 - **Extra-articular features** e.g. uveitis, psoriasis
- Be aware that **axial spondyloarthritis**:
 - **Can occur in people who are HLA-B27 negative** (this is important - about 10% of patients with ankylosing spondylitis will be HLA-B27 negative, and about 7-8% of Caucasian populations will be HLA-B27 positive with only ~5% of those having AS) this test should only be used in borderline cases
 - May be present despite no evidence of sacroiliitis on plain Xray

► Referral for suspected axial spondyloarthritis (AS)

Refer to rheumatology patients that have **back pain before the age of 45 years that has lasted longer than 3 months**, if 4 or more of the following are present (if exactly 3 present do HLA-B27 and refer if positive):

- Low back pain before age 35
- Waking during the second half of the night (sign of inflammatory pain)
- Buttock pain (can be a symptom of sacroiliitis)
- Improves with movement (again a sign of inflammatory pain)
- Improvement within 48 hours of taking NSAID
- First-degree relative with spondyloarthritis
- Current or past arthritis or psoriasis
- Current or past enthesitis (This is inflammation at a tendon/ligament or joint capsule insertion. It is characterised by pain and stiffness at tendon insertions and in SpA typically involves the big tendons e.g. Achilles and patellar tendons. Enthesopathy is a wider term covering both inflammatory enthesitis and mechanical or degenerative causes.

► Referral for suspected psoriatic arthritis and other peripheral SpA

- If new-onset inflammatory arthritis is suspected refer urgently (unless gout/pseudogout suspected)
- Refer people with dactylitis
- Refer people with **enthesitis without apparent mechanical cause** if any of the following:
 - It is persistent
 - It is in multiple sites
 - They have a first-degree relative with spondyloarthritis or psoriasis
 - Back pain without an apparent mechanical cause
 - Current or past uveitis
 - Current or past psoriasis
 - Gastrointestinal or genitourinary infection
 - Inflammatory bowel disease

► What about investigations?

The guideline has made clear that in primary care our focus is on making referrals based on clinical features. **Reliance on investigations may give false negatives.**

- **HLA-B27**
 - Only use if suspecting AS and exactly 3 of the presenting clinical features above are present - see above **Referral for suspected axial spondyloarthritis (AS)**
 - Do not rule out spondyloarthritis solely on the basis of a negative HLA-B27 result
- Do not rule out a diagnosis of spondyloarthritis if **CRP and ESR** are normal
- Axial spondyloarthritis may be present despite no evidence of sacroiliitis on **plain XRay**

► Management of spondyloarthritis

The new disease-modifying drugs (DMARDs) are the mainstay of treatment for spondyloarthritis, which our rheumatology colleagues will manage, but a couple of important points for us to remember in primary care:

- **NSAIDs** offer at the lowest effective dose and consider gastroprotection
- Non-pharmacological management: People with axial spondyloarthritis should be referred to **specialist physiotherapists** for a structured exercise programme and don't forget the importance of **aerobic exercise**

Long-term complications

- Be aware of the increased **skin cancer risk** with TNF-alpha inhibitors
- Review **cardiovascular risk factors**
- Consider **osteoporosis** assessment every 2 years (be aware DEXA measurements at the spine may be elevated, so hip measurements may be more reliable)
- Increased **fracture risk** with falls or trauma

KISS: Spondyloarthritis

NICE Feb 2017 NG 65

- **Spondyloarthritis refers to a group of inflammatory arthritic conditions** including ankylosing spondylitis (AS), psoriatic arthritis, reactive arthritis and enteropathic arthritis
- Picking up these conditions in primary care is difficult, but early identification and treatment with biological DMARDs improves outcomes significantly
- **Consider AS** if there is prolonged early morning back stiffness that improves with activity
- Refer to a rheumatologist for **suspected axial spondyloarthritis (AS)**:
 - If back pain presents **<45 years of age for >3 months** AND 4 or more of the following:
 - Low back pain <35 years old
 - Buttock pain (could be a symptom of sacroiliitis)
 - Waking during the second half of the night (sign of inflammatory pain)
 - Improves with movement OR within 48 hours of taking NSAID
 - There is a first-degree relative with spondyloarthritis
 - Current/past arthritis OR enthesitis OR psoriasis
 - If exactly 3 are present do HLA-B27 test and refer if positive
- Refer for **suspected psoriatic arthritis or other peripheral spondyloarthritis**:
 - If dactylitis is present
 - **Enthesitis** without apparent mechanical cause and ANY of the following are present:
 - Enthesitis is persistent OR in multiple sites
 - Back pain without an apparent mechanical cause
 - Current/past uveitis OR psoriasis
 - First-degree relative with spondyloarthritis or psoriasis
 - GI or GU infection OR inflammatory bowel disease
- **Normal investigations do NOT rule out spondyloarthritis** - ESR/CRP, HLA-B27 and plain Xrays can all be normal in spondyloarthritis; clinical features are more important
- **Management** includes NSAIDs, biological DMARDs and specialist physiotherapy
- **Complications and risks** include osteoporosis, CVD, fractures after falls/trauma

NB Learning Points: Spondyloarthritis (SpA)

- Consider AS in the under 45s with back pain > 3 months, especially if there is prolonged early morning stiffness that improves with moderate activity
- Enthesitis: Inflammation at tendon insertions, typically big tendons in SpA e.g. Achilles, patellar
- Consider SpA if unexplained enthesitis presents either recurrently or in multiple sites, or other risk factors e.g. uveitis, psoriasis, IBD
- Refer based on clinical features - investigations (ESR/CRP and plain XR) are often normal
- Refer early to rheumatology if you suspect spondyloarthritis
- **NB audit idea:** review the cases of low back pain in under 45s of >3 months duration

METASTATIC SPINAL CORD COMPRESSION

[BOA/BSR/RCGP Emergency MSK conditions 2020](#) [BMJ 2016](#) [NICE 2008](#)

Metastatic spinal cord compression (MSCC) is an oncological emergency. It occurs when a metastatic tumour grows into the epidural space and impinges on the dural sac and its contents. This leads to irreversible neurological damage (such as tetraplegia) depending on the level of compression. **Early diagnosis and treatment can prevent permanent neurological damage.** The NICE guideline on MSCC dates back to 2008, but is currently in the process of being reviewed for an update. Since then we have had a useful BMJ review article in 2016 and in 2020 a combined specialty guideline (including the BOA, BSR and RCGP).

KISS: Metastatic Spinal Cord Compression (MSCC): Assessment

[BOA/BSR/RCGP Emergency MSK conditions 2020](#) [BMJ 2016](#) [NICE 2008](#)

Background information:

- MSCC is caused by compression of the thecal sac by a tumour causing spinal cord pressure.
- **MSCC is a true spinal emergency.** Any delay can result in poorer functional outcomes.
- **Consider in ALL patients with a history of cancer and back pain or neurological signs/symptoms:**
 - In 75-80% of patients, there is a previous history of cancer (Breast, lung, prostate and renal most common), HOWEVER
 - In 20-25% of patients, spinal cord compression is the first presentation of malignancy.
- With an ageing population, likely to become more common.

Assessment:

- Always consider the diagnosis in patients with cancer.
- **The most common presenting feature is increasing back pain:**
 - Localised or generalised - can occur anywhere in the spine.
 - Can increase when lying down or straining.
 - Nocturnal pain.
 - Localised spinal tenderness.
- Characterised by progressive neurology and a sensory level
- Classically LMN signs at the level of lesion and UMN signs are below the level of the lesion.
- Heaviness or clumsiness of limbs may be an early sign.
- Progressive weakness occurs commonly.
- Sensory signs include dermatomal paraesthesia or anaesthesia.
- Autonomic involvement with bladder and bowel dysfunction is often a late and irreversible sign.
- It is a **time-dependent medical emergency** because ischaemia of the cord is nearly always involved and **rapid treatment can reverse ischaemia and improve neurological outcome.**
- A [combined speciality guideline was published in 2020](#) due to concerns about delays in diagnosis of some emergency MSK conditions (inc. MSCC), which highlighted a useful REDFLAGS mnemonic for symptoms suggestive of MSCC:
 - **R** = Referred back pain is multi-segmental or band-like.
 - **E** = Escalating pain which is poorly responsive to treatment (incl medication).
 - **D** = Different character or site to previous symptoms.
 - **F** = Funny feelings, odd sensations or heavy legs (multi-segmental).
 - **L** = Lying flat increases back pain.
 - **A** = Agonising pain-causing anguish and despair.
 - **G** = Gait disturbance, unsteadiness, especially on stairs (not just a limp).
 - **S** = Sleep grossly disturbed due to pain being worse at night.
 - A combination of red flag symptoms should increase suspicion of MSCC.

KISS: Metastatic Spinal Cord Compression (MSCC): Management

[BOA/BSR/RCGP Emergency MSK conditions 2020](#) [BMJ 2016](#) [NICE 2008](#)

Management:

- **All patients should be referred urgently for an MRI scan**, which should be done within 24 hours.
- Every cancer centre should have a metastatic spinal cord compression coordinator to refer cancer patients with suspected MSCC.
- **Urgent referral to MSCC co-ordinator within 24 hours if there is:**
 - Pain in thoracic or cervical spine.
 - Progressive lumbar spinal pain.
 - Severe unremitting spinal pain.
 - Spinal pain aggravated by straining, coughing or sneezing.
 - Localised spinal tenderness.
 - Nocturnal spinal pain preventing sleep.
- **IMMEDIATE referral to MSCC coordinator (or oncology) if:**
 - Neurological symptoms including root pain.
 - Any limb weakness or difficulty in walking.
 - Sensory loss or bladder or bowel dysfunction.
 - Neurological signs of spinal cord or cauda equina compression.

Treatment in secondary care:

- **All patients should be offered a loading dose of 16mg dexamethasone** (unless lymphoma)
 - Higher ambulation rates in patients who received dexamethasone prior to radiotherapy
 - 81% vs 63% at 3 months
- **Definitive management is surgical and/or radiotherapy.**
- NICE recommend referral for palliative DXT to patients with a poor prognosis.

Safety netting in patients with known cancer:

- **Metastatic spinal cord compression cards - “Stop Cancer Disability, Think Spine”**
- Patients with cancers which metastasise to bone are increasingly given an MSCC card warning them of symptoms and encouraging them to access urgent GP help if they have:
 - New back pain (in 95% for up to 2 months BEFORE signs of MSCC appear)
 - Increases in severity, may be worse on lying, straining or coughing.
 - Back pain which spreads to the front of the chest.
 - Progressive weakness, numbness or heaviness of limbs or difficulty in walking.
 - Bowel or bladder dysfunction.
 - Pins and needles in the limbs.

GREATER TROCHANTERIC PAIN SYNDROME

KISS: Greater Trochanteric Pain Syndrome

Based on [NICE-CKS 2016](#), [BJGP 2018](#), [BMJ 2018](#) & [J Orthop, Syst Review 2016](#)

■ Background

- Affects up to 25% of women aged over 40 and accounts for 20% of all hip presentations in primary care
- Formerly known as trochanteric bursitis, most cases are non-inflammatory and due to gluteal muscle tears or tendinopathy

■ Diagnosis

• History

- Typically lateral hip pain, worse with weight-bearing activity and at night which may radiate to the knee
- Medial hip pain or pain into the groin suggests hip pathology such as OA, avascular necrosis or femoroacetabular impingement syndrome

• Examination

- Point tenderness ('the jump test') over the greater trochanter
- Single leg stance (pain within 30 seconds of standing on one leg) & Trendelenburg test (pelvis dips on lifting the unaffected leg as opposed to staying horizontal or rising slightly)
- ROM usually preserved except at extremes

• Investigation

- Consider XRay to exclude OA or stress fracture
- If suspicious of stress fracture or avascular necrosis despite negative XRay refer for MRI

■ Management

- Reassurance and explanation: [Great PIL from patient.co.uk](#)
- Relative rest, ice packs & sleeping with a pillow between legs on the unaffected side
- Refer for physiotherapy
 - Physiotherapy-guided education and exercise classes are superior to wait-and-see at 8 weeks and also at 52 week
- Consider a corticosteroid injection (CSI)
 - Evidence shows that CSI gives good short-term analgesia for up to 3 months, maximising at 6 weeks and that 'blind' injection is as good as USS guided
 - At 52 weeks there is no difference in outcome between CSI and 'wait and see'
 - Consider CSI if very symptomatic to provide an 'analgesic window' to enable physiotherapy and rehabilitation
- Refer
 - Urgently if suspect cancer or red flags
 - If diagnosis in doubt or non-resolving despite physiotherapy and CSI

ELBOW PAIN

A guide to the assessment of elbow pain in primary care was published in the BJGP in 2015 ([BJGP 2015;65\(640\):610-612](#)). It was a useful paper to make us think beyond tennis/golfers elbow as the only causes of elbow pain and to consider other potential diagnoses. Having said that, tennis elbow is a condition we do see commonly, so it was welcome to see an *updated article on the assessment and management of tennis elbow recently in the BJGP* ([Br J Gen Pract 2018; 68 \(676\)](#)). As with the re-named 'greater trochanteric pain syndrome' (previously 'trochanteric bursitis'), evidence now shows tennis elbow to be tendinosis, rather than an inflammatory condition, and is increasingly being termed '*lateral elbow tendinosis*' (LET) rather than 'lateral epicondylitis'.

► *Assessment of elbow pain:* [BJGP 2015;65\(640\):610-612](#) [Br J Gen Pract 2018; 68 \(676\)](#)

■ **Consider red flags requiring urgent secondary care referral:**

- History of trauma, swelling, dislocation - fractures, dislocation, pulled elbow
- Unilateral swollen, red, tender joint +/- systemic symptoms - septic arthritis
- Rapidly increasing mass/swelling - malignancy

■ **Consider features suggesting causes other than tennis/golfers elbow:**

- Chronic progressive elbow pain with restriction of extension and end range of movements, painful clicking, catching, or locking of elbow, suggest osteoarthritis
- Bilateral elbow pain with stiffness and joint swelling, loss of full ROM, involvement of other joints, and systemic symptoms suggest inflammatory arthritis; 20–50% of patients with rheumatoid arthritis have involvement of elbows
- Assess for neurological symptoms - ulnar nerve radiculopathy is common in golfers elbow, osteoarthritis (50% of patients), and inflammatory arthritis
- Be aware referred pain from neck or shoulder can present as elbow pain

■ **Examination:**

- Consider assessing neck and shoulder for radiculopathy or referred pain
- Examine the joint for point tenderness, swelling/mass, and heat
- Tenderness to palpation just anterior to medial epicondyle suggests **golfers elbow**
- Ulnar nerve tenderness (Tinel's sign) can be elicited just posterior to medial epicondyle, indicating **ulnar neuropathy**
- Point tenderness anterior to lateral epicondyle is diagnostic of **lateral elbow tendinosis** (LET/tennis elbow); if LET suspected perform Cozen's test - resisted wrist extension with elbow extended and forearm pronated - test +ve if pain reproduced
- Tenderness localised to the area distal to radial head is diagnostic of posterior interosseous nerve compression/**radial tunnel syndrome**, an important differential to LET
- Check for active and passive ROM - loss of full extension suggests **elbow OA**; reduced pronation/supination together with tenderness over the radiocapitellar joint suggests OA at the radiocapitellar joint, another important differential to LET; ROM is not affected in LET

■ **Investigations:**

- Investigations are rarely required, and most diagnoses can be made clinically
- Do X Rays if bony pathology, arthritis or trauma present/suspected
- Blood tests (and X Rays) should be done if rheumatoid arthritis suspected (see our section on rheumatoid arthritis for more detail), but performing investigations should not delay urgent rheumatology referral

KISS: Tennis Elbow 'Lateral Elbow Tendinosis'[Br J Gen Pract 2018; 68 \(676\)](#)**Background:**

- Increasingly known as **lateral elbow tendinosis (LET)**, tennis elbow is a degenerative tendinosis (i.e. no inflammation present), caused by excessive loading in the context of individual risk factors (e.g. age, occupation, hobbies) with the most commonly implicated of the extensor tendons being the extensor carpi radialis brevis (ECRB)
- It is common (current estimated incidence 2.5 to 3.5 per 1000) and is most commonly seen in middle-aged patients with a peak incidence between 40 and 50 years
- Risk factors include those working in high-risk industries (involving loaded and repeated gripping and/or wrist extension activities), players of racquet sports, smoking and obesity

Diagnosis:

- LET classically presents with **insidious onset pain localised to the lateral epicondyle** of the elbow with radiation down the extensor aspect of the forearm
- On clinical examination, there is tenderness on palpation of the tendinous insertion at the lateral epicondyle
- There are a range of **clinical tests** to aid/confirm diagnosis; tests are +ve if pain reproduced:
 - **Cozen's Test** - resisted wrist extension with elbow extended/forearm pronated
 - **Maudsley's Test** - resisted extension of the middle finger
 - **Mill's Test** - passive wrist plantar flexion with elbow extension stretching the extensor tendons
- Important **differential diagnoses** to consider include:
 - **Osteoarthritis of the radiocapitellar joint** - pain on palpation of the joint line with reduced range of elbow pronation and supination; ROM is not affected in LET
 - Other differentials include **cervical radiculopathy** and **radial tunnel syndrome** (compression of the posterior interosseous nerve)
- X Rays are rarely required - only consider XR if bony pathology or OA suspected

Management:

- The majority of cases are self-limiting - **80% improve within 12 months**, but we need to warn patients the natural history of the condition is resolution over 6-12 months, not weeks
- **Exercise-based rehabilitation** is the cornerstone to successful management; this can be delivered through guided self-help ([click here](#)) and/or through referral to physiotherapy
- **Activity modification and relative offloading** are important; identify predisposing factors e.g. occupation, sport - reduce load (volume/intensity/frequency) & increase recovery time
- **Reduce pain** to facilitate rehab - topical NSAIDs; taping/strapping to offload the common extensor origin tendon; counterforce brace devices
- **Corticosteroid injections** should generally be avoided; although ~80% will gain short-term relief they are associated with high recurrence rates (>60%) and a significant delay in recovery; ~1:5 who have injections will have slower recovery than a 'wait-and-see' approach
- **Refer patients** with persistent pain and disability; secondary care options include autologous blood injections, platelet-rich plasma injections and extracorporeal shockwave therapy; surgical options are available but a recent RCT found that excision of the degenerative portion of the ECRB offered no additional benefit over and above placebo surgery

KNEE PAIN

KISS: Knee Pain

[CKS knee pain 2022](#), [BMJ 2015](#), [Sports Med 2012](#), [J Physio 2014](#)

Knee examination revision: [Click here](#) for a review. Aside from trauma and inflammatory conditions, *the most common causes in primary care are:*

Patello-femoral pain syndrome (PFPS)

- Gradual onset diffuse **anterior knee pain** behind or around the patella, often with crepitus.
 - Patients may describe clicking or grating behind the patella, or 'giving way'.
- Aggravated by ascending/descending stairs, or rising after squatting/sitting for prolonged periods.
- Very common: 11-17% of all knee presentations in general practice; over half of cases bilateral.
- Caused by **excessive loading of PFJ** - overuse (e.g. distance running), lower limb malalignment (e.g. femoral ante version or hyperpronation of foot), patellar tracking (e.g. imbalance of quads).
- Diagnosis is clinical; **anterior knee pain elicited in a squatting manoeuvre** is reasonably sensitive.
- Most cases resolve conservatively; there is evidence to support **physiotherapy**, **exercise therapy**, and a lesser extent, foot orthoses.
- Patellar taping alone is not recommended unless used in conjunction with physiotherapy-led exercise.
- [Click here](#) for good **patient information leaflet** including stretching exercises.

Osteoarthritis

- Activity-related persistent pain in >45s, morning stiffness <30 minutes, impaired function.
- **See the osteoarthritis chapter** for more detail on management.

Iliotibial Band Syndrome (ITBS) see KISS in the next chapter on ITBS.

Patellar Tendinopathy

- Overuse injury of the patella tendon leading to anterior knee pain also known as 'jumper's knee'.
- Typically insidious onset with tenderness over patellar tendon and lower pole of patella.
- **Refer to physiotherapy** for progressive strengthening & eccentric exercises.
- Advise that **return to activity can be slow** (weeks-months).

Osgood-Schlatter disease (OSD)

- Common cause of **anterior knee pain in adolescents** caused by traction of patella tendon on tibial tuberosity during forceful contractions leading to multiple small avulsion fractures.
- Pain is usually unilateral but can be bilateral in up to a third.
- Gradual onset pain & swelling below the knee; relieved by rest; exacerbated by running/jumping.
 - **Diagnosis is clinical** (routine knee x-ray not recommended):
 - Tenderness & swelling at the tibial tuberosity.
 - Pain provoked by knee extension against resistance.
 - **Examine hip** to make sure knee pain is not referred from hip (e.g. ?SUFE)
 - **Manage** with rest, ice and activity modification, but usually don't need to stop all sporting activity.
 - PIL with simple stretching exercises [here](#).
 - Reassure symptoms usually resolve with conservative measures. Physiotherapy if recovery is slow.
 - Resolution typically takes weeks-months but refer if symptoms persistent.

ILIOTIBIAL BAND SYNDROME

KISS: Iliotibial Band Syndrome

[BMJ 2019;364:l980](#), [Best Practice 2018](#)

- Most common cause of lateral knee pain in runners (7-14% affected), but also seen with other sports, esp. cycling, rowing, hiking, swimming
- Iliotibial band (ITB) is the tendinous continuation of the tensa fascia lata and gluteus maximus, attaching distally to the lateral patella and tibia
- Chronic inflammation is caused by excessive friction of the distal ITB on the lateral femoral epicondyle of the knee as it flexes to 30 degrees - known as the "impingement zone"
- **Presentation**
 - Pain and tightness typically occurs 2-3cm superior to the lateral joint line, at a predictable time or distance into exercise, and as it becomes more severe may be more continuous, radiate up or down the leg, and be felt with walking or sitting
 - Running on cambered roads or downhill is likely to exacerbate
 - Positive Ober's and/or Noble's tests (provoked pain) are indicative of ITBS:
 - For demonstrations click [here for Ober's](#) and [here for Noble's](#) tests
 - The BMJ suggests a combined test to increase differentiation of the cause:
 - Lay the patient on the unaffected side with the knee flexed to 90d
 - Apply direct pressure to lat fem condyle while flex/extending the knee
 - Presence of joint swelling, effusion, locking, giving way or +ve McMurray's indicate intra-articular pathology and make ITBS unlikely
 - Biomechanical abnormalities, such as over-pronation of the foot, limb length discrepancy or muscle imbalance may drive the condition
- **Management**
 - **Mild-moderate severity (the majority) can be managed with self-care advice**
 - **Activity modification** - reduction or avoidance of causative exercise, ideally avoid for 6/52; avoid training errors e.g. running downhill
 - **Stretching exercises** - key element to recovery and future prevention
 - **ITB band stretches** 5 repetitions, 3x/day until pain resolves
 - Consider **hip-strengthening exercises**, e.g. using a resistance band
 - BMJ does not discuss using **foam rollers**, often associated with ITBS treatments, while Best Practice suggests there is a role for their use alongside stretches
 - **Correct biomechanical abnormalities**, e.g. have a formal fit for trainers
 - Simple analgesia and ice for pain after exercise
 - **Gradual return to usual activity** after ~6-8/52
 - High-intensity sessions can improve fitness while limiting causative longer runs
 - 44% can return to sport by 8 weeks; 92% by 6 months
 - **Severe pain or refractory ITBS**
 - Referral to physio is advised
 - Steroid injections may be considered to facilitate rehab exercises
 - Rarely surgery is used in very persistent, severe cases despite all the above
 - Imaging is only usually recommended after 6/12 of unsuccessful physio
 - The BMJ have produced an excellent [patient information leaflet](#) with everything a patient needs to know about ITBS including appropriate stretching exercises

CARPAL TUNNEL SYNDROME (CTS)

KISS: Carpal Tunnel Syndrome

[BJGP2014;64;262](#). [CKS 2022](#)

Carpal Tunnel Syndrome is an entrapment neuropathy of the median nerve; the majority of cases are idiopathic but some secondary causes do need to be considered.

Assessment & Diagnosis:

- **Ask A)** Do you have numbness or tingling in your wrist, hand or fingers?
- **If yes, ask the following subsequent questions B):**
 - Do your symptoms spare your little finger?
 - Are the symptoms worse at night?
 - Do the symptoms wake you up at night?
 - Have you noticed your hand is weak e.g. dropping things?
 - Do you find shaking your hand, holding it or running it under warm water improves your symptoms?
 - Are the symptoms made worse by activities such as driving, holding a telephone, using vibrating tools or typing?
 - Have splints or injections helped with the pain in the past?
- **Examine:**
 - Wasting of thenar eminence.
 - Sensory loss in the distribution of median nerve.
 - Weakness of thumb abduction and opposition.
 - Provocation tests looking for pain and paraesthesiae or pain in median nerve distribution:
 - Phalen's test: +ve if flexing wrist for 60 secs causes symptoms.
 - Tinnel's test: +ve if tapping over the median nerve at solar surface of wrist causes symptoms.
- **Investigations?**
 - Nerve conduction studies rarely needed - refer if diagnostic doubt.
 - Consider secondary causes - e.g. hypothyroidism, diabetes.
- **The BJGP article (based on Primary Care Rheumatology Society guidance) suggests:**
 - If ≥ 3 of the subsequent B) questions positive, CTS diagnosed.
 - If ≥ 2 positive, CTS diagnosed if Phalen's test is also positive.
 - If <2 positive, consider an **alternative diagnosis** (e.g. cervical radiculopathy esp. C6/7, peripheral neuropathy, wrist arthrosis, tendonitis/tenosynovitis).

Management:

- Reassurance that 1/3 spontaneously resolve within 10-15 months.
- Consider work-related/exacerbating factors - consider adaptations if so.
- Do not prescribe NSAIDs or diuretics
- There is evidence to support **night splinting** and **steroid injections** in mild to moderate cases (absence of weakness or wasting of thenar musculature):
 - Splints can be purchased direct from pharmacies.
 - Steroid injections can be carried out in primary care if expertise is available.
- Refer for consideration of decompression surgery if symptoms are severe or constant, progressive motor or sensory deficit evident, or no improvement after 3 months of conservative management.
- [Click here](#) for patient information from Versus Arthritis.

ENDOMETRIOSIS

Not a new condition but our understanding of endometriosis continues to grow. This has made it a hot topic in the journals in the recent past with articles and research in the [BMJ](#), [Lancet](#) and [BJGP](#). After pressure from [the UK government NICE is starting a review](#) of its own guidance, while [European guidelines](#) were updated in 2022. So what's changing?

► *Defining endometriosis*

The classic definition is a condition defined by the presence of endometrial-like tissue outside the uterus. However, as a [2021 Lancet clinical review](#) explains this is a vast over-simplification. A heterogeneous disease, endometriosis is now considered a systemic condition and is associated with chronic inflammation, changes to metabolism in the liver and adipose tissue, and gene expression in the brain resulting in pain sensitisation and mood disorder. The mechanism for developing endometriosis remains unclear.

► *Diagnosing endometriosis*

There are clear issues with diagnosing endometriosis. 5-10% of females are affected by endometriosis during reproductive years. Diagnostic delay is reported by the BMJ ([2022;379:e068950](#)) as 8 years in the UK, and may be worse among adolescents due to the misconception that symptoms take time to develop after menarche. A [review in Women's Health journal](#) of this younger group found that endometriosis is common in this age group and present in 70% with dysmenorrhoea. At the other end of the spectrum, endometriosis-related pain may increase in the perimenopause and continue after menopause. It is also present in 30-50% of women requiring assisted reproductive treatment.

Diagnosis has always been complicated given the often non-specific features and imperfect diagnostic tools available. Traditionally, laparoscopy has been considered the gold standard investigation for diagnosis. However, new guidance from the European Society of Human Reproduction and Embryology challenges this notion. It states:

- *Laparoscopy is no longer the diagnostic gold standard and it is now only recommended in patients with negative imaging results and/or where empirical treatment was unsuccessful or inappropriate.*

Transvaginal ultrasound is considered the first line of investigation. It is useful in detecting endometriomas and excluding alternative pathology, however, it may miss superficial and deep endometriosis so a negative result cannot be relied on. More advanced US devices and techniques are improving sensitivity but are not in widespread use currently. MRI may also detect endometriosis lesions but is not considered superior to USS.

What about biomarkers? CA125 is not recommended by ANY guidelines. While levels may be elevated it lacks sensitivity or specificity and so does not help with our diagnostic process.

Does trialling treatment support the diagnosis? This was an idea raised by a number of GPs in qualitative research published in the BJGP ([2022;71\(710\):e668](#)). The title "**navigating possible endometriosis**" sums up the challenges, highlighting that delays in diagnosis do not simply represent lack of awareness - endometriosis is one of several possibilities that must be considered, sinister causes must be excluded, the process takes time, continuity is important, sharing decisions about investigations and referral keeping the woman's priorities central. The study reports **many GPs reported using a trial of treatment as part of the diagnostic process. No guidelines recommend this approach** - a positive response does not necessarily confirm endometriosis - but **the concept of a "working diagnosis of probable endometriosis" based on clinical features and initial tests (i.e. TVUSS) supports the notion of trialling treatment for pain without the need diagnostic laparoscopy.**

► *Managing Endometriosis*

A BMJ clinical review ([2022;379:e070750](#)) states:

- *The growing recognition that endometriosis-associated pain has a mixed pain phenotype (or occupies different points on a continuum) supports a personalized, multimodal, interdisciplinary treatment approach, which might include surgical ablation/excision of lesions, analgesics, hormonal treatments, non-hormonal treatments including neuromodulators, and non-drug therapies (or a combination of the above).*

What then is the evidence for different treatments?

- **NSAIDs/paracetamol**
 - Most women will try simple analgesia for pain management, although data to support this is limited. There is 1 small RCT (N=20) examining naproxen, which was found to be more effective than placebo for pain.
- **Hormone therapy**
 - Includes combined hormonal and progesterone-only methods, GnRH (gonadotrophin-releasing hormone) agonists & antagonists, and aromatase inhibitors are effective with similar levels of benefit.
 - CHC may be more effective when used continuously.
 - GnRH ag/antagonists are second line - they may be effective in women where standard hormone therapy has failed but have a higher side effect burden.
 - Aromatase inhibitors are reserved for refractory symptoms.
 - Hormone therapy is only effective when the patient is on it. Symptoms return on cessation.
- **Neuromodulator drugs**
 - Lack of a strong evidence base. TCAs, SSRIs and anticonvulsants may be recommended by specialists. Of note, gabapentin was recently shown to be no better than placebo for chronic pelvic pain without endometriosis.
- **Surgery**
 - There are a range of options depending on the type and site of endometriosis. Note recurrence rates for pain are around 21% at 2 years and 40-50% at 5 years.
 - Many women will be prescribed hormone therapy immediately after surgery in an attempt to maintain the benefits. This clearly would be not appropriate if the surgery is for fertility issues and ovarian suppression is not beneficial for later fertility. Likewise, evidence currently does not support surgery for fertility issues, only pain.
- **Non-pharmacological therapy**
 - Physiotherapy and psychology lack evidence to support their use but anecdotal reports suggest the former may be of use and the latter is effective for a range of chronic pain disorders so should be considered if felt appropriate.

Referral should be considered at all stages of the diagnostic and treatment pathway, depending on clinician concern, patient expectations, response to treatment and need for diagnostics.

NB Learning Points & Quality Improvement Ideas

- Endometriosis is a common condition with varied presentations in individuals, requiring a patient-centred approach which may involve different treatment options throughout life.
- Symptoms can start with menarche and age does not exclude endometriosis.
- Diagnostic laparoscopy is not routinely required and clinicians can make a "working diagnosis" of endometriosis with a trial treatment to manage pain, followed by referral if symptoms control is inadequate.
- Hormonal contraception is effective for many women - CHC can be used continuously to improve symptom control.
- Patient information can be found in the [NHS](#) and also [Endometriosis UK](#).

KISS: Endometriosis

[NICE NG 73 2017](#) [ESHRE guideline 2022](#) [BMJ 2022;379:e070750](#)

Definition

- Endometriosis is a condition defined by the presence of endometrial-like tissue outside the uterus
- It is now also considered a systemic disease with chronic inflammation
- Endometriosis can occur at any age from the point of menarche

Diagnosis

- **Diagnosis can be challenging - *it may be clinical or in conjunction with investigations***
- Consider if cyclical & non-cyclical features:
 - Dysmenorrhea, deep dyspareunia, dysuria, dyschezia, painful rectal bleeding or haematuria
 - Chronic pelvic pain
 - Shoulder tip pain, catamenial pneumothorax, cyclical cough/haemoptysis/ chest pain
 - Cyclical scar swelling and pain, fatigue, and infertility
- **Risk factors** include family history, early menarche & short menstrual cycle
- **Consider examination** - pelvic exam may not be informative for endometriosis but allows exclusion of other causes & may reveal visible vaginal lesions, tender nodularity, masses & reduced movement
- **Use imaging as part of diagnostic work up - 1st line: pelvic TVUS scan, MRI is alternative**
 - Note imaging cannot exclude superficial disease
- Do not use biomarkers such as CA125
- **Laparoscopy** - different guidelines take different approaches
 - No longer considered 'gold standard' as clinical assessment, imaging & trial of treatment may be sufficient. If the patient has a strong desire for referral then this should continue to be an option.
 - **NICE: consider laparoscopy** to diagnose endometriosis in women with suspected endometriosis, even if the ultrasound was normal
 - **ESHRE: if negative imaging results or where empirical Rx was unsuccessful/inappropriate, consider offering laparoscopy** for the diagnosis & treatment of suspected endometriosis
 - **BMJ: if "working diagnosis" of probable endometriosis & pain symptoms only, either offer short trial of analgesics or hormone therapy or refer for diagnostic laparoscopy**

Primary Care Management - Pain

- Think holistically, consider psychological support where necessary, discuss wishes & fertility priorities
- A multi-modal medical approach is likely to be required
- **Analgesia** - trial simple analgesia, paracetamol and/or NSAIDs, for up to 3 months
- **Hormone therapy** - combined or progesterone only (including IUS & implant) methods are options
 - GnRH agonists, GnRH antagonists, aromatase inhibitors are 2ndry care options for ongoing Sx
- **No evidence to support non-pharmacological treatments**, e.g. acupuncture, herbal meds
- **For ongoing symptoms, including while on treatment, or pelvic signs consider referral**
 - Surgical treatment (often at the time of diagnostic laparoscopy) may help with pain

Fertility

- Primary care therapies, e.g. hormones, are ineffective at improving or preventing fertility issues
- Operative laparoscopy may be appropriate & improve conception rates for some
- Medically assisted reproduction is an option

MENOPAUSE

Practices up and down the UK are having a surge in women consulting about menopause and HRT. Following Davina McCalls' Channel 4 programme: Sex, Myth and the Menopause many women may have identified troubling symptoms they had previously not associated with the menopause and there has been a resurgence in interest in medical treatment with HRT.

Perhaps then it is timely that the *British Menopause Society* in partnership with the Royal College of Obstetricians and Gynaecologists and the Faculty of Sexual and Reproductive Health amongst other professional organisations has recently published it's "**Menopause Practice Standards**".

These are important because, as the media has highlighted to the medical community, women with troubling menopausal symptoms have often been left wanting when it comes to advice and treatment. One might argue this is not surprising after safety concerns were raised by data from the Women's Health Initiative study, but these findings have been questioned.

1 **Menopause Practice Standards** [BMS July 2022](#)

The aim of these standards is to provide evidence-based recommendations on best menopause practice to support women. These standards are largely in keeping with the current NICE guidance on managing menopause. Perhaps the key message for clinicians comes from the very first standard:

- **Standard 1:** In women aged 45 years and over presenting with menopausal symptoms, the diagnosis of perimenopause or menopause should be considered based on their symptoms alone, without confirmatory blood tests unless uncertainty about the diagnosis

But the important point here is for **clinicians to understand the wide range of symptoms that may be driven by the menopause, or indeed in the perimenopause before periods cease**, which stretch well beyond simply vasomotor symptoms:

- **Cognitive symptoms and mood disorders**
 - Including low mood, labile mood, anxiety, irritability, loss of confidence, low self-esteem, difficulties with short-term concentration and memory ('brain fog'), and difficulties in multitasking
 - Guidance recommends menopause & perimenopause should be considered where mood disorders are experienced de-novo during the menopause transition in the absence of pre-existing depression/anxiety
- **Sleep disturbance, fatigue, tiredness, low energy levels, loss of sexual desire and libido, joint & muscle pains, headaches and genito-urinary symptoms** (which may present years later).

While all the standards are important, other key recommendations include:

- **Standard 2:** Women presenting with menopausal symptoms should be made aware of resources available for guidance and should be encouraged to seek help for managing their menopausal symptoms
 - See the Learning Point box for links to recommended menopause resources.
- **Standard 3:** Women aged 45 years and over who seek help for managing their menopausal symptoms should be offered treatment after information and support to make an informed decision about their management options
 - Encouraging a holistic approach including lifestyle measures, but noting that HRT is the recommended treatment for troubling menopause symptoms and should be offered if the patient wishes
- **Standard 4:** Women with genitourinary symptoms of the menopause should be offered vaginal oestrogen treatment and this can be continued long-term as required to relieve symptoms
 - 50% of postmenopausal women will develop GU Sx, low-dose topical oestrogen is safe & effective
- **Standard 5:** Women having treatment for menopausal symptoms should ideally have a review 3 months after starting treatment and should continue to be reviewed at least annually after that

- **Standard 6:** Duration of treatment should be individualised. No arbitrary limits should be placed on the dose of HRT, duration of usage or age of women taking treatment
 - Mean duration for menopause Sx is 7y but 20% will experience Sx for up to 15y
 - **Women continuing HRT over the age of 60 should have estradiol transdermally**

2 Raising Awareness of Menopause [BMJ 2022;377:e069369](#), [BMJ 2022;377:o1425](#)

With this current drive to increase awareness of menopause-related issues in the media, it is perhaps unsurprisingly that the BMJ sounded a note of caution on the [medicalisation of menopause](#). For several years it has run its Too Much Medicine campaign raising awareness of overdiagnosis and overtreatment.

It published a provocative BMJ Analysis "Normalising Menopause" article ([BMJ 2022;377:e069369](#)), written before but published after the huge surge in public interest and definitely running against the tide, arguing that medicalisation of menopause, which emphasises the negative aspects, may actually drive worsening of symptoms in the female population. It suggests "*changing the narrative by normalising menopause and emphasising positive or neutral aspects such as freedom from menstruation, pregnancy, and contraception, together with information about managing troublesome symptoms might empower women to manage menopause with greater confidence*".

The authors highlight data showing:

- Menopause experiences appear to differ widely between different countries and cultures, with expectations and experiences influenced by personal, family and sociocultural factors. For example, they report that African American women have earlier menopause and more troubling vasomotor symptoms than white American women, while Asian women in Korea and Vietnam report body and joint pains as the most troubling symptoms, and only 5% of Indonesian females suffered hot flushes.
- Negative views of menopause may increase the severity of symptoms. A systematic review suggests negative attitudes towards menopause predicted future troubling symptoms. Migration studies show women may experience troubling symptoms when moving to a country with negative views of menopause where they would not be in their home country.
- Poor support including limited public discussion or education about the menopause can drive negativity, as can experiences where women feel their symptoms have been dismissed by a HCP.

In the same edition of the BMJ, an editorial ([BMJ 2022;377:o1425](#)) written by the Chair and Chief Executive of the British Menopause Society balanced out the discussion by explaining how to help "women make informed choices". It quotes 75% of women report symptoms during the menopause and >25% describe these as severe. On average these last 7 years, but for 1/3 it may be longer. **The editorial concludes "nobody should be suffering in silence or feel that the effect of the menopause is not adequately recognised"**.

The Normalising Menopause paper received fierce criticism in the BMJ Rapid Responses. The problem is that for years women have lacked a voice to address menopause symptoms and the medical profession has often been a barrier rather than a gateway to effective treatment, leaving many women suffering unnecessarily from severe symptoms with a huge impact on their lives.

While the historic data around risks of HRT have been confusing, the latest data is clear that risks with HRT are small and outweighed by the benefits when given in women <60yo or within 10y of the menopause beginning. Risks can be further reduced or eliminated through sensible prescribing using transdermal oestrogen and micronised progesterone (e.g. utrogestan) or the Mirena coil. The main issue now is ensuring that HCPs understand this modern data and prescribing so that women can be adequately counselled.

Nevertheless, we also understand in general practice the importance of how information is communicated and how we approach a condition may have a powerful influence on a patient's experience of it. Encouraging women who need help to seek help while avoiding over medicalisation in those who do not need help is a complex area and one where it feels the medical community is unlikely to reach an agreement. Perhaps in these situations, wider discussion with society is the way forwards.

3 Risks and Benefits with HRT

As part of shared decision-making, women need to understand the risks and benefits of HRT.

► *All-Cause Mortality* [BJOG 2021](#)

With the majority of rhetoric surrounding HRT about breast cancer risk, it is helpful to step back and look at whether in total HRT has a positive or negative effect on mortality. This paper in the BJOG was a matched cohort study using UK primary care databases to examine 105k HRT users against 225k non-user controls to assess all-cause mortality.

Over a mean 13.5y follow up it was found in **women taking combined HRT there was a 9% lower risk of all-cause mortality** seen across users aged 50-65y, while oestrogen-only HRT was not associated with significant change.

This important finding raises the question: should HRT should be offered routinely to women at the point of menopause? This data along with several other studies in younger HRT users strengthens this argument, of course on a personalised basis considering patient wishes and individual risks, but currently, no guideline or national specialist body recommends this. There are vocal HRT advocates who have suggested women should go on HRT and take it for life. It should be noted that this approach has not been examined in the research and this study, in 'younger' women, followed up for 13 years, cannot help elucidate the issue.

► *Breast cancer risk* [Lancet 2019;394:1159](#), [MHRA 2019](#)

Breast cancer risk is largely driven by the progesterone component of HRT (which remains essential for endometrial protection from unopposed oestrogen in all women with a uterus). This Lancet paper from 2019 had lots of media attention as it showed an increased risk of breast cancer from HRT use that was double that previously reported. A very quickly published [MHRA report](#) provided more real-world figures for clinicians (summarised alongside [all risks here](#)):

- 63 / 1000 background rate of breast cancer in women aged 50-60 ('never users')
- HRT for 5 years in 40-50yr women leads to an additional...
 - 20 / 1000 cases by age 69 with continuous combined HRT
 - 14 / 1000 cases by age 69 with cyclical combined HRT
 - 5 / 1000 cases by age 69 with oestrogen-only HRT

However, there are caveats to this data. It is based on older types of progesterone - as a recent [review in the BJGP](#) points out newer options such as oral micronised progesterone or progesterone-releasing intra-uterine systems have a [lower or no risk of breast cancer, particularly when used for <5 years](#). Also, [Cancer Research UK](#) encourages us to keep any risk increase in context of wider patient factors - overweight and obesity, smoking and alcohol consumption are much bigger risk factors. [The RCOG and British Menopause Society](#) highlights *these data do not change how we manage women with troubling menopausal symptoms*, something which the MHRA agrees with.

► *VTE Risk* [BMJ 2019;364:k4810](#)

VTE risk is predominantly driven by oestrogen exposure but as this BMJ ([2019;364:k48100](#)) paper demonstrates this is dependent on the type of oestrogen and mode of delivery.

- Transdermal preparations (gel or patch) ARE NOT associated with VTE risk, while older oral HRT options are associated with a small risk increase.
- Transdermal preparations remain UNDER-USED with 80% of users still having oral therapy.

A review in the BJGP ([2018;68:499](#)) *argues that* transdermal oestrogen alongside micronised progesterone (there are no combination preparations) **may represent the optimal HRT regime and that it has a comparable cost to other HRT preparations**. Transdermal oestrogen plus IUS for endometrial protection would be another lower-risk alternative option.

KISS: Menopause

[NICE NG23](#), [CKS Menopause](#), [BMJ 2017;359:j5101](#), [MHRA 2019](#)

MHRA Principles on HRT for Menopause

- HRT should only be used to alleviate post-menopausal symptoms which adversely affect quality of life
- Continue only as long as the benefits outweigh the risks
- Careful appraisal of the risks and benefits should be undertaken before use and regularly during use
- For women it recommends:
 - Using HRT for the shortest duration necessary to help reduce overall risk
 - Taper treatment when stopping to avoid menopausal withdrawal symptoms
 - Low-dose vaginal oestrogen do not appear to increase risk where these are an option

Diagnosis

- **Diagnose menopause clinically** in otherwise healthy women >45yo with menopausal Sx:
 - Perimenopause based on vasomotor symptoms and irregular periods
 - Menopause if no period for >12 months and not using hormonal contraception
 - Menopause if symptoms in women with no uterus
- **Do not use investigations** (AMH, inhibin A or B, oestrogen, antral follicle count or ovarian reserve) to diagnose peri-/menopause in women >45yo
- Do not use FSH in women taking combined hormonal or high progesterone contraception
- **Consider using FSH** only in women 40-45yo with menopausal Sx or <40yo if menopause suspected - due to its variability, it is not useful during the perimenopause

Benefits and Risks of Menopause Treatments

- NICE updated its guideline in 2019 and now advises the use of the [MHRA risk tables](#) when discussing options with patients.

Treatment – short-term symptoms

- **Vasomotor**
 - **HRT first line** – discuss short and long-term benefits and risks
 - **SSRI, SNRI** and **clonidine** are alternative options as second-line therapies
 - BMJ recommends citalopram 10-20mg or escitalopram 10mg 1st line, venlafaxine ER 37.5mg-75mg 2nd line > paroxetine > gabapentin > clonidine
 - Also evidence for CBT, hypnosis and acupuncture for vasomotor symptoms
 - **St John's Wort, isoflavones** and **Black Cohosh** have evidence of benefit BUT:
 - Uncertainty about dosage, duration of effect, the variation of preparations and safety
 - Interactions (SJW > anticonvulsants, anticoagulants, CCBs, tamoxifen, etc.)
- **Psychological**
 - **CBT for anxiety** due to menopause; **CBT or HRT for low mood** due to menopause
 - SSRI/SNRIs are not recommended unless clinical depression
- **Urogenital**
 - **Offer vaginal oestrogen to women with urogenital atrophy** (even if on HRT)
 - Continue treatment as long as required to relieve symptoms
 - Discussion with a menopause expert is advised if:
 - Symptoms do not improve – dose may be increased
 - Systemic HRT is contraindicated – vaginal oestrogen may still be considered
 - Lubricants/moisturisers can be used for vaginal dryness (Replens + Sylk can be prescribed)

KISS: Menopause (continued)

Treatment - short-term symptoms continued

- **Altered sexual function (reduced libido)**
 - HRT is the first line but if not effective consider adding testosterone gel supplementation
 - Testosterone replacement is generally considered only appropriate where blood levels are low

Choice of HRT

- **Women with intact uterus = combined HRT**
 - *Perimenopause*: cyclical, monthly or 3 monthly; *post-menopause*: continuous/cyclical
 - *IUS* can be used to supply progesterone as an alternative to systemic progestogens – not examined by NICE but is a licensed and widely used option – [see CKS for details](#)
- **Women with no uterus = oestrogen only, continuous usually**
 - Use the lowest effective dose for the shortest possible time
 - Oral or transdermal (patches or gel) preparations available
 - Transdermal associated with no increased risk of cardiovascular events
 - If BMI >30 or increased VTE risk consider transdermal in preference to oral
 - For a detailed review of regimes, C/I and side effects see [CKS – menopause](#).
- **Review at 3 months, then annually thereafter** if no side effects, etc.
 - Unscheduled vaginal bleeding is common in 1st 3 months; after 3/12 it needs review
- **HRT may be stopped either gradually or abruptly**
 - A gradual reduction may reduce short-term symptoms, but has no long-term benefit

Refer if persistent troubling Sx or S/E, HRT is C/I or uncertainty about the most appropriate treatment

KISS: Management of Menopausal Symptoms in Women with a Hx of Breast Cancer [CG80](#)

- Stop HRT in women in whom breast cancer is diagnosed.
- HRT should not routinely be offered to women with a history of breast cancer.
- In exceptional circumstances, HRT may be offered – refer to a menopause clinic
- Fluoxetine and paroxetine may be used but avoid in women taking tamoxifen.
 - NB note: SSRIs are likely to have a class effect for benefit but others do not inhibit tamoxifen as significantly; venlafaxine is also used and may be the most effective
- Tibolone, progestogens & alternative therapies (e.g. isoflavones, Black Cohosh) are not recommended

KISS: Management of Menopausal Sx in Women with a High Risk of Breast Cancer [CG164](#)

- Women with an FHx of the breast should be informed about the increased risk with HRT including type and duration.
- If used it should be for the shortest duration, lowest dose, and oestrogen-only where possible.
- Alternatives should be considered for specific symptoms, e.g. vasomotor, osteoporosis.
- If early menopause and mod/high-risk HRT use should be confined to women <50yo.

KISS: Contraception in the Perimenopause [BJGP 2015;65:e630](#)

1. Combined hormonal contraception (CHC) - an effective option that should provide the same symptomatic benefit as HRT but we need to be aware of the UKMEC criteria

- Age 40 or over is UKMEC 2, so in the absence of other risk factors remains an option for older women, although prescription >50 years not advised by the paper
- Common C/Is include smokers >35y, hypertension, migraine with aura, complicated diabetes

2. Mirena plus oestrogen-only HRT - gold standard option

- Mirena is licensed as contraception and as the progestogen component of HRT
- 65% will achieve amenorrhoea by 1 year so offers a potentially bleed-free option
- If fitted for contraception or the progestogen element of HRT when <45yo the IUS should be changed at 5 years; if fitted purely for heavy bleeding it can be left as long as it proves effective

3. HRT plus progestogen-only contraception (POC)

- The FSRH currently recommend that combined HRT is still required despite POC use; however, some experts argue that progestogen pills or implant can be used in the same way as the Mirena is to provide endometrial protection.

KISS: Mental Health in the Peri-Menopause [NICE NG23](#) [BMJ Best Practice 2020](#)

Background:

- Mood changes and mental health disorders are common in the peri-menopause and early post-menopausal periods.
- However, there is limited evidence and a lack of consensus or recommendations as to how to assess and manage mental health problems during the peri-menopause.
- Menopause symptoms can complicate, co-occur and overlap with depressive symptoms, making assessment difficult.
- Most women who experience a major depressive episode during the peri-menopause have experienced a prior episode of depression, but there is obviously a huge spectrum of mental health illness that can present in the peri-menopause.

Assessment and Management:

- Assess for previous depressive episodes.
- Assess for suicidal ideation in women with severe depressive symptoms, as usual.
- Menopause-specific mood disorder scales/questionnaires do not exist, but other known screening tools (e.g. PHQ9) can be considered
- **NICE recommendations:**
 - Consider HRT to alleviate low mood that arises as a result of the menopause.
 - Consider CBT to alleviate low mood or anxiety that arises as a result of the menopause.
 - Ensure that menopausal women and healthcare professionals involved in their care understand that there is no clear evidence for SSRIs or SNRIs to ease low mood in menopausal women who have not been diagnosed with depression.
- **BMJ Best Practice:**
 - Short-term use of HRT may improve mood and depressive symptoms during the menopausal transition and in early menopause but study results are inconsistent.
 - Mood disorders often improve with HRT, although conventional antidepressants may be more effective in depression.
 - There is a lack of good-quality evidence relating to CBT, but given the low risk of harms and strong evidence in the general population, it is a reasonable option.

KISS: Premature Ovarian Insufficiency [NG23](#)

Diagnose premature ovarian insufficiency (POI):

- In the presence of menopausal symptoms, including absent or infrequent periods, AND
- Elevated FSH levels (>30mIU/ml) on 2 blood samples 4-6 weeks apart.

Treatment

- *Offer either COCP or HRT* unless contraindicated
- Treatment should be continued until at least the expected age of natural menopause (average age 52)
 - No evidence shows one is more effective than the other
 - HRT may have a beneficial effect on blood pressure compared to COCP
- Consider referral to a health professional with experience in POI
- If hormonal therapy is contraindicated, advise about bone and cardiovascular health and symptoms management

NB Learning Points & Quality Improvement Ideas: Menopause

- Menopausal symptoms can last for up to 12 years and significantly impair quality of life
- HRT is an effective treatment with the transdermal route safer than oral HRT
- Micronised progesterone appears to have lower risks than other progesterone types, and in combination with transdermal oestrogen may represent the optimal HRT regime
- Audit patients using HRT. What percentage of your population use transdermal HRT? Could more women have the potential benefits of transdermal HRT discussed and see if this alters their choices.
- Patient resources:
 - [Women's Health Concern](#) - patient-facing arm of the BMS
 - [Rock My Menopause](#) - patient-facing campaign of the Primary Care Women's Health Forum
 - [Daisy Network](#) - information and assistance of premature ovarian insufficiency

URINARY INCONTINENCE

Menopause has had a very high media profile recently, but it isn't the only condition affecting women which has not had the recognition deserved.

Both stress incontinence and overactive bladder can cause significant morbidity impacting the daily life of women of all ages. These conditions are extraordinarily common and as yet are rarely discussed with medical professionals. Two new articles from the BJGP ([2022;72\(717\):185](#)) and BMJ ([2021;375:e063526](#)) bring new insights to these problems.

1 Overactive Bladder [BMJ 2021;375:e063526](#)

While the management of overactive bladder has been well documented in national guidance ([NICE NG123 2019](#) - see KISS) this recent review in the BMJ provides clinicians with some practical tips for patients.

Prevalence of overactive bladder syndrome (OBS) is suggested at 20%, more commonly in women and increases with age. It is discussed infrequently (compared to its true prevalence) due to embarrassment or acceptance as a part of ageing.

There may be no clear cause but known drivers include excessive or inadequate intake (the latter due to concentrated urine irritating bladder mucosa), and carbonated or caffeinated drinks. The 'latch key' effect is also well documented: people needing to urinate as soon as they get home and put the key in the door; a similar effect is seen in a range of scenarios such as proximity to a toilet.

The paper highlights a range of differentials that should be considered and features that point away from OBS:

KISS: Differential Diagnosis of Overactive Bladder Syndrome [BMJ 2021;375:e063526](#)

Small & variable volume voids, normal fluid intake	Probable OBS
Small & fixed volume voids, normal fluid intake	Possible interstitial cystitis
Small volume voids, low fluid intake	Restrictive behaviour to control symptoms
Large volume voids	Usually large intake, consider DM, diabetes insipidus, hypercalcaemia, diuretics
Nocturnal polyuria	Possible CHF, also with ageing, sleep apnoea, also behavioural if restricting intake during day and increased in evenings (seen with certain jobs, e.g. taxi drivers, medics...)
Dysuria, haematuria, suprapubic pain	Not seen in OBS, consider UTI, STI, bladder Ca, interstitial cystitis
LUTS for only weeks > months	Consider bladder Ca, by the stage it causes Sx haematuria (visible or non-visible) is very likely to be present
Hesitancy, poor flow, incomplete bladder emptying	Possible reduced bladder contractility, can result in large residual volume which drives OBS symptoms
Dragging sensation, bulging	Possible prolapse or fibroids
Neurological Hx (e.g. CVA, Parkinson's, MS)	Possible neurogenic bladder

It is important to examine women to exclude a palpable bladder, abdominal mass, vaginal atrophy and pelvic prolapse. Urinalysis should be performed to exclude haematuria and infection.

► *Managing overactive bladder syndrome*

Given the significant limitations the condition can cause for people, [good patient information and support groups](#) are recommended. The authors also highlight the ['Just Can't Wait' card from the Bladder & Bowel Community](#) which shows businesses that this person has a medical reason to need the toilet quickly.

Conservative measures can be effective and durable - bladder training (min. 6/52), weight loss if obesity, stopping carbonated & caffeinated drinks, optimising fluid intake (~2-2.5L/day). Patients with mixed OBS and stress urinary incontinence can benefit from pelvic floor muscle training.

Vaginal atrophy can be an important barrier to improvement and in this group, vaginal oestrogen is indicated.

Pharmacological therapy may be tried where conservative measures have failed:

- **Anticholinergics are the 1st line treatment** but limited by their side effects - only 8-25% continue the treatment by 12 months. Selective anticholinergics such as solifenacin may cause fewer side effects.
 - They should be avoided in the elderly, frail and those with a high anticholinergic burden.
- Mirabegron, a β_3 -adrenoceptor agonist, is 2nd line in NICE guidance but may be a better choice in these groups if drug therapy is still desired.
 - Caution is required in hypertension, renal and liver impairment.
- A combination of mirabegron & solifenacin is supported by trial data but is not currently part of national guidance.

Secondary care options include further investigation to confirm the diagnosis, botox injections, sacral neuromodulation and, where these fail, percutaneous posterior tibial nerve stimulation.

2 **Stress Urinary Incontinence** [BJGP 2022;72\(717\):185](#)

As the authors point out, given stress urinary incontinence (SUI) affects around a third of women it is surprising that it "does not grab more attention". There appears an acceptance that this is a condition that 'just happens' to some women and is 'something to be managed', an attitude further entrenched by the normalisation of incontinence pads in retail stores. While an easy way to self-manage health issues is welcome, it should not come at the expense of education on genuine treatment options.

► *What are the causes of SUI?*

These are multifactorial and there may be multiple defects before SUI develops due to continence control consisting of several mechanisms. Childbirth, repetitive insults such as chronic coughing/constipation/etc., and high BMI may contribute. SUI is also common during pregnancy. SUI commonly affects two groups: around 20% of women aged 18-30, and aged 70+.

Prevention is achievable through pelvic floor muscle training (PFMT) and has shown to improve labour and delivery outcomes, and sexual function when undertaken during late pregnancy and postpartum, and could prevent 25% of SUI in primips. Data shows PFMT is also effective for established SUI. The paper argues there are many opportunities to discuss possible SUI with women and offer preventative education, such as ante- and post-natal visits.

► *Pelvic floor muscle training*

The key treatment is PFMT. NICE advises that women should undertake these exercises throughout life.

However, as the paper explained "research has found that women find it difficult to understand and perform PFMT", but that some basic, verbal guidance can help the majority.

Initially, women will need to undertake strengthening exercises and then they can move to maintenance. Regular PFMT can result in automatic contraction of the pelvic floor during activities when leakage may occur. It is also possible to use a pre-emptive 'squeeze and lift' before & during triggers of SUI, e.g. lifting, sneezing.

KISS: Practical Tips on Pelvic Floor Muscle Training BJGP 2022

- **Educate on correct pelvic floor muscle contraction**
 - To exercise these muscles squeeze your vagina then lift up inside
 - Hold the squeeze for as long as possible, counting the seconds, then release
 - Avoid breath-holding or squeezing the buttocks
 - The paper advises against stopping urine mid-flow as an exercise
- **Recommended regime**
 - Find how many squeeze, lift and holds the patient can do, and for how long
 - Aim for 2 sets of 8 maximal contractions with 8 seconds hold, 2-3x daily
 - Strengthening (and symptom improvement) may take 12-16 weeks
 - Maintenance: 2 sets of 8 maximal contractions with 8s hold, daily or 3x/wk min.
- **Counter-balance contraction (the Knack technique)**
 - Pre-emptive squeeze and lift before and during activities which may precipitate leakage

[Useful instructional video](#) from NHS Oxford University Hospitals is available free on youtube - the regime recommended there is very slightly different

► *What else can help?*

NICE suggests weight loss may be beneficial for women with overweight or obesity. If after 12 weeks of PFMT there is no improvement, referral to a community physio or continence service is recommended. This can then inform onward referral to specialist care if there is still failure to improve - NICE indicates further options include duloxetine and surgery.

NB Learning Points: Urinary Incontinence

- Avoidance of triggers and bladder training are the cornerstone of overactive bladder syndrome management, and pelvic floor muscle exercises for stress urinary incontinence. Weight loss in patients with obesity can also prove beneficial.
- Anticholinergics are 1st line drug therapy for OBS but limited due to side effects and should be avoided in the elderly, frail and those already with a high cholinergic burden. Mirabegron is a suitable 1st line alternative.
- Use of mesh procedures is currently on hold for women with SUI and POP, however, there are a range of effective non-surgical interventions that may be offered
- Women with surgery involving mesh and possible complications should be referred to a specialist centre for further assessment. The list is available [here](#).

NB Quality Improvement Ideas: Urinary Incontinence

- Perform a case review of your patients on anticholinergic medication. Are they receiving other medications with a high anticholinergic burden (e.g. TCAs, antihistamines)? Have anticholinergic side effects/symptoms been discussed with the patient? Should the anticholinergic be stopped?
 - Consider with hx of memory impairment, disorientation, instability, change in vision, etc.

KISS: Urinary Incontinence and Pelvic Organ Prolapse in Women: Management [NG123 2019](#)**Assessment**

- **Categorise the cause of urinary incontinence**
 - ***Stress, mixed, urgency/overactive bladder***
- Examination: include PV assessment of pelvic floor muscle contraction
- Urine dipstick +/- MC&S if positive leucocytes/nit
- Undertake a bladder diary for ≥ 3 days as part of the initial assessment
- Refer for bladder scan if possible voiding dysfunction or recurrent UTI
- Do not undertake urodynamic testing with SUI or stress-predominant mixed UI
- Imaging and cystoscopy are not routinely recommended

Referral

- Consider referral if persisting pain, palpable bladder, pelvic mass, associated faecal incontinence, suspected neurological disease, voiding difficulty, fistulae, previous continence surgery, previous pelvic cancer surgery or radiotherapy
- Refer via 2ww for women with haematuria or recurrent or persistent unexplained UTI

Non-surgical management of Stress UI or Mixed UI

- ***Offer a trial of supervised pelvic floor muscle training of at least 3 months as 1st line***
 - Involves at least 8 contractions TDS
 - Continue if beneficial
 - Electrical stimulation/biofeedback should not be used routinely but may aid those unable to contract their pelvic muscles themselves
- Do not offer intravaginal or intraurethral devices to prevent leakage for routine management
- Duloxetine is a possible second-line option for women who wish to avoid surgery

Non-surgical management of Urgency IU or Mixed UI

- ***Lifestyle measures***
 - Recommend a trial of caffeine reduction, fluid intake modification, weight loss if BMI >30
 - Offer bladder training for ≥ 6 weeks as 1st line
 - If not satisfactory benefit consider medication + bladder training
 - Do not offer transcutaneous sacral/posterior tibial nerve stimulation
 - Percutaneous post. tibial nerve stimulation may be used if continuing Sx and not using Botox
- ***Medication***
 - ***Offer intravaginal oestrogen in postmenopausal women with vaginal atrophy.*** Do not offer systemic HRT

KISS: Urinary Incontinence and Pelvic Organ Prolapse in Women: Management [NG123 2019](#)

Non-surgical management of Urgency IU or Mixed UI

■ **Medication - anti-cholinergics**

- Before starting, discuss the likelihood of benefit, common s/e, need for $\geq 4/52$ trial and uncertain long-term effects on cognition. Consider risks and cautions/contra-indications, e.g. dementia, poor bladder emptying, overall anticholinergic burden
 - Do not offer imipramine, propantheline or flavoxate
 - Do not offer oxybutynin immediate release to older women at risk of worsening health
- **Offer anticholinergic with a low acquisition cost**
 - If 1st medication is not well tolerated, offer an alternative with a low acquisition cost
 - Offer transdermal preparation if unable to use oral
 - **Mirabegron is an option** where antimuscarinic drugs are not tolerated/contraindicated or clinically ineffective ([TA290](#))
- **Desmopressin may be considered specifically to reduce nocturia.** Caution in women with CF, or $>65y$ + CVD or hypertension

■ **Invasive Procedures**

- Urodynamic testing will be required to demonstrate detrusor overactivity. If present:
- Botox injection may be considered. Women need to be willing to undertake intermittent catheterisation as this is a possible adverse effect. Benefits should last 6 months+.
- Percutaneous sacral nerve stimulation
- Severe and refractive cases may be offered augmentation cystoplasty or urinary diversion

Non-surgical management of Pelvic Organ Prolapse

- Lifestyle measures include losing weight if BMI >30 , minimising heavy lifting + preventing constipation
- Consider vaginal oestrogen if vaginal atrophy, including ring, if cream/pessaries difficult to use
- Consider a 16/52+ programme of supervised pelvic floor muscle training
- Consider a vaginal pessary with symptomatic prolapse. Should be removed at least once every 6/12.

Surgical management of Stress UI

- Current options include colposuspension or autologous rectus fascial sling, with mesh slings currently restricted. Intramural bulking agents are a further alternative.
- NICE have produced a [useful patient decision aid](#).

Surgical management of Pelvic Organ Prolapse

- Again NICE have produced a useful patient decision aid for [uterine](#) and [vaginal vault](#) prolapse.
- Options include vaginal hysterectomy and/or sacrospinous fixation or Manchester repair, mesh is currently off-limits in most areas, except in research settings.

CONTRACEPTION

Recently [published data in the BMJ Sexual and Reproductive Health journal](#) shows the changing trend in contraceptive prescription since 2000. In women aged 18-49 years old rates of combined hormonal contraception use have fallen from 26% to 14%. Meanwhile, use of the progesterone-only pill rose from 4% to 11% and long-acting reversible contraception from 4% to 6.5%, driven by the introduction of the longer-acting desogestrel and incentives for promoting LARCs.

Perhaps it also reflects the greater appreciation of the risks prescribing combined hormonal contraception with a shift towards safer prescribing, and consideration of the side effects different options can cause. This latter issue is made more complex by the diversity of COCPs on the market.

Nevertheless, oral contraceptives remain the most popular and prescribed option for women. A new paper in the BJGP ([2021;71\(712\):522](#)) provides a very helpful review of the differences between contraceptive pills and how we can harness this knowledge to help our patients and prescribe.

1 Prescribing the Oral Contraceptive Pill: Key Considerations [BJGP 2021;71\(712\):522](#)

While this paper fully discusses the relative merits of COCP and POP and how to minimise the risks, the real strength of this paper is to clearly show which of the dozens of options are linked to specific side effects and helps us think beyond our usual levonorgestrel starter for 10. As the authors point out, 50% of women discontinue oral contraception in the first year of use. This tells us that although it seems like a simple medical issue there are a number of challenges and considerations for clinicians and patients.

The authors consider 4 aspects to helping women choose the optimum oral contraceptive pill (OCP).

- **Consider if COCP or POP is most appropriate**
 - The [UKMEC has clear guidance](#) on when the risks outweigh the advantages for COCP use and the POP is a preferred option:
 - Age >45y or >35 + >15cigs/day, BMI >30, <6/52 post-partum, migraine with aura, PMHx or strong FHx of VTE, cardiac arrhythmias or impairment
- **Consider type & dose of oestrogen**
 - Most COCP contain ethinylestradiol, 20-35µg.
 - Starting with **lower doses is recommended to minimise the risk of S/E** (headaches, bloating & weight gain are most common). However, **lower doses have greater risk of unscheduled bleeding** (~2x for 20µg vs 30µg). The dose may need to be increased until unscheduled bleeding is controlled.
 - Natural oestrogen preparations are available (Qlaira, Zoely) but are expensive options and rarely recommended on the first few rungs of most medicines management guidance. Data on tolerability and side effects in this population is fairly limited. The latter has been associated with higher incidence of acne, weight gain and discontinuation vs standard COCP preparations.
- **Consider the COCP preparation and regime**
 - **COCP may be monophasic or multiphasic.** The latter aim to minimise side effects by varying the doses of hormone throughout the pack. This never really caught on in the UK and brings its own complications: varying levels of hormones may exacerbate PMS, require more careful adherence and cannot be used for extended pill regimes that have become commonplace in the past couple of years.
 - **Regime options include standard "traditional" 21/7 regimes, and tailored options such as shortened pill-free intervals (21/4 or 24/4), tricycling and continuous use.** See KISS in section 2 for more detail. While longer regimens increase the risk of unscheduled bleeding, having a range of options to tailor to the individual's needs and medical history is very welcome.
- **Consider the progesterone type**
 - There are 4 generations of progesterone, each with different characteristics. Even within each generation, there is considerable variation. This can make prescribing feel like a bit of a lottery, but as the authors explain, with a little knowledge of the common physiological effects of each we can harness these to effectively manage other symptoms (e.g. acne) and anticipate potential side effects.

- Physiological activity includes:
 - **Progestogenic**: breast tenderness, GI disturbance, bloating, ↑ appetite, tiredness, mood changes, skin changes
 - **Androgenic**: oily skin, acne, hirsutism, weight gain, mood changes
 - **Glucocorticoid**: mood swings, ↑ appetite, water retention
 - **Mineralocorticoid**: water retention, bloating
- Cyproterone (e.g. in Dianette) has an anti-androgen effect and is not primarily used purely as a contraceptive.

KISS: Characteristics and Indications for Progestones

[BJGP 2021;71\(712\):522](#)

Generation	Progestone	Physiological Activity	Best Option for the following conditions	Examples of branded options (not exhaustive)
1st	Medroxy-progesterone	Androgenic ++ Glucocorticoid +	Irregular or missed periods	Depot injection
	Norethisterone	Progestogenic ++ Androgenic +	Menorrhagia, adenomyosis	Brevinor, Norimin, Norinyl (Mestranol), Synphase (M)
2nd	Levonorgestrel	Progestogenic +++ Androgenic +++	Menorrhagia, adenomyosis	Microgynon, Rigevedon, Ovranette, Levest, Elevation, Maexeni, Logynon (M), TriRegol (M)
3rd	Desogestrel	Progestogenic +++	Dysmenorrhoea, endometriosis	E20: Bimizza, Mercilon, Gedarel; E30: Gedarel, Marvelon, Cimz
	Norgestimate	Progestogenic ++ Androgenic +	Moderate acne	Cilique, Lizinna
	Gestodene	Progestogenic +++ Androgenic +	Dysmenorrhoea	E20: Akizza, Femodette, Millinette, Sunya; E30: Femodene, Katya, Millinette, Akizza
4th	Drospirenone	Anti-androgenic Anti-mineralocorticoid	Acne, PCOS, PMS	Yasmin, Lucette, Yacella, Yiznell, Dretine
	Dienogest	Anti-androgenic	Endometriosis	Qlaira (M)
	Nomegestrol	Anti-oestrogenic (targeted at endometrium) Anti-androgenic (partial)	Menorrhagia, PCOS, PMS	Zoely

2 Tailored Pill Regimes [COCP Guideline FSRH 2019](#)

In early 2019 the FSRH updated its guidance on the COCP and perhaps the **biggest change is a recommendation about "tailored pill regimes"**. For years we have recommended a standard regime of 21 days of COCP, then a 7-day break. This evolved from the belief that having a withdrawal bleed mimicking women's natural menstrual cycles was important.

However, we now know that not only is **it not necessary to have withdrawal bleeds** - there is no difference in risk with continuous COCP use - but actually, there are **a number of important disadvantages of having a 7-day break**:

- Ovarian suppression is reduced and follicular development occurs during the hormone-free interval, especially in lower oestrogen pills (which we are using more of to reduce arterial risk), and missed pills in that first week after a hormone-free interval (HFI) or an accidental extension of the 7 days risks ovulation
- There may be adverse effects due to hormone changes, e.g. headache, mood change
- Withdrawal bleeding may be unpleasant (very much so if a history of dysmenorrhoea)

The traditional 21/7 approach appears increasingly outdated. *What other regimes could women use?*

Flexibility is the key here. Continuous use could be used, often suggested in conjunction with a 4-day pill-free break if the female develops breakthrough bleeding, or extended use (e.g. 63 or 84 days) with a 4-day break. Along with these newer regimens, the FSRH still recommends that standard use is an appropriate option.

The [FPA have updated their patient information leaflet](#) to include tailored regimes, and there are other [useful PILS from John Guillebaud and team](#) and from [Oxford University Hospitals](#).

KISS: Standard and Tailored COCP Regimes

[FSRH 2019](#)

Standard use 21 days pills/1 ring/3 patches 7-day hormone-free interval

Tailored use

- **Shortened HFI** 21 days pills/1 ring/3 patches 4-day HFI
 - *Indication: shorter bleeding time, good for women to ↓ period-related Sx but wish a withdrawal bleed*
- **Extended use (tricycle)** 63 pill/3 rings/9 patches 4 or 7-day HFI
 - *Indication: good for women with severe pain or mood changes (e.g. endometriosis/PMS) during or in lead up to menstruation*
- **Flexible extended use** Continuous use until breakthrough bleeding - then 4 days HFI
- **Continuous use** Continuous use without a break
 - *Indication: as per extended use, may further ↓ risk of Sx but ↑ risk of unscheduled bleeding*
 - Low-dose oestrogen COCPs (20mcg) are typically recommended for extended/continuous use

KISS: Common Questions with Tailored Pill Regimes

- **Are COCPs licensed for this method?**
 - No COCPs are licensed for tailored pill regimes, however, this method is endorsed by the FSRH
- **Is it safe to have no periods on the COCP?**
 - Yes - the bleed women get at the end of a pill pack is not a real period, it's a withdrawal bleed and not required for any specific bodily function. The safety data is very reassuring that tailored pill regimes have no increased risk over traditional regimes.
- **Why use a low-dose COCP for tailored pill regimes?**
 - Using extended or continuous regimes results in a higher number of pills taken and consequently higher amounts of the active products. Low-dose oestrogen COCPs (20mcg) therefore limit the exposure to oestrogen (from which is derived most of the COCP risks), resulting in a similar amount overall to a 30mcg dose on a standard 21/7 regime.
- **Does this have to be a COC pill?**
 - No - the FSRH details how to use patches or rings for extended/continuous use as well.
- **What do I do if I get breakthrough bleeding?**
 - Patients with 2 days of consecutive bleeding should omit the next 4 days of pills then restart daily pills. Patients need to take at least 14 days of pills between breaks. If bleeding continues for over 2 weeks they should be reviewed for underlying causes +/- a change of pill.

NB Learning Points and Quality Improvement Ideas: Contraception

- The UKMEC recommendations were updated in 2016 - while no huge change there is new guidance on bariatric surgery, transplant, cardiomyopathy, cardiac arrhythmias and anti-phospholipid syndrome
 - For the full FSRH UKMEC 2016 guidance [click here](#) or summary tables [click here](#).
 - For an 'at a glance' charts/leaflets of contraceptive options from the FPA [click here](#)
- For patient information on Tailored Pill Regimes [click here](#)
- For the latest Missed Pill guidelines [click here](#)
- Levonorgestrel COCPs have the lowest VTE risk and should be considered a 1st line option
- Women >40y have specific needs for contraception due to increased risks and menopause. FSRH have produced clear guidelines, summarised in our KISS below.
- Beware the risks of high dose norethisterone to delay menstruation - they equal the COCP
- If women on enzyme-inducing medications require emergency contraception, the IUD is the most effective option. If tablets are used a 3mg dose of levonorgestrel is recommended
- **Quality Improvement: audit women ≥50y in your practice**
 - *Are they still receiving the COCP? If so invite them in to discuss alternatives*
- **Quality Improvement: audit post-menopausal women using oestrogen-only HRT and Mirena**
 - *Is the Mirena <5 years old? If it is older then either the type of HRT or the coil needs to be changed.*

KISS: Emergency Contraception

[FSRH April 2019](#), [FSRH 2019: incorrect use of COC draft](#), [FSRH 2017: EC](#), [DSU 2016](#), [BMJ 2012](#)

Considerations:

- Copper IUD is the most effective method and should be offered a 1st line.
- Patients should be informed that *LNG and UPA are ineffective if given after ovulation*.
- UPA is considered more effective than LNG and is now considered 1st line oral EC
- The FSRH has [excellent algorithms to help decide on the best EC](#) for the patient (pg. ix & x)

Assessment:

- Include sexual history and details of contraceptive failure, date of LMP, chance of existing pregnancy/other episodes of UPSI in cycle, and calculate the expected date of ovulation.
- Discuss the need for ongoing contraception, and assess for risk of STIs.

Copper IUD – 20x more effective than oral EC

- Very effective prevents 99% of expected pregnancies and can be used for ongoing contraception.
- Works by direct toxicity so prevents fertilisation, also anti-implantation and anti-sperm effects.
- *Offer to all women attending for EC* even if presenting within 72 hours of UPSI.
- Ideally should be fitted at presentation, but can be offered later if a referral is required.
- Can be inserted up to 5days after the 1st episode of unprotected sexual intercourse **OR** up to 5 days after the estimated date of ovulation, whichever is later
- *Preferred option for women on enzyme-inducing drugs* - see [DSU '16](#)
- Can be removed at any time after the next period if no UPSI has occurred since.

Levonorgestrel/Levonelle (LNG)

- Prevents 52-100% of pregnancies, may have some effect up to 96 hours after intercourse but licensed for use within 72hours.
- **Mode of action** thought to be primarily inhibition of ovulation. Less effective just prior to and after ovulation, and when further episodes of unprotected intercourse occur.
- **Dose:** 1.5mg taken ASAP after UPSI, and within 72hours; repeat is vomiting within 2hours
- 3mg (2x1.5mg tabs) if on enzyme-inducing drugs (unlicensed - see [DSU '16](#)), or BMI >26 / >70kg
- No medical contraindications to use, will not interrupt an existing pregnancy and not known to have adverse effects on the fetus if pregnancy occurs.

Ulipristal Acetate (EllaOne, UPA)

- *More effective than LNG* ([Cochrane 2019](#)); 1st line if UPSI within 5/7 of expected ovulation
- **Mode of action:** Delays or inhibits ovulation, and has endometrial effects.
- **Dose:** Single dose of 30mg ulipristal acetate(EllaOne). Repeat if vomiting occurs within 3hrs.
- Binds to progesterone receptors, so additional method needed if on COC (for 14days), POP (9days), Qlaira (16days).
- May have reduced efficacy in women with BMI >30/ >85kg, not recommended to increase dose
- Not effective if on enzyme-inducing drugs within past 1/12, or gastric acid suppressants.
- Avoid in women with severe asthma controlled by oral steroids, and in hepatic impairment

For both Levonorgestrel and Ulipristal:

- Women should do a pregnancy test if expected period is >7days late, or is lighter than usual or they receive enzyme-inducing medication
- Either can be used more than once per menstrual cycle; avoid LNG within 5/7 of UPA
- Either can be used if there was also a UPSI early in the cycle as neither affects a pregnancy
- FSRH cautions that overweight women may have reduced the effectiveness of oral EC, esp LNG

KISS: Contraception in the Over 40s [FSRH](#)

Main issues in >40s

- While spontaneous pregnancy is rare >50y, effective contraception is required until menopause
- Pregnancy >40y is a higher risk for mother and baby
- Barrier protection should be considered to prevent STI even if contraception is not required
- Significant change to the bleeding pattern needs appropriate gynaecological assessment and investigation, even if using contraception
- Increased risk of CVD, obesity, breast and gynae cancer compared to women <40y

Suitability of contraception

- Copper coil, Mirena coil, all types of progesterone contraception including depot and potentially the COCP may be recommended depending on individual risk factors and age

Intrauterine contraception

- Copper coil - extended use until menopause is supported if inserted at ≥40y
- Mirena coil - extended use until age 55 if inserted ≥45y provided it is not being used for the progesterone component of HRT for endometrial protection (needs changing every 5y)
- Coils should be removed when they are no longer needed due to infection risk

Progesterone-only contraception (POC)

- The POP and implant do not ↑ risk of VTE, stroke or MI, nor affect bone mineral density (BMD)
- Depot can be used until 50y, then alternatives should be used. Depot injections cause an initial loss of BMD, but this is not repeated or worsened by the menopause.

Combined hormonal contraception (CHC)

- Can only be used up to age 50y, assuming a lower risk using UKMEC criteria
- Smokers should stop CHC aged 35y - at this age risk of mortality becomes clinically significant
- COCP should be levonorgestrel or norethisterone 1st line due to ↓ VTE risk, and ≤30µg ethinylestradiol due to ↓ VTE, CVD and stroke risk
- Can be used in extended or continuous regimes, also to control menstrual or menopausal Sx
- ↓ risk of ovarian and endometrial cancer, slightly ↑ the risk of breast cancer

When is contraception no longer needed?

- Most women can stop contraception aged 55 after this conception is exceptionally rare
- Menopause is a clinical diagnosis after 1 year of amenorrhoea, FSH is not required in most
- If on POC, women >50y can have FSH to check menopausal status if needed
- If on CHC or HRT, FSH is inaccurate - hormone tests cannot be used to base a decision to stop
- If ≥55y a woman does not want to stop, methods could be continued providing the risks/benefits have been assessed and discussed

Can HRT be used with contraception?

- HRT cannot be used as contraception!
- Mirena up to 5 years can be used as endometrial protection as part of an HRT regime
- POC can be used for contraception alongside HRT but not endometrial protection
- CHC (if eligible + <50y) can be used as an alternative to HRT to relieve Sx and prevent ↓ BMD

HEAVY MENSTRUAL BLEEDING

Such a common problem and treatment pathways are well established. But are they effective for women over the course of their whole reproductive lives? New research adds to our knowledge base here. Meanwhile, we still need to be mindful of restrictions placed on ulipristal due to concerns about potential liver injury.

1 Primary Care Management of Heavy Menstrual Bleeding [BJGP 2022;72\(725\): e857](#)

There is a wealth of evidence showing the benefits of non-surgical management of heavy menstrual bleeding and the IUS has revolutionised this condition. However, most trials are relatively short, whereas the condition can last decades. A new paper published in the [BJGP](#) aims to answer what happens to women over 10 years of medical therapy - how many continue, how many need surgery, and how many stop.

The **ECLIPSE study** was a primary care-based RCT comparing levonogestrel IUS to usual medical care (NSAIDs or hormones) for women aged 25-50y (mean age 41) with heavy menstrual bleeding. The study lasted 2 years but data has been collected via a patient questionnaire at 5 and 10 years since treatment began. 206 women out of the original 571 cohort responded at the 10-year mark. The results showed:

- 51% were post-menopausal, 15.5% premenopausal, 15.5% perimenopausal, 16.5% had hysterectomy
- Only 6% were still experiencing HMB
 - 29% of women had surgery for HMB over the 10 years (vs 20% at 5y, 10% at 2y)
 - 27% were still managed using IUS at 10 years
 - Between years 5 and 10, 42% reported not using any treatment for HMB
- The cumulative surgery free rate was 74% in the initial IUS group and 65% in the medical group
- There were significant and sustained improvements in QoL in both IUS and medical care groups

The authors conclude "*medical treatments for women with HMB can be initiated in primary care with improvement in quality of life and high likelihood of avoiding surgery 10 years later*".

2 Ulipristal Acetate for Fibroid Treatment [MHRA 2021](#)

Following reports of serious liver injury, the MHRA has restricted ulipristal for managing fibroids, marketed as Esmya (note no issues have been reported with EllaOne, the emergency contraceptive).

As a result, the MHRA have issued advice on Esmya, ulipristal for this indication:

- Esmya is indicated as intermittent therapy of moderate to severe uterine fibroid symptoms before menopause and when surgical procedures (including uterine fibroid embolisation) are not suitable or failed. It should not be used in women awaiting uterine surgery.
- It is now specialist initiation only, C/I in women with underlying liver disorders, and women need:
 - Baseline LFTs: do not use if LAT or AST >2x the upper limit of normal
 - 1st two treatment courses: monthly LFTs; further courses: LFTs prior to starting then PRN
 - End of each treatment course: LFTs after 2-4weeks
 - Stop if ALT/AST levels 2x the upper limit of normal, monitor closely, refer as appropriate
- Advise women about the features of liver injury and the need for urgent review if present

NB Learning Points and Quality Improvement Ideas: Heavy Menstrual Bleeding

- Women with heavy menstrual bleeding and additional symptoms should be offered an exam
- Drug treatment can be started without investigation in women with low-risk symptoms
- Levonorgestrel IUS is the first-line treatment for women with no identified pathology
- Audit your practice for patients receiving longer-term prescriptions of ulipristal for fibroids
 - Has a discussion re: liver injury been documented? Are they having monthly LFTs checks?

KISS: Heavy Menstrual Bleeding

[NICE NG88 2018](#), [BMJ 2015;351:h4887](#)

► **Initial assessment**

- History: discuss nature of bleeding, related symptoms (e.g. IMB), impact on QoL, comorbidities
- Exam if Hx of HMB & related symptoms - should be before investigations and IUS fitting
- Check FBC
- Screen for coagulation disorders if HMB since periods started + suggestive personal/FHx
- Other bloods (e.g. ferritin, hormones, TFTs) not routinely indicated, use clinical judgment

► **Further investigations**

- ?submucosal fibroids/polyps/endometrial pathology - refer for hysteroscopy +/- endometrial biopsy
- ?larger fibroids - pelvic USS
- ?adenomyosis - transvaginal USS preferred

► **Drug treatment, tailored to individual choice**

- **Levonorgestrel IUS:** first-line treatment
 - Supported by the ECLIPSE trial, but nearly 1/3 of women want it removed within a year
 - Advise about anticipated changes in bleeding pattern, wait for 6 cycles to see benefit
- **Non-hormonal options:** **tranexamic acid** (now available OTC as Cyklo-F or Femstrual – ask before you prescribe!) and/or **NSAID**
 - NSAIDs obviously preferred if dysmenorrhoea, equal efficacy
- **Hormonal options:**
 - **Combined OCP:** if no contraindications
 - **Progestogen:** cyclical may suppress menstruation
- **Ulipristal** use is currently not recommended for management of fibroids due to concerns over serious liver injury - current users should not have further prescriptions [DSU '18](#)

► **Surgical options**

- Uterine artery embolisation
- Uterine myomectomy - can preserve fertility
- Hysterectomy
 - In women with HMB alone and a uterus no bigger than a 10-week pregnancy, endometrial ablation should be considered before hysterectomy
 - Hysterectomy has a higher satisfaction rate than endometrial destruction or the IUS, so worth discussing if the IUS fails

MISCARRIAGE AND ECTOPIC PREGNANCY

General practice is usually the first place women seek help when they are unfortunate enough to have a miscarriage. Like the patient, the clinician may feel quite helpless - yes we can provide empathy and compassion and send the person onwards to the Early Pregnancy Unit but we also know that there is nothing that medicine can offer to alter the outcome for these women.

However, this changed in 2020 with the publication of the PRISM RCT, exploring the role of vaginal progesterone in preventing miscarriage in women with early bleeding during pregnancy. The results of these have now led to an important update to [NICE guidance on Ectopic Pregnancy and Miscarriage](#), published in November 2021.

► **PRISM RCT** [Health Technol Assess 2020;24\(33\):1](#)

This study aimed to clarify the signals from smaller, poor-quality trials which had suggested that progesterone supplementation, given its importance in the maintenance of a healthy pregnancy, may reduce rates of miscarriage in women with bleeding in early pregnancy.

This was a UK-based RCT across 48 hospitals, recruiting women aged 18-39y with early pregnancy bleeding. 4153 were randomised to either vaginal micronised progesterone 400mg bd or placebo from the time of assessment to 16 weeks gestation. The primary outcome was live births at ≥ 34 weeks.

■ **Results**

- Live birth rate = 75% in the treatment group and 72% in placebo, $P = 0.08$, a non-statistically significant result.
- However, the prespecified subgroups of women with prior miscarriage showed important improvements
 - 1 or 2 miscarriages: 76% progesterone vs 72% placebo
 - 3 or more miscarriages: 72% vs 57%
- No difference in adverse events between these groups.

► **NICE Ectopic Pregnancy and Miscarriage** [NICE NG 126, Nov 2021](#)

This study has led to changes in the NICE guidance for which it is important for primary care clinicians to be aware.

- **Offer vaginal micronised progesterone 400 mg twice daily to women with an intrauterine pregnancy confirmed by a scan, if they have vaginal bleeding and have previously had a miscarriage.**
- **If a fetal heartbeat is confirmed, continue progesterone until 16 completed weeks of pregnancy.**

This highlights the need for clinicians to ensure this option is discussed - many women after multiple miscarriages may have a sense of inevitability about the outcome of early bleeding - and facilitate timely onward referral for women with early pregnancy bleeding and prior history of miscarriage so they may be assessed and considered for treatment. The term 'offer' is a strong recommendation indicating solid data and that women in this situation may benefit from treatment and should be given it if they wish.

NICE notes that while the PRISM RCT demonstrates potential benefit in women with early bleeding and a history of miscarriage, data also shows that this treatment is not effective in women with bleeding and no previous miscarriage, nor as a prophylactic in women with recurrent miscarriage and no bleeding.

Elsewhere the guideline is essentially unchanged from the 2012 version.

KISS: Management of Ectopic Pregnancy

[NICE NG 126 Nov 2021](#)

- When assessing women of reproductive age be aware that they may be pregnant and symptoms & signs of ectopic pregnancy may resemble other common conditions, e.g. UTI, gastroenteritis
 - Have a low threshold for checking a urine pregnancy test

Referral

- *Haemodynamically unstable or concern about the degree of pain or bleeding?*
 - Refer directly to A&E
- *+ve pregnancy test + pain and abdo tenderness, pelvic tenderness or cervical excitation?*
 - Refer immediately to early pregnancy unit or OOH gynae unit
- *Bleeding or other features of early pregnancy complication + pain/gestation uncertain age or >6 weeks*
 - Refer to early pregnancy unit using clinical judgement to decide on the urgency
- *Bleeding <6/40 with no pain & no risk factors for ectopic (e.g. previous history)*
 - Use expectant management and advise:
 - Return if bleeding continues or pain develops
 - Repeat a urine pregnancy test after 7-10 days & return if it is +ve
 - A negative pregnancy test means that the pregnancy has miscarried
- Refer women with worsening symptoms or signs which could be ectopic pregnancy to EPU or OOH gynae using clinical judgement to decide if it should be immediate or within 24hr

Symptoms & signs of ectopic pregnancy

- Beware atypical presentations are common
- **Symptoms**
 - **Common:** abdominal or pelvic pain, amenorrhoea or missed period, vaginal bleeding with or without clots
 - **Additional:** breast tenderness, gastrointestinal symptoms, dizziness, fainting or syncope, shoulder tip pain, urinary symptoms, passage of tissue, rectal pressure or pain on defecation
- **Signs**
 - 'more common': pelvic tenderness, adnexal tenderness, abdominal tenderness
 - **Additional:** cervical motion tenderness, rebound tenderness or peritoneal signs, pallor, abdominal distension, enlarged uterus, tachycardia (HR >100) or hypotension (BP <100/60), shock or collapse, orthostatic hypotension

COMMON PROBLEMS IN PREGNANCY

In August 2021 NICE produced a new guideline on ante-natal care ([NG 201](#)). The vast majority of it was aimed at midwifery and obstetric services setting out the recommended service provision and scheduling for pregnant women, however, there was a useful section towards the end on 'common problems in pregnancy' which was a useful guide to troubleshooting the sorts of problems pregnant women commonly get including nausea and vomiting, and we which see regularly in routine (and emergency) primary care:

KISS: Common Problems in Pregnancy [NICE NG201 Aug 2021](#)

Heartburn:

- **General advice:** Consider raising the head of the bed; eat dinner well before bedtime; address lifestyle risk factors - smoking, alcohol, caffeine, chocolate, and fatty foods.
- Trial **antacid or alginate**.
- Some limited evidence **acupuncture** may be of help.
- **?PPI:**
 - Now commonly used and overall generally thought to be safe - Latest advice from [UKTIS here](#):
 - Overall safety data is reassuring.
 - However, need to note data is very limited for PPIs other than omeprazole.
 - [SPC](#) for omeprazole states it can be used in pregnancy.
 - There is some limited evidence PPIs may be effective in pregnancy but NICE assessed that the evidence was of low quality and not good enough to make recommendations about use, however, they do accept that *'PPI use in the first trimester is not harmful to the baby'*.

Symptomatic vaginal discharge:

- Advise women vaginal discharge in pregnancy is common and reassure them if no other symptoms.
- However if associated with itching, soreness, unpleasant smell or dysuria *consider* a vaginal swab if the cause is not clear (if the cause is clear e.g. typical thrush investigation may not be needed).
- If STI is possible refer for an STI screen.
- **Vaginal candidiasis:**
 - Use topical azoles e.g. clotrimazole or econazole - [PHE/NICE guidance](#) suggests 7 day courses are more effective than shorter courses.
 - Avoid oral azoles.
- **Bacterial vaginosis:**
 - Limited evidence to guide treatment regarding benefits and harms.
 - Oral or vaginal antibiotic treatments are BOTH recommended - usually oral metronidazole or vaginal metronidazole or clindamycin.
 - Click here for [PHE/NICE guidance](#) - note metronidazole 2g stat dose should be avoided in pregnancy.
 - Latest [evidence from UKTIS](#) on metronidazole states *'available data, which are almost exclusively based on oral exposure, do not indicate an increased risk of congenital malformation, low birth weight, intrauterine death, or neonatal complications with metronidazole use in human pregnancy'*.

Pelvic girdle pain:

- NICE advise referral to 'physiotherapy services' rather than a physiotherapist per se, as they note remote assessment and advice/management may be appropriate in some cases, negating the need for face-to-face appointments.
- This should include **exercise advice and/or a non-rigid lumbopelvic belt** - there is RCT evidence this can reduce pelvic girdle pain compared to exercise and general information alone.

KISS: Management of Nausea and Vomiting in Pregnancy**BMJ 2018;363:k5000, RCOG GTG 69, MHRA January 2020 NICE NG201 Aug 2021****Diagnosis:**

- **Nausea and Vomiting of Pregnancy (NVP):** Nausea +/- vomiting onset in 1st trimester where other causes are excluded.
- **Hyperemesis gravidarum (HG):** Protracted NVP + >5% pre-pregnancy weight loss, dehydration and electrolyte imbalance.
- **Red Flags:** impaired renal function, severe electrolyte disturbance, cognitive effects consistent with Wernicke's or central pontine myelinolysis, an indication of suicidal ideation.
- Consider the **psychological aspect of NVP** - up to 20% develop PTSD symptoms.
- Symptoms peak at W9-11, early treatments may appear to become ineffective but it is actually the condition getting worse and additional treatment may be required.

Investigations in primary care:

- Urine dip – 1+ ketones = ketonuria; send MSU; BMJ reports ketonuria is an unreliable marker.
- Bloods - U&E, FBC, blood glucose.
- USS – to confirm viable IUP, exclude multiple pregnancies and trophoblastic disease.

Treatment:

- Mild NVP should be managed in the community with anti-emetics.
- **Ginger and acupressure/puncture** are safe and may improve mild/moderate NVP.
- There is safety and efficacy data for antihistamines and phenothiazines so these are 1st line based on RCOG guidance, although none are licensed for use in pregnancy.
- NICE recommend we discuss pros/cons of the different antiemetics using their [summary table](#).
- **Doxylamine/pyridoxine** combination is the only licensed drug for NVP but NICE note the evidence to be of low quality and it did not show a convincing effect on symptom improvement for this drug.
- The RCOG recommend we consider **combinations of antiemetics** if single agents are ineffective.
- Be aware **extrapyramidal symptoms and oculogyric crisis can occur with phenothiazines and metoclopramide** – if this occurs stop these immediately. While metoclopramide is effective, given these risks it should be used only 2nd line and for max. 5 days.
- **Ondansetron** is effective but there is now evidence of a small increased risk of cleft palate (3 extra cases/10,000 births) if used in the 1st trimester although data on cardiac defects reassuring.
- **Consider treating reflux** - occurs in 80% with severe NVP - omeprazole is licensed for use in pregnancy and has reassuring safety data - [click here for latest from UKTIS](#) on PPIs.
- **Admit if:** Continued N+V and inability to keep down oral antiemetics, ketonuria and/or weight loss >5% despite oral antiemetics, red flags or if confirmed or suspected comorbidities.

Antiemetic options based on RCOG ([click here](#) for NICE summary table on pros/cons):

- **1st line:**
 - **Cyclizine** 50mg 8 hourly (PO or IM)
 - **Prochlorperazine** 5-10mg 6-8 hourly PO; 12.5mg 8 hourly IM; 25mg daily PR
 - **Promethazine** 12.5-25mg 4-8 hourly PO, IM or PR
 - **Chlorpromazine** 10-25mg 4-6 hourly PO or IM; 50-100mg 6-8 hourly PR
- **2nd line:**
 - **Metoclopramide** 5-10mg 8 hourly PO or IM (max 5 days)
 - **Ondansetron** 4-8mg 6-8 hourly PO
 - **Domperidone** 10mg 8 hourly PO (note **NOT recommended by NICE**)

VAGINAL DISCHARGE

Thrush is a hugely common cause of contact with primary care. As a new paper in the BMJ ([2020;369:m1995](#)) explains for a significant minority this can be frequent and troubling, affecting quality of life as much as conditions such as asthma and migraine. So how can we ensure we have the correct diagnosis and optimally treat these women?

Recurrent Vulvovaginal Candidiasis [BMJ 2020;369:m1995](#)

This new [BMJ paper](#), written by GPWSIs in gynaecology and based on the [latest BASHH guidance](#), sets out some useful tips for managing women with recurrent genital symptoms. It highlights that the public frequently self-treat recurrent episodes assuming it is thrush but 50% of the time the diagnosis is incorrect. The truth is we may be reaching for the clotrimazole a bit too often as well...

Background: Recurrent Vulvovaginal Candidiasis (RVVC)

- Affects 9% of 25-34yo women; 20% of women with 1 episode of candida will have 4+ over 1 year
- The negative impact on QoL is equal asthma, migraine, etc.
- There are >20 types of Candida that cause infection in humans:
 - Candida albicans causes 80%, C. glabrata causes most non-albicans candidiasis
- **Symptoms and signs:**
 - Vulval itching and non-offensive discharge (often curd-like but may be thin or absent)
 - Soreness, burning, superficial dyspareunia, cyclical symptoms
 - Erythema, fissuring, swelling/oedema, satellite lesions, excoriation
- **Causes and risk factors:**
 - Antibiotics (30% occurrence with oral or intravaginal), oestrogen exposure (hence more common in reproductive age), HRT, co-morbidities (e.g. diabetes), vaginal douching, non-breathable clothing, panty liners, sexual activity (vaginal dryness predisposes so water-based lubricants are recommended)
 - Both iron deficiency and combined hormonal contraceptives have conflicting data regarding risk

► How should we assess possible RVVC?

While it may be reasonable to empirically treat a single episode of presumed thrush, not every vulval problem is candida, so where symptoms have failed to resolve or recur further assessment is necessitated including examination and consideration of laboratory investigations.

The differential diagnosis includes (not an exhaustive list):

- Dermatitis/eczema, lichen sclerosus, other infections, vulvodynia, aerobic vaginitis, cytolytic vaginosis

The authors set out key findings of some of the more common diagnoses found on examination:

Symptom	Candidiasis	Lichen sclerosis	Vulvodynia	Contact dermatitis	Chronic lichen simplex
Erythema	+	+	+/-	+	+/-
Fissuring	+/-	+/-	-	+/-	+/-
Discharge	+	-	-	+/- (due to inflamed skin)	-
Oedema	+/-	-	-	+/-	+/-
Other	Satellite lesions	Pallor, atrophy	Cotton swab induced tenderness	Often exudate	Lichenification

The paper does not mention bacterial vaginosis in the differential diagnosis, presumably due to the markedly different presentation - as Clinical Knowledge Summaries states: "*BV is characterised by a fishy-smelling, thin, grey/white homogenous discharge that is NOT associated with itching or soreness*".

Other infections to consider include herpes simplex virus and trichomonas vaginalis.

► What investigations are recommended?

The authors recommend the following approach:

- **Take a vaginal swab for MC&S.** Specifically request for the subtype of candida and sensitivities.
 - If an examination is not possible or declined, self swab can be offered - NB: a UK case-control study ([BJGP 2017;67:e824](#)) showed vaginal self-swabs for candida and BV had high sensitivity (95.5%, 88.5%) and were a viable alternative to clinic-based swabs.
 - Note that asymptomatic colonisation occurs in 20% of women of reproductive age (and 30-40% during pregnancy), and is likely if a positive culture result is found with absent neutrophils on microscopy. Treatment is not required in the absence of symptoms or signs, or alternative cause of symptoms and simple colonisation.
- **Consider swabs for other STIs based on individual risk assessment** and clinical features.
- **Offer a blood test to exclude diabetes.** Consider checking for iron deficiency anaemia.

► When can we diagnose recurrent candidiasis?

The BASHH definition is:

- ≥4 episodes in 12 months
- 2 episodes confirmed by MC&S when the patient is symptomatic, of which 1 must be a culture

► How do we treat RVVC?

See KISS for full details of treatment. Interestingly, while intravaginal imidazoles and oral azoles are equally effective because fluconazole is 7-30x cheaper BASHH recommends this as first line. For RVVC this will mean weekly fluconazole 150mg for 6 months. Risks, however, are very low and no specific monitoring is recommended.

In past, many of us will have co-prescribed topical imidazole creams for a 'new' episode of candidiasis, however, BASHH does not suggest this as a strategy citing that clearance rates of ~80% with first-line oral or intravaginal treatments.

Many patients will already have tried various alternative treatments given their persistent and troubling symptoms. Probiotic use is common, although there is no good-quality data to support this approach. Likewise, dietary modification (e.g. reducing foods which use yeast) is not supported by data. A discussion around this can be a useful opportunity to discuss general lifestyle measures.

KISS: Management of Recurrent Vulvovaginal Candidiasis

BMJ 2020;369:m1995, BASHH 2019

Defined as:

- ≥4 episodes in 12 months
- 2 episodes confirmed by MC&S when the patient is symptomatic, of which 1 must be a culture

Treatment

- **General measures**
 - *Avoid potential triggers and focus on skincare:* "hypoallergenic regime" - avoid soaps, etc., wash with emollients as soap substitute
- **Candida albicans infection**
 - **Induction:** fluconazole 150mg one capsule every 3 days for 3 doses, then...
 - **Maintenance:** fluconazole 150mg one capsule weekly for 6 months
 - If fluconazole is C/I a topical imidazole for 7-14 days followed by either clotrimazole pessary 500mg weekly or oral itraconazole 50-100mg daily for maintenance period.
 - *Pregnancy:* oral treatments should not be used.
 - *Breastfeeding:* a single dose of fluconazole 150mg is safe but if higher or repeated doses are required then topical treatments should be used.
- **Non-albicans infections:** may respond to azoles but nystatin pessaries are most effective
 - Nystatin pessaries 100000units at night for 14 consecutive nights per month for 6 months
 - Check local formularies for availability - may be special order/secondary care issue only

Poor response

- Often due to a combination of factors including non-albicans species, azole resistance, or another/additional cause.
- Oral antihistamines (e.g. cetirizine 10mg daily for 6/12) can be helpful, especially in patients with atopy.
- Referral to gynae or GUM should be considered where symptoms are not controlled.

Partner treatment is not recommended if asymptomatic and treating males partners does not prevent recurrence in women.

NB Learning Points: Vaginal Discharge

- Self-taken low vaginal swabs have similar efficacy to clinician-taken high vaginal swabs and may provide a useful alternative for some patients
- Consider dequalinium to treat BV: an effective non-antibiotic topical vaginal antimicrobial antiseptic
- Simple litmus paper can clinch diagnosis without the need for other investigations
- Advise patients that environmental/lifestyle modification will help recovery:
 - Avoid soaps, irritating creams & sanitary towels; use loose cotton underwear.
- If symptoms deteriorate while using treatment for thrush, consider alternative diagnoses including contact dermatitis from the anti-fungal treatment
- Fluconazole should not be used in pregnancy, and with contraception if childbearing age

KISS: Management of Acute Candidiasis and Bacterial Vaginosis

[BASHH 2019, CKS BV 2018](#)

Candidiasis - treatment for asymptomatic carriers is generally not recommended

■ Acute candidiasis

- BASHH recommends oral fluconazole 150mg single dose as the preferred regime
- If C/I clotrimazole pessary 500mg single dose if the preferred alternative
 - In severe candidiasis the treatment used should be repeated on day 4
- BASHH does not suggest the addition of externally-applied imidazole cream but comments that low-potency corticosteroid creams possibly accelerates symptomatic relief in severe candidiasis.

Bacterial Vaginosis

- The commonest cause of abnormal vaginal discharge in women of reproductive age
- Can occur and remit spontaneously, caused by an **overgrowth of mixed anaerobic organisms** that replace normal lactobacilli. This leads to an **increase in vaginal pH >4.5**.
- Empirical treatment is reasonable if classical features and low risk for STI or alternative diagnosis
- Investigations
 - Testing pH is useful - >4.5 (usual vaginal pH is 3.5-4.5) is suggestive however it is not specific and is also seen with trichomoniasis. Vaginal swab can be taken if doubt persists
- **First line treatment: oral metronidazole** (400mg bd for 5-7days, or 2g single dose)
 - Consider dequalinium as an alternative to antibiotics - similarly effective and low rates of side effects, and avoids excessive antibiotic exposure ([DTB 2017;55:54](#))
 - Intravaginal metronidazole gel (0.75%) od 5/7 or clindamycin cream (2%) od 7/7 are alternatives
 - Testing/treatment of male partners is not indicated; consider for female partners.
- BV in pregnancy is associated with poorer outcomes. If symptomatic treat with oral metronidazole 400mg twice daily for 5-7days. Avoid higher doses. If asymptomatic discuss with obstetric team whether treatment is required.
- **Recurrent BV**
 - Reconsider diagnosis. Examine and take swabs for culture. Discuss contributing factors.
 - If treating consider prescribing an alternative treatment option to that previously used, e.g. intravaginal treatment if oral metronidazole previously used.
 - For frequently recurring, proven episodes (e.g. 4+ per year) despite appropriate management, metronidazole gel used as suppressive therapy is an option if the clinician has suitable experience in managing the condition, otherwise discussion with gynae or GUM is advised.

BREAST PAIN

This common problem frequently results in a referral to specialist breast clinics, but does mastalgia imply serious underlying pathology? A new BJGP Clinical Practice review ([BJGP 2020;70:419](#)) examines the causes of breast pain and an approach to assessment in primary care.

► Is mastalgia associated with breast cancer?

This is a primary concern of patients. Recently we have had very reassuring evidence from [BJGP 2022; 72 \(717\)](#):

- This was a large prospective cohort study of 10,830 women referred to secondary care with breast pain (18%), lumps (62%), nipple symptoms (4%) and other symptoms (15%).
- Of the 1972 women with breast pain, the breast cancer incidence was 0.4% vs ~5% in all other groups.
- Given the background incidence of breast cancer is ~0.8% in asymptomatic screened women the conclusion of this study is that there is **no association between breast pain and breast cancer**.

► What are the causes of breast pain?

Breast pain can be split into three groups: cyclical, non-cyclical and extra-mammary.

1. Cyclical

- Two-thirds of breast pain with true mastalgia
- The mechanism remains unclear
- Worst in the week prior to menses then improves

2. Non-cyclical

- One-third of true mastalgia
- Does not follow menstrual cycle
- Causes include large pendulous breasts, cysts, pregnancy, thrombophlebitis, trauma, Hx of breast surgery
- Also acute causes such as mastitis and abscesses - usually accompanied by erythema, swelling, focal pain, etc. Inflammatory cancer may have similar signs but usually, pain is absent and there is a lack of improvement with antibiotics

3. Extra-mammary

- Reported as the most common type of breast pain referred to secondary care.
- MSK pain, costochondritis, radicular pain secondary to OA. T3-5 supply the breast, T4 the nipple.
- May be linked to a wider differential as well, e.g. pleuritic pain, IHD, gall bladder disease.
- We should look to make a positive diagnosis of extra-mammary after exclusion of true breast disease by clinical assessment.

► How should we assess women with breast pain?

KISS: Assessment of Women with Breast Pain [BJGP2020;70:419](#)

History

- Characterise type of pain, identify cyclical elements, separate extra-mammary causes
- Exacerbating or relieving factors?
- Other breast symptoms: lump, breast changes, nipple discharge, axillary symptoms.
- FHx? DHx, e.g. hormonal therapies?

Examination

- Key is excluding signs of malignancy. Also assess for mastitis, cysts. Ask re: pregnancy risk.
- Chest wall tenderness? Best assessed in lateral semi-recumbent position: moves the breast away from chest wall, allowing more accurate palpation. Localised, reproducible pain confirms the Dx.

KISS: Assessment of Women with Breast Pain [BJGP2020;70:419](#)

Patterns to aid discrimination between causes of breast pain:

Cyclical	Non-cyclical	Chest wall pain
Starts 1-2/52 prior to menses, improves with menstruation	No temporal relationship to menses	Unilateral, activity related
Dull, heavy, or aching	Constant or intermittent	Very lateral or medial in the breast
Bilateral	Unilateral and variable	Reproducible pain with pressure on causative area on chest wall
Poorly localised, extends to axilla		

► How should we manage breast pain?

Breast pain often resolves without intervention but in persisting cases, there are supportive and interventional options which can be undertaken in primary care.

■ Supportive Measures

- *Careful examination to exclude palpable abnormalities, explanation and support can be therapeutic, including stressing the lack of association between pain and cancer.*
- *Acknowledging the impact such symptoms can have on an individual is helpful and further aids reassurance.*
- The author's counsel that *a good-fitting bra is essential* and may entirely alleviate the pain (we will let particularly male GPs consider the best way to approach this subject).
- *NICE CKS do not recommend dietary modifications* due to lack of evidence but the paper suggests some may benefit from avoidance of caffeine and foods containing other methylxanthines (e.g. chocolate) and increasing dietary soya.

■ Interventional measures

- *Topical NSAIDs should be considered a first-line treatment.*
- Oral simple analgesia may be of benefit.
- *Evening primrose oil has no proven efficacy compared to placebo.* The paper does not recommend it routinely.
- Oral contraceptives have traditionally been considered linked to cyclical breast pain but NICE CKS quotes there is little evidence of a temporal link to initiation and data showing they may actually help ease pre-menstrual pain, so stopping/switching existing treatments is not advised.
- It also does not advise diuretics, antibiotics, tibolone, pyridoxine or vitamin E.
- Bromocriptine is used with apparent good effect (NB: unlicensed, may be better initiated by 2dry care)

► When should women be referred?

Women should be referred if 1) the pain is affecting quality of life or sleep AND 2) has been ongoing for >3 months despite first-line treatment.

The goal of referral is to access more specialist measures (e.g. tamoxifen, SSRIs) rather than due to concern over cancer risk, although imaging is often undertaken in secondary care to reassure the patient.

NB Learning Points & Quality Improvement Ideas: Breast Pain

- Breast pain is not associated with cancer. Examination and reassurance may be all that's required.
- Pain may be cyclical or non-cyclical or extra-mammary originating from the chest wall.
- A well-fitted bra, topical NSAIDs and simple analgesia may all be effective treatment options.
- Pain affecting quality of life which has failed to respond to treatment for 3/12 merits referral.

PERINATAL MENTAL HEALTH

The RCGP report that up to *1 in 5 women and 1 in 10 men have mental health problems during the perinatal period, half of which go undiagnosed*. This can have obvious consequences for bonding between parent and baby, but also with relationships and long-term development of the child. Recognition and appropriate treatment can help to prevent this.

While clinicians are becoming more aware of depression and anxiety during pregnancy, as a very recent paper in the BMJ ([2019;367:l6574](#)) highlights, we may not consider perinatal OCD, a surprisingly common condition. Regardless of the nature of mental illness, many women who become pregnant while be taking psychotropic medication. A new review in the DTB ([2019;58:8](#)) discusses the best approach to these medications during the perinatal period.

1 Intrusive Thoughts in Perinatal Obsessive-Compulsive Disorder [BMJ 2019;367:l6574](#)

This interesting paper (co-authored by Stephanie DeGiorgio from NB) highlights how common and troubling OCD can be in and around pregnancy. 2-4% of women are affected, which is more common than background rates. Fathers may be affected as well.

Often the condition can be managed in primary care but is often missed as new parents often hide how they feel due to fears of stigma and shame, and it requires careful distinction from other perinatal mental health disorders to effectively assess risk.

What's normal? What's not?

Intrusive thoughts are a normal phenomenon and most of us will have these time to time (90%). These flash into our head, are not significantly personally meaningful, are easily dismissed and we do not act on them.

However, in people with OCD, this thought process is perceived as a threat or danger, driving anxiety and compulsions to reduce the sense of threat or responsibility. This can severely impact on day to day activity.

Do other mental health conditions drive intrusive thoughts?

Intrusive thoughts are associated with a range of mental health disorders, not just OCD. Postnatal psychosis and severe depression need to be excluded. Also consider PTSD, tokophobia (fear of pregnancy) and GAD. The latter has features which may overlap with OCD but is characterised more by a chronic worry which flits between subjects rather than a persistent focus on single issues as with OCD.

In depression and (obviously) psychosis intrusive thoughts are usually 'ego-syntonic' - the woman may feel she wants to act on them - and is associated with increased risk to mother and/or child.

Interestingly, the thoughts linked to OCD are 'ego-dystonic' (meaning not fitting with the person) and even though there may be thoughts focusing on harming the baby (possibly accidentally, e.g. by not caring for them adequately) this type of thought process does not indicate risk. The paper reports no known recorded cases of patients with OCD acting on their intrusive thoughts towards others. Clinicians misunderstanding these thoughts, often due to misdiagnosis, can lead to problems, for example, if they result in a safeguarding referral which is actually unnecessary. And while harm to others is very unlikely we need to remain mindful of, and assess for, suicidal thoughts and intent.

However, while direct safety is rarely an issue with perinatal OCD, there can be serious secondary risks. The actions taken to deal with the compulsions may lead to neglect and poor bonding.

How should we manage perinatal OCD?

KISS: Management of Suspected Perinatal OCD [BMJ2019;367:l6574](#)

Providing a diagnosis of OCD, with appropriate explanation, can be therapeutic for the patient.

- **Requires a sensitive discussion about intrusive thoughts**
 - Many may not freely disclose the nature of their thoughts, especially if they are not familiar with the clinician doing the questioning. The patient may feel more comfortable simply talking broadly about their "intrusive thoughts of harm"
 - The paper provides some example of questions to ask:
 - *"Sometimes people experience distressing thoughts or images of harm coming to their baby, accidentally or deliberately. Do you have those?" "Most parents have occasional thoughts that their baby may become ill or even die. Do you have lots of these kinds of thoughts?" "What do you feel you want to do when you have these thoughts?"*
- **Assess the degree of impairment due to intrusive thoughts**
- **Refer**
 - *Refer for CBT in primary care in mild and moderate cases*
 - For severe cases refer to a specialist perinatal mental health team.
 - Interventions should start in <6 weeks (NICE recommendation)
 - Very severe cases may be referred on to maternity units
- **Escalate safety concerns where appropriate**
 - Given people with OCD do not act on their intrusive thoughts, thoughts of deliberate harm in this group does not automatically necessitate social services referral.
 - Where uncertainty exists discuss with the practice safeguarding lead and mental health team.
- **Consider medication**, e.g. SSRI, especially if comorbid depression and/or anxiety. It is reasonable to initiate this in primary care as per NICE guidance for OCD ([CG31, 2005](#)).
- **Encourage the patient to use their own support systems and signpost to information:**
 - Best Beginnings Baby Buddy App - free app for all parent, but includes info on OCD - [click here](#)
 - The [RCGP Perinatal Mental Health Toolkit](#) has links to dozens of resources for clinicians and parents
 - Break Free from OCD - self-help book available free via the [Reading Well programme](#)

2 Prescribing for pregnancy in women with mental health diagnoses [DTB 2019;58:8](#)

Many women taking psychotropic medication for a pre-existing condition have concerns that the treatment may affect the developing fetus during pregnancy or be present in breast milk after birth. This new DTB paper reviews common medications in women with mental health issues and the available data on potential adverse effects on fetus or infant.

The authors remind us of the guiding principles of prescribing for any patient: using the lowest effective dose for the shortest duration necessary with monotherapy if possible. In reality, many of these medications will be required before, during and well after the pregnancy. We should also bear in mind that due to metabolic and circulatory changes during pregnancy, higher doses of some medications may be required.

Regarding safety data, we must consider that there is a background congenital malformation rate of 2-4% and studies are often subject to bias or confounding which may influence the results and explain historic concerns for some drugs.

KISS: Medicines for mental health diagnoses during pregnancy [DTB 2019;58:8](#)

- **SSRIs**
 - **ADVICE:** *women do not need to discontinue SSRIs during pregnancy or for breastfeeding.* Conflicting data regarding ↑ risk of congenital malformation with in utero exposure.
 - V small increased risk of persistent pulmonary hypertension of the new-born if used >20/40
 - Transient neonatal withdrawal syndrome in up to 30%, usually within 8-48h post-birth and resolves by 72h. Usually mild, which only requires observation. F/up data at 8/12 shows no persistent issues.
 - Breastfeeding - some drug passes to milk. Sertraline has least but all SSRIs considered suitable.
- **Quetiapine, olanzapine, aripiprazole**
 - **ADVICE:** *continue as required, screen for gestational DM. Considered ok to breastfeed.*
 - Quetiapine appears safe throughout gestation, limited data for others but 1st-trimester exposure appear ok.
- **Risperidone**
 - **ADVICE:** *in women with mental illness who are well controlled, benefits may outweigh risks of changing.*
 - Increased risk of congenital malformation (5%v3%).
 - May have withdrawal features after birth termed poor neonatal adaptation syndrome. Needs monitoring.
- **Lithium**
 - **ADVICE:** *no advice to stop/reduce dose. Check levels frequently, aim for the upper level of the therapeutic window due to increased risk of relapse post-delivery. Breastfeeding is usually not advised due to high levels.*
 - Possible small increased risk of cardiac defect with 1st-trimester exposure. Later gestation use may be linked with polyhydramnios.
 - Lithium levels require more frequent monitoring during pregnancy, increased dose often required
- **Sodium valproate**
 - **ADVICE:** *if conception while on drug continue medication and refer for urgent specialist assessment.*
 - Very high risk of congenital malformation (40%). Not to be prescribed to women of childbearing age
- **Benzodiazepines**
 - **ADVICE:** *no recommendation to stop during pregnancy. If breastfeeding required specialist input.*
 - No ↑ in congenital anomalies but possible ↑ in spontaneous miscarriage. Risk of neonatal withdrawal can occur 2-3/7 after birth. Needs observation.
 - Specialist input required for breastfeeding due to increase in neonatal adverse events. Short-acting preferred.

Perinatal Mental Health: Learning Points and Quality Improvement Ideas

- Perinatal OCD affects 2-4% of women during or after pregnancy. Ask about the presence of intrusive thoughts. Refer mild to moderate cases for CBT in the community, more severe cases should be referred to a specialist perinatal mental health team.
- Depression & anxiety are common in the antenatal & post-natal periods - SSRIs are largely safe during pregnancy & breastfeeding. Advice is not to stop them where they are helping with mental ill-health.
- Antidepressant use during pregnancy is associated with a small increased risk of children developing ADHD ([BMJ 2017;357;j2350](#)) and autism ([BMJ 2017;358;j2811](#)). Previous studies may have overestimated this risk.
- For mild and moderate depression psychological therapies are the first line
 - Do you know what services are available locally? What is access like?
- Women with severe mental illness should be managed by a specialist perinatal mental health team.
- The [RCGP Perinatal Mental Health Toolkit](#) is a valuable resource for information on services, management and specific issues such as suicide risk mitigation and personal illness as well

KISS: Management of Ante- and Post-natal Depression and Anxiety

BMJ 2016;352:i15547, NICE 2015 CG 192, SIGN 2012 CG127

Postpartum blues - >30%, **PN depression** – 10%, **PN psychosis** – 0.1%, but 1 in 2 if Hx of previous PN psychosis or bipolar

Assessment

- **Assess women at booking and post-natal 6-week check**
- Ask all women about previous mental health problems, perinatal illness and family history
 - Consider referral to perinatal mental health team as early as possible
 - The principal risks factors are depression or anxiety during pregnancy, past psychiatric history and poor social support, particularly the lack of a supportive partner.
- **Screen for depression and anxiety using:**
 - During the past month have you often been bothered by 1) feeling down, depressed or hopeless, and 2) having little interest or pleasure in doing things?
 - During the past month have you 1) been feeling nervous, anxious or on edge, and 2) not been able to stop or control worrying?
- If either is positive consider using the [Edinburgh Postnatal Depression Scale](#) and/or [GAD-7](#) scale for further assessment, in conjunction with clinical judgment
 - Tiredness, disturbed sleep, reduced libido are common perinatally but may mask more significant depressive symptoms, so ask about other symptoms particularly:
 - Feelings of worthlessness and hopelessness about the future; early morning waking before the baby, severe insomnia, poor appetite, weight loss

Referral - to specialist mental health services:

- There is a sudden change in behaviour or mental state/suspected postpartum psychosis – immediate assessment (<4hrs)
- History or current severe mental illness
- Current mild>mod mood disorder and 1st degree relative with Hx of bipolar or psychosis
- Not responding to treatment

Mild and Moderate Depression

- Women with persistent subthreshold depressive symptoms or mild>mod depression consider facilitated self-help and/or cognitive therapies
 - Include specific interventions if impairment of mother-infant relationship
 - If already taking an AD NICE suggests consider gradual withdrawal and initiation of psychological therapies; the BMJ would suggest probable continuation
- In mod>severe depression consider:
 - High-intensity psychological intervention, anti-depressants or both
- If a history of severe depression in past and presents with mild depression during pregnancy
 - Consider SSRI, SNRI or TCA
- **Antidepressants**
 - Data to guide us is generally poor; equally absence of negative data does not equal safety
 - Paroxetine and venlafaxine have higher risk of withdrawal symptoms in mother and the baby when used near delivery
 - For breastfeeding if starting a new AD sertraline and TCAs have the lowest levels present in milk; SIGNS reports fluoxetine, citalopram and escitalopram levels are very high and should be avoided if possible; doxepin should not be used

HYPERTENSION IN PREGNANCY

A fundamental part of good antenatal care is regular checks of the mother-to-be's blood pressure, always being mindful about the risk of pre-eclampsia. In 2019 NICE updated its guidance on hypertension in pregnancy ([NG133](#)), with several recommendations which are helpful for clinicians in general practice and the inclusion of the recent advancement in the assessment of pre-eclampsia: PIGF-based testing ([DG23](#)).

1 PIGF-based testing for Pre-eclampsia [NICE DG23](#)

This guidance was originally published in 2016 but will have been off the radar for most primary care clinicians. It recommends the use of PIGF-based testing between 20 - 34+6 weeks gestation to help rule out pre-eclampsia in women where it is suspected.

PIGF, or placental growth factor, is a protein linked with the development of new blood vessels in the placenta. Around 26-30 weeks levels of the protein peak, but in pre-eclampsia, this may be inappropriately low due to placenta dysfunction. The protein (or other related compounds) can be measured in the mother's blood and there are a range of commercially available tests.

NICE reviewed data which showed these are useful rule-out tests for pre-eclampsia, less useful to rule it in as interpretation of positive results is difficult, and overall a cost-saving for the NHS. These tests are now widely used in the maternity unit. In the latest NICE guideline on the management of hypertension, there are clear indicators of when testing is recommended.

2 Hypertension in Pregnancy: Diagnosis and Management [NICE NG133](#)

This guideline has recommendations for the whole spectrum of hypertension in pregnancy including pre-eclampsia and so understandably many of the recommendations are aimed at secondary care. However, in reality, the specialist follow-up for most women with gestational hypertension is lacking and reviews and decisions are often taken by the GP, so it is useful for us to have an understanding of management.

The most important areas for us are: when to treat chronic or gestational hypertension, what to use, what targets to aim for and how does treatment alter in the post-natal period. All the recommendations are summarised in the KISS below.

NB Learning Points & Quality Improvement Ideas: Hypertension in Pregnancy

- PIGF-based testing is a new secondary care-based technique to effectively rule out pre-eclampsia.
- Target blood pressure for hypertension in pregnancy is <135/85. Titrate antihypertensives if necessary.
- Postnatally treatment can be reduced when BP is <130/80.
- Audit women with gestational hypertension - did they have appropriate f/up?

KISS: Hypertension in Pregnancy [NICE NG133](#)

Reducing the risk of hypertension in pregnancy

- Educate women on features of pre-eclampsia, e.g. severe headache, visual issues, severe pain below ribs, vomiting, sudden swelling of the face, hands or feet
- **Advise women to take 75-150mg aspirin daily** from 12/40 until birth if high risk or >1 mod risk factor:
 - High risk: HT in previous pregnancy, CKD, autoimmune disease, T1 or T2DM, chronic hypertension
 - Mod risk: 1st preg, >40y, last preg >10y ago, BMI >35, FHx pre-eclampsia, multi-fetal pregnancy
- Check urine for proteinuria at antenatal appointments
 - 1+ protein is +ve (trace is ok) - proteinuria alone is a poor marker of pre-eclampsia risk
 - Dipstick interpretation generally hard, automated is better and advised if in hospital-setting
 - If +ve check ACR (+ve if ≥ 8) or PCR (+ve if > 30) - do not use first-morning void as it's less accurate

Managing Chronic Hypertension (CHT)

- More common due to obesity and shift to pregnancy at older ages
- **Offer referral to a specialist** to discuss risks and benefit of treatment
- **Offer lifestyle advice** on weight management, exercise, healthy eating and lower salt intake
- **Offer aspirin 75-150mg daily** from 12 weeks until birth
- **Offer PGI-based testing at 20-35 weeks if suspected pre-eclampsia**
 - e.g. sustained sBP > 160 , $\uparrow$ creat, $\uparrow$ ALT, $\downarrow$ plts, p. oedema, other signs, fetal compromise
- **If already on anti-hypertensives:**
 - ACEi and ARBs $\uparrow$ risk of fatal abnormalities/death and alternative Rx will be necessary
 - Stop within 2/7 if a woman becomes pregnant and offer alternatives
 - Risks are thought to be in 2nd/3rd trimester, data suggests no $\uparrow$ risk in 1st but data is limited
 - Thiazide / thiazide-like diuretics may also $\uparrow$ risk of abnormalities/complications - switch
 - Other antihypertensives have shown no increased risk but again data is limited
 - **Continue treatment if safe or switch unless BP $< 110/70$, or symptomatic hypotension**
- **If BP $> 140/90$ and not on treatment, offer antihypertensives - target = $< 135/85$**
 - Consider labetalol 1st line, nifedipine 2nd line, methyldopa 3rd line

Managing Gestational Hypertension (GHT)

- **Refer for specialist assessment**, they can assess for additional risk factors which need managing
- **BP $\geq 160/100$? Admit.** Note, bed rest is no longer recommended as a treatment.
- **Offer antihypertensive treatment and investigation if BP 140/90 - 159/109**
 - Target $< 135/85$. Review 1-2x/wk with BP, urine dip, weekly FBC, E+C & LFTs
 - Listen for fetal heart every A/N appt, USS at diagnosis then 28, 32 + 36 weeks
 - PIGF-based testing if suspected pre-eclampsia
 - Labetalol $>$ nifedipine $>$ methyldopa

Postnatal Management

- For CHT and GHT check BP D2, D3-5, then as clinically indicated; GP/specialist r/v at 2/52 & 6-8/52
 - **If BP $< 130/80$, reduce anti-hypertensives**
 - Duration of antihypertensive use is often about the same as the duration used during pregnancy
 - **If using methyldopa stop + switch within 2/7 of delivery** - linked to PN depression
 - If Hx of pre-eclampsia pt should be managed by hospital until BP controlled and BTs normalising
- **Antihypertensives**
 - Can transfer to breast milk but small amounts - ask the mother to monitor for signs of hypotension
 - **Offer enalapril to treat hypertension postnatally**, will need blood monitoring
 - **If black African/Caribbean family origin consider nifedipine or amlodipine** (if previously helpful)
 - Avoid diuretics and ARBs if breastfeeding/expressing, otherwise follow NICE HT guidelines

MESH IMPLANTS

One of the biggest controversies of the past few years and a really hot topic in the past 12 months has been the *use of 'mesh' for the treatment of stress urinary incontinence (SUI) and pelvic organ prolapse (POP)*. A [parliamentary review](#) reviewed the problems with mesh procedures and as a result, use of mesh is currently restricted around the UK for both POP and SUI.

The use of mesh has become commonplace and replaced traditional organ repair surgery. 'Mesh' is a broad term used to describe a number of different types of manufactured biological or synthetic implantable devices, usually made from polypropylene. The mesh often shrinks after implantation - by up to 50% after 4 weeks - and may become contaminated with bacteria - found on 1 in 3 removed meshes.

While for many women operations involving mesh are successful and problem-free, complications are well established. Data appears fairly reassuring for SUI, but for prolapse surgery the frequency of serious problems are only now, arguably far too late, becoming apparent.

Severe pain, dyspareunia and tissue erosion may result, necessitating surgical excision, made particularly difficult as mesh is designed to encourage local fibrosis to aid tissue support.

Many women have developed significant and often chronic complications due to mesh complications, with the official national reported figures likely to be a gross under-estimate. Unfortunately, as the complications were not widely understood, they were often missed in women, labelled with chronic pelvic pain (and the negative connotations associated with such as diagnosis) even as mesh eroded through their vaginal wall or abdomen.

► *What does the data show?*

There are currently 100000 litigation cases globally relating to mesh use. A very recent BMJ paper ([2017;359:j5515](#)) explains where things went wrong.

There are currently dozens of mesh products available. Unfortunately, not all mesh devices are the same: pore size, fibre configuration, coating and surface area all vary between products. Assessment of medical devices has always been an area of concern. Many new devices make it to market by stating substantial equivalence to existing proven products, both in Europe and the US. Data can be as limited as 'a critical narrative review'.

Lack of long-term data on mesh devices was recognised by Cochrane in 2005. Gradually evidence has accrued showing erosion rates as significantly higher than originally thought (~10%) for prolapse operations. Dyspareunia was also common (9%). Rates of complications for POP were significantly higher in mesh treatments than with traditional repair.

A telling finding by the BMJ paper was that when in 2012 the FDA requested 33 manufacturers to conduct new safety studies, 66% ceased market distribution of the mesh product instead...

Major reports from [NHS England](#) and a [Scottish Independent Review](#) indicate the future for mesh devices: the latter advised *mesh should not be used pelvic organ prolapse operations*, and NICE have stated mesh should only be used in a research context. In SUI following a [parliamentary review](#), there is now *immediate suspension of the use of surgical mesh throughout the UK* until a number of conditions are met to mitigate the potential risks. The latter recommendation was to the dismay of the British Society of Urogynaecologists who felt the decision was not based "on any scientific logic".

NICE [guidance published in 2019](#) but added a [supplemental paper](#) later in the year restricting the use of mesh in SUI and POP following an extension to the 'pause' on mesh use recommended by the government.

► *What do we need to do in general practice?*

Knowledge of the risk of complications with mesh-based operations and how they present is essential so that we can help women who develop problems.

NHS England provides simple guidance for GPs, that can be found [here](#). While the symptoms are common to many abdominal and gynaecological conditions, when present in the context of a mesh operation patients should be referred to the appropriate specialist gynae or urology team (the British Association of Urological Surgeons has produced a [national list](#)) and the adverse incident reported to the MHRA via the [Yellow Card system](#).

KISS: Symptoms and Signs of Mesh-Related Complications

[NHSE](#)

- Irregular vaginal bleeding
- Pelvic pain or swelling
- Dyspareunia
- Recurrent or new bladder and bowel symptoms
- “Prickling” feeling or pain in the vagina, worse on exercise.
- Buttock and leg pain
- Erosion of tape into vagina, bladder or bowel
- Tenderness on palpation of mesh
- Recurrence of prolapse or incontinence
- Vaginal adhesions

NB Learning Points and Quality Improvement Ideas: Mesh Implants

- Stress urinary incontinence operations involving mesh appear fairly safe, but significant complications in prolapse repair using mesh is common, ~10%
- Mesh repair of prolapse is no longer recommended and for SUI currently prohibited
- GPs and practice nurses need to be aware of the symptoms of mesh-related complications including abnormal vaginal bleeding, pelvic pain, dyspareunia, new bladder/bowel symptoms and abnormal examination findings
- Complications should be reported via the [Yellow Card system](#) and patients referred to a local specialist centre - [click here for list](#)
- For a simple but effective piece of quality improvement, review your practice population for women who have had mesh surgery
 - Put an alert on their notes so that clinicians are aware if they present with these symptoms to consider mesh complications

Patient Information

- The British Association of Urological Surgeons has [good information on stress urinary incontinence operations](#) including details frequency of complications
- For PILS on a variety of gynaecological conditions including non-surgical options [click here](#)
- In Scotland, there is also a dedicated NHS line which women affected by mesh implants can call themselves to report symptoms - [click here](#)

POLYCYSTIC OVARIAN SYNDROME

PCOS incidence is increasing with rising obesity rates. Associated long-term risks include cardiovascular disease and type 2 diabetes. An updated [international guideline for PCOS focused on the assessment of adolescents in 2018](#). The problem is that ovulatory dysfunction and irregular menstruation is common in the pubertal transition, which makes clinical assessment more complex, coupled with a high incidence of multi-follicular ovaries at this stage making ultrasound an unreliable marker. The result has been possible over-diagnosis of the condition and unnecessary treatments and anxiety.

KISS: Diagnosis of PCOS in Adolescents

[International guideline PCOS 2018](#) [CKS 2022](#)

- In adolescents both hyperandrogenism (e.g. hirsutism or acne) *and* irregular menstrual cycles are required to make the diagnosis.
 - Irregular menstrual cycles are defined as:
 - Normal in the 1st year post-menarche as part of the pubertal transition.
 - > 1 year of irregular cycles post-menarche: > 90 days for any one cycle.
 - > 1 to < 3 years of irregular cycles post-menarche: < 21 or > 45 days.
 - > 3 years of irregular cycles post-menarche to perimenopause: < 21 or > 35 days or < 8 cycles per year.
 - Primary amenorrhea by age 15 or > 3 years post-thelarche (breast development).
- USS scan is no longer recommended for diagnosis in women within 8 years of menarche.
- Do blood investigations as per adults (see below).
- Adolescents with features of PCOS but who fail to meet the diagnostic criteria should be considered at 'increased risk' of PCOS and reassessed at or before full reproductive maturity (defined as 8y post-menarche). Includes those with features prior to starting COCP, persisting features and significant weight gain in adolescence.

KISS: Polycystic Ovarian Syndrome: Diagnosis in Adults

[CKS 2022](#), [RCOG GTG No33](#)

- Suspect PCOS if these features are present:
 - Infertility, oligomenorrhoea, amenorrhoea.
 - Hyperandrogenism - Hirsutism, acne, alopecia or male pattern baldness.
 - Central obesity, acanthosis nigricans, or +ve family history.
- Investigations:
 - Blood tests - take a hormone profile to assess androgen status & exclude other causes:
 - TSH, prolactin, LH, FSH, testosterone, sex-hormone-binding globulin (SHBG).
 - Free Androgen Index (FAI) = $\frac{\text{total testosterone}}{\text{SHBG}} \times 100$; FAI gives a calculated free testosterone level, which compensates for the low SHBG level seen in PCOS which can affect the total testosterone result. Normal FAI is less than 5, but laboratory dependent.
 - An elevated FAI is suggestive of PCOS but it may also be normal.
 - Testosterone level >5nmol/l, signs of virilization or rapidly progressive hirsutism should raise suspicion of more sinister causes e.g. androgen-secreting tumours, Cushing's disease, Congenital adrenal hyperplasia...needs referral
 - Transvaginal USS – not required if already has scored 2 out of 3 (e.g. oligomenorrhoea and ↑ FAI).
- Diagnosis requires 2 out of 3 criteria:
 - Oligo- or anovulation.
 - Clinical and /or biochemical (raised Free Androgen Index) evidence of hyperandrogenism.
 - Polycystic ovaries on USS.

KISS: Polycystic Ovarian Syndrome: Management[CKS 2022, RCOG GTG No33](#)**General:**

- Offer lifestyle advice, and monitor for long-term effects.
- **Regular exercise and weight control** (RCOG advise at least 30 minutes/day average) and healthy low-calorie diets for women who have a raised BMI and PCOS).
 - Obesity, particularly central obesity worsens insulin resistance.
 - ↓ weight can lead to return of ovulation, ↑ fertility, ↑ SHBG and ↓ circulating insulin reducing the risk of T2DM, ↓CVD risk and improve hirsutism.
- Offer annual CVD risk assessment inc. fasting glucose/HbA1C, fasting lipids and BP.
- Screen for cardiovascular risk factors and obstructive sleep apnoea.
- Do not initiate insulin-sensitising drugs (e.g. metformin) in primary care:
 - There is no consensus on who should be referred for ?insulin sensitising drugs but consider for women with more severe oligo/amenorrhoea or impaired glucose tolerance/fasting glycaemia.

Managing oligo- or amenorrhoea to prevent endometrial hyperplasia:

- Initially **assess endometrial thickness** to exclude hyperplasia or carcinoma if prolonged amenorrhoea (less than 1 period every 3 months):
 - Induce withdrawal bleed (14/7 oral progesterone e.g. medroxyprogesterone 10mg OD) then USS.
 - If fails to shed, thickening (>10mm) or unusual appearance, refer urgently for sampling.
- **Options to prevent endometrial hyperplasia** (options will depend on underlying sx/risk factors):
 - Oral contraceptive (which can be continued e.g. low dose COC).
 - Cyclical progestones e.g. medroxyprogesterone 10mg OD for 14 days, every 1-3 months.
 - The Mirena IUS will also offer protection.

Hirsutism: (For management of acne see dermatology section)

- Give advice about cosmetic measures, and if ineffective combine with:
- **COC if no C/I.** Traditionally dianette has been recommended but alternative low-dose COCs have better safety profiles e.g. Eloine or Mercilon; consider tricycling for 3 months with 4-day breaks.
- **Refer if:** severe, of recent onset or rapidly progressive or if primary care options are ineffective.
- Secondary care options: high-dose cyproterone acetate, spironolactone, flutamide and finasteride.

Infertility and pregnancy associated with PCOS:

- **Infertility** - Assess for alternative causes as usual. Strongly advise weight loss if appropriate, and refer if no conception after 1 year (note local pathways may differ). **Metformin?** Latest [Cochrane review 2020](#) did not show any clear evidence of it increasing live birth rates.
- **Pregnancy:**
 - We should assess glucose tolerance/screen for diabetes, lipid profile, BMI and BP before conception and optimise risk factors.
 - Pregnant women should have an OGTT test before 20 weeks gestation and at 24-28 weeks.
 - May be at increased risk of other pregnancy complications such as pre-eclampsia, and should be monitored more regularly during pregnancy with a low threshold for referral.

PRENATAL TESTING

New technologies are being introduced into UK antenatal care to provide better accuracy with antenatal screening and rule out pre-eclampsia in suspected cases.

1 Non-invasive Prenatal Testing

2018 will see a significant change to antenatal screening, with a new test being introduced in England to assess for Down, Edward's and Patau's syndrome.

The technique is *non-invasive prenatal testing, or NIPT* has been available in the USA and some other countries in 2011, initially at significant cost. It has been used within the NHS for years for fetal Rhesus D typing but now a wider range of conditions can be tested.

With falling costs and a growing evidence base to support its antenatal screening role, it now becomes a viable option within the NHS. The technique and its related issues were neatly summarised in a recent BJGP editorial ([2017;67:298](#)).

► What is NIPT?

When cells undergo apoptosis (programmed cell death) fragments of DNA are released into the bloodstream. This is termed cell-free DNA (cfDNA). During pregnancy, cell-free fetal DNA (cffDNA) from the placenta passes into maternal plasma, accounting for around 10-20% of DNA fragments in the woman.

This can be analysed via a sample of the pregnant woman's blood. Cell-free fetal DNA is detectable from week 4-5 of gestation, but specific conditions only become testable during later weeks. For example, Down's Syndrome can be tested from 10 weeks gestation.

DNA can be tested in different ways.

For trisomies Down's, Edward's and Patau syndrome, there are subtle increases in the cffDNA of the relevant chromosomes compared to those without a trisomy. While this is accurate, it is not sufficiently so to be considered diagnostic and further testing, e.g. amniocentesis, is still required.

For other conditions, e.g. single-gene disorders, such as achondroplasia and Apert syndrome, or for sex determination (useful for X-linked conditions such as Duchenne's), the DNA can be directly tested. This is diagnostic and is termed NIPD - a non-invasive perinatal diagnosis.

► Why is this better than existing screening?

Currently, throughout the UK women are offered the first trimester combined test and quadruple test in the second trimester. For Down's the detection rate is around 85-90%, so up to 1 in 6 cases will be missed, and a false positive rate to 2-5%.

A recent systematic review and meta-analysis, published in the BMJ Open ([2016;6:e010002](#)), confirms the advantages of the newer technique.

- >100 papers were included to assess NIPT for Down, Edwards and Patau syndromes
- Pooled sensitivity was 99.3%, 97.4% and 97.4% respectively
- Pooled specificity was 99.9% for all 3 trisomies
- For 100000 pregnancies this would translate to
 - 417, 89 and 40 cases identified of Down, Edwards and Patau
 - 94, 154 and 42 false-positive results

A prospective study in the NEJM ([2015;372:1589](#)) compared NIPT for Down Syndrome to first-trimester combined screening and found:

- Out of 16000 unselected pregnancies
 - False-positive rate was 0.06% for NIPT vs 5.4%
 - Positive predictive value was 80.9% for NIPT vs 3.4%

NIPT has been trialled within the NHS in 8 participating maternity units, with data published in the BMJ last year ([2016;354:i3426](#)). This was useful to see how it could be implemented into the NHS and what likely benefits would be realised.

- NIPT was offered to women with a Down syndrome risk on combined tests of >1:1000
- They found if NIPT was offered to women with a risk of >1:150 (the threshold for chorionic villus sampling or amniocentesis) there would be:
 - 195 cases detected
 - 3368 fewer invasive tests (each with a 0.5% risk of miscarriage)
 - 17 fewer procedure-related miscarriages

In a cost analysis, it was found that NIPT used at 1:150 risk threshold would be cost-neutral compared with current practice, assuming NIPT cost <£256. In fact, it is estimated the test would be around £100 within the NHS.

► *How will NIPT be used in the NHS?*

So the [National Screening Committee](#) has recommended that NIPT be included as part of antenatal screening in England, not to replace triple/quadruple testing, but as a “contingent” test.

Women with a risk of Down greater than 1 in 150 will then be offered NIPT before invasive tests. While not diagnostic, given its very high specificity there will be fewer false positives so less unnecessary invasive procedures.

Interestingly the NSC found that if NIPT was used to replace the combined test it would cost an extra £105 million and result in more invasive tests than the chosen option.

► *Are there any controversies?*

The NIPT is a better tool than existing screening tests. However, when the decision to introduce it was announced there is significant discussion in the popular press about the fear that more women would choose to terminate a pregnancy with Down syndrome.

In reality, NIPT is simply another part of the antenatal screening process and is optional. It gives better information to allow for a more informed decision. The BJGP editorial highlights one paper which showed 31% of women with Down predicted on NIPT continued the pregnancy versus a national rate of only 8% under existing screening.

2 PIGF-based Testing in Pre-eclampsia

Second, we have an interesting new test recommended by NICE ([DG23](#)). Placental growth factor is involved in placenta angiogenesis. PIGF continue to rise throughout pregnancy until weeks 26-30, and abnormally lows indicate placenta dysfunction and are seen in pre-eclampsia.

Currently, pregnant women are suspected of pre-eclampsia if they have raised blood pressure, proteinuria and/or clinical symptoms (e.g. headache or oedema). Until now there are no tests to rule out pre-eclampsia so many women are admitted to hospital weeks while being closely monitored.

Two new PIGF immunoassay tests have now been endorsed by NICE - Triage PIGF and Elecsys immunoassay sFlt-1/PIGF ratio (the latter would benefit from a catchier name...), used in conjunction with clinical assessment for women with suspected pre-eclampsia between 20 and 34+6 weeks gestation.

The tests are quantitative and in the 4 studies, NICE reviewed a variety of different cut-off points were examined. As you would expect low cut-offs improve sensitivity while higher ones improve specificity.

Data shows the tests are effective RULE OUT tools, with very high negative predictive values at the cut-offs chosen by NICE. Both have an NPV of ~99% for ruling out preeclampsia in the following 1 week, and an NPV of ~94-95% to rule out pre-eclampsia over 2-3 weeks.

The positive predictive values are less impressive and NICE do not recommend using the tests to rule in pre-eclampsia given the current data.

NICE conclude "PIGF-based tests could be used to safely rule-out pre-eclampsia" and in women in early gestation, the test was highly cost-effective due to the significant reduction in unnecessary bed days.

NB Learning Points: Prenatal Testing

- NIPT has improved sensitivity and specificity for Down, Edwards and Patau detection
- It will be used as a contingent test, offered to women with a risk of >1:150, and will reduce the number of invasive tests needed for women
- Roll-out in England will be in 2018/19, with the rest of the UK yet to confirm its implementation
- PIGF testing may be used in women with suspected pre-eclampsia and provides a reliable rule-out test, resulting in fewer unnecessary bed days for those women

GROUP A STREPTOCOCCUS

One of the biggest medical news stories in the past year has been that of Group A Strep. In the last few weeks of 2022 general practice saw its busiest period ever due to fear of the infection and runs on antibiotics in pharmacies throughout the UK.

While it is easy for medics to feel that the media and parents have exaggerated the risks, as **the BMJ points out this is a condition which can affect "previously healthy children with a short prodrome and rapidly progressive course"**.

What's different about this outbreak is the timing and the response, both from the media and medical community. There has been a flurry of publications as a result, including a joint statement from the [RCGP](#), [RCPC](#) and [RECH](#), also the [UKHSA](#), [NHSE](#) and [BMJ](#).

► *What is Group A Strep infection?*

Group A Streptococcus (GAS) or Streptococcus pyogenes can result in a wide range of infections, predominantly of the throat, skin and respiratory tract, e.g. tonsillitis, pharyngitis, scarlet fever, impetigo, erysipelas, cellulitis, and pneumonia. It is spread via respiratory droplets and direct skin contact, as well as direct contact with contaminated objects.

Invasive GAS is when the bacterium is isolated from normally sterile areas of the body, e.g. lung, joints or blood. Skin lesions are the most commonly identified risk factor, especially linked with chickenpox, while late pregnancy/post-partum and older age are other risk factors.

► *Epidemiology* [UKHSA](#)

Putting the current crisis in context, from 19th September 2022 to 1st January 2023 there were >35 thousand notifications for scarlet fever compared to 4192 during the same period in 2017-18, and 30k total notifications for that entire year.

While the focus during the current outbreak has been on children, of the 151 deaths reported so far only 15% has been in <10yos and older people are at much higher risk with a significantly increased case fatality rate.

What is unusual about this outbreak is the timing. Scarlet fever typically peaks from March/April. The BMJ editorial explains the cause and severity of this outbreak is multifactorial, including lack of immunity in children due to limited exposure during the pandemic (immunity to GAS builds with repeated contact hence adults rarely get scarlet fever or GAS tonsillitis), and although not a new strain the current prevalent subvariant (M1UK) does have increased transmissibility and expresses the emm1 gene which is associated with more severe disease.

Rates of iGAS are also raised compared with usual, tracking the levels of scarlet fever albeit at much lower levels.

► *Presentation*

Due to the wide variety of infections GAS can cause, presentation is understandably variable. Most is predictable based on the system affected, with sore throat being the more common but scarlet fever and iGAS also merit particular attention.

Strategies for managing sore throat are well established in general practice. FeverPAIN and Centor have been the recommended tools by NICE for several years. In the current pandemic, it was felt that the unusually high rates of GAS would result in under-detection using the FeverPAIN score (usually around 65% of people with a score of 4 or 5 have GAS) and so NHSE recommended clinicians consider antibiotics at a score of 3 or more. It still seems quite hard to achieve even this score with the FeverPAIN tool...

Group A Strep Infection Presentation [NHSE](#)

Scarlet fever

- Clinical diagnosis, notifiable disease
- **Initial symptoms** are of non-specific illness, most commonly sore throat, fever (usually >38°C), headache, nausea & vomiting
- Rash appears 12-48h after onset of Sx
 - Typically on chest/torso first, then spreading to other parts of the body
 - Fine, red rash with sandpaper texture - this can help with diagnosis in patients with darker pigmented skin where the erythema is harder to discern
 - Often there is facial flushing with "sunburnt" cheeks and relative sparing around the mouth
- Other features which may be present include fatigue, lymphadenopathy, "strawberry tongue", skin desquamation of fingertips, toes and groin as the rash resolves
- Throat swabs are currently only recommended where there is diagnostic uncertainty or antibiotic resistance is a concern

Invasive Group A Strep

- Not necessarily resulting from scarlet fever, de novo presentations occur
- Includes pneumonia, cellulitis, meningitis, sepsis, septic arthritis, necrotising fasciitis, and streptococcal toxic shock syndrome
 - Empyema in children has been an unusually common feature of the current outbreak
- More common in extremes of age (very young or old), chickenpox, diabetes, immunosuppression, cancer, alcoholism, IVDU, late-stage pregnancy and post-partum females
- Likely to present unwell - look for signs of sepsis and refer immediately if concerns. NHSE recommends IV antibiotics even prior to transfer should be considered (as we would normally in such cases)

► Management of GAS

Invasive GAS is a medical emergency and requires immediate transfer to hospital and consideration of pre-hospital antibiotics if delays or very severe illness.

Scarlet fever should be treated promptly with oral antibiotics to minimise illness and reduce transmission.

- **Phenoxymethylpenicillin 1st line**
 - If unavailable amoxicillin, macrolides, flucloxacillin & cefalexin are alternative agents in decreasing order of preference
 - **Macrolides are 1st line in penicillin allergy** - clarithromycin is currently recommended by NHSE
 - Macrolide resistance is currently around 10%
 - **Treatment duration is 10 days for scarlet fever (5 days for 'sore throat')**
 - There may be difficulties getting liquid medicine, there is [a useful PIL on how to help children swallow tablets](#).
- **Children should avoid school for at least 24h after antibiotics are started or 2/52 if no antibiotics**
- Counsel patients on general management such as rest, fluids, good hygiene to reduce spread, antipyretics for fever or discomfort and to seek review if they become more unwell
 - **Safety netting advice can be found [here](#).**

Complications of scarlet fever, particularly serious ones, are uncommon: [DTB 2017;55:9:102](#)

- **Suppurative complications, e.g. otitis media, peritonsillar abscess, sinusitis**
 - Tend to occur early; due to local or haematogenous spread

- **Non-suppurative (autoimmune) complications:** e.g. rheumatic fever, glomerulonephritis
 - More common in the untreated, but may occur regardless. Thankfully rare - rheumatic fever occurs in <1/100000 in the UK. Persisting renal issues are rare post-glomerulonephritis
- **Invasive GAS infection**
 - Affects 3/100000 in UK, and not necessarily as a direct result of scarlet fever

► **Who needs prophylaxis?**

Antibiotic prophylaxis is recommended for high risk individuals in close contact with GAS. This should be undertaken by the local Public Health team after notification. The BMJ counsels that most secondary cases occur within 1-2 days of exposure to the index case.

Close contact is defined by NHSE as: prolonged contact with the case in a household-type setting during the 7 days before onset of symptoms and up to 24 hours after initiation of appropriate antimicrobial therapy in the index case.

High risk individuals include:

- Pregnant women from ≥37 weeks gestation;
- Neonates and women within the first 28 days of delivery
- Older household contacts (≥75 years);
- Individuals who develop chickenpox with active lesions either seven days prior to
- Onset in the iGAS case or within 48 hours after commencing antibiotics by the iGAS case, if exposure is ongoing

NB Learning Points & Quality Improvement Ideas

- There is a continuing high level of GAS infections in the community
- Prompt antibiotic therapy is indicated for suspected scarlet fever
- iGAS remains uncommon but is an important diagnosis not to miss - assess unwell patients for signs of severe illness/sepsis
- Scarlet fever and iGAS are notifiable disease
- Public health will consider whether contacts require prophylaxis
- For safety netting info re: scarlet fever [click here](#)
- For information on helping children swallow tablets [click here](#)

SEPSIS

It has been 6 years since [NICE guidance on sepsis](#) was published. We were all reminded of it in the winter of 2022 with the Group A Strep outbreak where there have been a number of linked deaths. Invasive GAS is a serious illness with variable presentations but all may cause sepsis, so as [UK government guideline](#) reminds us using existing non-specific tools including the NICE Sepsis guideline is advisable.

► How is sepsis defined?

In *2016 new international definitions for sepsis were published* ([JAMA 2016;315:801](#)) which already change the way in which we consider the condition:

- **Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection.**

This is a change from the previous definitions from 1991 and 2001 (used by the UK Sepsis Trust in its 2015 guidance), which stated *sepsis was SIRS in the presence of infection* (“systemic inflammatory response syndrome”: 2+ of T>38°C or <36°C, HR> 90bpm, RR >20bpm, WBC >12 or <4).

Interestingly the international task force “unanimously considered as unhelpful” the previous use of 2 or more SIRS criteria to identify sepsis. They highlighted that SIRS may simply be an appropriate response to an assault on the body, whereas sepsis now implies a response that is uncontrolled and causing damage. Of note however, is that the concerns were also “unanimously” aimed at secondary care patients, many of who may have SIRS without infection, or do not meet the criteria despite infection and adverse outcomes.

Indeed, diagnosis using the new definition (using the SOFA score – Sequential Organ Failure Assessment) still cannot be made in primary care as it necessitates a variety of investigations.

► Sepsis: recognition, diagnosis and early management [NICE NG51](#), [BMJ 2016;354:i4030](#)

Covering primary and secondary care, and all ages, NICE builds on guidance from the UK Sepsis Trust. It reflects the new international definition of sepsis so does not require the diagnosis of SIRS but uses a familiar traffic light system to help ratify the risk of sepsis (low, moderate to high or high risk) and provides recommendations on early management.

The BMJ summary of the guideline includes an [excellent infographic](#) that makes the assessment very clear and straight forward including all the vital sign thresholds for every age group.

While it aims to provide the most recent evidence-based guidance it acknowledges that in many areas good data is limited – much is based on low-quality data and expert opinion.

The key message is for clinicians to ask “**could this be sepsis**” when people present with infection in the same way we think about MI when a patient develops chest pain. It recommends:

- **Assessment of people with any suspected infection to identify:**
 - What is the source of the infection?
 - Is there an increased risk of sepsis?
 - Are there indicators for clinical concern, particularly abnormal vital signs?

► *How should we manage remote assessment?*

NICE leaves this open to interpretation. It suggests we assess for risk factors of sepsis when using remote assessment and attempt to identify indications of clinical concern such as new-onset abnormalities in behaviour, circulation or respiration. This will guide us to whether the patient requires face-to-face assessment and with what urgency.

KISS: Risk Factors for Sepsis

- Very young (<1y), older people (>75y) or frail
- Impaired immune systems, e.g. on chemotherapy, diabetes, splenectomy, sickle cell, long-term steroids, immunosuppressants
- Surgery or invasive procedure in the past 6 weeks
- Any breach of skin integrity, e.g. cuts, burns
- Misuse of drugs intravenously
- Indwelling lines or catheters
- Pregnant women, post-natal/termination/miscarriage in past 6 weeks - confers a 50% increase in risk compared to non-pregnant women ([BJGP 2017;67:185](#))

► *Does this change face-to-face assessment?*

Fundamentally the guideline simply reflects what we all do on a daily basis. Assessment of “risk of sepsis” can be made using simple basic observations without the requirement for additional tests (such as white blood cells, glucose or lactate suggested in other, older guidelines).

- *Check temperature, heart rate, respiratory rate, blood pressure, level of consciousness, oxygen saturation, skin appearance and urine output in young people and adults*
- *All of the above excluding BP plus capillary refill time in children <12 years*
 - The risk thresholds are subtly different from those used in Advanced Paediatric Life Support
 - BP can be checked in younger children if facilities exist
 - The guideline acknowledges that facilities for O2 sats in the very young may not be available

One of the most interesting changes from the UK Sepsis Trust guidance is that ***raised temperature is no longer a risk criteria except in infants < 6 months.***

This is because sepsis can present with or without fever, so it is not a reliable marker for the severity of illness. Indeed certain groups (young children, older people, immunosuppressed) are less likely to mount a fever but are at higher risk of serious illness. Hypothermia remains an important marker of illness.

► *How should we manage suspected sepsis?*

Severe sepsis is a time-critical emergency. For every hour delay in a patient receiving antibiotics, there is an 8% increase in mortality. NICE use risk categories to guide management in patients with suspected sepsis.

These correspond to ***low, moderate to high or high-risk categories***

- See the KISS summary at the end of the chapter for all the features in each category
- ***High Risk*** – any 1 criteria should prompt referral to hospital for emergency medical care
- ***Moderate to High Risk*** – use clinical judgment to ascertain if patient can be managed in the community
 - Consider physiology, co-morbidities, social (support, will they manage meds)
 - Consider a planned second visit, written handover for colleagues and specific safety netting advice

- **Suspected sepsis but absent moderate>high or high-risk criteria**
 - Use clinical judgment and standard protocols to guide management
 - Give safety netting advice including how and when to access further medical review
- The BMJ summary recommends where a definitive diagnosis/source cannot be identified a structured reassessment should be repeated

► *How should we manage patients in remote settings?*

Data shows that by following a simple management strategy for treating sepsis outcomes can be dramatically improved (NNT = 4.6 to prevent 1 death). This has been termed the “Sepsis 6”. One of the key goals is to administer IV antibiotics within 1 hour of diagnosis of severe sepsis.

In remote settings, there may be more than 1-hour delay for transfer. Here it is appropriate to administer IV antibiotics for patients with high-risk criteria in primary care. Where meningococcal disease is suspected (fever and purpuric rash) this should be IV benzylpenicillin.

Giving oxygen is also recommended for all practices pre-ambulance where oxygen saturations are low. We should aim for:

- 94-98% for most patients; 88-92% if at risk of hypercapnic respiratory failure (e.g. COPD)

“Sepsis 6”

- Administer oxygen and maintain sats >94%
- Take blood cultures and consider infective source
- Administer IV antibiotics
- Consider IV fluid resuscitation
- Check serial lactates
- Commence hourly urine measurement

NB Learning points: Sepsis

- Sepsis is a major cause of death and a time-critical condition. Prompt diagnosis and management in primary care will save lives.
- We need to “think could this be sepsis” and then check vital signs to assess a patient’s risk
- Low-risk patients still need appropriate safety netting advice
- All patients with high-risk sepsis need high-flow oxygen and immediate ‘blue light’ transfer
- The BMJ have a [great infographic](#) with all the key vital sign limits at all ages
- NHS Scotland have produced a [useful app](#) for the key features of sepsis

NB Quality Improvement Ideas: Sepsis

- Develop a practice guideline, with clear clinical pathways, for the recognition and management of sepsis in the practice
- Organise some in-house teaching and training based on this material
- Ensure oxygen therapy available and considered for all patients with high-risk sepsis
- Audit a selection of your practice population presenting with infection
 - Have all the appropriate vital signs been checked and documented?
 - Has the level of risk been assessed and documented?

INFECTIOUS DISEASES

KISS: Sepsis NICE Guidance

Stratifying risk of severe illness or death from sepsis:

Assessment: “COULD THIS BE SEPSIS?”		Risk Category		
		HIGH	MODERATE to HIGH	LOW
RESPIRATORY				
Resp rate (breaths per minute)	<1y	60+	50-59	No higher risk criteria met
	1-2y	50+	40-49	
	3-4y	40+	35-39	
	5y	29+	24-28	
	6-7y	27+	24-26	
	8-11y	25+	22-24	
	12y+ and adults	25+	21-24	
O2 sats	<5y	<90%	<91% air	
	5-11y	<90%	<92%	
	12y+ and adults	New need for O2 to maintain sats >92% (or 88% in COPD)	-	
Other	<5y	Grunting, apnoea	Nasal flaring	
CARDIAC				
Heart rate (beats per minute)	<1y	160+ (or <60)	150 – 159	No higher risk criteria met
	1-2y	150+ (or <60)	140 – 149	
	3-4y	140+ (or <60)	130 – 139	
	5y	130+ (or <60)	120 – 129	
	6-7y	120+ (or <60)	110 - 119	
	8-11y	115+ (or <60)	105 - 114	
	12y+ and adults	>130 (or <60)	91 - 130	
	Pregnancy	-	100 - 130	
Blood pressure	12y+ and adults	SBP <90 or>40mmHg below normal	SBP 91 - 110mmHg	
	≤11y	No recommendations		
Cap Refill Time	≤11y	No higher risk criterion	3sec+	-
	12y+ and adults	No recommendations		
Urine Output	≤11y	-	Reduced urine output	-
	12y+ and adults	No PU in 18hrs, or <0.5ml /kg/hr if catheter	No PU in 12-18hrs, or 0.5-1ml/kg/hr if catheter	-
TEMPERATURE	<3m	38°C+	<36°C	-
	3-6m	39°C+	<36°C	
	>6m	No higher temp	<36°C	
SKIN	All ages	Mottled or ashen appearance, cyanosis, non-blanching rash	Signs of potential infection at surgical site or breakdown of wound	Normal colour <5y; no non-blanching rash
OTHER	≤11y	-	Leg pain, cold hands / feet	-



KISS: Sepsis NICE Guidance continued

		Risk Category		
		HIGH	MODERATE to HIGH	LOW
BEHAVIOUR				
	<5y	No response to social cues	No smile	Content and smiles
		Appears ill to HCP	Wakes only with prolonged stimulation	Stays awake or awakens quickly
		Does not wake, or if roused dose not stay awake	Decreased activity	Strong normal cry or not crying
		Weak high-pitched or continuous cry	Parent/carer concern child is behaving differently from usual	
	5-11y	Objective evidence of altered behaviour or mental state	Not behaving normally	Behaving normally
		Appears ill to the HCP	Decreased activity	
		Does not wake or if roused does not stay awake	Parent or carer concern that the child is behaving differently from usual	
	12y+ and adults	Objective evidence of new altered mental state	New onset of altered behaviour/mental state	Normal behaviour
			Acute deterioration of functional ability	
			Impaired immune system	
			Trauma, surgery or invasive procedures in the last 6 weeks	

Management:

- **Any high-risk criteria = Red flag suspected sepsis**
 - Refer all patients
 - 999 ambulance and pre-alert secondary care
- **Any moderate to high-risk criteria = Amber flag suspected sepsis**
 - Use clinical judgment to decide whether to refer or treat in the community
 - Refer all patients if:
 - Definitive diagnosis unclear
 - Unsafe to treat in community
 - Under 18 and immunocompromised
 - Low threshold to refer to those with sepsis risk factors or multiple amber flags
 - All patients being referred require 999 ambulance
 - If treating in the community always safety net carefully, explaining how and when to seek further medical review. Useful patient information leaflets:
 - [Child sepsis safety net PIL](#)
 - [Adult sepsis safety net PIL](#)

MINOR ILLNESS

Over the years research has demonstrated the limitations of antibiotics for a range of minor illnesses. New research extends that concept even further and data demonstrate effective alternative therapies. But first, can we close the debate around whether treating a fever due to infection is better or worse for patients?

► Fever therapy in febrile adults [BMJ 2022;378:e069620](#)

It is routine for clinicians to suggest patients with fever take antipyretic medication in an attempt to control the fever but there is considerable uncertainty about whether this strategy is actually optimal. Many scientists have remained proponents of the theory that the natural defence of mounting a fever may improve the clearance of infection.

In this new BMJ paper ([2022;378:e069620](#)), a systematic review and meta-analysis of 42 relevant trials involving >5k adult patients, the authors attempted to clarify whether there is an effect on mortality with the use of antipyretics in critically and non-critically ill individuals with either infectious or non-infectious fever.

The data showed:

- **Antipyretics did not affect the risk of death and serious adverse events.**

Good news then that we are not harming patients, but *does it actually help* in any way?

This is an even more difficult question to answer as only one trial assessed quality of life (which showed no difference between treatment and controls, although one study is insufficient to draw meaningful conclusions), and meta-analysis could not clarify on the risk of non-serious adverse events.

So, antipyretics appear to do no harm, but the existing data can't tell if it actually helps. I suspect anyone who has had a fever and tried some paracetamol or ibuprofen will have their own opinion.

► Probiotics for preventing URTI [Cochrane 2022](#)

Probiotics have been tested in most body systems - here is the turn of the respiratory tract. Cochrane reviewed 23 papers, N = 6950, examining the effects of probiotics on rates and duration of URTI. Studies included generally used 1-2 strains for at least 3 months. They found:

- The number of people having ≥ 1 URTI per year was reduced by 24%, and ≥ 3 URTIs per year was 41% compared to placebo or no treatment
- Reduced number of people using antibiotics for URTIs by 42%

The limitations of these findings are the strength of the evidence. Cochrane was 'moderately confident' that numbers of ≥ 3 URTIs and antibiotic use were decreased, but low confidence in other findings. And there are still unanswered questions which are important: does the type and dose of probiotic matter? It feels like we are still a way off in our understanding about the role probiotics can or should play in health.

► Anaesthetic ear drops for acute otitis media [NICE NG91 2022](#)

NICE reckons that pain relief is parents main priority when their child has acute otitis media. Most parents will try oral paracetamol and/or ibuprofen before consulting with us, and often these fail to adequately control the pain. Is there a better option?

NICE has just updated its guidance on management of acute otitis media with a recommendation to:

- Consider ear drops containing an anaesthetic and analgesia for pain if:
 - An immediate oral antibiotic prescription is not given and
 - There is no eardrum perforation or otorrhoea
- Review treatment if symptoms do not improve within 7 days or worsen at any time.

Why has NICE made this recommendation?

Firstly, NICE believes the main issue for parents is to manage their child's pain. Secondly, there is data, albeit somewhat limited in a number of participants, showing benefit of anaesthetic/analgesic ear drops above and beyond oral analgesic in otitis media, and that this approach reduced antibiotic use.

- A [Cochrane review showed a NNT = 5 for a 50% reduction in pain at 10 minutes with ear drops](#) from 5 RCTs (n=391) comparing active ear drops to placebo in children aged 3y and over without perforation and receiving oral analgesia.
- More [recent data shows that using this approach significantly reduces subsequent antibiotic use](#) by around 90% without an increase in adverse events, albeit in low numbers of participants. n = ~100
- There is no data examining the use of such ear drops in children receiving antibiotics, so concomitant use is not recommended by NICE.

Secondly, a licensed product is now available in the UK: *Otigo* is an ear drop preparation containing phenazone, a non-steroidal anti-inflammatory analgesic, and lidocaine. While this is not the same medication used in all the studies reviewed, NICE felt it likely there is a class effect and it is reasonable to use this treatment for this indication.

► Antibiotics for LRTI in children in primary care - ARCTIC PC [Lancet Sept 2021](#)

Observational data shows that antibiotics are prescribed for at least 40% of children with a 'chest infection', usually diagnosed due to a cough and largely upper respiratory in origin. In this group antibiotics are ineffective. However, most of us would consider a lower respiratory tract infection to be more serious and prescribing rates would be even higher. But do antibiotics make a difference in LRTI in children?

A new primary care-based double-blind, randomised placebo-controlled trial examined the benefit of antibiotics (amoxicillin 50mg/kg for 7 days) in children aged 6 months to 12 years with acute uncomplicated LRTI for <21 days. Children with likely pneumonia or who were severely unwell were not enrolled, also if a non-infective origin (e.g. hayfever) was suspected.

The **primary outcome was duration of moderately bad or worse symptoms**. Secondary outcomes included severity of various symptoms (e.g. cough, SOB, feeling unwell, fever), total symptom duration and side effects. The results showed:

- Of 432 children, median durations of moderately bad or worse symptoms were similar between both treatment and placebo groups
- No benefit seen in important subgroups including children with chest signs, fever, SOB or chest rattle
- Side effects were similar between groups & significant complications were low irrespective of treatment

The authors conclude that: "*amoxicillin for uncomplicated chest infections in children is unlikely to be clinically effective either overall or for key subgroups in whom antibiotics are commonly prescribed. Unless pneumonia is suspected, clinicians should provide safety-netting advice but not prescribe antibiotics for most children presenting with chest infections.*" This provides us with more confidence in our management when assessing children with possible lower respiratory tract infection that holding off antibiotics as we would with clear URTI is unlikely to do harm.

NB Learning Points & Quality Improvement Ideas: Minor Illness

- Acute cough and bronchitis are not helped by inhaled/oral bronchodilators/steroids, mucolytics, and possibly not by codeine antihistamines or decongestants. However, honey, guaifenesin and some cough suppressants may be of use.
- NICE have produced [a webpage collating all their recent antimicrobial guidelines](#)
- Antibiotics are largely ineffective for most minor illness and fail to reduce most complications
- Vitamin D may reduce the frequency of URTI but is most effective in those with vit D deficiency

KISS: Acute Cough

[NICE 2019, NG 120](#)

- Acute cough is usually self-limiting due to a viral upper respiratory tract infection, gets better in 3-4 weeks without antibiotics; acute bronchitis is due to infection of the lower respiratory tract, which is usually viral but can be bacterial; and both can have other infective or non-infection causes
- If <5yo with cough and fever use [NICE guideline on children <5y with fever](#)
- If suspected pneumonia, use [NICE guideline on pneumonia in adults](#)
- Refer patients for further assessment if features of more serious illness (e.g. sepsis, PE, etc.)
- **Treatment**
 - Give advice on likely duration, self-care and safety net with when to seek medical help
 - ***Do not offer oral/inhaled bronchodilators or steroid*** (unless known airways disease, e.g. asthma) ***or mucolytics***
 - ***Do not routinely offer an antibiotic for an acute cough or acute bronchitis***, explain why it is not needed, e.g. makes little difference in cough but can cause side effects
 - If the patient has had a CRP test refer to the guidance on pneumonia in adults (see chapter)
 - ***If systemically very unwell or at higher risk of complications***
 - After face-to-face review, ***if very unwell offer immediate antibiotics***
 - If person with acute cough is at ***higher risk of complications consider immediate or backup antibiotics***
 - Higher risk includes:
 - Those with pre-existing co-morbidity such as significant heart, lung, renal, liver or neuromuscular disease, immunosuppression or cystic fibrosis
 - Young children born prematurely
 - >65y with ≥2 of the following, or >80y with ≥1 of the following:
 - Hospitalisation in <1y, diabetes, heart failure or current oral steroids
- **Self-care**
 - ***“Be aware that some people may wish to try the following self-care treatments, which have limited evidence of some benefit for the relief of cough symptoms:***
 - **Honey** (not in <1yo), **pelargonium**, cough mixtures with guaifenesin or cough suppressants except codeine
 - Limited evidence suggests antihistamines, decongestants and codeine do not help cough symptoms
- **Antibiotics**
 - Adults: 1st line doxycycline, alt 1st line: amoxicillin, clarithromycin, erythromycin; all 5/7
 - Children: 1st line: amoxicillin, alt 1st line: clarithromycin, erythromycin, doxycycline; 5/7



KISS: NICE - Acute Sore Throat

NG84

Assessment

- In patients with acute sore throat assess symptoms using either FeverPAIN or Centor scores
 - ***FeverPAIN - scored 0-5; 1 point for each of:***
 - Fever in last 24h, Purulence on tonsils, Attend rapidly (<3d from onset), severely Inflamed tonsils, No cough or coryza
 - For a more nuanced assessment use the [FeverPAIN tool](#)
 - Score 0-1 = 13-18% risk of Strep; 2-3 = 34-40%; 4-5 = 62-65%
 - ***Centor - scored 0-4; 1 point for each of:***
 - Tonsillar exudate, tender anterior cervical lymph nodes, history of fever (>39°C), absence of cough
 - Score 0-2 = 3-17% Strep; 3-4 = 32-56%
 - Note the Centor was developed in US Emergency departments and was only tested in adults; FeverPAIN was developed in UK general practice and includes children >3yo
- FeverPAIN 0-1 or Centor 0-2 = do not offer an antibiotic
- FeverPAIN 2-3 = consider no antibiotic or a backup prescription
- FeverPAIN 4-5 or Centor 3-4 = consider immediate antibiotic or backup prescription
 - ***Back-up scripts - advise use if no improvement in 3-5d or symptoms worsen***
 - ***For all methods safety net*** by advising medical review if symptoms worsen rapidly or significantly, or the person becomes very unwell
 - Consider alternate diagnoses, features of serious illness or Abx resistance
- In patients who are systemically very unwell, have features of more serious illness or who are at high risk of complications:
 - Offer immediate antibiotic prescription, refer to the hospital if severe infection/complications

Treatment

- ***Self-care***
 - Paracetamol or ibuprofen for pain/fever, fluids, medicated lozenges (benzocaine, flurbiprofen) may help
 - No evidence supports non-medicated lozenges, throat sprays or mouthwashes alone
- ***Antibiotics (adults)***
 - ***1st line: phenoxymethylpenicillin 500mg QDS or 1g BD 5-10 days***
 - This mirrors the recommendation in the latest [PHE common infection guideline](#)
 - ***Penicillin allergy or intolerance: clarithromycin 250mg - 500mg BD 5 days or erythromycin 250-500mg QDS or 500mg-1000mg BD 5 days***
 - See BNF for child regimes and dosing in hepatic/renal impairment, etc.
 - Erythromycin is preferred to clarithromycin in pregnant women

KISS: Acute Sinusitis [NICE NG79](#)

Common features

- **Adults:** nasal blockage/congestion/discharge, dental/facial pain/pressure, ↓ sense of smell
- **Children:** often non-specific URTI features, e.g. nasal blockage, discoloured discharge, cough

Factors making bacterial infection more likely:

- Bacterial cause may be more likely if several of the following are present:
 - Symptoms >10/7, purulent nasal discharge, several localised unilateral pain (especially over teeth/jaw), fever, marked deterioration after initial milder phase (“double sickening”)

≤10 days of symptoms

- ***Do not offer an antibiotic prescription***
- Give advice about usual course (2-3/52), antibiotics are not needed, managing symptoms with self-care, seeking medical help if symptoms worsen rapidly or significantly, fail to improve at 3/52, or they become systemically unwell
- If symptoms worsen rapidly or significantly reassess: consider DDx e.g. dental infection, and assess for more serious illness

~10 days or more of symptoms

- ***Consider prescribing high-dose nasal corticosteroid*** (i.e. mometasone 200mcg BD - unlicensed dose) **for 14 days** for adults and children >12yo
 - May improve symptoms but unlikely to affect duration; also can cause systemic effects
- Consider no antibiotic prescription or a backup antibiotic prescription
 - If offering backup script give advice (verbal and written) on self-care, not using the Abx immediately but after 7/7 or if symptoms worsen, seeking help if not effective/tolerated
 - If ineffective consider previous antibiotic use and possible resistance

Patients who are systemically very unwell, have symptoms/signs of more serious illness or at high risk of complications

- Offer immediate antibiotics
- Refer to hospital if severe systemic infection, intra-/peri-orbital complications (e.g. peri-orbital cellulitis, displaced eyeball, double vision, etc.) or intracranial complications (e.g. swelling over the frontal bone, possibly meningitis, severe frontal headache, focal neurology)

Self-care

- Consider paracetamol or ibuprofen for pain and/or fever as appropriate
- Nasal saline or decongestants may be tried but lack of evidence to show they are effective
- No evidence for oral decongestants, antihistamines, mucolytics, steam or warm face packs

Antibiotic choice - treat for 5 days

- **1st line:** phenoxymethylpenicillin; co-amoxiclav if very unwell, risk of complications, etc.
- **Pen allergy/intolerance:** doxycycline, clarithromycin (preferred alternative in children), erythromycin (in pregnancy)
- **2nd line:** if symptoms worsen after 2-3/7 of 1st choice abx: co-amoxiclav
 - If penicillin allergy consult a local microbiologist

KISS: NICE - Acute Otitis Media [NG91](#)

Self-care

- AOM usually settles in 3 days but may last 7-8 days - symptoms can usually be managed with self-care
- Offer paracetamol or ibuprofen for pain - equally effective, NNT 6-7 for no pain at 48hr
- Decongestants and antihistamines are not helpful

Management - uncomplicated AOM

- Consider no antibiotic prescription or a backup antibiotic prescription
 - If using back up script recommend use if symptoms significantly worsen or no improvement in 3 days, and seeking help if remains unwell or Abx not tolerated
- Give advice about usual course, self-care, why an antibiotic is not required and seeking help if symptoms deteriorate rapidly or significant, do not improve after 3 days, or very unwell
- Anaesthetic ear drops in addition to oral analgesia may improve pain in children, NNT = 4 for 50% reduction in pain at 10mins. Avoid if perforation, otorrhoea or prescribing antibiotics
 - Phenazone 40mg/g with lidocaine 10mg/g 4 drops 2-3x daily for 7d. Licensed med = "*Otigo*"

Management - AOM and otorrhoea at any age or <2yo with bilateral AOM

- Consider either a backup or immediate antibiotic prescription

Management - systemically very unwell, more serious illness, or high-risk of complications

- Offer immediate antibiotic prescription +/- further investigation and management
- Refer for emergency care if symptoms/signs of sepsis or acute complications, e.g. mastoiditis, meningitis, cranial nerve palsy

Choice of antibiotic - treat for 7 days

- **1st line:** amoxicillin, 2nd line: co-amoxiclav if worsens after 2-3 days
- Pen allergy: clarithromycin, erythromycin
- For child dosages see [helpful infographic from NICE](#) or [BNF](#)



URINARY TRACT INFECTIONS

The most common of primary care problems has recently had new data published on an effective non-antibiotic prophylactic agent, complications of long-term nitrofurantoin, time to symptom resolution with and without treatment, and updated NICE guidance on assessing possible UTI in children.

1 Antibiotic consumption and time to recovery [BJGP 2022;72\(725\):e882](#)

The authors of this new research published in the BJGP note that the average duration of UTI symptoms is 10 days. Which is quite a lot! They note that uncomplicated UTI may be treated with empirical antibiotics, but also there is a growing number of women who manage without where possible. This study asks the question, **how long does recovery from UTI take with antibiotic use** (rather than Abx prescription, which may not be used) **compared to no antibiotics**, and what effect does the duration of antibiotic have on recovery?

The team used data from the POETIC trial, a point-of-care study conducted across 4 European countries including England and Wales between 2012 and 2014, which followed women with UTI for 14 days.

- N = 300 women consumed Abx, N = 80 did not consume Abx, mean age 49y
- Median time to recovery was 7 days with Abx use, 9 days without
- 84% on Abx reported recovery during the 14 days vs 63% without Abx
- There was no difference in recovery time comparing 3 days of Abx to more than 3 days

They conclude that *consumption of antibiotics for uncomplicated UTI shortens recovery time, but acknowledge that the difference may not be considered large by many women*, especially with milder symptoms, and in this group may help to inform shared decision-making about antibiotic use.

2 Methenamine hippurate for UTI prophylaxis [BMJ 2022;376:e068229](#)

Antibiotics are an effective option for preventing recurrent UTI. NICE recommends considering daily antibiotic therapy, but also mentions non-antibiotic options of D-mannose or cranberry in non-pregnant women, despite lack of evidence. It does not mention methenamine hippurate, a non-antibiotic urinary antiseptic. Various small studies have suggested benefit in UTI prevention but debate has continued. Could a new study in the BMJ help to clarify?

This was a UK-based open-label, randomised, non-inferiority trial. It recruited 240 women aged ≥18yo with recurrent UTI requiring prophylactic treatment. Patients received either antibiotics (nitrofurantoin, trimethoprim or cefalexin OD) or methenamine (BD) for 12 months. The primary outcome measure was absolute difference in incidence of symptomatic, antibiotic-treated UTI during treatment, with non-inferiority set as less than one episode per person year difference.

- Incidence of antibiotic-treated UTI
 - Antibiotics: 0.89 episodes per person-year
 - Methenamine: 1.38
 - Absolute difference was 0.49, meeting the pre-set non-inferiority level
- Adverse events occurred in 24% of women on Abx and 28% on methenamine and were mostly mild.

The study also examined the rates of UTI in the 6 months following treatment. Prior to starting treatment, almost 90% of women had 4 or more UTIs in the previous year. After treatment, the rates were 1.2 for the antibiotic group and 1.7 for methenamine, both comfortably lower than pre-trial rates.

The linked editorial ([BMJ 2022;376:o533](#)) agrees with the author's conclusions: *this new research increases the confidence in which methenamine hippurate can be offered as an option to women needing prophylaxis against recurrent UTI*.

3 Nitrofurantoin: adverse effects with long-term use [DTB 2022;60:4:59](#), [BJGP Open 2021](#)

This recent [BJGP Open paper](#), discussed further in the [DTB](#) and in an [NB Blog by Rob Walker](#), highlights the lack of awareness amongst clinicians for the potentially serious adverse effects relating to nitrofurantoin.

NFILD, nitrofurantoin-induced interstitial lung disease, comes in a number of forms. It can occur acutely with a hypersensitivity reaction that can develop within 1-2 weeks, or chronically with fibrosis developing after months to years of exposure, which may be [at least partially reversible on cessation](#). Liver injury is also known to have occurred. Prevalence of severe adverse effects is estimated at 0.2% in controlled trials, and between 0.02 to 1.5 events per 1000 user from observational data.

The BNF recommends for both children and adults taking NFT: "on long-term therapy, monitor liver function and monitor for pulmonary symptoms, especially in the elderly", although frustratingly it fails to clarify what is long-term, nor method and frequency of monitoring.

One [medical defence company](#) recommends LFTs and a check for respiratory symptoms at least every 6 months, and advise patients seek urgent review if they develop breathing problems.

In a questionnaire to GPs and urologists (the vast majority of responses were from GPs) the BJGP Open paper found that 42% were unaware of the potential for hepatotoxicity and 28% for lung toxicity. Over half did not report measuring baseline lung or liver function, and less than 1 in 10 'always' undertook monitoring.

When the researchers examined practice-level data they found actual practice was even worse than what was reported - any form of baseline monitoring was absent in 74% of patients on long-term NFT.

So, NFT is a good medicine but it can cause uncommon but important adverse events. Monitoring is advised and we as the clinicians need to be aware of the recommendations, but to protect our patients and ourselves.

NB Learning Points & Quality Improvement Ideas: Urinary Tract Infection

- [NICE guidance suggests considering delayed antibiotics for women with simple lower UTI and minor symptoms](#) and use of simple analgesia and fluids to manage those symptoms
- Pregnant women, men and children with lower UTI should receive immediate antibiotics
- [Trimethoprim use with ACEi and ARB increases the risk of hyperkalaemia and AKI](#)
- [Nitrofurantoin use long-term is associated with lung and liver adverse effects](#) and requires at least 6 monthly monitoring
- [Prompt treatment with antibiotics is recommended for UTIs in the >65s](#) - deferred or no antibiotics was associated with a 7-8x increase in bloodstream infections
- Methenamine hippurate and D-mannose are effective non-antibiotic options for managing recurrent
- **Audit patients receiving long-term nitrofurantoin**
 - Did they have baseline and then 6 monthly assessment of liver and lung function?
 - If not organise review & discussion on adverse events including need for r/v with new lung Sx.
- **Audit post-menopausal women with recurrent UTI**
 - Have they been offered a trial of topical oestrogen? Are there red flags requiring further Ix?

KISS: Diagnosis of Urinary Tract Infection [PHE Nov 2020](#) (adults), [NICE 2022](#) (children)

Women <65yo

- Consider 1) sepsis and 2) vaginal/urethral causes
 - 80% with new vaginal d/c do not have UTI - STI, irritants or post-menopausal atrophy?
- **Are there any of 3 key symptoms?**
 - **Dysuria, new nocturia, cloudy urine**
 - **2 or 3 present: diagnose UTI** (74-82% have a +ve culture), consider MC+S, treat as per NICE
 - **1 present: perform urine dipstick** to improve accuracy (68% +ve culture)
- Nitrite +ve or leucocyte + blood +ve: UTI likely, manage as above
- Nitrite -ve and leuc +ve: UTI is equally likely as alternative diagnoses
- Consider time of specimen (morning is more accurate), send for MC+S, consider immediate abx or backup abx depending on the severity
- Nitrite, leucocyte & blood -ve: UTI is less likely, no MC+S, consider DDx
- **Nil present:** are there other severe symptoms e.g. urgency, frequency, haematuria, suprapubic tenderness? If so, check urine dipstick; if no, UTI less likely
- Always send an MC+S in pregnant women

Men <65yo

- As well as the usual UTI causes consider acute prostatitis (see Men's Health section)
- Dipsticks are unreliable at excluding infection; although PPV is very high if +ve nitrites & leucocytes.
- Always send an MC+S

Women and men >65yo

- Dipsticks are unreliable in this age group (due to asymptomatic bacteriuria in up to 50%)
- Base decisions on the clinical picture
- UTI likely if:
 - New dysuria, or ≥2 of fever, new frequency/urgency/incontinence/delirium/suprapubic tenderness or visible haematuria
 - Always send an MC+S if possible prior to any antibiotics; manage as per NICE



KISS: Diagnosis of Urinary Tract Infection [PHE Nov 2020](#) (adults), [NICE 2022](#) (children)

Children

- <3 months old, refer for urgent assessment
- >3 months:
 - Test urine if they have features that increase the likelihood of UTI, consider testing if absent
 - ↑ likelihood of UTI e.g. dysuria, frequency, new bedwetting, foul smell, darker/cloudy urine, ↓ fluid intake, fever, shivering, abdo pain, loin/suprapubic tenderness, CRT >3s, PMHX of UTI
 - ↓ likelihood of UTI e.g. no dysuria, nappy rash, breathing difficulties, abnormal ear/chest signs, fever of known alternative cause
 - Single features are not very predictive, multiple features make Dx more likely
 - Do not routinely test urine if features of infection other than UTI. If they remain unwell and have diagnostic uncertainty consider testing.
 - If collecting aim for sample ASAP (<24h) and before initiating antibiotics
 - Interpret dipstick findings depending on age:

	Age 3m to 3y	Age 3y+
Leuc +ve, Nit +ve	Send MC+S, start Abx	Start Abx, send MC+S if PMHx UTI or risk of serious illness
Leuc -ve, Nit +ve	Send MC+S, start Abx	Start Abx if testing on a fresh sample, send MC+S
Leuc +ve, Nit -ve	Send MC+S, start Abx	Send MC+S, only start Abx if good clinical evidence of UTI
Leuc -ve, Nit -ve	No Abx, consider MC+S	Assume no UTI. Explore for other causes of illness.

- If 3m to 3y and dip -ve, send MC+S if 1 or more: ? upper UTI, intermediate/severely unwell, <3mo, recurrent UTI, no response to Rx in 24-48h, clinical features + dipstick do not correlate
- A clean catch urine is preferable - in infants, using the “Quick-Wee” method is highly effective, using SP stimulation with a cold-soaked gauze - see [BMJ for details](#)



KISS: Lower UTI antibiotic prescribing

[NICE NG 109, Oct '18](#)

- Give advice on *managing symptoms with self-care* to all people with lower UTI
 - Consider paracetamol for pain, or if preferred/suitable ibuprofen
 - Advise on adequate intake of fluids
 - No evidence was found to recommend cranberry products or other common over-the-counter treatments such as alkalinisation agents
- **Non-pregnant women with lower UTI**
 - **Consider a backup antibiotic or an immediate antibiotic** taking account of:
 - Severity of symptoms, risk of developing complications, previous micro results, previous antibiotic use which may lead to bacterial resistance, patient preference
 - If giving a backup antibiotic recommend use if symptoms fail to improve within 48h or if they worsen at any time; discuss potential side effects and safety net
 - If sending an MC+S and antibiotics have been prescribed review choice when results returns, change if resistant and symptoms are not already resolving
- **Pregnant women, men and children with lower UTI**
 - Obtain midstream urine and send for MC+S (and/or dipstick for children)
 - **Offer immediate antibiotics**; adjust according to micro results
 - Take account of previous susceptibilities and antibiotic use
- **Asymptomatic bacteriuria is not routinely treated**, except in pregnant women, who should receive immediate treatment based on MC+S results
- Refer patients to hospital if they have a severe systemic infection and children following current NICE guidance on fever in <5s or UTI in <16s
- **Antibiotics:** (take account of available MC+S)
 - **Non-pregnant women**
 - **1st line:** nitrofurantoin MR 100mg BD for 3 days (eGFR ≥45) OR
 - **1st line:** trimethoprim 200mg BD for 3 days if low risk of resistance
 - E.g. no use in >3/12, previous MC+S show sensitive (and not used it), a younger person in an area with low local resistance
 - **2nd line:**
 - Nitrofurantoin (if not used)
 - Pivmecillinam 400mg initial dose then 200mg TDS for a total of 3 days
 - Fosfomycin 3g single dose sachet
 - **Pregnant women**
 - **1st line:** nitrofurantoin MR 100mg BD for 7 days (avoid at term)
 - **2nd line:** amoxicillin (if sensitive on MC+S), cefalexin, or d/w micro
 - **Men**
 - **1st line:** trimethoprim 200mg BD 7 days, or nitrofurantoin MR 100mg BD 7/7
 - **2nd line:** consider alternative diagnosis (e.g. prostatitis) and culture results
 - **Children <3 months:** refer immediately to the acute paediatric team
 - **Children >3 months**
 - 1st line: trimethoprim (if low resistance) or nitrofurantoin 7/7
 - 2nd line: nitrofurantoin (if not used), amoxicillin (if sensitive), cefalexin

KISS: Management of Recurrent UTIs

[NICE NG112](#), [BMJ 2017;359:j5193](#), [PHE 2017](#), [BMJ 2013;346:f3140](#)

Recurrent UTI in otherwise healthy women is part of a natural and expected disease process and does not indicate failure of treatment or the need for complicated investigations

- 30-44% of women with acute cystitis will have a recurrence, often within 3 months
- Recurrence within 2 weeks is a relapse and indicates treatment failure or persistent focus for infection, e.g. renal stone

Definition:

- ≥ 2 infections in the past 6 months, or ≥ 3 in one year
- However, any second UTI requires consideration and an appropriate management plan

Investigations

- Clinical presentation of recurrent UTI is the same as for sporadic UTI
- **Urinary culture is not usually required** to make the diagnosis but may help to confirm infection (excluding other pathology e.g. overactive bladder, interstitial cystitis) and guide antibiotic choice
- **Cystoscopy and imaging have limited roles** – diagnostic yield is low; further investigation/referral may be indicated in the presence of renal stones, persistent macro- or microscopic haematuria, history of urinary tract surgery, known tract abnormalities, immunocompromise, etc.
- Refer or seek advice on Ix and Mx in men, recurrent upper UTI, recurrent lower UTI of unknown cause, pregnant women, children, and refer via cancer pathway if red flags.

Management

- Recurrent episodes can be treated with the same antibiotics as a single UTI
 - Personal resistance can develop and persist for up to a year
 - No evidence suggests that the duration of treatment should be $>3-7$ days course
- Advise on behavioural and hygiene measures and self-care
 - E.g. avoid dehydration, avoid delaying habitual or post-coital urination, wipe front to back after defecation, avoid douching and occlusive underwear
- In non-pregnant women consider single-pill antibiotic prophylaxis to a specific trigger (e.g. intercourse); if ineffective consider daily antibiotic prophylaxis
- In pregnant women, men & children consider daily prophylaxis with specialist advice
 - Review every 6/12, advice on side effects, resistance and managing acute UTI
- In post-menopausal women consider topical oestrogen therapy
- Antibiotics: 1st line either trimethoprim or nitrofurantoin (single dose or daily strategies)
 - Trimethoprim: 200mg stat as single-dose or 100mg at night daily
 - Nitrofurantoin: 100mg stat as single-dose or 50-100mg at night daily
 - Amoxicillin and cefalexin are 2nd line
- Non-antibiotic treatment
 - **Methenamine hippurate**: 1g BD for prophylaxis
 - **D-mannose**: over the counter - beware, counsel patients that it's a sugar!
 - **Oestrogen**: topical post-menopausal women; in oral has no role in prevention
 - **Chinese herbal medicine**: slightly dubious data but very small trials have been positive
 - Cranberry, probiotics, urinary alkalinisation and immunostimulants either lack data or fail to show benefit



KISS: Pyelonephritis Antibiotic Prescribing

[NICE NG 111, Oct '19](#)

- Refer to the hospital if features of severe infection; consider admission if v. dehydrated or unable to take oral fluids/medicine, pregnant, or high risk of complications
- Send a urine MC+S prior to starting antibiotics
- Self-care with paracetamol +/- low-dose opioid if required for pain; maintain hydration
- Offer an antibiotic
- Reassess if symptoms worsen or fail to improve within 48hrs
- **Antibiotics:**
 - **Non-pregnant women and men - 1st line:**
 - Cefalexin 500mg BD or TDS for 7-10 days
 - Co-amoxiclav 625mg TDS for 7-10 days (only if culture result available + sensitive)
 - Trimethoprim 200mg BD 14 days (again, only if culture result available + sensitive)
 - Ciprofloxacin 500mg BD 7 days (but consider safety issues)
 - **Pregnant women - 1st line: cefalexin**
 - **Children <3 months:** admit!
 - **Children >3 months:** if managing in the community, cefalexin 1st line, co-amoxiclav if sensitive on culture
 - **2nd line:** consult local micro team

KISS: Catheter-associated UTI Antibiotic Prescribing

[NICE NG 133, Nov '19](#)

- Be aware that *catheter-associated UTI is a symptomatic condition*
- *Asymptomatic bacteriuria occurs in nearly all within 1 month of insertion and does not require treatment*
- Consider removing, or if not possible, changing the catheter as soon as possible if in place for >7 days. Do not delay starting antibiotics for this.
- Obtain a urine sample for MC+S from the catheter prior to starting antibiotics
- Offer an antibiotic
 - **No upper UTI sx**
 - **1st line:** NFT 100mg MR BD if eGFR >45, trimethoprim 200mgBD, or amoxicillin 500mg tds (if proven sensitive) - all 7 days
 - **2nd line:** pivmecillinam 400mg stat then 200mg TDs, total 7 days
 - **Upper UTI sx**
 - Same as for upper UTI guidance, see above
- Refer/seek advice if bacteria resistant to oral Abx, recurrent infections, pregnant, v. unwell

KISS: Acute Prostatitis

[NICE NG 110](#)

- See Prostatitis chapter in Men's Health
- Quick reference: **exclude serious illness, obtain MC+S, offer oral Abx**
 - **1st line:** Cipro 500mg BD or cefalexin 200mg BD 14/7 (or trimethoprim if others C/I)



MONKEYPOX (MPOX)

The current outcome of monkeypox, now rebranded as mpox, came to the UK in May 2022 amongst concerns it could spread far and wide. Thankfully to date, the numbers have been limited with 3700 cases in the UK to December 2022. Most of us will never see a case in general practice, but it's important we understand how it presents and how it is currently managed if we do.

► *What is mpox?*

Mpox was first described in 1958 in colonies of monkeys kept for research. It is a DNA virus that belongs to the orthopoxvirus genus which also includes smallpox and cowpox.

► *Who gets mpox?*

The first case was reported in a human in the Democratic Republic of Congo in 1970. Since then, infections have generally occurred in central and western Africa with occasional cases in other parts of the world linked to international travel or imported animals.

However, while the index case in the UK outbreak is thought to be linked with travel to Nigeria, subsequent spread has occurred locally. The UK Health Security Agency reports that transmission is primarily within interconnected sexual networks with very low levels of transmission outside of these.

A [case report in the NEJM](#) of >500 cases across 16 countries shows that 98% of persons with infection were gay or bisexual men, mostly white and middle-aged, although the UKHSA stresses that **anyone can get mpox**.

► *How is it transmitted?*

Mpox is transmitted through close contact with an infected person or animal, or with material contaminated with the virus. The NEJM case report shows that 95% of transmission in the current outbreak is linked to sexual contact, although it is close contact rather than sexual contact that appears important.

► *How does it present?*

Symptoms usually arise 5-21 days (median 7) after exposure and include fever, headache, muscle aches, backache, lymphadenopathy and fatigue. The rash normally appears 1-3 days after the onset of fever, usually starting on the face before becoming more widespread. It can look like chicken pox or syphilis and goes through several stages, including vesicles, pustules and scabs, which then drop off. Given the link to sexual contact, the UK Health Security Agency (UKSHA) is warning people to be especially suspicious of genital rashes, found in three-quarters of cases.

While classically mpox presents with multiple lesions the NEJM report that 2 in 3 have <10 lesions and 1 in 10 only show a single lesion genital lesion, making it easily confused with syphilis.

The illness is generally mild and self-limiting and typically lasts 2-4 weeks. The case fatality ratio has been suggested between 3-6% by the WHO but there have been no deaths reported in the UK since this outbreak began. Likewise, no deaths were reported in the 500+ patients in the NEJM case series, although 13% of patients in the NEJM paper were hospitalised. This was mostly for pain management (severe ano-rectal pain is a common feature) but serious complications were rare.

► *Can it be treated?*

Most people will require no treatment beyond conservative measures and then self-isolation, avoiding other members of the household. Current UK guidance recommends self-isolation until:

- No fever for 72h, no new lesions for 48h, all lesions have scabbed, no lesions in the mouth, any lesions on the face/arms/hands have scabbed, the scabs have fallen off and fresh skin has formed underneath.

Adults with suspicious symptoms should call the local health clinic for review, and children their GP. **Mpox is now a notifiable disease** and so Public Health should rapidly be in the driving seat.

Treatments are available - including smallpox vaccine, cidofovir and tecovirimat. These are reserved for people with severe disease or high risk of severe disease. Tecovirimat and cidofovir are antiviral drugs shown in animal studies to be effective for orthopoxvirus disease, but human data is lacking and there is no efficacy data for mpox. Smallpox vaccine may be used as pre- or post-prophylaxis and is up to 85% effective at preventing mpox. Prior vaccination to smallpox may result in milder disease.

A targeted vaccination programme has begun in the UK with a second dose being becoming available in late 2022 after initial supply limitations. Current groups recommended for vaccination include healthcare workers caring for people with confirmed mpox, gay, bisexual and other men who have sex with men, and close contacts of confirmed mpox. It is given as 2 doses, at least 28 days apart.

Most clinicians in general practice are unlikely to come across a case of mpox, but if a patient with possible monkeypox requires face-to-face assessment barrier precautions are recommended and rooms are decontaminated after use.

NB Resources: Mpox

- [UK Government guidance and background information](#)
- [WHO Mpox Fact sheets](#)
- [NHS Patient Info](#)
- [WHO Recovering at home infographic](#)

INFLUENZA

Every year general practice assesses a huge number of patients with likely influenza. When circulating levels increase [national guidance recommends anti-influenza treatment](#) in higher-risk patients presenting within 48hrs from the onset of symptoms. These recommendations have been in place for over a decade. And yet most of us have never prescribed a neuraminidase inhibitor in our patients, even when they clinically are highly likely to have influenza.

Why not? What stops us? It may be concern over genuine benefits of these drugs - the majority of published data on the subject comes from the drug's manufacturers. There is deep mistrust in such data among clinicians. It may be concern over the side effects of the medications given what appears to be only modest benefit.

But just like as the tide comes in, you can guarantee each winter there will be a flu outbreak. So impartial, primary-care orientated data would be welcome to clarify whether treatment is actually of any benefit.

Oseltamivir for influenza-like illness in primary care [Lancet 2020;395:42](#)

This new paper published in The Lancet is an unashamedly pragmatic primary care trial of patients presenting with clinical features of influenza. It was funded by the European Commission's Seventh Framework Programme.

It was not placebo-controlled but randomised 3,266 patients aged 1 year or more from across 15 European countries to either usual care or usual care + oseltamivir. Patients presenting beyond 48h from onset were still eligible for treatment. Patients were analysed as low, medium or high-risk groups for adverse effects. Patients had oral and nasal swabs taken for influenza PCR but clinicians were not aware of the results and did not influence treatment options or group allocation.

The primary outcome was time to recovery, defined as a return to usual activities without significant symptoms. Secondary outcomes included rates of hospital admissions and pneumonia although the study was underpowered to assess these accurately.

■ Results

- Mean duration of illness = 6.5 days
- **Oseltamivir-treated patients recovered on average 1 day quicker** than those in usual care alone
- **Patients in higher-risk groups** (older, more unwell, comorbidities, longer duration of symptoms) **benefited more improving 2-3 days earlier**
- Conversely young, healthy patients with low severity symptoms had minimal benefit: average of 0.7 days shorter duration from an average total duration of 5.1 days
- Interestingly, around 50% had confirmed influenza but improvements were the same between those and negative cases suggesting the medication has a mechanism beyond simply inhibiting influenza virus replication
- Benefit appears independent of duration of illness and initiation of treatment, so patients presenting after 48hr of illness still derived the same benefit as those receiving it <48hrs
- **Serious adverse events showed no difference between the groups. New nausea and vomiting was marginally higher in the oseltamivir group (21% vs 16%)** but less so than in previous data.
- Secondary analysis showed no statistical difference in hospital admissions and cases of pneumonia
- Conclusion: primary care patients with influenza-like illness treated with oseltamivir recovered one day sooner on average than those managed by usual care alone, and older, sicker patients with co-morbidities and longer previous symptom duration recovered 2-3 days sooner.

This real-world study provides a number of interesting and novel findings.

- Patients presenting after 48hrs of symptoms benefit from treatment to the same degree as earlier presenters, especially for those more unwell with protracted symptoms. This challenges current NICE/public health guidance on how to use neuraminidase inhibitors.
- Patients with flu-like symptoms but without influenza still benefited, suggesting we don't need to be too concerned about having the diagnosis correct. Having said that, no one is suggesting these drugs from the common cold...
- The side effects appear more modest than perhaps clinicians have previously considered, which improves the benefit:risk ratio, especially in higher-risk groups.

So this data demonstrates antivirals for influenza in a more positive light than perhaps many of us have considered in the past, with reasonable symptomatic improvement in those at higher risk and minimal risk of adverse effects. However, this study did not (and was not designed) show benefit in serious complications from flu such as hospital admission and pneumonia. Ultimately the clinician and patient together will have to decide whether they consider the benefits meaningful enough.

► *What is the public health guidance?*

It is first important to note that in primary care antiviral medicine for influenza should only be used once public health bodies issue notice that the pathogen is circulating in the community. The only exception would be in proven influenza outbreaks in “at risk” people who live in long-term residential or nursing homes. Their recommendations are based on [NICE TA 168 \(2009\) on antivirals](#) and are materially unchanged for 2021/22 ([UK Gov](#)).

Once that notice is issued they recommend:

- **In suspected or proven cases with “uncomplicated influenza”** (e.g. no pneumonia)
 - **Healthy people:** usually no treatment, unless considered high risk of complications
 - **At risk population (aged >65y, chronic disease, pregnant women or <2w post-partum, children <6m, BMI>40):** oseltamivir 75mg BD 5/7, ideally within 48hrs, off-license after 48hrs at clinician’s discretion
 - **Severely immunosuppressed (including those with pred 40mg+ for 1/52 in past 3/12):** prescribe antivirals, seek local guidance as recommendations vary with circulating strains
- **Post-exposure prophylaxis** (to start within 48hrs, >48hrs on specialist advice only)
 - **Healthy people:** no prophylaxis
 - **At risk population:** oseltamivir PO 10/7 (unless known resistance)
 - **Severely immunosuppressed:** antivirals, depends on circulating strain, 10/7
- Safety net regarding complications of influenza and re-presenting if condition worsens

NB Learning Points: Influenza

- New data shows there is benefit of using oseltamivir in addition to usual care in patients with flu-like illness, although in lower-risk groups the benefit is modest
- There is evidence that antivirals may be useful as post-exposure prophylaxis in at risk groups.
- We should encourage pregnant women to have the flu vaccination for the protective effects in the infant after birth - rates of flu-like illness reduce by 66% and hospitalisations by >80%. [Paediatrics 2016;137:e20152360](#)

NB Quality Improvement Ideas: Influenza

- Audit patients prescribed anti-viral medications
 - Did prescription follow the recommendations? Did patients report adverse effects?

CLOSTRIDIODES DIFFICILE

Awareness of the drivers of C. diff has improved dramatically over the past decade and it is increasingly uncommon to see a true case of primary community C. diff. thanks to our switch away from the broadest spectrum antibiotics. Nevertheless, when we do see a case it can be severe and having a clear management strategy is important. [NICE has now published evidence-based guidance](#) on the best management of C. diff.

There are two interesting recommendations in this guideline:

- **Consider faecal microbiota transplant for recurrent episodes**
 - FMT has been shown to be effective whether delivered via [colonoscopy or oral capsules](#)
 - Presumably, this will be specialist-led rather than the GP sorting it out themselves...
- **Do not advise people taking antibiotics to take pre- or probiotics to prevent C. difficile infection**
 - Many of us might tell patients to consider probiotics to reduce the risk of adverse effects from antibiotics.
 - [Cochrane](#) found high dose probiotics (≥ 5 billion CFUs per day) ARE effective at reducing antibiotic-associated diarrhoea in children, NNT = 9, using *Lactobacillus rhamnosus* or *Saccharomyces boulardii* - there is uncertainty about other types of probiotic
 - [Meta-analysis published in J Clin Gastroenterology 2021](#) showed a relative risk reduction in antibiotic-associated diarrhoea of 38% when probiotics were taken early in antibiotic treatment
 - NICE found [1 meta-analysis](#) showing probiotics statistically significantly reduce the incidence of C. difficile infection in inpatients but not in outpatients or those in mixed settings. General adverse events may be reduced.

So it may still be worthwhile advising probiotics alongside antibiotic treatment to reduce general diarrhoeal adverse events, but prevention of C. difficile is less likely to occur.

KISS: Clostridioides Difficile Infection: Antimicrobial Prescribing

[NICE NG199 July 2021](#)

Assessment

- First episode or recurrence? Severity? Individual risk factors e.g. frailty, age, co-morbidities?
- Review existing antibiotic use - stop unless essential - if essential can it be switched to lower risk abx?
- Review medications - PPIs, laxatives, SADMAN meds (e.g. NSAIDs, diuretics, ACEi, SGLT2i)

Treatment

- Offer an oral antibiotic to adults with C. diff - consider prompt specialist advice prior to Rx
 - 1st line: Vancomycin 125mg orally QDS for 10 days
 - 2nd line: Fidaxomicin 200mg orally BW for 10 days
- In children treatment should be initiated by, or after, advice from a specialist
- Advise on maintaining hydration, preventing spread, safety net
- Reassess if symptoms do not respond as expected or worsen
- Refer to hospital if severely unwell, consider if risk of complications or recurrence is high
- Consider faecal microbiota transplant for recurrence with 2 or more previous episodes

Prevention

- Do not offer antibiotics.
- Do not advise people taking abx to take prebiotics or probiotics to prevent C. diff. infection.